# The canonical problem of time as a category error:
an empirically forced cosmic time, deparametrization, the dissolution of frozen dynamics,
and a graviton quantization the de Sitter horizon closes without a free parameter—the seam where $\hbar$ enters gravity—in Cosmological Relativity

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**Abstract**

In the canonical formulation of general relativity the Hamiltonian is a sum of constraints that vanish on the physical phase space: no preferred time survives, and the generator of evolution annihilates physical states rather than evolving them. This is the canonical "problem of time." We argue that it is not a technical defect awaiting a technical repair, but the canonical symptom of a category error—treating the four-dimensional manifold of *occurrences* as the thing that *exists*, the block universe.

The companion framework paper [3] introduced, as an ontological commitment of Cosmological Relativity (CR), a global cosmic time that general relativity's formalism admits but does not supply. Here we show what that commitment does to the canonical formalism. The decisive premise is not a new calculation but a selection: the bare Arnowitt–Deser–Misner (ADM) machinery propagates data along an *arbitrary* hypersurface and settles nothing, whereas CR selects a preferred cosmic foliation—forced empirically by the isotropy of the cosmic microwave background (CMB) and structurally by the horizon geometry of collapse [4]. On that foliation the lapse recovers its meaning as the metrical rate of the existing layer's advance, $d\tau = N\,dt$; the constraint, deparametrized along cosmic time, is solved for a *true* Hamiltonian that generates the advance; and quantum evolution in cosmic time is unitary. The frozen constraint and the true Hamiltonian are the same canonical content under two readings, differing only in whether the manifold is granted existence.

The deparametrization is, as a technique, entirely standard; what CR adds is not a mechanism but a *warrant*. The multiple-choice problem—that many candidate internal times exist, that inequivalent quantizations follow from different choices, and that the formalism prefers none—has force exactly when the clock is chosen for formal convenience internal to the theory, and loses it when the clock is fixed from outside the formalism by ontology and observation. It is dissolved, not solved within the space of formal clocks.

We exhibit the structure first in flat Friedmann–Lemaître minisuperspace, where the deparametrized Hamiltonian reproduces the flat-ΛCDM expansion—the $\sinh^{2/3}$ law established in [3]—and then lift it to the closed-$S^3$ layer's propagating sector: the transverse-traceless$^{\mathsf{L}\hat{2}0}$ graviton modes form a discrete tower that deparametrizes to a unitary evolution in cosmic time, and project into a fundamental observer's frame as the flat-ΛCDM graviton, with matter density appearing only as the bend of the slicing rather than as ontological content. The external clock is no convenience: an internal area clock separates the dynamics at $\Lambda = 0$ and would serve, but at $\Lambda > 0$ it no longer does, and the absolute cosmic time is forced.

In that sector the quantization moreover closes *without a free parameter*. The scale-factor Hamiltonian carries deficiency indices $(1, 1)$ independently of operator ordering—the inverse-square coefficient at the origin attaining $\gamma = \frac{1}{4}$ across the natural ordering family, strictly below the essential-self-adjointness threshold $\frac{3}{4}$—so a single boundary condition at $a = 0$ remains; and that point is not a generic half-line endpoint but the de Sitter cosmological horizon, whose Hartle–Hawking thermal state, at the surface gravity $\kappa = 1/\alpha$ the substrate itself fixes, selects the Friedrichs extension. The deficiency freedom is thereby not merely

parametrized but closed, and closed without a free parameter: $\hbar$ enters the gravitational sector at the seam, scaled by $\Lambda$ alone.

With the tower coupled, the boundary coefficient is promoted to an operator straddling the threshold and the same thermal regularity supplies the condition fibre by fibre, so that what remains open is not the boundary condition but the *definition of the interacting tower. That is a different thing from a residual freedom in the quantization, and its shared character with every interacting field theory does not settle it*: the divergence is measured here rather than characterised—quartic at a leading constant fixed by the propagating-component count rather than assumed; logarithmic at a coefficient the tower's own spectrum settles exactly; and, where the shear breaks the degeneracy among the quadratic invariants, one further counterterm whose Weyl-squared coefficient is twice a real scalar's—and it is carried as this programme's own open frontier.

The problem of time is thereby *dissolved* as a category error, not solved by an addition to the formalism, in exactly the way the inference that gravitational collapse completes at finite exterior time falls to a refusal of the same reification [4]—the shape of the two errors being one, while the horizon's refutation is carried by general relativity's own causal structure and needs no part of this correction.

# 1 Introduction: the problem of time, as standardly posed

The canonical quantization of general relativity confronts a difficulty that has resisted resolution for half a century. In the ADM decomposition [16], spacetime is foliated by spatial hypersurfaces and the Einstein–Hilbert action is recast in Hamiltonian form. The result is a fully constrained system in the sense of Dirac [17]: the total Hamiltonian is a sum of the scalar (Hamiltonian) constraint $\mathcal{H}$ and the momentum constraint $\mathcal{H}_i$, each weighted by the lapse $N$ and shift $N^i$, and each vanishing on the physical phase space. The lapse and shift are not dynamical; they are Lagrange multipliers enforcing the constraints. Promoting the scalar constraint to an operator yields the Wheeler–DeWitt equation $\hat{\mathcal{H}}\Psi = 0$ [18]: the physical state is annihilated by the generator that should evolve it. The wavefunction of the universe does not appear to depend on any time parameter at all.

This is the problem of time [19, 20]. Its sharpest statement is that the formalism contains no preferred time variable, that every spatial foliation of a given spacetime is related to every other by a gauge transformation generated by the constraints, and that "evolution" in the coordinate label $t$ is therefore pure gauge—a re-slicing of one fixed four-dimensional object, carrying no physical content. The many proposed responses—identifying an internal time among the canonical variables before quantization, coupling a material clock, treating observables as relational "evolving constants," or abandoning a fundamental time altogether—each carry well-known costs [19, 20, 23, 24], to which we return in Section 8.

Our claim is that the difficulty is conceptual before it is technical, and that the conceptual diagnosis is precise rather than vague. The frozen formalism is the canonical expression of a single metaphysical commitment: that the four-dimensional manifold is the object that exists. We will argue, following the ontological foundation of CR [13, 3] and the necessity assembled in Section 2, that this commitment is a category error—the confusion of what *happens* with what *exists*—and that it is empirically false, the universe having furnished, in the CMB, direct evidence of the very cosmic time the formalism declines to recognize. When that error is declined, the same canonical equations describe a real cosmic time all along.

We stress at the outset the register of this paper. We do not propose a modification of general relativity, a new term in the action, or an additional structure bolted onto the canonical formalism. The equations are those of Einstein and of ADM, unchanged. What changes is the reading. The frozen constraint and a true Hamiltonian generating cosmic-time evolution are, as we will show, the same content under two readings, differing only in whether the manifold of occurrences is granted the kind of existence that belongs to the evolving three-dimensional world. The problem of time is not solved here. It is dissolved.

The paper proceeds as follows. Section 2 earns the premise the argument needs—that a preferred cosmic time is real—on two independent grounds, one conceptual and one empirical. Section 3 reads the unchanged ADM formalism under the two ontologies, and Section 4 isolates why the move is a *selection* rather than a construction: the bare machinery settles nothing, and everything turns on whether a foliation is distinguished to be selected. Section 5 carries out the deparametrization, structurally and then in flat minisuperspace, marking the boundary between the two. Section 6 lifts it to the closed-$S^3$ layer's propagating sector—the transverse-traceless graviton tower, advanced unitarily in cosmic time—and there the quantization closes: the scale-factor Hamiltonian's deficiency-$(1,1)$ freedom, ordering-independent, is fixed by the de Sitter horizon's own thermal state, without a free parameter, leaving open not the boundary condition but the definition of the interacting tower. Section 7 states the verdict negatively, and Section 8 places the argument against the internal-time, relational, and timeless programmes, where the multiple-choice objection is met not with a mechanism but with a warrant.

## 2 The necessity: a cosmic time that is real, not posited

The canonical move of this paper rests on one premise: that there is a preferred cosmic time, and that it is a real feature of the world rather than a convention chosen for convenience. This premise is not assumed. It is forced, on two independent grounds—one conceptual, one empirical—which we summarize here; the full argument is developed in [13, 14, 15] and stated as the necessary-and-sufficient augmentation of general relativity in [26].

**The conceptual ground: occurrence is not existence.** To exist is to endure—to persist through an interval. The three-dimensional world exists in this sense, and so do the things in it. An *event*, by contrast, is something that *happens*: it occurs at a place and an instant and is gone. A worldline is the continuous succession of events associated with an enduring thing, and four-dimensional spacetime is the manifold of all such events—the totality of what happens. The persistent confusion about time, traceable from Heraclitus and Parmenides through to modern eternalism, is the treatment of occurrences as though they were existents: the reification of the event-manifold into a thing that exists [13]. Once spacetime is imagined to exist, every moment along a worldline is imagined to exist on equal footing, the distinction between past, present, and future is declared an illusion, and time is said not to pass. But that conclusion was smuggled into the premise. Granting existence to the manifold of occurrences is not a discovery about time; it is a misuse of the word "exist." The coherent picture is the one in which an evolving three-dimensional world exists and advances, and events happen in the course of that advance, mapping to a four-dimensional spacetime that represents them without being a further existent of the same kind.

In CR this distinction is structural. What exists is the layer $\mathcal{S}_t$—a real three-sphere evolving in a real de Sitter substrate [3, 8]. The four-manifold $M$ is the representational record of the events that happen as the layer advances. The layer's existence and advance are not optional gloss on a fundamentally four-dimensional object; they are the ontology, and $M$ is its shadow.

**The empirical ground: the CMB measures the cosmic rest frame.** That a preferred cosmic time might be real but undetectable is the situation relativity leaves open: local experiments cannot distinguish a true state of rest, because the relativity of synchrony—a general kinematic consequence of one's chosen rest convention, independent of the light postulate—and the round-trip symmetry of local signalling together hide one's motion. Einstein's identification of simultaneity with frame-relative synchrony was, on this much, a philosophical choice among two empirically equivalent options, not a deduction; treating the local inaccessibility of a cosmic rest frame as evidence of its nonexistence is a modal fallacy [14].

The universe, however, did furnish the missing evidence. The cosmological redshift of the CMB is isotropic to high precision, and the primordial temperature anisotropies at the level of one part in $10^5$ have been preserved across the entire history of expansion [25]. Had the expansion rate varied locally with the inhomogeneous matter distribution, the accumulated redshift along different lines of sight would differ by far more than is observed—by some three orders of magnitude, as we establish quantitatively in the companion paper [26]—and the isotropy therefore certifies that cosmological space has expanded uniformly throughout cosmic history. But to speak of a three-dimensional space expanding uniformly *through time* already presupposes a cosmic time, a cosmic rest frame, and an associated simultaneity. The same data fix our own velocity relative to that frame through the CMB dipole. The cosmic rest frame is thus not a large-scale average or a coordinate convenience but one of the better-measured structures in cosmology. Synchrony is relative; simultaneity is absolute [14, 15].

**The two grounds are one correction.** These are not two arguments but two faces of one. The relativity-of-simultaneity premise is precisely what licenses the reification of the block: if there is no objective "now," the four-dimensional manifold is all there coherently is, and the layer's privileged advance has no referent. Conversely, an empirically real cosmic "now" restores the layer as the existent and demotes the manifold to its record. Declining the block (the conceptual correction) and admitting the cosmic rest frame (the empirical correction) are the same move. The equations of relativity are untouched by it; only the interpretation shifts—from "if two events appear simultaneous they must be simultaneous" to "apparent simultaneity is a local matter with no bearing on the global cosmic simultaneity that cosmology establishes empirically" [15].

This is the premise the canonical argument needs, and it has been earned rather than assumed: there is a preferred cosmic foliation, and it is real.

## 3 The canonical formalism, read two ways

We now read the standard canonical formalism under the two ontologies. Nothing in this section alters the equations.

In the ADM form, a spacetime $(M, g)$ is foliated by spacelike hypersurfaces $\Sigma_t$ with induced metric $h_{ij}$ and conjugate momentum $\pi^{ij}$, and the line element is

$$ds^2 = -N^2\,dt^2 + h_{ij}\,(dx^i + N^i\,dt)(dx^j + N^j\,dt), \tag{1}$$

with lapse $N$ and shift $N^i$. The action becomes $S = \int dt\,d^3x\,\big(\pi^{ij}\dot{h}_{ij} - N\mathcal{H} - N^i\mathcal{H}_i\big)$, and variation of $N$ and $N^i$ enforces the constraints $\mathcal{H} = 0$, $\mathcal{H}_i = 0$. The total Hamiltonian $H = \int d^3x\,(N\mathcal{H} + N^i\mathcal{H}_i)$ is a sum of constraints; on the constraint surface it vanishes.

**The frozen reading.** Suppose the existing object is the four-dimensional manifold itself. Then a choice of foliation is a choice of how to slice one already-existing thing into a stack of three-dimensional pictures, and a different foliation is a different slicing of the same thing. No slicing is distinguished; the relation between any two is a gauge transformation generated by the constraints; and since the generator of $t$-evolution is a constraint, it moves a physical state nowhere. Quantum-mechanically, $\hat{\mathcal{H}}\Psi = 0$: the state is timeless. This reading is internally consistent. Its content is exactly the content of the premise: *if* the block is what exists, time does not pass, and the formalism correctly reports that there is nothing for a Hamiltonian to do. The frozen formalism is not a mistake of calculation. It is the faithful canonical image of the block.

**The reading CR adopts.** Suppose instead, with CR, that what exists is the evolving layer $\mathcal{S}_t$, and that $M$ is the record of the events that happen as it advances (Section 2). Then a foliation is not a slicing of a pre-existing block; it is a choice of how to coordinate the advance of the existing world, and one such foliation—the cosmic one—tracks that advance rather than cutting across it. On this reading the lapse is not a bookkeeping multiplier but the metrical amount by which the existing layer advances per unit of the time label,

$$d\tau = N\, dt, \tag{2}$$

with the shift $N^i$ carrying the foliation's non-orthogonality to the layer. The constraint $\mathcal{H} = 0$ is still an equation; but now it is read as the relation that, once solved for the momentum conjugate to the cosmic time, yields the generator of the layer's advance—a true Hamiltonian. The same symbols, the same constraint surface, the same dynamics; a different answer to the single question of what exists. This lapse and shift are the very levels the cosmological development computes on: the lapse is the foliation stacking rate the observable expansion rides—set by the geometry, the density the content that set rate carries, read off the clock rather than sourcing it—and the shift is the synchronization through which that expansion is *observed* as distance and redshift, with the leaf's bend the matter it carries [9, 33]. That a plasma process rides the one rate or the other is then fixed by what the process is—a process running in the content reads the leaf's rate, a comoving separation read across leaves the stacking rate [34]—not chosen: the same selection of the existent's foliation that turns the frozen constraint into a true Hamiltonian here fixes, one level down, which rate a recombination-era or a nucleosynthesis-era computation must read—and reifying the shift's synchronous slicing as the existent is the block error, met at the level of a rate.

## 4 The selection is the whole of the move

It is worth isolating why the foregoing is a recognition rather than a new construction, because this is the crux of the paper.

The ADM machinery propagates canonical data along whatever foliation one supplies. It is indifferent to which foliation that is; it has no preferred time of its own, and in this exact sense the bare formalism contains no resolution of the problem of time and no obstruction to one. Everything turns on what is fed to it. If nothing distinguishes a foliation, the constraint is all there is and the state is frozen. If a foliation is distinguished, the constraint deparametrizes along it and a true Hamiltonian appears. The canonical formalism is the same in both cases; what differs is whether a preferred foliation exists to be selected.

CR supplies exactly that, and supplies it from outside the bare formalism, on the grounds of Section 2: the cosmic foliation is real, forced empirically by the CMB and conceptually by the occurrence/existence distinction. It is moreover the foliation that the rest of the programme independently singles out. The metric-singularity structure of the event horizon determines a unique limiting causal orientation in collapse, generically non-orthogonal to any spacelike slice, and this is the orientation CR reassigns as cosmic time [4, 3]; the representational freedom in choosing among Lorentzian metrics on the fixed manifold is a genuine gauge symmetry, organized by the description groupoid [12], but the cosmic foliation is not one of its orbits to be quotiented away—it is fixed by the ontology the gauge freedom is defined over.

The consequence is methodological as well as physical. There is no separate "canonical machinery" result to be proved here; the deparametrization that follows is standard once a clock is in hand. The content of this paper is that CR's cosmic time is the physically correct clock, on grounds that are ontological and empirical rather than formal, and that the canonical face of the existing physics is therefore a true Hamiltonian rather than a frozen constraint. The move is the selection. Everything else is reading the textbook on it.

*The same selection settles what a mass charge could be.* The ADM mass is defined relative to an asymptotic time translation, and exists only where the geometry supplies one—which asymptotically-de Sitter spacetimes do not, there being no global timelike Killing vector, so that no conserved charge is well defined there at all [8]. *On the reading taken here that absence is not a deficiency of the geometry but a misplacement of the question*: the time this construction runs on is not recovered from an asymptotic symmetry but selected and *measured*, so a quantity whose definition waits on an asymptotic Killing vector is waiting on the wrong thing. It is the same move as the paper's own—what the formalism cannot supply from inside, the selection supplies from outside, and the question that presupposed the missing structure lapses with it.

## 5 Deparametrization and the true Hamiltonian

We now carry out the reading. The structural statement is independent of any model; the explicit minisuperspace realization that follows is an illustration of it, and we are careful to mark the boundary between the two.

**The structural statement.** Let the cosmic time of Section 2 be the distinguished foliation parameter, and let $p_\tau$ denote the momentum conjugate to it. The scalar constraint, linear in $p_\tau$ after the cosmic clock is isolated, takes the deparametrized form

$$p_\tau + H_{\mathrm{phys}}\big(q^A, p_A\big) = 0, \tag{3}$$

where $(q^A, p_A)$ are the remaining (true, gauge-invariant) degrees of freedom of the layer and $H_{\mathrm{phys}}$ depends on the cosmic time only through them. Equation (3) is the same constraint that, read with no preferred foliation, gives $\hat{\mathcal{H}}\Psi = 0$. Read with the cosmic clock selected, it is solved for the generator of advance: classically $dq^A/d\tau = \partial H_{\mathrm{phys}}/\partial p_A$, and on quantization

$$i\,\partial_\tau \Psi(q^A, \tau) = \hat{H_{\mathrm{phys}}}\,\Psi(q^A, \tau). \tag{4}$$

This is a genuine time-dependent Schrödinger equation generating unitary evolution in cosmic time, not the frozen Wheeler–DeWitt constraint. The foliation is fixed—by the ontology, not by a gauge choice—so there is a true time and a true Hamiltonian. The frozen constraint and the true Hamiltonian are (3) read two ways; they are the same content, differing only in whether the manifold is granted existence. Deparametrization here is not a trick that restores a time that was really absent; it is the formal face of a time that was never lost, because the cosmic foliation was real all along.

**A zero-duration boundary, and what it does to the evolution.** The cosmic time of Section 2 is the real part of the companion framework's complex cosmic time, and that framework's closed contour contains one segment—the *lift*, joining the collapse's comoving turnaround to the branch point at which a collapse becomes a cosmology—along which *the real part does not advance* [3]. The consequence for (4) is immediate and worth stating plainly, because it is unusual: cosmogenesis occupies *no cosmic time at all*, so the evolution operator across it is $U(\Delta\tau = 0) = \mathbb{1}$. The true Hamiltonian has no interval in which to act there. The geometry nonetheless changes across the segment—the areal radius climbs from the turnaround to zero and the expansion rate is carried from zero to divergent—so the change is not generated by $H_{\mathrm{phys}}$ but is the segment's own analytic content, the contour read at another point of itself. *The beginning is therefore not a first instant of the evolution this section constructs; it is a boundary of zero duration within it*, and that is why nothing can be lost across it: no evolution acts, so none can be non-unitary.

We mark the limit of the claim. What is established is the classical statement—the foliation parameter does not advance on that segment, so no interval of $\tau$ separates the two legs—together with its immediate consequence for the evolution operator. *What the deparametrization*

*does not supply is a quantization of the Euclidean segment itself.* The state on either side is related by the identity in cosmic time; whether the imaginary-time stretch admits a quantum treatment in its own right, and what that treatment would say about the $\{q^A, p_A\}$ across it, is not addressed by this deparametrization and is not claimed by it. *The gap is named rather than managed, and its shape is definite. The silence is structural, not a matter of refinement*—an evolution parametrized by cosmic time cannot describe a segment of zero cosmic duration, so no sharpening of $\hat{H}_{\rm phys}$ reaches it. What the deparametrization cannot supply is an account of an imaginary-time stretch across which the classical configuration changes while the foliation parameter does not, and that is the natural point of contact between this construction and Euclidean approaches to the quantum beginning. *The classical account of the beginning is complete, and the quantum account is not the deparametrization's to give*: the step below gives its kernel, and the framework carries the segment's quantum reading on from there [3].

**The propagator across the lift is Euclidean.** The evolution (4) is generated in the framework's complex cosmic time, and on the lift that time runs along the imaginary direction: $\tilde{\tau} = \text{const} + is$, so $\partial_s = i\,\partial_{\tilde{\tau}}$ and (4) becomes $\partial_s \Psi = -\hat{H}_{\rm phys}\Psi$. The kernel carrying the state across is therefore not the unitary $U = e^{-i\hat{H}_{\rm phys}\Delta\tau}$ but

$$K = e^{-\hat{H}_{\rm phys}\,|\Delta\eta|}, \tag{5}$$

a Euclidean kernel—real, contracting, and not unitary—with $|\Delta\eta|$ the lift's imaginary conformal-time interval. This is consistent with $U(\Delta\tau = 0) = \mathbb{1}$ rather than in tension with it: *no cosmic time elapses, and the kernel is not an evolution in cosmic time.*

*Two things follow, and the first is a check the construction did not have to pass.* A mode of frequency $\omega$ is damped by $e^{-\omega|\Delta\eta|}$ under (5); the framework's independent classical reading of the same segment damps a mode carrying $e^{ikc_s\eta}$ by $e^{-kc_s|\Delta\eta|}$ [3]. With $\omega = kc_s$ these are the same expression, term for term. *The classical selection rule—frozen content crosses, oscillating content does not—is the Euclidean kernel, read classically.* Second, and this is the content the classical reading could not supply: a Euclidean kernel run for a long imaginary interval *projects onto the ground state*, suppressing each excitation by $e^{-(E_n - E_0)|\Delta\eta|}$. *The scope of that statement should be fixed carefully, because the two sectors it separates are not symmetric.* The kernel acts on *oscillatory* content; a frozen, zero-frequency mode is a fixed point of it and passes unchanged. And at the crossing no mode is oscillating: on the contracting leg $aH$ grows without bound as $r \to 0$, so the comoving horizon shrinks to zero and every mode exits it and freezes before the branch point [34]. *So the projection does not select the frozen content that crosses—it has nothing to act on there.* What it acts on is the oscillatory content of modes while they were sub-horizon, which it annihilates; the ground-state projection is the statement that this content, and not the frozen amplitude, is what the segment removes.

*The projection is not exact, and where it fails is a scale rather than a caveat.* At $|\Delta\eta| \simeq 1.7 \times 10^4\,\text{Mpc}$ the damping reaches $e^{-2}$ only at $k \sim 2 \times 10^{-4}\,\text{Mpc}^{-1}$, so modes below that wavenumber are not strongly suppressed and cross with residual excitation. That wavenumber subtends $\ell \approx 3$ on the last-scattering sphere, which is where the framework's independent transmission calculation places the boundary of the filter's reach [3]. The two routes agree on the scale as well as on the factor.

*The obvious objection is that a Euclidean kernel needs a Hamiltonian bounded below, and that Euclidean gravity notoriously lacks one—the conformal factor entering the Einstein–Hilbert action with the opposite sign, so that the Euclidean action can be driven arbitrarily negative.* That objection does not reach this construction, and the reason is structural rather than fortunate. The conformal-factor problem arises when the path integral ranges over the conformal factor of the metric. Here it does not. The substrate's scale is $\alpha$ and is fixed—not chosen, but required, as the unique maximally symmetric structure carrying no unforced modulus [27, 8]—so there is no conformal mode to integrate over; and the propagating gravitational degree of

freedom of the layer is the *transverse-traceless* shear of Section 6, which is precisely the sector the conformal mode is absent from. Mode by mode that sector is (12), a harmonic oscillator, whose Hamiltonian is bounded below. The areal radius along the lift is likewise confined, running between the comoving turnaround and the branch point on an interval fixed by the progenitor mass, with the substrate curvature finite throughout [3, 35]: there is no runaway direction for the kernel to diverge along.

*And the state the kernel projects onto is not a new posit but one this paper has already selected on independent grounds.* Section 6 finds the scale-factor Hamiltonian to have deficiency indices $(1, 1)$—a one-parameter family of candidate quantizations—and closes it, without stipulation, by observing that the boundary at $a = 0$ is the de Sitter horizon, that its surface gravity is $\kappa = 1/\alpha$, and that the regular Euclidean state at that $\kappa$ leaves nothing to choose. *That is the same condition the kernel* (5) *enforces*: regularity in the Euclidean continuation. *And that regularity is available only because of what the causality companion establishes about the boundary.* A regular Euclidean state exists at a horizon only if the Euclidean continuation is smooth there—free of a conical defect, curvature finite. The horizon at $a = 0$ is a *metric* singularity, at which the spatial measure collapses while the curvature stays finite, and not a curvature singularity [4]; had it been the latter there would be no smooth Euclidean section, no regular state to select, and the extension would remain unfixed. *The distinction that dissolves the information paradox is the same distinction that makes this quantization unique.* The extension-fixing condition and the lift's kernel are one requirement met twice, once at the horizon and once across the beginning.

*One approximation is carried and its size is bounded below rather than left open.* The kernel's frequency is not constant along the segment—the layer's scale varies with the path parameter—so the projection is adiabatic rather than exact, and the residual at the largest scales quantified above is one face of that. *The ceiling on what it can transmit is computed in the adiabatic correction below*, and it is small enough that nothing above turns on it. *The segment's variational treatment is given below.* The relation between the configuration variables $\{q^A, p_A\}$ on the two sides is described through the kernel's action on the modes, and the segment is exhibited there as a solution of a variational principle in its own right.

The segment rides the marginal congruence $(\mathrm{d}r/\mathrm{d}\tilde{\tau})^2 + f = 1$—the $\gamma = 1$, $J = 0$ case of the effective-potential form $(\mathrm{d}r/\mathrm{d}\tau)^2 = \gamma^2 - V_{\text{eff}}$ with $V_{\text{eff}} = f$ derived in [1, Eq. (4.24)]—which is $\frac{1}{2}\dot{r}^2 + V = \frac{1}{2}$ with $V(r) = \frac{1}{2}f(r)$. Continuing $\tilde{\tau} = is$ sends $\dot{r} \mapsto -i\,r'$ and the Lorentzian action $S = \int \mathrm{d}\tilde{\tau}\,[\frac{1}{2}\dot{r}^2 - V]$ to $S = -iS_E$ with

$$S_E = \int \mathrm{d}s\,\Big[\tfrac{1}{2}r'^2 + V(r)\Big], \tag{6}$$

whose stationary points obey $r'' = +V'(r)$—motion in the *inverted* potential, the standard signature of a Euclidean solution—with first integral $\frac{1}{2}r'^2 - V = -\frac{1}{2}$. The lift's closed form $r(s) = -(2M\alpha^2)^{1/3}|\sin(3s/2\alpha)|^{2/3}$ satisfies both: the first integral is $-\frac{1}{2}$ identically along the segment, and $r''$ agrees with $V'(r)$ to the precision of the check[P10R11]. *So the imaginary-time segment is not a formal device but a solution of a variational principle*—an instanton connecting the Lorentzian turning point to the branch point.

Its action converges. Near the branch point $V \simeq -M/r$ while $r \propto s^{2/3}$, so the integrand of (6) grows as $s^{-2/3}$ and is integrable; numerically $S_E \to 1.487$ in units $\alpha = 1$ on the forced member. *The segment therefore carries a finite weight rather than a divergent one*, which is what a saddle-point treatment of the beginning would require of it. *Two normalisations are in play, and the relation between them can be settled rather than flagged.* The functional (6) carries a unit kinetic term; the gravitational action carries the metric factor and the prefactor implied by (9). They share a solution—the contour's law $(\mathrm{d}r/\mathrm{d}\tilde{\tau})^2 = 2M/r + r^2/\alpha^2$ is the Friedmann equation with $a = r$, the identification $2M \leftrightarrow (8\pi/3)\rho_d$ being just $M = (4\pi/3)\rho_d$—so the instanton is one trajectory read in two measures, and only its *action value* differs.

That value follows without further input. On shell the reduced action is $S = \int p_a \, \mathrm{d}a$, and inverting (9) gives $p_a = (3/4\pi)a\dot{a}$; continuing $\tilde{\tau} = is$ yields $S = -iS_E$ with

$$S_E^{\mathrm{grav}} = \frac{3}{4\pi} \int \mathrm{d}s \; r \, r'^2 = \frac{3}{4\pi} \int \mathrm{d}s \; r \, \big[ f(r) - 1 \big], \tag{7}$$

using $r'^2 = f - 1$ on the segment. The integral converges rapidly and gives $S_E^{\mathrm{grav}} = -0.0481 \, \alpha^2/G$ on the forced member[P10R10].

*Its sign is the informative part, and it is not a pathology.* A negative Euclidean gravitational action is the Hartle–Hawking sign: the no-boundary de Sitter action is likewise negative, $-\alpha^2/8G$ in this convention, yielding an enhanced rather than suppressed weight [30]. The present value is the same sign and the same order, smaller by a factor of order two, which is what a segment of the contour rather than a full hemisphere should give.

The sign traces to the segment lying on the $r < 0$ branch, where $p_a < 0$ while $r$ increases. *So the beginning's weight is of the no-boundary type, obtained here from a contour the construction already possessed rather than from a boundary condition imposed on the path integral. One register point must be made here, because the vocabulary invites a reading the construction refuses.* The segment's parametrisation runs imaginary, and the kernel and action computed above are the kernel and action of that parametrisation; *they are not the Euclidean objects of a Wick-rotated spacetime.* The geometric-core paper states the guard generally: the construction reaches a real manifold through imaginary instruments, and the continuations it uses—the embedding coordinate, the equatorial seam, the cosmogenesis reassignment—are *real analytic continuations* on a spacetime that is Lorentzian throughout, not Wick rotations [7]. The de Sitter horizon's Gibbons–Hawking state, by contrast, *is* a Euclidean continuation, and that paper marks it as distinct in kind from these. *So the two must not be conflated*: the thermal register in which $\hbar$ is fixed by the period $\beta = 2\pi\alpha$ is not the register the segment's action belongs to, and dividing that action by that $\hbar$ would be joining two continuations the construction keeps apart. What the action is, is a real integral along a real curve of a real Lorentzian geometry, whose parameter happens to run imaginary over one stretch; what it is not is an exponent awaiting a quantum of action. *Its comparison with the no-boundary sign above is a comparison of signs and magnitudes, not an identification of frameworks.*

**The adiabatic correction, and where it fails.** The kernel projects exactly only if $\hat{H}_{\mathrm{phys}}$ is constant along the segment, and it is not: the tensor tower of Section 6 carries frequencies $\omega_n = \mu_n/a$, and $a$ varies along the lift, diverging in $\omega$ as the branch point is approached. The exact suppression exponent is therefore $\int \omega_n \, \mathrm{d}s$ rather than $\omega_n |\Delta\eta|$, and the first question is whether it converges at all. *Two different questions are in play here and it is worth keeping them apart, because only one of them is what the word "adiabatic" names.* An adiabatic expansion is controlled by $|\mathrm{d}\omega/\mathrm{d}s|/\omega^2$, and along this segment that quantity is $\propto s^{-1/3}$ and *diverges* at the branch point—so the adiabatic invariant $E/\omega$, which is what an adiabatic approximation conserves, is not conserved through it. *The suppression is nonetheless finite, and for a different reason*: the exponent $\int \omega_n \, \mathrm{d}s$ converges because $\omega \propto s^{-2/3}$ is integrable, which is a statement about the action integral and not about slow variation. *So the correction here is semiclassical rather than adiabatic*—the companion cosmology paper names the same object correctly, as a WKB form whose adiabaticity parameter is of order unity [34][L14]. It does: near the branch point $|r| \propto s^{2/3}$, so the integrand grows only as $s^{-2/3}$ and

$$\int_0^{\pi\alpha/3} \frac{\mathrm{d}s}{|r(s)|} = 3.3387 \tag{8}$$

on the forced member[P10R9]. *The correction is not small and it runs the conservative way*: the constant-frequency estimate using the turnaround value gives 1.4396, so the exact exponent is larger by a factor 2.32 and the suppression is *stronger* than the naive reading of (5) suggests.

*Whether the projection is adiabatic is a separate question with a definite answer.* The adiabaticity parameter $|\mathrm{d}\omega/\mathrm{d}s|/\omega^2$ is $C/\mu_n$ with $C$ running from 1.72 near the branch point to 0.16 near the turnaround, so the approximation is controlled by the harmonic index alone. With $\mu_n^2 = n(n+2)$, $n \geq 2$, this gives 0.61 at $n = 2$, 0.44 at $n = 3$, and 0.16 by $n = 10$. *The projection is therefore adiabatic for all but the lowest few harmonics, and degrades to order unity only at $n = 2$ and $n = 3$.* That the approximation should fail exactly where the tower is coarsest is expected rather than surprising—there are no modes below $n = 2$ on $S^3$—and it locates the correction rather than leaving it as a caveat. *We record, without claiming it, that the harmonic indices at which the treatment loses control are the lowest ones.*

*What that loss of control costs is bounded, and the bound is one-sided because of the tower's own floor.* The suppression falls monotonically with the harmonic index—$7.9 \times 10^{-5}$ in amplitude at $n = 2$, $2.4 \times 10^{-6}$ at $n = 3$, and by upwards of an order of magnitude per step thereafter—*so the mixing the adiabatic projection neglects can only carry amplitude into more-suppressed modes.* And there is no mode below $n = 2$ to carry amplitude the other way: *the least-suppressed mode has no source beneath it*, so the bound is structural rather than estimated. Taking the whole $O(\epsilon)$ fraction out of $n = 2$ at the worst case leaves that mode more suppressed, not less, and the mode it enriches does not reach the unmixed ceiling. *The ceiling on transmitted power is therefore the unmixed $n = 2$ value,* $7.9 \times 10^{-8}$—and below $10^{-3}$ even on the naive exponent, which is the weaker estimate. *So the residual is located, bounded, and reaches nothing observable*[P10R14]. The relation between $S^3$ tensor harmonics and observed multipoles is developed in the companion framework [3], whose perturbation-spectrum treatment carries the source spectrum of a closed $S^3$, discrete by degree, projected to the sky through the *flat* distance slicing rather than a closed one—the flat/closed decoupling carried by the non-synchrony. *That projection is that paper's; this section's own contribution to it is the harmonic-side adiabaticity above*, controlled by the harmonic index alone—so the two meet at the lowest harmonics, where that paper's lowest physical mode and this section's loss of control coincide.

**An explicit illustration, and its limits.** To show the structure is not empty we exhibit it in the simplest cosmological model, flat Friedmann–Lemaître minisuperspace. With scale factor $a$, conjugate momentum $p_a$, cosmological constant $\Lambda$, and a pressureless (dust) reference fluid serving as the cosmic clock in the manner of Brown and Kuchař [21], the deparametrized gravitational Hamiltonian reduced from the Einstein–Hilbert action (in units $G = 1$, fiducial comoving volume unity) is

$$H_{\mathrm{phys}}(a, p_a) = \frac{2\pi}{3}\,\frac{p_a^2}{a} - \frac{\Lambda}{8\pi}\,a^3, \tag{9}$$

equal on shell to the conserved comoving dust energy[P10R1]. Hamilton's equation $\dot{a} = \partial H_{\mathrm{phys}}/\partial p_a = 4\pi p_a/3a$, together with $H_{\mathrm{phys}} = \rho_d$ on shell, gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3}\,\frac{\rho_d}{a^3} + \frac{\Lambda}{3}, \tag{10}$$

the flat-ΛCDM Friedmann equation. Its solution is the $\sinh^{2/3}$ law

$$a(\tau) = A\,\sinh^{2/3}\!\left(\tfrac{3\tau}{2\alpha}\right), \qquad \alpha = \sqrt{3/\Lambda}, \tag{11}$$

with the single scale $\alpha$ of the de Sitter substrate; one verifies directly that this $a(\tau)$ satisfies (10) identically. This is precisely the expansion history that the companion framework paper derives, by a wholly independent route, as the SdS expansion law of the layered construction [3]: the deparametrized generator reproduces, in cosmic time, the same flat-ΛCDM evolution that the null-boundary correspondence delivers geometrically. Promoting (9) to an operator on $L^2$ of the half-line and applying (4) gives unitary evolution of the cosmological wavefunction in cosmic time.

Two qualifications attach to this illustration. First, it is a minisuperspace toy: a homogeneous truncation with a dust clock, not the full Schwarzschild–de Sitter object of CR, and the operator ordering of $p_a^2/a$ and the self-adjoint extension on the half-line are genuine technical questions not settled by the reduction. The illustration establishes that the deparametrized structure is consistent and reproduces the right cosmology; it does not by itself constitute the canonical quantization of the full theory. That next object—the lift to the closed-$S^3$ layer and its propagating degrees of freedom—is carried out in Section 6. Second—and this is a feature rather than a gap—the CR beginning is the branch point, a divergence of the geometry read over the areal chart on a substrate whose curvature is finite there, which the bead's contour passes through rather than ending at [4, 3]. The usual motivation for quantizing minisuperspace, to resolve an initial singularity, therefore does not arise. The point of (4) is not singularity avoidance but the demonstration that, on the selected foliation, the canonical generator evolves rather than annihilates.

## 6 The closed-$S^3$ lift: the graviton sector on the layer

The minisuperspace illustration of Section 5 truncates the layer to its scale factor alone; it carries no propagating degree of freedom and so cannot exhibit what the true Hamiltonian generates beyond the background expansion. The full layer does carry such a degree of freedom—the transverse-traceless shear of its spatial geometry, the graviton—and the lift of the deparametrization to it is the substantive canonical content. We carry out that lift here. Throughout, the foliation parameter is the same absolute cosmic time of Section 2: the comoving congruence of the layer, which the companion framework paper identifies with the cosmic rest frame the CMB measures and with the $S^3$ family of the null-boundary correspondence [3]. It is one congruence, carrying the flat-ΛCDM and the closed-$S^3$ slicings as two synchronizations of itself, not two frames.

The layer is best represented by the background geometry on which the cosmology is built—the reassigned de Sitter geometry, distinct (as a Lorentzian metric) from the Schwarzschild–de Sitter cosmology it underlies but sharing its manifold and foliation [3]. Its closed synchronous slicing is the evolving round three-sphere of radius $a(T) = \alpha \cosh(T/\alpha)$ in cosmic time $T$. A transverse-traceless perturbation of the spatial metric decomposes into the tensor harmonics of $S^3$, $h_{ij}(T,x) = \sum_n \phi_n(T)\, Y_{ij}^{(n)}(x)$, with the $Y_{ij}^{(n)}$ the standard transverse-traceless rank-two harmonics of the unit three-sphere and Laplace eigenvalues $n(n+2) - 2$, $n \geq 2$. *The frequency the reduction gives is not that eigenvalue*: the second variation of the spatial curvature carries a non-derivative contraction alongside the gradient term, $R_{ikjl}h^{kl} = -Kh_{ij}$ pointwise for any traceless symmetric perturbation on a maximally symmetric section, and it displaces the eigenvalue by $+2K$ with no dependence on which harmonic is displaced, so the tower's frequencies are $\mu_n^2 = n(n+2)$—in the labels $m = n+1$ below, $\mu^2 = m^2 - 1$[P10R37]. *That the displacement is the same at every level is a computed fact and not an extrapolation from the lowest*: reducing the action again at a level whose harmonic has components varying in all three directions—so that the gradient terms absent at the floor are present—returns the displaced value and not the eigenvalue, the two differing by exactly two with every step exact, while the same pipeline in the same arithmetic reproduces the floor's own value[P10R38]. *And the two parametrisations separate the object from its presentation exactly*: the difference between them is the same constant at both levels, while the difference between the levels is the same in both parametrisations. The second-order Einstein–Hilbert action in the transverse-traceless sector reduces, mode by mode, to that of a harmonic oscillator with time-dependent mass $a^3$ and frequency $\mu_n/a$,

$$S_n = \tfrac{1}{2}\int dT\, a^3\Big[\dot{\phi}_n^2 - \frac{\mu_n^2}{a^2}\,\phi_n^2\Big], \qquad \pi_n = a^3\dot{\phi}_n. \tag{12}$$

Read on the absolute foliation exactly as in Section 5, the constraint deparametrizes and (4)

becomes

$$i\,\partial_T \Psi = \hat{H}_{\rm phys}\,\Psi, \qquad \hat{H}_{\rm phys} = \sum_n \Big[\frac{\hat{\pi}_n^2}{2\,a(T)^3} + \tfrac{1}{2}\,a(T)\,\mu_n^2\,\hat{\phi}_n^2\Big], \tag{13}$$

a unitary evolution in cosmic time of a tower of time-dependent oscillators[P10R2], one per tensor harmonic. This is the lift the toy lacked: the abstract gauge-invariant degrees of freedom $(q^A, p_A)$ of Section 5 are here concrete—the graviton modes $\phi_n$—and the deparametrized generator advances them rather than annihilating them. Each mode is a Schrödinger oscillator on $L^2(\mathbb{R})$ and is manifestly self-adjoint; the closed topology of the layer enters as the *discreteness* of the tower, in contrast to the continuous spectra of the flat minisuperspace and planar reductions.

The same tower has a second reading, in the frame of a fundamental observer. Under the null-boundary reassignment the observer's worldline is one of the reassigned null rulings, and the layer's expansion read along it is the areal radius $a(\tau) = (2M\alpha^2)^{1/3}\sinh^{2/3}(3\tau/2\alpha)$ in the observer's proper time $\tau$—the flat-ΛCDM scale factor of the framework paper, the expansion leg of the *cosmogenetic bead* the framework proves one closed curve [3]. The cosmic time deparametrized here is the very parameter along that bead: real on the expansion leg the observer inhabits, it is what continues off the real axis, through a bounded contour, along the bead's lap into the antecedent collapse. The tower projected into this frame is (13) with $a(T)$ replaced by $a(\tau)$, the discrete index passing to the continuous wavenumber of the observer's locally flat slices. What distinguishes the two readings is the bend: the observer's Friedmann readout is

$$\Big(\frac{\dot{a}}{a}\Big)^2 = \frac{2M}{a^3} + \frac{1}{\alpha^2}, \tag{14}$$

carrying a pressureless matter term $2M/a^3$ that is absent from the background geometry's pure-Λ expansion, $(\dot{a}/a)^2 \to \alpha^{-2}$. The matter the observer attributes to the graviton's background is thus the bend of the slicing—a feature of the projection, not of the layer—exactly the reclassification the framework makes for matter content generally [3, 9]. The graviton sector therefore exhibits, in one object, both faces of the layered reading: the discrete unitary tower on the ontological background, and the continuous flat-ΛCDM spectrum a real observer measures, with the matter density appearing only in the latter.

This realizes concretely the loop drawn in Section 7 with the dynamical content. The propagating degree of freedom is the transverse-traceless shear of the evolving layer, advanced by a true Hamiltonian on the absolute foliation—the same structure the companion dynamics paper exhibits for the confined polarized wave [11], here for the cosmological sector and the full tensor tower. The necessity of the external clock is sharpest here: an internal (area) clock is available at $\Lambda = 0$ and would serve, but at $\Lambda > 0$ it no longer separates the dynamics, and the absolute cosmic time is forced. Of the two background-sector questions, the first is settled in structure. The operator ordering and the half-line self-adjoint extension of the scale factor (Section 5) are one coupled question: in the geodesic coordinate $x \propto a^{3/2}$ the kinetic operator carries an inverse-square term $\gamma/x^2$ at the origin whose coefficient, across the natural ordering family, attains a maximum $\gamma = \frac{1}{4}$—strictly below the essential-self-adjointness threshold $\gamma = \frac{3}{4}$—so the operator is limit-circle at $a = 0$ for every ordering, while the $-\frac{\Lambda}{8\pi}a^3$ term maps to the marginal $-x^2$ and is limit-point at $a \to \infty$, forcing no condition there. The scale-factor Hamiltonian therefore has deficiency indices $(1, 1)$ independently of the ordering [31, 32]: a one-parameter family of self-adjoint extensions fixed by a single boundary condition at $a = 0$. The ordering ambiguity cannot render the quantization unique; the residual reduces to the physical choice of that boundary condition. That choice is fixed, for the free scale-factor sector, by the geometry of the boundary itself. It is worth naming what kind of closure that is, because the corpus has a criterion for exactly this and this is its sharpest instance. A one-parameter family of self-adjoint extensions[L10] is a one-parameter family of *candidate quantizations*—distinct physical predictions indexed by a $\lambda$ that the formalism does not pin. That is the structure the epistemic reading excludes: a

structure carrying an unforced parameter is not a single world but a family, and answers "what is the world?" with a family rather than a world [27]. What follows below is not a stipulation that removes the family but the substrate *requiring* its own configuration: the boundary at $a = 0$ is the de Sitter horizon, its surface gravity is $\kappa = 1/\alpha$, and the regular Euclidean state at that $\kappa$ leaves nothing to choose. *The distinction that makes this the criterion's instance and not merely an analogy is worth keeping: $\lambda$ indexes distinct worlds, so it is an ontological family and inadmissible as it stands, whereas a parameter that merely records an unknown datum of one world—the progenitor's mass, say—is an epistemic gap and not a family at all.* The point $a = 0$ is not a generic half-line endpoint but the de Sitter cosmological horizon—the smooth, finite-curvature metric singularity at which the comoving congruence's past null ruling closes, the substrate curvature staying finite while the comoving ruler collapses [4, 9, 3]. A Killing horizon of surface gravity $\kappa$ carries a thermal (Hartle–Hawking) state [29], the unique state regular in the Euclidean continuation and free of a conical defect at the horizon; for the de Sitter cosmological horizon $f = 1 - r^2/\alpha^2$ gives $\kappa = \frac{1}{2}|f'(\alpha)| = 1/\alpha$, regularity period $\beta = 2\pi/\kappa = 2\pi\alpha$, and the Gibbons–Hawking temperature $T = \hbar\kappa/2\pi k_B = \hbar/2\pi\alpha k_B$ [28]. Imposed on the scale-factor wavefunction, Euclidean regularity is exactly the condition that retains the regular indicial branch $x^{1/2+\nu}$, $\nu = \sqrt{\gamma + \frac{1}{4}} < 1$, and excludes the irregular admixture $x^{1/2-\nu}$, whose presence is a conical defect—an off-temperature state. This is the Friedrichs extension, and it is a single boundary condition: the deficiency-$(1, 1)$ freedom is thereby not merely parametrized but *closed*, and closed without a free parameter, since the de Sitter horizon fixes one temperature. The physical choice the ordering ambiguity left open is the horizon's own thermal state, and $\hbar$ enters the gravitational sector here, at the seam, scaled by $\Lambda$ alone—the quantum instance of the substrate's constant ledger, in which the fundamental constants enter as unit gauges over the single scale and the one place a free quantum parameter could sit is closed by the horizon's own thermal state [7]. *And that continuation is not a device brought in to close the freedom; it is the substrate's other real form.* The global Wick rotation $x_0 \mapsto ix_0$ carries $\mathrm{dS}_5 = SO(5, 1)/SO(4, 1)$ to the compact $S^5 = SO(6)/SO(5)$, and the two are the real forms of the one complex group $SO(6, \mathbb{C})$ [2, 7]. *So the register in which $\hbar$ enters is the same substrate read on its compact face*, and the horizon—where the period is $\beta = 2\pi\alpha$—is where the two faces meet: the extension is fixed not by an extra assumption but by the geometry the Lorentzian reading already carries, seen from its other side. This determines the free scale-factor sector; the coupled sector below is where the open structure remains. *And what remains there is one of three parts, the other two being settled.* The boundary operator is bounded below and its spectrum is computable branch by branch, both shown below; what is not settled is the ultraviolet definition of the mode sums. *That the remainder is that part and not another is worth stating, because it bounds what any calculation on the free spectrum can reach*: a deformation of the free frequencies is answerable exactly—no multiplicative renormalisation moves the logarithmic coefficient, and the one mass-like displacement that removes it is the curvature offset itself—but *whether the coupling acts on the sum that way cannot be settled ahead of the sum's own definition*, which is the open item. *So the free-spectrum method reaches exactly as far as it reaches*, and the remainder is not a finer calculation of the same kind[P10R7]. The second background-sector question—the couplings of the tower to the scale-factor sector beyond quadratic order—sharpens this structure rather than threatening it. Deparametrizability is not at risk: the cosmic clock is a Brown–Kuchař dust fluid (Section 5), whose momentum enters the scalar constraint linearly whatever the graviton content, so the constraint solves for a true Hamiltonian at every order of the coupling—which enters $H_{\mathrm{phys}}$ as interaction terms between $a$ and the modes, not as an obstruction. The free tower above evolves on $a(T)$ as a fixed classical background, which is why each mode is manifestly self-adjoint on $L^2(\mathbb{R})$; the coupling question is what happens once the scale factor is itself quantized and back-reacts, and there it acts at the $a = 0$ boundary of the scale-factor half-line. Each mode's kinetic term $\pi_n^2/2a^3$ is, in the geodesic coordinate, the inverse-square operator $\pi_n^2/2x^2$ at the origin, so the boundary coefficient is promoted from the c-number $\gamma \leq \frac{1}{4}$ of the free scale

factor to an operator $\hat{\Gamma}$ on the tower whose spectrum straddles the $\frac{3}{4}$ threshold. *At leading order that spectrum is bounded below by $\gamma$ and unbounded above—the operator is a sum of squares, displayed below—but the decomposition that follows uses only that both sides of the threshold are occupied, and not that the spectrum has a floor; whether the complete $\hat{\Gamma}$ is bounded below is not assumed here* [P10R12]*, though it does in fact follow: the full inverse-square coefficient is positive on non-degenerate metrics, so $\hat{\Gamma} = \gamma + c\sum_n \hat{\pi}_n^2$ retains its floor at $\gamma$ beyond leading order, and the question of a spectrum below $-\frac{1}{4}$—where $\nu = \sqrt{\Gamma + \frac{1}{4}}$ would leave the reals and no regular branch would exist—has no object. The straddle itself is a computed fact*[P10R13]*: the spectrum does occupy both sides of $\frac{3}{4}$, since $\hat{\Gamma} = \gamma + c\sum_n \hat{\pi}_n^2$ with $c > 0$ has $\operatorname{spec}\hat{\Gamma} = [\gamma, \infty)$, and $\gamma \leq \frac{1}{4} < \frac{3}{4}$ places spectrum strictly below the threshold while the unboundedness above places spectrum strictly above it—the negation requiring $\gamma \geq \frac{3}{4}$, three times its own bound. That threshold is a different one from $-\frac{1}{4}$, deciding a different property: $\frac{3}{4}$ separates limit-point from limit-circle and so decides whether a boundary condition must be chosen at all, while $-\frac{1}{4}$ decides whether a regular branch exists to choose. How the straddle falls does not bear on the closure below, which is supplied fibre by fibre and so cannot be broken by the size of the sub-threshold set* [9]. As $\hat{\Gamma}$ commutes with the radial part, $-\partial_x^2 + \hat{\Gamma}/x^2$ decomposes as a direct integral over its spectrum: essentially self-adjoint where $\hat{\Gamma} \geq \frac{3}{4}$—the graviton momentum makes the origin limit-point and removes the boundary freedom—and limit-circle where $\hat{\Gamma} < \frac{3}{4}$. The single boundary condition of the free scale factor is thereby replaced, in the coupled theory, by a self-adjoint condition supported on the sub-threshold graviton subspace, its deficiency that subspace rather than a line. At leading order $\hat{\Gamma} = \gamma + c\sum_n \hat{\pi}_n^2$; but the cubic and higher self-interactions enter at the same inverse-square order at the origin ($\pi_n^2\phi_m/a^3$ in kind)—the potential sector's own cubic being regular there, and so reaching the trace without reaching the boundary coefficient—so the complete boundary coefficient is the singular part of the interacting graviton Hamiltonian, and the *ultraviolet definition* of the tower sums is the open frontier—the interacting tower as a defined theory, met here at its boundary face. *The spectrum is not open with it*: it is computed branch by branch below, and which branch is a question the construction has been shown not to answer internally. *Two things can be said about that spectrum without settling the frontier, and both bear on the decomposition. First, it is computable branch by branch: on the natural ordering family the minimum of $\hat{\Gamma}$ is $\frac{1}{4}$ under normal ordering and $\frac{3}{4}$ under symmetric ordering with one mode occupied, and the difference between the two is exactly the zero-point quantum of that mode—which is why the threshold sits at $\frac{3}{4} = \frac{1}{4} + \frac{1}{2}$, the free boundary coefficient plus one such quantum, rather than at a value the construction chose. Second, the two orderings answer the paper's own question oppositely: under normal ordering the origin remains limit-circle and the boundary freedom survives, while under symmetric ordering it becomes limit-point from the first occupied mode upward. The decomposition below is untouched by the choice, since it uses only that both sides of the threshold are occupied and both orderings occupy both sides; but the physical content is not, and the choice is not one this construction can make internally. Asking which ordering is asking whether the graviton tower's zero-point energy gravitates at the horizon—the cosmological-constant problem in local dress, reached from inside the boundary coefficient rather than imported. That the construction makes no such choice is an exhaustion rather than an omission.* The ordering is inert in the bulk—normal and symmetric ordering differ by the c-number zero-point $\frac{1}{2}\hbar\omega_n$, a global phase under the deparametrized evolution—and becomes physical only in the boundary coefficient, so any selector would have to act there; and each candidate the construction offers fails to, for a structural reason: the horizon's thermal state and the seam's characteristic structure act *downstream*, closing the extension given $\hat{\Gamma}$ (the regular branch $x^{1/2+\nu}$ exists for either ordering); the substrate isometry, positivity, the single-scale ledger and covariance under configuration redefinition each respect *both* orderings; and the deparametrization, which supplies a preferred vacuum and so makes normal ordering available, does not make it mandatory—and, by solving the constraint rather than imposing it, removes the anomaly-freedom lever a Wheeler–DeWitt quantization would have used. What is left is

not a residual freedom the quantization failed to close but a single physical datum—whether vacuum energy gravitates—localized precisely as one computable quantity: an epistemic gap of the kind the construction admits, not an ontological family of the kind it excludes[P10R21]. The boundary condition on that sub-threshold subspace, however, is supplied by the same principle that fixed the free sector. Thermal (Hartle–Hawking) regularity is Euclidean smoothness at the horizon mode by mode: across the direct integral it imposes the regular branch $x^{1/2+\nu}$, $\nu = \sqrt{\hat{\Gamma} + \frac{1}{4}}$, on each sub-threshold fibre at the one horizon period $\beta = 2\pi\alpha$ (the surface gravity $\kappa = 1/\alpha$ belongs to the background horizon, not to the graviton content, and so is common to every fibre), and asks nothing of the limit-point fibres $\hat{\Gamma} \geq \frac{3}{4}$. The self-adjoint condition on the sub-threshold subspace is therefore the de Sitter horizon's thermal state at every order of the coupling, exactly as in the free sector; what remains open is accordingly not the boundary condition but the *definition of the interacting tower*—the spectrum of $\hat{\Gamma}$ and the ultraviolet definition of the tower sums—the first of which admits a verdict here, and it is affirmative[P10R15]. *The DeWitt form carries exactly one negative direction and it is the conformal one*, which is why the Wheeler–DeWitt equation is hyperbolic; on transverse-traceless momenta that term drops and the form is $\mathrm{tr}((g\pi)^2)$, equal to $\mathrm{tr}(S^2)$ for the symmetric $S = L^{\mathsf{T}}\pi L$ with $L$ the Cholesky factor of $g$, hence positive definite for every non-degenerate spatial metric. *So the cubic term's apparent unboundedness is an artefact of truncation*: $\pi^2(1 - \lambda\phi + \cdots)$ is the expansion of $\pi^2/(1 + \lambda\phi)$, whose full coefficient is positive wherever the metric is non-degenerate.[9] *The truncated form is easily misread, because a truncation of a positive function reads as an operator that has lost its floor*: the first-order term $-\lambda\phi\pi^2$ sampled alone does go arbitrarily negative, while the resummed coefficient on the same points never falls below $\gamma$. *The deparametrization that yields a true Hamiltonian is the same move that yields a bounded-below one, both being the removal of the conformal direction*—though the positivity speaks only of the interior, the degenerate boundary being what the thermal condition above is for. What remains genuinely open is accordingly the ultraviolet definition of the tower sums—a different thing from a residual freedom in the quantization at the boundary, and not settled by being shared with every interacting field theory. *And what remains open is narrow, because the coefficient's response to the coupling is settled even though the coupled sector is not.* The logarithmic coefficient is the $1/m$ term of $d(m)\mu(m)$, hence an exact functional of the spectrum, so the deformations that could discharge it can be enumerated without solving the coupling: an overall rescale $\mu^2 \to (1+\epsilon)(m^2-1)$ returns $L = (15/4)\sqrt{1+\epsilon}$, which vanishes only where every frequency in the tower does. *So no multiplicative renormalisation of the frequencies discharges the log*—and that is the class the leading back-reaction lives in, the coupling acting through $a$ and through $\hat{\Gamma}$, whose instantaneous expectation is the same sum rescaled. A mass-like shift $\mu^2 \to m^2 - 1 + \delta$ returns $L = -(\delta - 1)(\delta + 15)/4$, so the one such discharge is $\delta = 1$: there $\mu_n = n + 1$ and $d(m)\mu(m) = 2m^3 - 8m$ is a polynomial, the $1/m$ term not cancelling but *not existing*—and $+1$ is exactly the offset that makes these frequencies non-integer, being the net of the Laplacian's $-3$ and the curvature term's $+2$. *The enumeration is one formula and does not depend on where the offset is put*: at $\mu^2 = m^2 + u$ the coefficient is $-u(u+16)/4$, vanishing at $u = 0$ and at $u = -16$ alone.[P10R17] *That is a characterisation, and a measurement and a structural remark sharpen it.*[9] *The measurement first.* The tower is a discrete sum rather than a momentum integral—the harmonics run from $n \geq 2$ with no zero mode and no soft region—so the infrared is regulated by the compactness of the layer and needs nothing. The ultraviolet is untouched by that: with $\mu_n \sim n$, $\langle\hat{\pi}_n^2\rangle \sim n$ in the instantaneous ground state, and a degeneracy growing as $n^2$, the shell contribution goes as $n^3$ and the sum *diverges quartically*—the ordinary zero-point degree of a field in four dimensions. *The third of those inputs is a formula and not only a scaling*: since $S^3 = SU(2)$ is parallelizable, a frame-indexed field of frame-spin $s$ is $L^2(SU(2))\otimes V_s$, Peter–Weyl returns level-$j$ totals of 1, 3 and 5 times $(2j+1)^2$ for $s = 0, 1, 2$—the component counts—and the transverse-traceless part is the two extreme summands, exactly two fifths of the symmetric-tracefree total. Organised by eigenvalue, *the degeneracy is* $2(n-1)(n+3)$,

ten at the floor $n = 2$, so the shell contribution is $2n^3$ and the leading constant is settled rather than assumed[9]. *And the constant is the propagating-component count rather than a universal*: the scalar tower's counting function grows with leading coefficient $\frac{1}{3}$ against the tensor tower's $\frac{2}{3}$, so the growth is $n^2$ per propagating component and not per tensor rank. *So "standard" is the right word and it is a number. And the leading term's counterterm is one this framework already carries.* The quartic piece of the one-loop vacuum energy is independent of mass and of field—it survives at $m = 0$ and is the same for every species—so it is a constant vacuum energy, and *the counterterm a constant vacuum energy requires is a cosmological-constant term.* That is the framework's single dimensionful constant, absorbed into the one observed curvature with no bare-$\Lambda$-versus-vacuum split [7]; and since $\ell_P$ is a gauge-combination rather than a second physical length, the cutoff is not smuggling a scale in either.

*The subleading term is measured on the same spectrum rather than estimated.* Written in the labels $m = n + 1$ that carry $\mu^2 = m^2 - 1$ and $d = 2(m^2 - 4)$, the spectral zeta at the origin is $\zeta(0) = 10$ *exactly*—the binomial expansion terminating there, so the value is an identity and not an asymptotic estimate, and it is a functional of the degeneracy alone, the frequency entering at $s = 0$ as unity whatever its offset—and the zero-point sum's logarithmic coefficient is $\frac{15}{4}$, the $1/m$ term of $d\,\mu = 2m^3 - 9m + \frac{15}{4}m^{-1} + \frac{7}{8}m^{-3} + \ldots$, which a hard cutoff carrying no zeta function reproduces. *And the scale factor does not enter it*: the free tower's frequencies are $\mu_n/a$, so $a$ factors out of the sum identically and the coefficient's independence of the expansion is exact rather than adiabatic—*which is also why the coupled sector, and not a time-dependent free one, is where what remains lives*[P10R3].

*The structural remark is about the terms after it, and it is sharper than the leading case because their counterterms are not cosmological-constant terms.* Only the *quadratic and logarithmic successors* carry the mass, which is to say only they renormalise masses and couplings; in the standard curved-space expansion the first goes with the Ricci scalar and the second with curvature-squared invariants, and a curvature-squared coupling is not an entry in this framework's ledger. *That last statement is two claims and not one, and the useful thing is that the criterion separating them does not depend on the operator dimension at all.* The framework's dimensionful content is one length, and the gauge length is a combination over it rather than a second physical length; so a coefficient carrying any power of length is admissibly the substrate scale to that power times a pure number, and it introduces a second length *only if that pure number is left undetermined. The question at every dimension is therefore the same question*—whether each coefficient is a number the construction determines—*and this section already holds one instance of it settled*, the shear's own coefficient being exactly twice a real scalar's and therefore a count of propagating modes rather than a number chosen to fit. *What the dimension ladder decides is how many such numbers there are, which is the renormalisability question and not the single-scale one*[P10R41]. *The reason that is not fatal here is a property of the backgrounds this construction admits rather than of the divergences.* On a maximally symmetric geometry of radius $\alpha$ every curvature invariant is a pure power of $1/\alpha^2$ [7], so $\int\sqrt{g}\,R^2$, $\int\sqrt{g}\,R_{\mu\nu}R^{\mu\nu}$, $\int\sqrt{g}\,R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ and $\int\sqrt{g}$ are not four independent functionals but four multiples of one; and the substrates the construction admits are a *one-parameter* family, the $\alpha$-foliation, every descent step above the last changing the scale and nothing else [7, 5]. *So the counterterm basis is one-dimensional on the admitted family*, and a divergence of any degree requires exactly one counterterm where a generic theory requires three.

*And that reduction does not in fact need the family, which matters because the family's own flatness does not survive second order in the shear.* In four dimensions $C^2 = E_4 + 2\big(R_{\mu\nu}R^{\mu\nu} - R^2/3\big)$ is an *algebraic identity on any metric*, with $E_4$ the Gauss–Bonnet density: the curvature-squared invariants are dependent because the dimension is four and not because the background is symmetric, *so no order in the shear can separate them.* What the admitted backgrounds then supply is the rest of the step rather than the whole of it: on an exact vacuum-$\Lambda$ space the Ricci terms are constants and $C^2 = E_4 - 8\Lambda^2/3$ exactly, the Gauss–Bonnet part shifting the on-shell

action by a topology-fixed constant and the Ricci remainder being what the field equations trade for the stress tensor. *So a change of subtraction point is a cosmological-term renormalisation*, reached from an identity rather than from a property a deformation could remove[P10R23]. *The restriction is the on-shell one and it is the whole restriction*: off shell $C^2$ is an independent invariant and nothing here bounds it.

*What that argument does not reach is stated here rather than left for a reader to find, because it is the whole of what remains.* A degeneracy among counterterms is a property of the background class: the functionals are distinguished by their response to *varying* the background, and on a one-parameter family they cannot be told apart. *The first thing to check is therefore which background the tower is defined on, and it is this section's own*: the closed synchronous slicing $a(T) = \alpha\cosh(T/\alpha)$ of §5, whose Ricci scalar is the constant $12/\alpha^2$. *That geometry is exactly de Sitter, so the degeneracy holds on the very background the free tower uses*, and the argument above is not an appeal to a nearby object.

*Where it genuinely stops is elsewhere, and the section has already named the place.* The free tower evolves on $a(T)$ as a *fixed classical* background; the coupled sector is what happens once the scale factor is itself quantized and back-reacts, and that is precisely the regime the boundary operator $\hat{\Gamma}$ belongs to. *The natural worry there is that a counterterm basis is a statement about a class of fixed backgrounds, so that once the scale factor is quantized there is no fixed background left to state it on. That worry does not survive being checked, and what replaces it is sharper.*

The collapse of the three quadratic invariants is not a consequence of maximal symmetry: in four dimensions their deficit $\mathrm{Riem}^2 - 2\,\mathrm{Ric}^2 + \frac{1}{3}R^2$ is exactly $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$, every Friedmann geometry is conformally flat for *every* scale factor, and on such a geometry the Gauss–Bonnet[L17] combination is an exact total derivative, $\sqrt{g}\,(R^2 - 4\,\mathrm{Ric}^2 + \mathrm{Riem}^2) = 24\,\mathrm{d}(\dot{a}^3/3 + \dot{a})/\mathrm{d}T$.

So $\int\sqrt{g}\,\mathrm{Ric}^2$ and $\int\sqrt{g}\,\mathrm{Riem}^2$ are both fixed by $\int\sqrt{g}\,R^2$ up to a boundary term whatever $a$ does: *no scale factor breaks it, because no scale factor can make an FRW geometry anything but conformally flat*[9]. *And because that is an identity holding pointwise in $a(\cdot)$ rather than an evaluation on a chosen class, it survives superposition and descends to the quantized sector as an operator relation*: the coupled sector never required a class of fixed backgrounds, because the statement was never made by evaluating on one. *What maximal symmetry did buy is a weaker and different thing, and that half does not survive*: on a constant-curvature background terms of *different* dimension—$\int\sqrt{g}$, $\int\sqrt{g}\,R$, $\int\sqrt{g}\,R^2$—are proportional as well. *What that half actually needs is weaker than a constant-curvature background and weaker than the* cosh: it is exactly that $R$ be $\sqrt{g}$-almost-everywhere constant, and two things that are not the cosh keep it so. A cosmological term with radiation gives $R = 4\Lambda$ exactly, radiation being traceless; and *the free tower's own bare zero-point energy is exactly radiation-like*—its energy is $\mathcal{S}/a$ with $\mathcal{S}$ free of $a$, so the density goes as $a^{-4}$, the pressure is a third of it, and the trace vanishes identically. *So the naive back-reaction of the quantized tower breaks nothing*, and both cases serve as null controls that measure the numerical floor rather than assuming it.

*What does break it is the logarithm, which is to say the very constant at issue.* The spectral $Z(s)$ has a pole at $s = -1$ of residue $\frac{15}{4}$, so the renormalized energy carries a $\ln a$—and a $\ln a$ spoils the exact $a^{-4}$ that made the bare sum traceless. The trace is then $\Theta = \rho - 3p \propto \frac{15}{4}a^{-4}$, *independent of the renormalization scale*: it is scheme-independent, which is what an anomaly is, and is why the split between the constant and the logarithm being a convention does not reach it. So $R$ acquires an $a^{-4}$ piece and is constant *if and only if* that residue vanishes. *The three integrals then span three dimensions rather than two*, at a ratio thirty-five decades above the floor the null controls measure, with the departure signed by the identity $I_2 I_0 - I_1^2 = I_0^2\,\mathrm{Var}_{\sqrt{g}}(R) \geq 0$ and growing as the square of the residue: *the logarithm's counterterm is a genuinely new entry, and the one ultraviolet constant becomes observable once the scale factor is quantized*[P10R24].

*That is the first of a sequence, and saying what the sequence is says what defining the interacting tower consists of.* It is the rank of the counterterm basis, one rank per operator dimension, taken on the class the construction admits—which, the scale factor being quantized,

is not a class of fixed backgrounds—together with, at each dimension, the data that settles that many numbers. *Two entries of that sequence are computed.* At dimension four the three quadratic invariants have rank two of three pointwise on a general scale factor, adjoining the Gauss–Bonnet combination leaves it at two—*so that combination is already inside the pointwise span and the reduction to one functional is the total-derivative step rather than a second degeneracy*—and as functionals on the one-parameter family the rank is one, which the trace anomaly then opens by an exact Gram identity. *The vanishing of the Weyl invariant there is an identity in the scale factor rather than an evaluation*, and it is read against a control that separates them: the same combination on an anisotropic metric with three different rates does not vanish.

*At dimension six there is no degeneracy at all, and that is the finding.* Five dimension-six scalars are pointwise *independent* on a general scale factor—no relation whatever—while on the one-parameter family two of them vanish identically and the rest are pure powers, so the rank there is one. *So the collapse at dimension six is entirely the one-parameter evaluation, and at dimension four it was not.* At dimension four the collapse rests on an identity holding pointwise in the scale factor, which is why it descends to the quantized sector as an operator relation; *at dimension six there is no such identity to descend*, so the half of the argument that already fails under quantization is the whole of the collapse there—and the fixed-background reading buys strictly less at the higher dimension, not more. *What that count is a fact about is said before it is used*: it is read off a chosen five-element subset, so it is a fact about those five—*but it is a lower bound on the full basis's rank, and a lower bound is the direction the argument needs*, completion being able only to raise it.

*And the narrower of the two questions this raises is not closer than the other; it is finished.* Whether the coefficient the free spectrum settles survives the coupling is answered by the same counting, and the counting is fixed to an observable rather than to a way of writing the field. The expansion is that of the vacuum energy in the ratio of the gauge length to the scale factor and in no other ratio; a counterterm of operator dimension $2k$ contributes at an order in that ratio equal to $2k-4$ exactly, and reading that backwards, second order admits one operator dimension and no other, a logarithm carrying no power of the scale factor and so riding at the same order. *The interaction therefore enters at second order and cannot populate the zeroth at any order*, while the free tower populates the zeroth alone—its large-label weight carrying no logarithm of the label, checked against a weight that does carry one so that the test discriminates. *And the warrant for that is the vacuum energy being a spectral invariant rather than any count of vertices*, which matters because a vertex count is a property of the split: the framework offers two lengths, so a redefinition off the grading is dimensionally admissible and generates vertices carrying none of the gauge length—*exhibited on a free oscillator, where a canonical point transformation leaves every spectral quantity fixed and yet reads as a genuinely interacting truncation*, and with a control that returns a non-zero shift on a genuine quartic so that the silence on the generated one is a finding. *So that coefficient lives at a power no order of the coupling reaches, and it survives unconditionally rather than conditionally, on a warrant that no change of variables can move*[P10R41].

*The divergence does reach second order in the coupling, and it does so by an identity rather than by a search that failed to find a reason for it to vanish.* At that order two channels contribute. Every cubic here is non-resonant—no matrix element is diagonal in the energy—so the generator of the redefinition it induces exists term by term, and the redefinition has a *unique* quartic completion whose own second-order shift is exactly zero. *What survives is therefore the actual quartic's deviation from the completion the cubic itself generates*, which is a difference of two expectation values and not a residue of either. *And no symmetry, parity or count of modes can remove it*: one channel is a sum of non-negative numerators over positive denominators, the whole is even in the cubic coupling, a purely cubic interaction already gives a non-zero value, and a total derivative changes no spectrum. *Vanishing requires one coincidence between the two*

*couplings, of codimension one—and that locus is exactly where the interaction is a redefinition and there is nothing to renormalise. The parity escape is closed by this section's own arithmetic*, the cubic overlap being a determinant that does not vanish on a traceless shear. *The instrument is calibrated in both directions*: exactly zero on two unitary conjugations, with both channels individually non-zero and equal, and the textbook value on a genuine pair[P10R44].

*That ratio is settled, and it is settled by reading the construction rather than by arguing about it: it is taken from the world.* The two objects share no argument—the substrate scale is a curvature radius of the four-geometry and has zero derivative with respect to every quantum and gravitational gauge, while the gauge length is not a length on any space at all but a combination of conversion factors whose dimension is a length by unit-counting and has zero derivative with respect to the curvature—*so no relation among the inputs of either can fix their ratio.* Calling the second a gauge combination therefore fixes that it introduces no new *scale*; it fixes neither a value nor a functional form. *And the place the quantum constant enters is not the place a value is fixed*: the horizon's thermal condition defines a temperature and so arrives with its own new quantity, adding one equation and one unknown, whose net constraint on what the ledger already holds is zero. *The separating test is an arithmetic*—the ratio has non-vanishing derivative with respect to every measured input, where the shear's coefficient has vanishing derivative with respect to all of them, being a mode count[P10R42].

*So the affirmative branch is a second number owed and not a second length introduced, and the single-scale ledger stands either way*, an undetermined dimensionless number not being a scale. *What it costs instead is an entry in a different ledger this construction also keeps*—the count of dimensionless constants spent—*which stands at one for these very sums. And a determination is not merely absent but declined, which is the honest form of the statement*: because the substrate horizon's entropy is exactly a multiple of the squared ratio, determining the ratio would be determining that entropy as a pure number, and the object required is a state count for that horizon. The construction takes the horizon's temperature and never its entropy, on the stated ground that a temperature carries a dimension where the entropy is a cross-register pure number. *This is therefore a single-scale ledger with one number read from the world*, and that is stated here rather than left to be inferred.

*And the size of what the affirmative branch would add is bounded without computing it, by the geometry rather than by an estimate.* The scale factor of the admitted family has a positive *minimum*—its only stationary point is the turnover, where the second derivative is positive—so the expansion parameter is largest exactly there and nowhere else, and it is bounded above by the product of the curvature with the squared gauge length at every epoch, a number of order $10^{-122}$. *So the case in which the scale factor's own powers undo the suppression does not exist*, and it does not exist because the background turns over rather than because the epochs happen to cooperate[P10R43]. *And the new number is separately visible rather than degenerate with the first*: the zeroth-order counterterm reaches the trace through the logarithmic term of the trace formula at one power of the scale factor and the second-order term through the other term of the same formula at another, so the two are distinguishable by where they appear.

*And then the counterterm it would need collapses, because the substrate background is not merely conformally flat but exactly maximally symmetric.* That is computed from the metric's own connection rather than assumed: every component of the curvature is the maximally symmetric form with sectional curvature one third of the cosmological constant. *So every component of the Weyl tensor vanishes, and with it both cubic Weyl contractions—the pair that carries pure gravity's own two-loop counterterm*—and all eight algebraic invariants of the next dimension are exact rational multiples of the cosmological constant cubed, of rank one among themselves. *And the covariant derivative of the curvature vanishes identically here in every component*, so the collapse is not confined to the Weyl part: *the whole derivative sector at that dimension vanishes as well*, and what is left to count is the algebraic part alone. *And that argument does not stop at one dimension.* Any invariant carrying an explicit covariant derivative

contains that vanishing factor, so *the derivative sector is identically zero at every dimension*; and at operator dimension $2k$ the algebraic invariants are polynomials of degree $k$ in a curvature that is itself the maximally symmetric form, hence *rational multiples of the $k$-th power of the one input. Verified at the next dimension up on ten invariants, each homogeneous of the right degree and of rank one among themselves, and then for an arbitrary rational combination of them. The scope belongs in the same sentence as the result*: that is a rank over the *values these invariants take on this background*, where the rank sequence of the counterterm basis is a rank over the admitted class, and the two are different objects[P10R45]. *The control separates that from an evaluation*: the same eight on an anisotropic metric are of rank two together with these, with both cubic Weyl contractions non-zero and a non-vanishing curvature gradient, returning the quadratic invariant's own value quoted earlier in this section.

*Which settles the coefficient's status without computing its value, and that is the useful half.* For an *arbitrary* rational combination of the basis, the residue has vanishing derivative with respect to every quantum and gravitational gauge and is homogeneous of degree three in the cosmological constant. *So whatever the computation returns is a rational number times the one input cubed, and the count of dimensionless constants spent stays at one*[P10R44]. *The scheme dependence of the finite part is shown rather than asserted*: a logarithmic mode sum regulated at two cutoffs an octave apart carries the same logarithmic coefficient and finite parts differing by a constant, so the finite part is a convention and the coefficient is not.

*And what that subtraction has to absorb is one double sum over pairs of labels, which is fewer sums than the order of the calculation suggests.* Because the identity above sets a difference of two *quartic* operators against each other, a free expectation of either is a sum over two contractions and not over intermediate states: *the identity has already performed the sum a third-order reading would have left open.* The index set closes from below on its own, the degeneracy vanishing at the label that would otherwise have to be excluded by hand, and the two field variances multiply to the minimum-uncertainty product exactly—*which is the check that they were not scaled independently of each other.*

*That sum diverges, and it diverges two powers above the free tower's own quartic.* The action being second order in derivatives supplies exactly three quartic structures, and their large-label weights put the leading behaviour at the sixth power of the label cutoff. *So the zeroth-order subtractions cannot reach it, which is the case that matters*: a divergence the existing subtractions absorb would have changed nothing, and this one requires a subtraction of its own. *Two independent routes agree on where it lands*—the mode count in labels and the momentum cutoff carrying the squared gauge length both give an energy density at the next operator dimension, which is where the dimensional bookkeeping had already put it, so the counting and the bookkeeping are not the same argument twice. *And the quantity the rate turns on is computed rather than bounded, so the rate is exact.* The object the double sum actually reaches is the *level-summed* overlap, the field variances depending on the level alone—and on a space whose isometry group acts transitively, with each level a single irreducible representation, that sum is a constant tensor by homogeneity and Schur's lemma. *The completeness relation is verified exactly in every component on a basis checked orthonormal*, and at the one level this section owns the density is a computation rather than a theorem, the frame-constant harmonics summing to exactly the degeneracy. *The control separates the identity from the basis*: dropping a single element leaves the components disagreeing with the projector and the trace short by one. *So the level-summed overlap is the product of the two degeneracies times a pure number*—and that product is exactly the degeneracy already counted, so the residual factor has no dependence on either label at all. *The fall-off exponent is zero where the reversing threshold was three, checked from both sides, and the sixth power is therefore exact rather than conditional*[P10R46].

*And it never reaches a four-label overlap at all, which is what makes the rate factorise.* The quartic vacuum expectation of a multi-mode free field equals its Wick sum with no departure,

and because the two-point function is diagonal that sum is supported on one or two distinct labels and never on three or four—*computed from the mode operators rather than assumed. So the pairings differ in how the indices are arranged and not in how many labels they carry*, each submitting to the same level-summed identity with a different pure number, so an off-diagonal arrangement moves the coefficient and not the rate. *The two-label term being a product of two single-label moments is what lets the rate factorise, and that is derived here rather than assumed*[P10R46].

*So the subtractions are three and not two, and the third is the one this row was opened over.* The quartic and the logarithm sit at zeroth order with dimensionless coefficients; the third sits at second order, at the next operator dimension, with a coefficient that is a rational multiple of the one input. *And the same argument gives the whole sequence rather than one entry of it*: at every operator dimension the basis takes values of rank one on this background, in rational multiples of the corresponding power of the one input, with the derivative sector absent throughout.

*What the section holds is therefore the subtraction's whole form, and the conventions it leaves behind are three, necessary, and countable.* They are the finite part of the sixth-order coefficient, the finite part of the cosmological one, and *the scale inside the logarithm*—and the asymmetry between them is what everything below turns on, the third being the only one that is a *scale* rather than an additive constant and the only one whose action on the energy density carries a power of the scale factor rather than none. *They are necessary rather than listed*: leaving any of the three unsubtracted leaves an infinite limit, and subtracting all three leaves exactly the finite part, so the count of conventions is the count of structures that have to be fixed. *And the logarithm's own coefficient is not among them*: an octave in the subtraction point moves the finite part by exactly that coefficient times a logarithm of two and moves the coefficient not at all, and the same shift acts identically on every member of the family—*so a measured difference fixes the shift's action and never a point of it*, which is why the coefficient is a computed rational and not a convention.

*Two of the three reach nothing observable, and the reason is the collapse above rather than a cancellation.* The three move the energy density along three directions, of which the first two are pure constants here because every counterterm's value on this background is a rational multiple of a power of the one input. *So the rank is two of three and the kernel is exhibited rather than counted*: an explicit combination of the sixth-order and cosmological conventions shifts the energy density by exactly zero, identically in the scale factor. *The convention this row added is therefore rank one with the cosmological one—new as a subtraction and not as a number taken from the world. The control puts the degeneracy where it belongs*: give the sixth-order counterterm a value that does not collapse and the rank is three with no kernel at all, so the degeneracy is this background's and not the rank test's.

*And the one observable this section names is untouched by a grading rather than by the cases tried.* The trace of a term carrying an inverse power of the scale factor is that term times four minus the power exactly, *so the fourth inverse power is the whole kernel of the trace and the zero is there alone.* The anomaly is that coefficient, and it has vanishing derivative in all three conventions: the two additive ones move only the trace's *constant* part, which is the cosmological term the framework absorbs into the one observed curvature, and the third moves nothing at all, being traceless. *Controls at the neighbouring powers return non-zero*, so the vanishing is radiation's own property and not the formula's—*and that the constant part does move is what makes this a grading and not a silence. And the tower's spectrum is untouched at fixed scale factor*, a convention being a multiple of the identity there: every spacing exactly unchanged while the total energy shifts, against a control that is not a multiple of the identity and moves every spacing.

*The third convention does reach an observable, and that observable is the one this section already records the mode sums as spending.* The logarithmic counterterm's coefficient *splits*:

a divergent part fixed at the anomaly, which is scheme-independent, plus a finite part which *is* the subtraction point's convention. *So the constant is neither downstream of the three nor one of them—it contains exactly one*, and that the anomaly is scheme-independent while the constant is a convention are both true of the same object, separated by where each appears. *This section's one spent dimensionless constant is that convention*, and saying so is what makes the accounting closed: *the dependence is on a constant already paid for, and nothing new is owed*[P10R47].

*And one number makes the third convention and the observable direction together.* Set the anomaly to zero and the logarithm, its convention and the fourth-power direction all go at once, the rank falling to one and every convention absorbing into the single constant. *So the anomaly buys exactly one observable direction and no more*—the two ranks differ by one—which is this section's own shape holding for the conventions and not only for the counterterm: *the saving mechanism and the thing it saves us from are one number. That also fixes the reach of an earlier result rather than correcting it*: the finding that no observable depends on the subtraction point rests on a one-dimensional counterterm basis, which is the reading on the one-parameter family and is right there, *while the anomaly makes the basis two-dimensional once the scale factor is quantized*—so that finding does not reach this sector, and the protection here comes from the ledger entry instead.

*Every rank above is a rank over the values these objects take on the substrate background, so whether any of it survives the scale factor becoming an operator is a question in its own right—and it is one object rather than several.* Every counterterm value on this family is a function of the *single* operator, and for functions of one operator a combination vanishes as an operator exactly where the function vanishes on the spectrum: *the operator-valued basis and the basis of values are the same object*, with equal ranks and, normalised, the identical kernel. *So a rank descends here with no appeal to linearity at all. The expectation-value reading adds nothing*, an expectation of a function of one operator being phase-independent and so seeing the spectral measure alone, against a non-commuting control whose phase derivative does not vanish. *And the reading in which the family itself is quantized has no object*: the curvature radius is a coefficient of the Hamiltonian rather than an observable of it, so its operator carries a one-point spectrum on which every function of it is a multiple of the identity—*which is this section's own superselection result, exhibited as an arithmetic rather than quoted.*

*On that reading the descent is free at the order this section's back-reaction is semiclassical, and it fails at the next one.* Pure radiation leaves the curvature scalar at exactly four times the cosmological constant, so the fourth inverse power there is the logarithm's alone; and along that deformation every invariant of the higher dimension shifts along the *same* function, so the rank of one descends at first order. *At second order the rank is two*, the breaking carried by the eighth inverse power and by squared logarithms. *The control separates the two*: putting a sixth inverse power into a single invariant gives rank two already at first order, so the first-order agreement is the deformation's property and not the test's.

*Item by item, and they do not fail together. The rate survives at every order*—the weights carry no scale factor, so a state contributes one moment multiplying the sum—*and so do both counts*, of subtractions and of conventions. *What is order-dependent is the per-dimension rank alone. And the kernel survives in number while rotating*: with the anomaly present the free direction necessarily carries the logarithm's own convention, in a computed ratio whose classical limit is the kernel above. *So the finding that the added subtraction costs no observable survives the descent, with the absorption now into the constant already spent*—the same coefficient appearing for the fourth time in this section, which is what a structural quantity does[P10R48].

*And the ranks above are the whole sector's rather than a chosen subset's, which is a mechanism and not a larger count.* At a maximally symmetric point the gradient of every realizable invariant is a multiple of the identity, so the two independent curvature directions are indistinguishable there; and *the realizable cubics are exactly the subspace that condition cuts out*, which

the invariants already computed span. *The control separates the realizable from the merely algebraic*: a monomial that violates the condition raises the rank, and it is not an invariant this construction produces.

*The semiclassical reading of the scale factor is then not a choice but a consequence, and the order it truncates has a size.* The expansion parameter's maximum is the geometry's own: one stationary point, at the turnover, with positive second derivative, so the parameter is largest exactly there and equals the product of the curvature with the squared gauge length—*of order* $10^{-123}$ *on the measured inputs, and its square of order* $10^{-245}$. *And the truncation follows rather than being imposed*: an independent second-order source would move every second-order coefficient while leaving every first-order one untouched, and the coefficients here carry no such amplitude—*so the rank the iterated first-order deformation gives bounds the construction's own second order from below rather than being it*, and the two are separated below, where the genuine source is computed and the bound turns out to be attained.

*At that order the rank is two of three, and the second direction is a single combination of the eighth inverse power with its squared logarithm—and it brings no fourth convention.* The weights carry no scale factor, so a deformed geometry enters the mode sum as one multiplying moment and the divergent structures stay the three already named: *the accounting survives with a rank change and nothing more*[P10R49].

*And the coefficient that has now arrived four times in this section is one object rather than four agreeing numbers.* It is the residue of the tower's spectral series at the pole the logarithm's subtraction sits on, and the partial sum's logarithm, the trace's fourth inverse power and the curvature scalar's return it as computed objects—*all four vanishing together on a control weight built without the term that produces it. So the saving mechanism, the observable direction, the surviving kernel and the anomaly are one quantity seen from four sides*, which is what a structural number does and is why this section names it once.

*And that pole is one entry of a table the whole scheme turns out to be.* The spectral series of the weight carrying the frequency has simple poles at arguments four, two and zero with residues 2, $-9$ and $\frac{15}{4}$; the series of the weight carrying its reciprocal has poles at two and zero with residues 2 and $-7$. *Each pole contributes its residue over its own location to the corresponding power of the cutoff, checked pole by pole*; the logarithmic coefficients are the residues at the origin *exactly*, so the anomaly is an entry of this table rather than a separate object; and the second-order sum's leading coefficient is a *product* of two residues. *So the three subtractions, the anomaly and the value-rank are all statements about the pole data of one series, and the coefficients this section names are polynomials in its residues. The subleading integer powers mix, and that is a property of partial sums and not of the spectrum*—the leading weight's own partial sum already carries the second power—which is this section's division between a coefficient and a convention seen from the other side.

*And the vertices those coefficients are made of are available in closed form at the level this section owns, which turns the outstanding arithmetic from a calculation into a sum.* The lowest harmonics are constant in the frame, so a perturbation there is a *left-invariant* metric on the three-sphere—and for those the geometry is known to all orders rather than to the order one expands to. The extrinsic curvature gives $K_{ij}K^{ij} - K^2 = -6H^2 + 6(\dot\beta_+^2 + \dot\beta_-^2)$ exactly in the two shear parameters $\beta_\pm$, the anisotropy velocities summing to zero so that volume and shear separate with no cross term and the isotropic limit returns the closed Friedmann Lagrangian; and the three-curvature, a fixed quadratic in the metric's three eigenvalues at unit determinant and equal to six on the round metric, expands as

$$R^{(3)} = 6 - 48\,(\beta_+^2 + \beta_-^2) + 160\,(\beta_+^3 - 3\beta_+\beta_-^2) - 336\,(\beta_+^2 + \beta_-^2)^2 + \dots, \tag{15}$$

*the cubic exactly the hexagonal invariant and the quartic exactly isotropic, so one number carries it. And the calibration is what identifies this as the section's own level rather than a neighbouring problem*: the ratio of the first two coefficients is eight, which is this tower's frequency-squared

at the floor, and the degeneracy the same reduction carries is ten, which is the tower's own degeneracy there—*two numbers the spectrum fixes, arriving from the metric.*

*One of the three quartic structures that reduction exhibits is a parameterisation rather than an interaction, which is this section's own redefinition identity arriving from the other side.* In the exponential variable the kinetic term is *exactly* quadratic—no momentum-squared-times-field-squared vertex at any order—where in the linear one the same term carries both a cubic and a quartic, and the map between the two is a point transformation of unit Jacobian. *So that structure is the completion the cubic itself generates, and in the variables where it is absent the physical shift is carried by the potential's quartic and the cubic alone.*

*The shift at this level is then a number rather than a structure.* Computed exactly in a two-mode Fock space, it is $\frac{4}{27}(63\mu^2 - 200)\mu^{-4}$ times $\hbar^2$ times the gravitational coupling over the comoving volume and the cube of the scale factor, which at this level's own frequency is $\frac{19}{27}$ of that combination and is *positive*, the quartic beating the cubic. *And the threshold belongs in the same sentence as the value, and it is a criterion rather than a number*: the shift is positive exactly where twice the quartic's coefficient exceeds the cubic's squared coupling divided by the frequency-squared, which at this level's own vertex numbers puts the turning point at a frequency-squared of 200/63. *That figure and the tower-wide criterion derived below are in one convention, and they agree.* The criterion is a ratio of two energies, so it is *invariant* under a change of convention—rescaling the field carries the couplings with it, and matching the two kinetic terms fixes the rescaling with nothing left over—and the criterion below, evaluated on this reduction's own two-mode sector, returns exactly the ratio this reduction banks, with no parameter fitted between them. *What separates the two numbers is the object and not the normalisation*: two frame-constant modes against a whole level summed over its degeneracy. The level's quartic is not a proportional slice of the sector's, *and the factor between their two ratios drifts across the tower*—which a convention factor cannot do. *Both return the same sign.*[P10R67P10R50] *The criterion is kept in that undivided form, because dividing it through by the quartic's coefficient reverses it wherever that coefficient is negative*—the two forms then disagreeing in sign and not in size, so the divided form carries a positivity condition it does not state. *And keeping it undivided is what makes the quartic's coefficient decide whether the coupling matters at all*: were that coefficient negative the criterion would fail at every level for any coupling whatever, the squared coupling being a square and every level's frequency-squared at least eight. *It is not negative*—the passage from the action to the Hamiltonian, written out below, returns it positive at every level—*so the criterion is that twice it, times the frequency-squared, exceed the squared coupling*, which is satisfied at vanishing coupling and fails at large. *The sign therefore waits on the coupling only where the coupling exists, and on half the tower it does not exist.* The transverse-traceless multiplet at each level is a pair of conjugate representations of the isometry group, fixed two independent ways—its dimension is the degeneracy already in use, and its Casimir less the isotropy group's is the eigenvalue already in use—and *the number of invariants in its triple product is exactly zero at every even level*, two at the first two odd ones, and eight at every odd level above them. *So the level's own cubic vertex does not exist at any even level, in any basis and under any truncation*, the squared coupling vanishing there identically; *and with the quartic's coefficient positive the criterion holds at every even level on that coefficient alone. The back-reaction's sign is settled positive on half the tower with no bound, no estimate and no two-point object. Two consequences the selection rule was not fitted to return with it*: at the lowest odd level the chiral half carries exactly one cubic invariant, which is the count this construction banked independently for that multiplet, and no level carries an invariant vector at all—*which is why the cubic's one-quantum amplitude cancels identically*, read above as an arithmetic and here as a selection rule.

*At the odd levels the coupling does exist, and it is not an integral at all.* A level's harmonics are matrix elements of one representation, so a triple overlap of three of them is a product of two recoupling coefficients, and *summing over the degeneracy each factor contributes exactly*

*one*: the degeneracy sum of squared same-level triple overlaps is an algebraic identity with no integral and no two-point object anywhere in its statement. *A second route returns the same number through the object that looked like the obstruction*—the level's kernel is a character, so the double integral is a class function of the relative element and collapses to a single class integral, exactly, the character squared integrating to one always and *the character cubed integrating to the number of invariants. So the sum is the cube of the degeneracy, over the volume, times that number*, and *the vanishing at even levels is this one formula's vanishing case* rather than a separate result.

*Which locates the obstruction more precisely than a wall in front of the value.* The same quantity has a representation carrying no two-point object, and on a group manifold the kernel route closes as well: *what the coincidence limit stands in front of is a route and not the object. And a bound that sums the levels by completeness is loose by exactly one power of the label*—such a bound being the fourth power of the degeneracy where the object is the third—*so the coupling's own growth is of the seventh degree against the target's eighth* in the level-summed convention, and the ratio falls like one over the label rather than tending to a constant[P10R60P10R59].

*And the coefficients that ratio is built from are written down, the first of them not being recoupling data at all.* The vertex's own weight factorises into a pointwise part and a recoupling part, and the pointwise part is the cubic term of the substrate's curvature integrand on transverse-traceless jets—the third-order companion of the second-order identity above. On such jets it is

$$\big[\varepsilon^3\big]R^{(3)}\big[e^{\varepsilon H}\big] = \tfrac{1}{6}\,\mathrm{tr}\,h^3+\tfrac{1}{6}h_{ab}(e_a h_{cd})(e_b h_{cd})-\tfrac{1}{6}h_{ab}(e_c h_{ad})(e_c h_{bd})+\tfrac{1}{4}h_{ab}(e_c h_{ad})(e_d h_{bc})+\tfrac{1}{6}h_{ab}h_{cd}(e_a e_c h_{bd}), \tag{16}$$

obtained on twelve jets and confirmed on six drawn independently of them. *The basis closes at rank six of nine, and the deficiency is exactly three identities that hold on transverse-traceless jets*—one structure vanishing by tracelessness, one relating the derivative-free invariant to the second-derivative structures, and one coinciding pair—*so Eq.* (16) *is a representative of a three-parameter family and is reported as one*, every member returning the same value on every jet. *Two facts about its shape are derived rather than assumed*: no parity-odd cubic structure can exist at all, three indices on the symbol and two on the derivatives against six on the tensors being eleven, which cannot pair; and at frame-constant $h$ every derivative structure vanishes and the identity collapses to the single cubic invariant the constant multiplet carries.

*The recoupling part is then exact, and it is a function of the level rather than a constant.* The degeneracy-summed square of a same-level triple overlap is, in each channel, the product of the three right dimensions with one squared $9j$ symbol—reached by summing over every right label, projection and Clebsch–Gordan coefficient before any $9j$ is named. Summed over the admissible channels it runs $126/125$ at $m = 3$, $444/1715$ at $m = 5$, $1441/7875$ at $m = 7$, strictly decreasing to the limit $81/640$; *the variation is of order $m^{-2}$ about a finite non-zero value, so it moves the constant and leaves every exponent where it was. And the channel count arrives out of the same structure*: the eight assignments of three harmonics to the level's two irreducible parts, six of them failing the triangle rule at $m = 3$ and $m = 5$ and all eight admissible and non-vanishing from $m = 7$ upward—the count read above as a selection rule, recovered here as arithmetic on a different object.

*Which derivatives the identity carries is the question the assembly turns on, and the jet variables themselves answer it.* They are the covariant gradient and its second covariant derivative, not the frame derivative, and the difference is not a convention: assembled with the frame derivative the level's recoupling sum *depends on which member of the three-parameter family is used*—$120575/6$ against $19775/6$ at the lowest odd level—while with the covariant derivative it is *invariant in all three directions and at every level. And the reason the distinction is easy to miss is exhibited rather than asserted*: in the transversality condition the two connection terms cancel identically, one against the trace of the structure constants and the other against the perturbation's own symmetry, *so transversality cannot see which derivative it was written with*

*and the cubic vertex can.*

*The frame this is computed in is derived from the substrate rather than adopted.* Of the four combinations of invariant frame and generator sign, exactly one carries the structure constants the construction already banks; it is orthonormal for the unit-radius round metric in these coordinates and returns the banked volume $2\pi^2$—*two independent ties to the same geometry.* On it the frame derivative acts on a harmonic's right index by a *constant* spin-one generator, in each of the three directions, and the two sides are inequivalent everywhere, so the banked structure constants are what force the choice. *Transversality in that frame is one explicit linear condition, and the transverse-traceless content is its kernel*: dimension $4j + 2$ at left spin $j$, with right Casimir $j'(j' + 1)$ for $j' = j \pm 2$—*the two-part multiplet read above as a selection rule, arriving here as the kernel of a map. The amplitude is proportional to a single recoupling symbol on the right labels at every admissible triple of a channel*, so each channel's degeneracy sum is one constant with no basis or phase convention entering, *and the two mirror channels contribute exactly equally at every level*—a structure the assembly did not put in.

*The split into a derivative-free part and derivative parts is then not a property of the vertex.* On transverse-traceless jets the family's own null direction expresses $\operatorname{tr} h^3$ in the second-derivative structures, *so which part carries what moves with the representative and only the total has a degree in the label. That total is of the seventh degree against the target's eighth, so the ratio falls like one over the label*, the label times it settling between 11 and 13 from the second odd level upward where the lowest carries 24—*which is this section's first statement about the tower's tail rather than about one level of it.*

*What the comparison then needs is its own absolute scale, and that is a derivation rather than a convention.* Assembled through the per-mode Lagrangian the section already uses, the criterion reduces to a statement in which the gravitational coupling and the scale factor *cancel exactly*, and its left-hand side is the quartic target identically at every label—*a closure no fitted factor could produce.* Every ingredient is read off oscillator matrix elements rather than adopted: the second and fourth vacuum moments fixing the scale, the three-quantum amplitude, the polarisation factor checked on random symmetric tensors at three and four modes, and the multi-mode three-quantum norm. *The volume enters by a calibration and not by a choice*: rebuilding the second-order level sum from its own pointwise identity together with the gradient closure returns the banked value *with no volume in it at all*, while a triple product of orthonormal harmonics carries one inverse square root of the volume—*and that asymmetry is the whole content of the chain. So the criterion is a pure number*: the vertex's level-summed square against the quartic target must stay below $18V = 36\pi^2$ on the substrate's own volume $2\pi^2$.

*Against that threshold the residue is empty, and the sign is positive on the whole tower.* The largest of the computed ratios is $175/22$ at the lowest odd label, *below the threshold by a factor above forty-four*, and every higher level falls further below it—to a factor of 131 at the next, 206, 277, 346 and 414 at the ones after. *So the quartic term dominates the cubic at every odd level, and at every even level the coupling vanishes identically by the selection rule above*: the sign is *positive everywhere and by a wide margin. That is the object this section has been carrying since the mode sums were first named, and the answer is uniform.*

*What a uniform sign then does to the sum itself is the opposite of what its steadiness suggests, and the summand is what settles it.* The quartic energy a level contributes is $(5m^6 - 41m^4 + 88m^2 - 16)/\big(15(m^2 - 1)\big)$ in units of the gravitational coupling times $\hbar^2$ over the cube of the scale factor—*positive at every level and growing as the fourth power of the label*—and because the criterion's ratio stays strictly between zero and one everywhere, the net a level contributes is bounded *below* by thirty-nine fortieths of that. *So the partial sums over the tower are bounded below by a divergent series, and the mode sum has no finite value. The decay established above is the ratio's and not the summand's*, and the ratio's falling to zero makes each level's net *closer* to the divergent quartic term rather than smaller. *And a sign that does not alternate forecloses*

*the one route to a finite value that needs no regulator*: conditional convergence is available only to a series whose terms change sign, so uniformity settles the sign and closes that route in the same stroke. *This is the first statement this section has about the summand rather than about a ratio of two of them*, and what it establishes is that the definition the section owes must come from a regulator.

*And the construction's own regulator reaches it, which is settled by exhibiting the continuation rather than by an estimate.* The regulator is the one the free tower already uses—the spectral sum over the tower's own frequencies, evaluated by the same terminating expansion and calibrated against that tower's banked value before being applied to anything new. *In the regulator's own variable the interacting summand is exact*: three polynomial terms and one simple pole, whose pole sits at the label 1 and so outside a tower beginning at 3. *So the fourth-power growth is not the obstruction the question anticipated.* What is new against the free case is that the summand is a *rational* function, so its large-label expansion does not terminate—*but that whole non-terminating tail is the single pole*, which the regulator sums by telescoping rather than by continuing it, and which converges absolutely at the point in question. *The continuation therefore exists, each of its four pieces separately finite*, and it returns the exact rational $-\frac{4}{3}$.

*What it costs is one subtraction more than the free case, and the cost is counted rather than asserted.* The subtractions a spectral regularisation spends are its summand's non-negative powers of the label: the free summand carries two, and this one carries three. *The extra one is the constant term*, which the free summand cannot have at all, being odd in the label to every order. *So the interacting sum is defined, and the count of three against two is a fact about that summand: a count is not a charge on the ledger*—what a ledger entry answers to is a pole, as the free tower's own does, whose counterterm the section has already identified with the curvature-squared invariants, *and which of these subtractions answers to one the next paragraphs settle rather than the count. What is not claimed at this point is the renormalised value*: the count is three against two, and the count is the claim. *And the extra subtraction is not a separate invariant but a separate power of the label*: the whole per-level energy is the gravitational coupling times $\hbar^2$ over the cube of the scale factor, times a function of the label alone, *so all three subtractions sit at one power of the scale factor and it is the label's power that distinguishes them*[P10R68P10R67P10R64P10R63P10R62].

*Which dimension that subtraction sits at is then forced rather than chosen, and three routes fix it at the sixth.* The interacting quartic energy is the Planck length over the scale factor, squared, times the free tower's own scale, so it enters at second order—*and the section's own order rule, that dimension $2k$ contributes at order $2k-4$, admits one dimension there and no other. The power of the scale factor gives the same answer independently*: an energy of this kind sits at $a^{3-2n}$, so the free tower's $a^{-1}$ is $n=2$, which is why its counterterm is the curvature-squared one, and this sum's $a^{-3}$ is $n=3$—*the rule calibrated on the free case's known dimension before being used on the new one. And the section had already placed it*: the density above falls as the sixth inverse power of the scale factor, at the operator dimension the mode count and the dimensional bookkeeping put it, *and what was missing was only the link from the summand's constant term to that entry.*

*It is not a volume term, and that settles which way the ledger reads.* A volume term is the cosmological one, which sits six powers of the scale factor away on the other side, *so the subtraction is a new entry and not one the ledger already carries. And on the class this sector works on it is not degenerate with the free tower's either*, which is computed rather than inherited: degeneracy would require one functional to be a multiple of the other, and $a^{-1}$ and $a^{-3}$ are distinct monomials once the scale factor is itself quantised. *The same test separates the two regimes rather than always returning independence*—at fixed background every invariant is a multiple of every other and the two *are* degenerate, which is this section's own one-dimensional statement, *and the asymmetry between the dimensions is the section's own as well*: at dimension

four the collapse rests on an identity pointwise in the scale factor and so descends to the quantised sector, where at dimension six there is no such identity to descend[P10R69].

*The quartic's entry at that dimension is empty, and what settles it is a parity rather than a cancellation.* What exposes the ledger is not a count of subtractions—*those a cutoff trades for the analytic continuation rather than pays*—but a *pole*, which is why the free tower's own counterterm is tied to a logarithm. The spectral function here has poles only at half-integer argument, the underlying zeta's single pole being reached only through an *odd* power of the frequency; *the free weight carries exactly one such power and meets a pole, while the quartic summand regularised above is a rational function of the frequency* squared—*even in the frequency—and misses every one.* At the physical point all four of its arguments are integers, where the spectral function is regular and each term's residue vanishes separately. *So for that summand there is no pole, no logarithm, no scale dependence, and nothing for a counterterm to absorb*: its coefficient is *zero.*

*And the instrument that returns that zero is calibrated on the case whose answer is known, and shown to discriminate.* The same residue formula applied to the free tower's own weight at its own physical point returns that tower's banked logarithmic coefficient exactly, rebuilt from the expansion's coefficients and nothing else; and a hypothetical odd power of the frequency *does* produce a pole, *so the zero is a property of the summand's parity and not of the instrument. A zero coefficient is zero for every scalar at that dimension*, so *this* contribution is well defined without any choice of representative—and that convenience is the quartic's alone, the cubic's non-zero coefficient below requiring one. *And the count of three subtractions is a fact about the summand rather than a charge on the ledger*, a cutoff's bookkeeping traded for the continuation rather than paid[P10R70].

*That parity is not an accident of the assembly it was read off, and its shape is what fixes how far the conclusion reaches.* A contribution built from $N$ vertices of total degree $P$ carries the frequency to the power $-P/2-(N-1)$, and with this order's own bookkeeping, $P=j+2N$ at order $j$, that power is $1-j/2-2N$: *the entire dependence on the number of vertices is $-2N$, which is even, so the parity is a function of the order alone and not of how the order is assembled.* Four different assemblies of the fourth order—a single sextic, two quartics, a cubic with a quintic, and four cubics—return the same parity, each computed rather than argued, and the rule reproduces the case whose answer is already banked, $j=0$ giving an odd power and with it the free tower's logarithm. *That power alternates with $j/2$*—odd at zeroth order, even at second, odd again at fourth—and the order rule inverts to $2k=j+4$ with exactly one dimension admitted per order, *so each even order answers to one dimension and the assembly cannot be what decides it. What the counting therefore delivers is not the answer but its location*: the total parity is the assembly's power together with the vertex coefficients' own, and since the first is fixed by the order, *everything turns on the second, one order at a time. And the question was posable without the higher orders being constructed at all*: the assembly's half follows from the vertex counting and the perturbative scaling, not from the value of any higher vertex[P10R71].

*The two halves of second order carry opposite parities, and the one that decides the entry is the cubic's.* The counting turns on whether a vertex coefficient is even in the frequency, and the quartic's is: the summand regularised above is a function of the squared label alone. *The cubic's is odd, and settling it settles the entry.* Its recoupling sum closes in two orbits, each internally degenerate and each exact—the extreme pair at $9(m^2-1)(9m^2-25)(9m^2-1)^3/65536$ and the mixed sextet at $(m^2-25)(m^2-9)(m^2-1)^2(9m^2-1)/65536$, both over $m^3$—so the sum is an *even* polynomial of degree ten divided by $m^3$, and is therefore odd in the label. *One form carries every level with no case split*: the two-channel levels are not a second regime, $(m^2-9)(m^2-25)$ vanishing at exactly the two lowest of them.

*So the sum meets the pole the quartic misses, and at dimension six the ledger pays.* The assembly's remaining factor is an integer power of the squared frequency, so the parity rides entirely on the recoupling sum; the summand's powers of the label are all odd, the continued

sum therefore reaches the underlying zeta at odd integer arguments, and one of those is its single pole. The coefficient of the inverse label is $-15405/8192$—*not zero*—giving a logarithm and with it a required counterterm, of residue $-15405/32768$ over the cubic's own odd levels and $-15405/16384$ over the tower. *And the instrument is discriminating because it recovers the other half's answer as a control rather than citing it*: run on the quartic summand, whose powers of the label are even throughout, the same construction returns residue zero. *What the two halves therefore exhibit is one selection rule read in its two parities*, and second order carries a pole contributed by the cubic and not by the quartic[P10R72].

*And the parity the pole turns on is proved rather than read off those levels, by a symmetry of the channel list itself.* The reflection that sends the label to its negative is the Casimir-preserving involution applied to both of the vertex's labels, and *it exchanges them*; the channel list is closed under the relabelling that induces, and so is each orbit, the swap carrying the count of one label to three minus itself so that the two orbits are exactly its blocks. *The recoupling coefficient is invariant under it and the degeneracy factor is odd*—each dimension being the label displaced by two, which the reflection negates—so each channel's term at the reflected label is minus the swapped channel's term, *and a sum over an orbit closed under the swap is odd in the label.* The result holds with the orbit coefficients left as free symbols: *it needs the orbit's closure and not the equality of the channels within it*, so the mirror coincidence noticed earlier—two channels contributing exactly equally, which the assembly did not put in—is this reflection, with the equality its stronger half. *So the parity holds at every level rather than at the nine computed ones, and the pole no longer rests on a fit*[P10R74].

*One scope belongs with this and it is the closed forms themselves, which the same symmetry does not reach.* They are established on nine levels, six fitted and three held out, with a discriminating control—an odd ansatz of the same coefficient count fails on those same held-out levels—*and they are not proved from the recoupling algebra. Two routes to proving them were taken and neither closed*: an a-priori degree bound needs the vertex contraction evaluated, which is the work it was meant to avoid, and the factored shape's rewrite into products of consecutive integers is an exact identity that yielded no derivation. *What the parity needs is proved and what the forms would add is warrant rather than content*, the residue's value being unchanged either way. The pole's *existence* needs only that the remaining factor carry an integer power of the squared frequency; the residue's *value* rests further on the assembly. *What is not claimed at this dimension is the renormalised value*, the coefficient being what is computed here and not the constant a scheme assigns it.

*That assembly is derived rather than carried, and deriving it moves its coefficient while leaving the form the parity needs.* For a totally symmetric cubic on degenerate modes the exact second-order shift carries *two* invariants rather than one—the squared vertex and the square of its own trace, at weights two and nine—read off oscillator matrix elements, calibrated against the textbook single-mode result before the unknown was touched, verified on a generic two-mode vertex with four free coefficients, and separated by a traceless three-mode control that leaves only the first. *And the trace channel is absent here at every level, by a count rather than an assumption*: the pair contraction vanishes unless the free label is zero, and the same identity on the remaining indices makes a trace require a full singlet, which no channel at any level is. *So the assembly is the squared vertex alone*—and because both invariants carry the same power of the frequency, an integer power of the squared frequency, *the parity above is untouched and the cubic summand still reaches the pole. The turning-point criterion is untouched as well*, its own overall factor cancelling in the ratio of two energies that the criterion is[P10R73].

*And a non-zero coefficient asks a question a zero one does not, which this background cannot answer.* While the coefficient vanished, the entry was well defined for every scalar at that dimension and no representative had to be chosen; *a coefficient that does not vanish is the coefficient of something, and this section names the invariant at dimension four and not at dimension six. What blocks the naming is the same degeneracy that made the count easy.*

The five dimension-six scalars are pointwise independent on a general scale factor, and on the one-parameter family two of them vanish while the rest are pure numbers—so matching the residue there is one equation on three surviving directions, *and its solution set carries two free coefficients*. The degeneracy is exhibited at its sharpest: *each of the three alone reproduces the residue after a single rescaling*, so the matching does not narrow the representative to a subset, let alone to one. *A rank of one makes representatives indistinguishable rather than forced*, and naming the invariant therefore needs a datum this background does not carry—one that separates those three. *What the family does decide is the small affirmative half and it is kept*: the two derivative scalars vanish on it identically, so neither is the sole carrier. *And the count is a lower bound rather than an evaluation*, read off a chosen five-element subset, which is the direction this statement needs—completing the basis can only raise the pointwise rank and so only deepen the degeneracy[P10R73].

*And the two halves settle a question about the regulator itself, which is worth stating because it is a choice the section has been making silently.* Continued in the squared frequency, the cubic's inverse cube of the label expands into half-integer powers, so it meets the spectral function's poles at *infinitely many* arguments, one per order of that expansion, and the residue there is an infinite sum; continued in the label it is a single term. *Continued in the squared frequency the quartic's is likewise the single term, its powers being integers there.* So the two continuations agree on whether there is a pole and differ in how much work it takes to read its residue, *and which one is the right instrument is decided by the summand's parity rather than by the passage's habit*[P10R72].

*The cubic carries the non-resonance the identity above requires as an arithmetic rather than an assumption*—its one-quantum vacuum amplitude cancels identically, the two channels being equal and opposite, so it reaches the vacuum only at three quanta. *And the energy density it implies falls as the sixth inverse power of the scale factor*, which is the operator dimension the mode count and the dimensional bookkeeping already put it at, arriving here from the vertices themselves[P10R50].

*The construction's own second order is reached at this level, and reaching it is what turns the rank there from a bound into an equality.* The reduction by itself has no second-order datum: its three eigenvalue logarithms satisfy exactly one relation, so the geometry is exhausted by the scale factor and the two shear parameters, and where the first-order system determines its one unknown uniquely the second-order one returns a *family*, the free state parameter still in the answer. *What supplies the datum is the interacting ground state, which at this level is a finite object.* The cubic's first-order correction to the field variance vanishes identically by parity—it reaches only odd states where the variance reaches even ones—while the quartic's is twice its coupling and linear in it, so the interacting variance departs from the free one at *first* order in the coupling. *And it is resolved rather than inferred*: the truncated two-mode Hamiltonian's lowest eigenvalue tracks that prediction at three couplings with the residual falling as the next order, and the state's own variance departs from the free value by more than twice its agreement with the prediction. *The source's power is then computed rather than chosen*—restored to physical units the correction carries a second-order energy density falling as the sixth inverse power of the scale factor, the same operator dimension the vertices already gave.

*With that source in place the rank at second order is still two, and the reason is this section's own gradient result.* Across every realizable invariant the source's second-order contribution is exactly twice the first-order pattern with the opposite sign—*the same direction in invariant space*—so it adds a function and not a direction. *So the bound the iterated deformation gives is attained*: the second-order rank at this level is an equality and nothing is missing from above[P10R51]. *The scope belongs in the same sentence*: the equality is at the level whose vertices are exact and whose state is finite-dimensional, and the state is the perturbative one—a cubic potential being unbounded below, the diagonalisation means what it says exactly where it agrees with the expansion, which is what the three couplings establish.

*The quartic's algebraic part is settled at every level, and by the level this section owns.* On a symmetric three-by-three matrix the difference between the two algebraic quartic structures—the fourth trace and half the square of the second—factorises with the matrix's own trace as a factor, so it vanishes wherever that trace does. *That is an identity in the entries and refers to nothing else*: it holds at every point of every field, however the field depends on position. *And the tower it applies to is the transverse-traceless one*, whose perturbation is traceless at every point and whose tracelessness is linear, so the collapse holds on any superposition, within a level or across levels—*which also fixes where the tower starts, the label's floor being where the transverse-traceless degeneracy first becomes positive rather than a convention. The identity does not run backwards*: a vanishing trace is sufficient and not necessary, the remaining factor having its own roots, so tracelessness here comes from the construction and not from the identity. *With the algebraic quartics spanned by one structure, the frame-constant family sweeps the invariants independently and determines that structure's coefficient outright*, at a single rational for the whole tower, validated against the full reduction on two constant directions and exact at every order including the cubic that one of them cannot see.

*What is not computed is therefore the quartic's* derivative *sector, which the frame-constant level cannot see because its harmonics have no gradients—and it now has a size rather than a name.* Enumerated at a point over the admissible one-jet of a transverse-traceless field, with the pairs tracelessness and transversality forbid excluded, the full contractions of two derivatives with two fields span *eight* independent invariants. A level supplies one relation among their integrals, exhibited on the harmonic rather than assumed, and the only other candidate total derivative reduces to that same relation by the identity above—*so at most seven survive as integrated invariants*, eight being the exact pointwise count and seven an upper bound with one relation exhibited rather than the absence of others proved[P10R52]. *What a level can determine is far less than either number, and by the same argument that made the level-summed overlap exact.* The isometry group acts transitively, so a sum over a complete basis of a level of any *bilinear* in the harmonics has the same coincidence limit at every point; and that constant tensor is invariant under the stabiliser of the point, which for the three-sphere leaves only the metric and the volume form to build it from. *The three bilinears a level defines are accordingly tiny*: the undifferentiated pair is one-dimensional and fully determined, being the transverse-traceless projector normalised by the degeneracy; the mixed one is one-dimensional and *orientation-odd*, forced to be so because the first is constant and the volume form is the only odd invariant available; and the doubly-differentiated one is two-dimensional with one direction fixed by the eigenvalue equation, so it carries a single free number.

*Over the whole set of contractions the level-summed values therefore have rank three, and then parity takes one more.* A level is the sum of two helicity representations of *equal* dimension, identically in the label, so the orientation-odd bilinear cancels over a full level and the rank falls to two—*of which one is fixed by the level's own labels. So a level carries one unknown number and not seven. The scope belongs in the sentence*: that is the rank of the *level-summed values*, so it bounds what a level-summed measurement can determine and leaves the covariant basis at its eight pointwise structures. *And one such reduction is already in hand*, the second anchor carried to fourth order returning the derivative sector as exact rationals at second and fourth order against a control that returns exactly zero at the floor at every order, *where the gradients are absent.*

*That one unknown number is not free either, and what closes it is a second eigenvalue the harmonics carry.* Computed from the connection in coordinates, *the harmonics are eigenstates of the curl* as well as of the Laplacian, with the two eigenvalues related so that the curl's is the label itself and its sign is the helicity—checked at a level whose frame components vary and at one whose components are constant, so it is not an artefact of either case. *Two relations follow.* The odd pairing of a curl with an undifferentiated harmonic cancels over a full level because the two helicities have equal dimension, which *derives* the parity step above rather than arguing it;

and the pairing of two curls is helicity-blind and is an even contraction of the two-derivative sum with two volume forms, *so it solves for exactly the number the eigenvalue trace had left free.* The fully determined sum satisfies both relations identically, component by component.

*So both level-summed coordinates are closed functions of the label alone, with no free constant in either*—and both vanish at the label below the tower's floor and not at the floor itself, *the degeneracy's own starting point reappearing in the level-summed values rather than being imposed on them. Which settles how far a level-summed route can reach, and the answer is not far*: the span is two-dimensional at every level and both of its coordinates are functions of the label, so the coordinate pairs of any number of levels span the same two dimensions and a further level returns a point in that plane. *At most two combinations of the covariant coefficients are determinable from levels, however many levels are spent—so the covariant expansion is not the cheaper route but the only one that reaches them*, and it is now a route to level-independent numbers whose tower sums are already closed in the label. *The scope is again in the sentence*: what that argument closes is the label dependence, the coefficients themselves being level-independent numbers that no level-summed measurement separates more than two combinations of—*and they are reached instead by the expansion below*[P10R54].

*The coefficients are closed too, as exact rationals, by a route that takes no measurement at all.* In the left-invariant frame the structure constants are constant—computed from the frame fields rather than assumed—so the metric components are the identity, no index is raised, and the quartic coefficient of the curvature functional under an exponential deformation is a universal polynomial in the deformation and its first two derivatives with rational coefficients. *So the coefficients are the solution of a linear system over the rationals on exact jets*, with no configuration, no integral and no level entering it.

*The basis closes rather than opening.* Enumerated on transverse-traceless jets, the two-derivative sector admits 372 admissible contractions of pointwise rank 8, the second-derivative sector 672 of rank 4, and the purely algebraic sector 60 of rank 1; their union has rank 12, and the 945 available total derivatives span 5 directions, all of them inside that union. *So the integrated space has dimension exactly seven*—an equality where the count above gave only an upper bound—*and the second-derivative sector is entirely removable by parts*, the five divergences taking all four of its directions beyond the two-derivative sector and exactly one of that sector's eight, which is what the drop from eight to seven always was. *Dropping transversality alone raises the two-derivative sector's rank from eight to eleven, so the reduction is the constraints and not the enumeration.*

*And two structural facts came out of the enumeration rather than being put into it. The algebraic quartic is not a separate unknown*: the union's rank does not rise when the algebraic sector is adjoined, because the curvature commutator fixes the antisymmetric part of the deformation's second derivative to the deformation itself—*so the natural split into one algebraic coefficient and a set of derivative ones is not a split this geometry has.* And the integrated space is reached by the two-derivative sector alone.

*The seven coefficients are solved on twelve jets and then verified pointwise on every jet not used to solve them*, and they carry a single denominator with seven different numerators:

$$
\begin{aligned}
-5/48 &\quad \nabla_a h_{bc}\,\nabla_a h_{bc}\,h_{de}h_{de}\\
+1/8 &\quad \nabla_a h_{bc}\,\nabla_a h_{bd}\,h_{ce}h_{de}\\
+1/24 &\quad \nabla_a h_{bc}\,\nabla_a h_{de}\,h_{bc}h_{de}\\
+19/192 &\quad \nabla_a h_{bc}\,\nabla_b h_{ad}\,h_{ce}h_{de}\\
+7/48 &\quad \nabla_a h_{bc}\,\nabla_d h_{ab}\,h_{ce}h_{de}\\
-53/192 &\quad \nabla_a h_{bc}\,\nabla_d h_{ae}\,h_{bc}h_{de}\\
+1/8 &\quad \nabla_a h_{bc}\,\nabla_d h_{be}\,h_{ac}h_{de}
\end{aligned}
$$

*Three independent anchors return.* Twelve frame-constant configurations reproduce the algebraic quartic exactly *through the covariant basis, where the derivative terms do not vanish*, so

it is not the same computation restated; the position-dependent harmonic that anchors the second order returns its known fourth-order value from a pointwise identity fitted on random rational jets; and a value computed from the identity in advance for a superposition matches that configuration's own curvature integral, computed afterwards.

*And this extends the cap above rather than escaping it, which is what says why the route had to be this one.* The per-configuration route caps too, and lower: the whole frame-constant sector has measurement rank *one* over any number of configurations, and six transverse-traceless configurations spanning both available levels reach rank *four—against seven unknowns. So no route that measures configurations reaches these coefficients at any expense*, and what makes the expansion reach them is that it is an identity and never takes a measurement. *The scope is in the sentence*: what is closed is the coefficients of the quartic two-derivative sector in a named basis[P10R55].

*Put through the level reduction, those seven contractions close both orders in the tower's own two variables and in nothing else.* Every one of their level-summed values lies in the three invariant structures above, and the combination the seven rationals pick out uses both surviving coordinates with non-zero weight—*so it needs the whole two-dimensional level object rather than collapsing onto one of its coordinates*, and the weight standing on the orientation-odd structure is non-zero too, *so the parity argument that set that structure to zero is load-bearing here and not decorative*: without it the level object would carry a third number and the rank above would not be the operative count for this combination. *Reading the second order required its own pointwise identity*, which the same machinery supplies—the second-order coefficient of the curvature is a fixed combination of the deformation's squared gradient and its square, *with the mixed contraction's coefficient exactly zero*—solved on jets, verified on six held out, and checked against the frame-constant value and the harmonic's second-order anchor, neither of which it was fitted to. *With that, both level sums are closed in the degeneracy and the frequency alone*:

$$\begin{array}{rl} \text{second order} & -\frac{1}{4}\, d\, \mu^2 \\ \text{fourth order} & -\frac{1}{120}\, d^2\, (5\mu^2 + 4) \end{array}$$

*No third quantity appears in either, both vanish at the label below the tower's floor and neither at the floor itself*—the degeneracy's own starting point arriving in the values a third time—*and both are strictly negative at every level, so the fourth-order level sum carries the sign of the second-order one everywhere. The scope is the integrand*: that is the sign of the level-summed coefficients of this curvature functional, and not the sign of a back-reaction.

*And the tower sum factorises, which is the structural fact that makes the assembly a product rather than a double sum.* Taking the two factors from different levels, every monomial is of degree one in each level's numbers, because every contraction pairs a harmonic with itself—*so the sum over the tower is a product of single sums and not an irreducible double one*, and setting the second level equal to the first returns the level object above exactly, the diagonal rather than a different object. *Summed unweighted to a cutoff the two orders differ by two powers of it*, which is the same gap the free tower's quartic divergence and the order-$\lambda^2$ sum already showed. *And the physical weighting is carried, with the gap surviving it untouched.* The free vacuum puts one inverse frequency on each propagator, and that factor has no closed-form sum, so the leading terms are bracketed rather than fitted—between the label less one and the label, compared on their squares so that no root is ever compared, both bounds being exactly summable. *Both return the same leading power and the same leading coefficient in every sum involved*: the weighted second order at the fourth power of the cutoff and the weighted fourth order at the sixth, *so the gap is two powers weighted where it was two unweighted, and the weighting moves both ends alike. And a control arrives that was not designed*: the weighted second order lands on the same power as the free tower's quartic divergence, read independently and earlier, *so the weighting is the tower's own and not one chosen here*—with both weighted leading coefficients negative, as every level's value is.

*What is left is one passage, and naming which one is the result.* Carrying a level sum to the vertex coefficient the criterion above uses takes four steps: the level sum to a per-mode coefficient; the pairing combinatorics, which the level sum has already performed and which must not be applied twice; *the passage from the action to the Hamiltonian, which carries a sign*; and the mode normalisation with its powers of the scale factor. *The third is where a sign can turn, and it is written out here.* With the lapse unity, the deformation traceless so that the determinant and hence the measure carry no dependence on it, and the kinetic scalar's second-order part computed rather than assumed—*it is a quarter of the squared velocity's trace, and positive*—the Legendre transform returns a potential that is the curvature integral with a single power of the scale factor and an overall minus. *The sign is then read off a frequency the construction already holds rather than chosen*: the resulting equation of motion carries the expansion's own friction and a frequency whose square is the label's, matching the tower's banked value, where the opposite choice would make every level of the tower unstable. *So the vertex coefficient is positive at every level*, which is the sign this construction's second-order computation already carried, *and the apparent disagreement between that and the negative level-summed coefficient closed above resolves with neither quantity having been wrong—the passage between them was simply never written. And the pairing count is spent rather than repeated*: the level object is built from three terms, and those are the three pairings of four factors, so applying the combinatorial factor again would treble the answer[P10R56][P10R57][P10R58].

*And the same argument says which object the sign is a property of, by showing that the obvious candidate is not one.* The level-summed cube of the shear is not invariant under an orthogonal recombination of a level's own basis—exhibited by rotating an orthonormal pair inside its span and watching the sum move—*so it is not a property of the level at all*, which is sharper than reading the vanishing third order as a property of one direction. *The squared coupling that the sign criterion actually uses is built from pairwise contractions and is invariant*, so the level-summed sign is a well-posed question and the object that answers it is well posed. *Its value is not claimed, and where the machinery stops is part of saying so*: that coupling is a sum over squared *triple* overlaps, pairwise in each index but with each pairing joining two different integration points, because a vertex is an integral of a product of three harmonics. *So it is an integral over two points of the level's bitensors where everything above used their coincidence limit*, and the argument that closed the count does not reach it[P10R53][P10R54].

*And the form holds at the order this construction operates at while the rank at the order beyond it is now an equality rather than a bound—so what carries an order attached is everything past the second, whose parameter is the cube of a number of order* $10^{-123}$*, and the binding scope is accordingly the level rather than the order. The distinction is kept because a definition that holds at one order of an expansion and is silent at the next is a definition with an order attached, however small the next order is. This row exists because a predecessor was struck on an inference of just that shape—a result true where it was computed, read as true where it was wanted—and a quantity being negligible is not the same as a scheme being specified.*

*The shape of that is worth stating plainly, because it is not an accident of this tower.* On a fixed background the one constant is unobservable *because* of a degeneracy—and the degeneracy is switched off by exactly the constant it was hiding. *Were there no logarithm, R would stay constant and the counterterm would stay invisible; but then there would be no constant to hide.* The saving mechanism and the thing it saves us from are the same number, so the sums cannot be carried into the coupled sector free of charge by that route: *the counterterm basis is one-dimensional at fixed background and two-dimensional once the scale factor is quantized. And it does not rest on the substrate scale being indefinite*, which is the reading the statement invites and which fails twice over. A superselection rule forbids *coherence* between sectors and not an *ensemble* over them, and the three integrals are read off the ensemble linearly, so an ensemble over distinct curvature radii spans three dimensions inside a superselected theory and with no anomaly at all; and the question has no object as put, since $\alpha$ is a coefficient of the physical

Hamiltonian rather than an observable of it, so the corresponding operator is a multiple of the identity with a one-point spectrum, and a central operator with one point decomposes nothing. *Neither route is the one taken here*: the rank is three already at a single curvature radius, which is where the $a^{-4}$ anomaly puts it, and the operator the Ricci scalar depends on commutes with nothing in the coupled algebra. *Centrality is a relation to an algebra*, and the theory's one genuine superselection label is the boundary operator—central in the radial algebra, labelling the boundary condition and not the background scale[P10R25]. *The back-reaction here is semiclassical and the interacting theory is not built*; what higher orders could do is narrow, since restoring the degeneracy would need the total trace exactly constant—a pure cosmological term—and an $a^{-4}$ anomaly is not one. *And the mode in which the constant is observable is fixed, because no state resolves the curvature it is read off.* At this order the excitation trace vanishes on the *diagonal* of the adiabatic basis, which is the harmonic virial theorem and not an operator identity: with $\omega_n = \mu_n/a$ the excitation energy is $\mathcal{S}/a$ with $\mathcal{S}$ free of $a$ *at fixed occupation*, so the pressure is a third of the density in every stationary state and in every mixture of them. *Written in the field operators themselves the trace carries two terms and not none*—$\pi_n^2/a^6$ and $\mu_n^2\varphi_n^2/a^2$, the second a potential-like term from the tower's own gradient energy, with $\mu_n^2 \geq 8$ for every mode so that it is neither zero nor small—and their combination is $a^{-4}(a_n^2 + a_n^{\dagger 2})$: *purely off-diagonal, non-zero on any squeezed state, and untouched by normal ordering*, which has no contraction to take. *The conclusion the diagonal statement was carrying survives and no longer rests on it.* That combination is an *inverted* oscillator, and an inverted oscillator has no normalizable eigenvector at any real eigenvalue: in the momentum representation the null equation is solved by $\sqrt{p}\,J_{\pm 1/4}(p^2/2\mu a^2)$, both solutions regular at the origin and neither square-integrable at infinity, the modulus squared falling as $1/p$ with the envelope exponent holding at $-1/2$ exactly at every real eigenvalue. *So the curvature has no point spectrum, and it has none without any term being absent*—which is the stronger warrant, since it does not turn on an absence[P10R32]. *The spectrum is purely continuous and two-sided*: the same operator puts spectrum below $4\Lambda$ as well as above it.

*And the obstruction is the anomaly coefficient itself, which is this result's shape for the third time and from a third direction.* The commutator of the curvature with the momentum conjugate to the scale factor gives $\mathrm{Var}(R)\,\mathrm{Var}(p_a) \geq \frac{\hbar^2}{4}\langle R'\rangle^2$, saturated to a third of a per cent by narrow admissible states, and *the right-hand side vanishes exactly when the residue does*—which is the case in which the curvature is constant and there is nothing to resolve. *So the same number that makes the counterterm observable is what forbids any state from sharpening the curvature the counterterm is read off*, and the constant is observable *distributionally* rather than as an ordinary sharp-valued observable. That is unconditional rather than weakened: there are no special states to except, the ordinary reading being excluded by a theorem and not by a missing calculation. *One consistency there was not put in by hand*: the expectation of $a^{-4}$ converges at the origin only above a threshold in the boundary index, and the horizon's own thermal condition clears it on either ordering—*the boundary condition that closes the deficiency is the same one that makes the curvature's expectation exist*[P10R26]. *What the coupling is genuinely needed for is smaller than that, and it is smaller than the order of the promotion suggests.* At higher order the cubic and above put genuine tower operators into the trace, and neither the bound nor the level-set argument turns on their absence. *The commutator is an algebraic identity and not a dynamical one*: $[f(a)\otimes T, p_a] = \mathrm{i}\hbar f'(a)\otimes T$ for *any* tower operator $T$, commuting or not and ordered any way, since $p_a$ differentiates the scale-factor function alone and the tower factor rides through—so the inequality holds verbatim in form at every order, and its form is blind to the ordering. *And pointwise cancellation is excluded by the trace rather than found to be absent*: the trace of an energy term $h(a)\otimes T$ is $(h + ah')T/2\pi^2 a^3$ exactly, so a term $c\,a^{-n}\ln^m a$ contributes $2\pi^2\Theta = c\,a^{-(n+3)}\big[(1-n)\ln^m a + m\ln^{m-1} a\big]$, and the counterterm's own $a^{-4}$ is the member of that family at $n=1$, $m=1$. *The whole family is closed under the operation rather than annihilated by it, and closure in one direction only is what settles the question.*

From a term at $n = 1$ the highest power surviving in $\partial_a\Theta$ is $\ln^{m-1} a$ with coefficient exactly $-4mc$, while a term one logarithm higher reaches both $\ln^m a$ and $\ln^{m-1} a$: so the terms chain *downward* in the logarithm and never upward, the coefficient system over log powers is lower triangular with diagonal $-4m$, and $\partial_a\Theta \equiv 0$ therefore forces every coefficient to vanish. *The leading logarithm has no partner*, and that is why nothing cancels the residue at any $(n, m)$—the pure-power case, where $n = 1$ kills the coefficient outright, is this statement at $m = 0$. *What a tuned coefficient would do is relocate the obstruction rather than remove it*—at exactly twice the residue a second-logarithm term annihilates the constant coefficient the counterterm sits in and leaves the next logarithm up at sixteen times the residue, with the curvature's variance at that same state eleven decades above the measured floor[P10R28]—*but that relocation is a fact about the trace formula's closure and not about this theory, because the term it needs does not exist here.* The relocation must sit at the power of $a$ the counterterm sits at, since that is what lets it reach the constant coefficient at all. *And in the reduced theory the only scales are $a$ and the Planck length*, so the vacuum energy is $a^{-1}\sum_j f_j(\ell_P/a)^j$ against a polynomial in $\ln(a/\ell_P)$: the power is $1 + j$, one vertex pair costs $\ell_P^2$, and the interaction therefore enters at $j \geq 2$. *So the counterterm's own power is populated at $j = 0$ alone*—by the free tower, whose large-label weight $2(m^2 - 4)\sqrt{m^2 - 3}$ is a pure Laurent series and carries no logarithm of the label, so the pole there is simple and the logarithm is capped at the first power. *The second logarithm is therefore unavailable at every order of the coupling and not merely at the lowest*, and asking the two families to cancel each other instead has no solution, one coefficient being unable to annihilate two different powers of $a$ at once. *The cubic's own sum carries no logarithm either*: the three-harmonic overlap on the spatial section obeys an exact multiplet sum rule, weight $n_1n_2n_3$ on the triples satisfying the triangle inequality with odd perimeter, and every term of the soft expansion has the form $j(j^2 - 3)^p$—an odd polynomial terminating above the first power when $p$ is an integer, an even-power series otherwise—so the inverse first power, which is what a harmonic number would occupy, has coefficient exactly zero at every order[P10R29]. The level-set argument likewise survives on every fibre that exists: wherever the cubic's tower operators are simultaneously diagonal, the curvature is multiplication by $4\Lambda + \sum_j c_j a^{-n_j}$, whose derivative has no zero while $r > 0$, whose level sets clear to a polynomial of degree six and are therefore finite, and a direct sum over such fibres manufactures no point spectrum from parts that have none[P10R27]. *A tuned coefficient can make the right-hand side vanish, and that is not sharpness*: the bound goes vacuous while the curvature's variance stays five decades above the measured floor, and two distinct powers of $a$ agree at isolated points rather than on a set of positive measure. *And the correlated case is a question about one operator family rather than about states.* Because the trace equation is *algebraic* in the matter trace—$R = 4\Lambda + \kappa\Theta$ carries no derivative of the scale factor—the curvature commutes with it exactly, so the curvature is a direct integral of $4\Lambda + \sum_k \kappa_k a^{-m_k} T_k$, one operator on the excitation factor at each scale. *A constant eigenvalue branch then forces its own value*: every power is negative, so the family tends to $4\Lambda$ and a branch constant on a set of positive measure can only be $4\Lambda$ itself—leaving the condition that $\sum_k \kappa_k a^{-m_k} T_k$ be singular for almost every scale. *At finite truncation that is exactly the singularity of the excitation operators' own pencil*, a condition carrying neither the scale nor the couplings, because with distinct powers no two monomials of the determinant can meet on the curve the scale traces; and the pencil is non-singular at every truncation computed, in exact arithmetic, for the two non-commuting operators the cubic supplies and for three at the powers where they collide[P10R30]. *The one place the truncation speaks for itself runs toward the affirmative rather than away from it*: at odd truncation a ladder matrix is singular and the coupling system does acquire solutions, but every one of them switches off at least two couplings, and the operators the tower actually carries have purely continuous spectrum and so no null vector at all. *And in the tower the criterion is not a limit of determinants at all but an ordinary differential equation*—one equation in the momentum representation, whose order is the number of powers of the field the trace carries. *Which powers it carries is not a free choice,*

*because the parametrisation of the spatial metric is not one*: a cubic term with two powers of momentum is present or absent according to how the shear is written, and only one way of writing it separates the shear from the volume. In the volume-preserving parametrisation the trace of the spatial metric's logarithmic derivative is the expansion alone, so the extrinsic-curvature invariant carries no shear–volume cross term; the volume element is the scale factor's own cube, so the cosmological term carries no shear at all; and both statements are exact rather than leading order. *There the momentum-weighted cubic vanishes identically*—the cubic term of the shear's kinetic trace is the trace of a commutator—*and it vanishes at every level rather than for a single mode*, the first kinetic correction being a quartic. The parametrisation linear in the perturbation gives the same term a non-zero value, and that is the whole of the difference between them[P10R36].

*So the cubic the construction carries is the spatial curvature's own, and it reaches the trace by an identity rather than a computation.* For a constant frame metric the scalar curvature is a closed form in the elementary symmetric functions of the perturbation, and about the round sphere it is $6 - 2\,\mathrm{tr}\,h^2 - \frac{10}{3}\,\mathrm{tr}\,h^3$ to that order, with the volume element contributing nothing and the first-order term absent because the first curvature perturbation of transverse traceless data vanishes on a maximally symmetric section. *And the cube's trace is three times the determinant for any traceless symmetric matrix, so the cubic term is a determinant rather than an integrand.* The closed form answers to three checks that share no machinery: a coordinate Ricci scalar at two metrics, its own covariance under rotations of the frame, and the standard anisotropy potential of the closed homogeneous cosmology, which it reproduces exactly through cubic order with both cubic terms included.

*At flat order that term does not reach the single-mode problem*: for one wave vector transversality is algebraic and makes that vector a null eigenvector of the polarisation, so the determinant—and with it the whole term—vanishes identically across the flat polarisation space, lifting only for two wave vectors together, *and what would put a third power of one excitation into the equation sits exactly on that degenerate set. On the closed section it does reach it, and the overlap is a determinant rather than an integral.* Transversality there is differential rather than algebraic, and the lowest transverse traceless harmonics—identified by their own eigenvalue and degeneracy as the tower's lowest level, and transverse across the whole five-parameter multiplet at once—have constant components in the left-invariant orthonormal frame. *So the mixed components are a similarity transform of those constants, the determinant is theirs pointwise, and the overlap is that determinant times the volume of the section, with no integration performed at all*—non-zero, and non-zero for the ordinary reason that a determinant is[P10R34]. *That the determinant is basis-dependent does not reach any ratio of two such overlaps*, and the reason is representation-theoretic rather than computational: the symmetric cube of the five-dimensional multiplet contains the trivial representation exactly once, so every cubic overlap at that level—whether it couples one excitation or three, and whether it comes from the kinetic or the potential side—is a multiple of one symmetric trilinear form, and in the ratio of two of them the form cancels and a pure number remains.

*So the operator content is three structures and the null equation is third order, with no first-order term in it.* At that order the operator is regular: the third power of the field is a third derivative with a constant coefficient, so there is no singular point anywhere on the line and the whole question is at the two ends; and the cubic needs no symmetrisation of its own, being a power of one symmetric operator and leaving an exact total derivative as its Lagrange difference. *The counting at the two ends is a genuinely different problem at odd order*—there the two deficiency indices need not agree, and a first-order operator on a half-line is the standing example of one that admits no self-adjoint extension at all—*but the mechanism that separates them needs the spectral parameter to change the asymptotic count between the two half-planes, and here it enters neither the characteristic exponent nor the prefactor.* At the dominant balance the three branches carry real parts $\pm\frac{\sqrt{3}}{2}|\beta|^{1/3}$ and zero, and on the marginal branch

the modulus is exactly $p^{-2/3}$, which is below the square-integrability threshold by exactly one sixth and is settled by an exact solution rather than an estimate of order. *Two of the three are square-integrable at each end, the far end being the same equation under one change of signs, so the indices are equal and a self-adjoint realisation exists*—a circle of them[P10R35]. And three coefficients, less one rescaling and one overall factor, leave a *single* dimensionless invariant, a fixed fractional power of the scale factor and strictly monotone in it.

*What a count like that is not is a property of the trace operator, and this is the place to say so.* A redefinition of the field by its own square moves the two cubic couplings against each other while generating quartics, so a cubic truncation is not closed under it and the ratio of the two couplings takes every value, both signs included. *What the truncation does fix is one combination of them, and that combination is non-zero*: the cubic is really present, it is the curvature's, and the equation really is of third order, while the ratio is a feature of where the expansion was cut and not a datum about the operator. *So the criterion at third order is a question about one real variable.* The rescaling identity is exact, so the whole family carries a single dimensionless invariant—a fixed fractional power of the scale factor times the squared frequency—and whether zero is an eigenvalue survives both the rescaling and the positive prefactor, while that power is a diffeomorphism of the half line, so a set of positive measure in the scale factor is one in the invariant.

*What it is not is a question about the operator family, and that is what makes it answerable.* No self-adjoint realisation of this operator has a domain independent of the invariant: the witness is the marginal branch itself, which the operator annihilates to square-integrable order while the second derivative carries it out of the space, and a self-adjoint domain must contain the deficiency subspaces. *On a boundary triple the analytic object is a finite Weyl function rather than the family.* Zero is an eigenvalue of a realisation precisely where the determinant of that realisation's parameter minus the Weyl function vanishes—a one-by-one object where the deficiency is one and three-by-three where it is three, hence one scalar equation in one real variable in either case—and the realisation is held fixed as a finite-dimensional datum while the domain is free to move[P10R38]. *And both halves of that hypothesis hold, for a structural reason rather than a technical one.* Transform the family and the third-order term becomes a *potential* while the multiplication operator becomes the *kinetic* one: the object is unitarily a second-order Schrödinger operator with a cubic potential, for which the Weyl function is a classical object—and no rescaling absorbs the transformation, because the scaling that normalises the cubic term is forced to the identity. *Real-analyticity then follows from the equation's own coefficients*: they are of degree one in the invariant and the boundary data do not depend on it, so the fundamental system is entire, and reality of the equation carries the analyticity to the real axis. *And that the function is not constant is proved rather than measured*: an exact two-solution identity gives its derivative in the invariant as an integral of a square over the half line divided by a square, hence *strictly* positive at every value—so there is no interval of constancy and no sub-family on which it is constant, and the half that could have failed for a boundary reason fails for no reason at all[P10R39].

*The same transformation explains why this route reaches where the other two could not, and the explanation is not that it is cleverer.* The obstruction that stopped them is representation-independent—the weighted integral over the whole line diverges in the transformed picture exactly as it did in the original—*but the half line is the opposite case*, and that is the entire point: there the recessive solution decays faster than any exponential and is dominated exactly, so the same weight is integrable. *The half-line solution object exists precisely where the whole-line operator object does not*, and no operator family, form domain or parameter-independent domain appears anywhere in the statement. *The dependence on the choice of boundary triple splits, and it splits the useful way*: non-constancy is independent of that choice unconditionally, a change of triple being a constant Möbius map of non-vanishing determinant, while analyticity on a given set is not, the same map having a pole a choice can move onto the real axis—so the

statement is proved for the triple constructed and transferred wherever the denominator does not vanish.

*So the criterion closes, and it closes more sharply than the second-order case did*: monotonicity gives *at most one* value of the invariant at which zero is an eigenvalue—a single scale factor rather than a discrete set, and outside the measured window none at all—so the condition cannot hold on a set of positive measure and the curvature has no eigenvector at this order either. *Two things bound that, and both are the construction's own.* The deficiency count, re-derived here from the endpoint thresholds, comes out at the lower of the two values a scalar second-order expression on a line admits, so the larger case is not reached rather than being disposed of. *And the whole statement is the criterion at the cubic truncation*, whose count is a property of that truncation and not of the trace operator—which is what the section says two paragraphs above, and is the reason this closure is a closure of the third-order question and not of the row.

*And there is a second place the degeneracy ends, at fixed background, which is the shear.* The deficit *is* $C^2$; for an anisotropic shear over an isotropic expansion the invariant is $C^2 = 2\sum_i(\dot\sigma_i + H\sigma_i)^2$ at second order in the amplitude[P10R4]—so it carries two derivatives more than the shear itself, and for a mode of frequency $\omega$ it grows as $\omega^2\sigma^2$, which is what makes it ultraviolet-sensitive at all; and the propagating content of this tower is the transverse-traceless shear. *So the tower's own shear is a calculation and not a question about meaning*—entering at second order in the mode amplitude, with the sub-leading heat-kernel coefficients as its instrument. *That calculation is run, and its answer is smaller than the question sounds*: for a transverse-traceless perturbation the first variation of the Ricci scalar vanishes identically and $\sqrt{g}$ is unchanged, so $R^2$ at second order in the amplitude is $2\bar R$ times $R$ at that order—*pointwise, with no integration by parts*—and $\int\sqrt{g}\,R^2$ is accordingly a multiple of the Einstein–Hilbert term rather than a new structure.

With the Gauss–Bonnet combination contributing no field equation, the three quadratic invariants leave $\int\sqrt{g}\,R^2$ and $\int\sqrt{g}\,C^2$, of which only the second is new: *the shear therefore costs exactly one new counterterm, and it is the Weyl-squared one*[9].

*And its coefficient follows from what the tower is, without a second-order spectrum.* Each mode is a harmonic oscillator of mass $a^3$ and frequency $\mu_n/a$, which is term for term the reduction of a minimally coupled massless scalar on this background, so the tower is *two* such degrees of freedom carrying the transverse-traceless spectrum. The $C^2$ coefficient is a property of the field content rather than of the background, which is why it is fixed without one: for a single real scalar the entire $C^2$ content of the second heat-kernel coefficient [6] sits in $2\,\mathrm{Riem}^2 - 2\,\mathrm{Ric}^2$—independently of the curvature coupling, which is why the answer needs none—and that combination is $3C^2 - E_4$, giving $1/120$ in units of $(4\pi)^{-2}$ and therefore $\frac{1}{60}$ for the tower[P10R8]: *exactly twice a real scalar's. The count is of physical modes, and that is the reading on which the number stands*: the constraints here are solved rather than gauge-fixed, so no ghost sector enters and no covariant spin-2 tabulation applies. The overall normalisation by which the coefficient enters a divergence is convention-dependent; the ratio is not.

*Two things about that entry are settled and belong beside it. First, the two helicities are not identified by any symmetry the layer connectedly carries*: on $S^3 = \mathrm{SU}(2)$ the connected isometry group is $(\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R)/\mathbb{Z}_2$, transverse-traceless rank two carries $|j_L - j_R| = 2$, and each level is $(j, j+2) \oplus (j+2, j)$ with $2j+1 = n-1$—whose dimensions sum to exactly the $2(n-1)(n+3)$ derived above[P10R5]. *They are inequivalent irreducibles, and the map exchanging them is the exchange of the two factors, an orientation-reversing isometry*—so a parity-odd term is not excluded by the geometry, and whatever excludes it does so through the state. *Second, a topological term contributes no field equation, so a coefficient multiplying one is not read off any dynamics*; its only remaining home is the horizon entropy, on which this construction is deliberately uncommitted [7]. *So whether such a term is an entry in this ledger is not a separate question from that one.*

*One caveat belongs in the statement rather than after it*: the dimension-four scalars also include the parity-odd Pontryagin density, which is not a curvature-squared invariant of the kind counted above and is a total derivative, but which can be non-zero at second order in the metric perturbation and reverses with the handedness—*so the zero is a property of the mode chosen and not of the geometry* [11]. *Which modes give the zero is a question about the configuration and not a general rule*: on a plane-wave pair the density goes with the relative phase of the two channels, vanishing where they are a quarter-cycle apart and largest where they are in step, so a prescription reaching for one handedness may be reaching for the case that vanishes.

*And what that scopes, first of all, is this computation.* A counterterm answers to a parity-odd divergence of the effective action, which is a property of the theory and the state; *it is not a property of a basis mode.* The two helicities span the same space in either polarisation basis, and on a parity-symmetric state the two contributions cancel whatever an individual basis element returns. *So a mode-level evaluation cannot fix what counterterm the theory needs, in either direction* — the caveat bounds its own instrument rather than deferring the question to a choice of polarisation.

*Two things would have to be settled before a parity-odd term could be scored at all, and they are independent. Whether the tower is chiral on the corpus's own criterion* — which is not that a reflection exists but that *no connected action completes it*, a condition applied to the radiating sector and not to this one. *And whether a total derivative is a ledger entry.* The density above contributes no field equation, so it is not a coupling in the sense the constant-counting audit means, and the ledger counts no counterterm coefficient [7].

The propagating sector itself is a clean, unitary, deparametrized evolution—a true Hamiltonian generating the graviton on the layer, in cosmic time.

*What the coupled sector establishes is one claim, and it is worth stating as one.* Once the scale factor is quantized, the graviton tower's renormalized zero point makes a single ultraviolet constant physically observable, and the construction has no route left by which to hide it again. The constant is carried by a residue of the spectral zeta function that is an exact functional of the free spectrum: no multiplicative renormalisation of the frequencies removes it, and the one mass-like displacement that does is precisely the curvature offset which makes those frequencies non-integer in the first place. The logarithm that residue produces has no partner to cancel against, the coefficient system over logarithmic powers being lower triangular with non-vanishing diagonal, so the family is closed downward and never upward. The interaction cannot reach the power of the scale factor at which the counterterm sits, because in the reduced theory that power is populated at zeroth order alone, by a free tower whose large-label weight is a pure Laurent series, and the cubic's own overlap sum carries no logarithm of the label. And the curvature operator that results has no eigenvector—not because it is a function of the scale factor alone, which it is not, the trace carrying the tower's momentum and its gradient energy as genuine operators, but because the operator it carries is an inverted oscillator, whose point spectrum is empty at every real eigenvalue and stays empty under normal ordering.

*And the trace carries a third power of the field as well, from the spatial curvature rather than the kinetic term, so that step is made twice and by different arguments.* At quadratic content the operator is an inverted oscillator with empty point spectrum. At third order it is regular, self-adjoint realisations exist, and the couplings run along a single monotone invariant—and there the eigenvector question is settled by the one object whose analyticity does not require a domain to be carried along that invariant, a finite Weyl function, which is real-analytic and strictly monotone. *The second closure is the sharper of the two*: strict monotonicity allows at most one value of the invariant rather than a discrete set, so the condition fails on a set of positive measure by a wider margin than the first argument gave. *Both are scoped by the same thing, and it is stated rather than implied*: the third-order statement is the criterion at the cubic truncation, whose deficiency count is a property of that truncation and not of the trace

operator. *So what remains open is one thing, and it is not a category*—and it is the completion of a definition rather than a verdict this construction could fail, the question it reduces to having been settled above in both of its arms. *The interacting tower's divergence structure does reach second order in the coupling*, and it does so by an identity rather than by a search that failed to find a reason for it to vanish. *What follows from that is the definition's incompleteness and not the ledger's refutation*: the affirmative branch is a second dimensionless number owed and not a second length introduced, so the single-scale ledger stands either way—an undetermined dimensionless number not being a scale—and that number is already counted, the ledger of dimensionless constants spent standing at one for these very sums, while what the branch adds is bounded by the background's own turnover and the counterterm it would need collapses on an exactly maximally symmetric substrate. *So the one open thing is the ultraviolet definition of the interacting tower's mode sums*—with the renormalised value at the paying dimension unclaimed and the invariant it is the coefficient of unnamed, the degeneracy on this background leaving the representative indistinguishable rather than forced. *So the two extension questions this sector meets pair, and the pairing is the clearest thing it has produced about its own shape*: the one at the origin of the scale factor is closed, without a free parameter, by the de Sitter horizon's own thermal state; the one at unbounded momentum is open, and it is not a new object standing beside the construction but the object the construction was opened to address.

# 7 Dissolution, not solution

The result of Sections 4–5 is best stated negatively. We have added nothing to the formalism. We have not introduced a new time variable into general relativity, nor a matter field whose role is to be a clock, nor a modification of the constraint algebra. We have identified, on external grounds, which foliation is physically real, and read the existing constraint on it. The problem of time, on this account, was never a problem internal to the formalism; it was the formalism faithfully reporting the consequences of an ontological premise—the block—that is both a category error and empirically false.

This is why the appropriate verdict is dissolution rather than solution, and the distinction is not merely rhetorical. A solution would add structure to make a defective formalism work. A dissolution removes a mistaken premise and finds the formalism was working all along. The four epistemic rules that govern such corrections [15, 27]—resist the naïve identification of appearance with reality; prefer a reading on which the observed structure is a structural necessity rather than a tuned permission; consolidate phenomena under one principle; treat accumulating patches as a red flag—each favour the CR reading over the frozen one. The cosmic time is a structural necessity of the ontology and an empirical datum, not a tuned addition. The reading consolidates: the correction that turns the frozen constraint into a true Hamiltonian refuses the very reification the completed-horizon inference also rests on, both inferences granting existence to a manifold of occurrences that no observer's present ever contains [4, 15]. *The two share a diagnosis and not a premise.* The completed horizon is withheld by general relativity's own causal structure, for every admissible exterior slicing and with no ontology among the hypotheses, so what this reading supplies there is the account of why the error was available rather than the result that removes it. *And the shared diagnosis is the cited paper's own rather than a reading imposed on it*: the causality paper names the existence–occurrence distinction as the one it turns on—that the horizon *occurs* without ever *existing* on a finite exterior slice—and records that it reaches that distinction from standard general relativity alone [4]. *The one move that would have called for an ontological refusal is not an exterior-time claim at all*: a defender of the completed horizon may grant that no finite exterior time contains it and hold instead that it exists on the maximally extended geometry. *At that one site the refusal would be weaker than what is already available*—the extension's stacked horizon points display the causal ordering of events the metric assigns no separation between, so there is nothing there for an ontology

to withhold existence from$^{\mathsf{P10R75}}$. One ontological correction, one canonical payoff, and one standing error of the literature it explains.

The reading also closes a loop with the dynamical content of the programme. The true Hamiltonian generates the advance of the layer $\mathcal{S}_t$; the geometry of that advance is what the operator and range papers read directly, where the matter content of a layer is the bend of its spatial cut off the vacuum profile and gravitational radiation is the transverse-traceless shear of the evolving layer, up to the wall at which a symmetric layer can no longer carry the wave and ordinary inhomogeneous evolution begins [9, 10]. The canonical face (a Hamiltonian that advances the layer) and the geometric face (matter and radiation as features of the advancing layer's intrinsic geometry) are two descriptions of the one evolving three-dimensional world. The cosmogenesis re-expansion is one such advance: the companion cosmogenesis paper [33] runs the collapse-and-re-expansion thermal history—the layer's advance along cosmic time, generated by this true Hamiltonian rather than frozen—through a nuclear network to the primordial light-element abundances. The companion foundation roots these faces in a single augmentation of general relativity [26]; the present paper supplies the canonical one, and with it the recognition that CR's distinctive contribution to quantum gravity is not a new dynamics but an absolute cosmic time—real, forced, and already implicit in the formalism—that turns the constraint into a generator. This dissolution is the canonical face of the substrate's maximal symmetry gathered in the geometric core [7]: the problem of time is identified in the constraint-algebroid construction as the base-dependence of the substrate's coset metric [5], and the selection of the forced foliation is what deparametrizes it away—the structural and the canonical sides of one dissolution.

# 8 Relation to existing approaches

The deparametrization carried out in Section 5 is, as a technique, entirely standard, and it is important to be clear about what is and is not new here. Kuchař's survey [19] classifies responses to the problem of time into those that locate an internal time before quantization, those that quantize first and seek time afterwards, and those that abandon time outright; Isham's review [20] maps the same terrain. The construction of Section 5 sits squarely in the first class, and its minisuperspace realization uses a dust clock of exactly the Brown–Kuchař type [21]. We claim no novelty in the mechanism.

What CR adds is not a mechanism but a warrant. The standard and well-known objection to internal-time approaches is the multiple-choice problem: many candidate internal times exist, inequivalent quantizations follow from different choices, and the formalism provides no principle to prefer one [19, 20]. This objection has force precisely when the choice of clock is a matter of formal convenience internal to the theory. It loses that force when the clock is fixed from outside the formalism by ontology and observation. CR's cosmic time is not selected because it makes the equations tractable; it is selected because the occurrence/existence distinction identifies the evolving layer as what exists, and because the CMB measures the corresponding cosmic rest frame (Section 2). The multiple-choice problem is thus not solved within the space of formal clocks but dissolved by the recognition that one clock is physically real and the others are not. This is the same pattern as the rest of the paper: a difficulty that is genuine so long as a mistaken premise is held, and that disappears with the premise.

The timeless and relational programmes—Page and Wootters' conditional probabilities [22], Rovelli's relational quantum mechanics and evolving constants of motion [23], Barbour's denial of fundamental time [24]—take the opposite path with full consistency: they accept that no preferred foliation exists and reconstruct the appearance of time from correlations within a fundamentally timeless structure. We do not claim these programmes are internally flawed. We claim that the premise they share—that there is no objective cosmic now—is the very premise the CMB refutes [26], and that once a real cosmic time is admitted, the elaborate reconstruction

of temporal appearance from timeless correlations is solving a problem that no longer arises. The choice between the programmes is, in the end, the choice of Section 2: whether to grant the manifold of occurrences the existence that the evidence assigns to the evolving layer.

# 9 Conclusion

The canonical problem of time is the block universe in Hamiltonian dress. Read the four-dimensional manifold as the existent, and the constraint correctly reports that nothing evolves; read the evolving three-dimensional world as the existent and its record as the manifold, and the same constraint, deparametrized along the cosmic time that the universe has let us measure, is a true Hamiltonian generating a unitary advance that reproduces the observed flat-ΛCDM expansion. Lifted beyond the background to the layer's transverse-traceless sector, that same Hamiltonian advances the graviton tower of the closed-$S^3$ layer unitarily—the propagating degree of freedom a fundamental observer reads as the flat-ΛCDM graviton, with matter the bend of its slicing rather than an ontological content. The equations never changed. What changed was the answer to a single question—what exists—and that answer is not a matter of taste but one the occurrence/existence distinction and the isotropy of the CMB settle together. The problem of time was the canonical symptom of a category error; read without it, the formalism was describing a real cosmic time all along. Read so, the dissolution is one of a family: the same distinction that turns the frozen constraint into a true Hamiltonian is shown, in the companion framework paper's first synthesis, to dissolve a wider set of general relativity's standing problems together—the problem of time among them, and the unitary evolution established here underwriting the resolution of the black-hole information paradox on the realised, globally hyperbolic spacetime [3]. That places the canonical face within the framework's one synthesis: the maximally symmetric de Sitter substrate the framework reads at once as general relativity's solution space [10], as the discrete $CPT$ and charge structure, and as the gauge sector on its conjugate real form, is read here on its *canonical* constraint—where the deparametrized generator is the layer's true Hamiltonian, and the quantum of action enters the gravitational sector at the seam, scaled by $\Lambda$ alone [7].

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# Appendix R Computational Receipts

Each computation marked R in the text is verified by a runnable receipt in the `receipts/` directory that ships with this work; status OK = verified (traced, run, output matches the claim). This appendix is generated from the receipt index, not maintained by hand.

**P10R1** `P10_minisuperspace_friedmann` [OK]
*sec:deparam* (deparametrized minisuperspace Hphys=(2pi/3)p_aˆ2/a-(L/8pi)aˆ3 generates Friedmann (adot/a)ˆ2=(8pi/3)rho_d/aˆ3+L/3 via Hamilton's equations). *Computes:* adot=4pi p_a/3a; Hphys autonomous=conserved; eliminate p_a -¿ Friedmann; dust aˆ-3; Lambda-sign control *Bound:* the true-Hamiltonian structure is non-empty *Run:* `python3 receipts/P10_canonical_time/P10_minisuperspace`

**P10R2** `P10_graviton_lift` [OK]
*sec:lock* (closed-Sˆ3 graviton lift: tensor-harmonic action S_n=1/2 int dT aˆ3[phidotˆ2-(muˆ2/aˆ2)phiˆ2] Legendre-transforms to Hphys_n=piˆ2/2aˆ3+1/2 a muˆ2 phiˆ2; mode eq phiddot+3(adot/a)phidot+(muˆ2/aˆ2)ph *Computes:* pi=aˆ3 phidot; Legendre H; EL mode eq; Hamilton eqns combine to it; flat-a oscillator control *Bound:* unitary cosmic-time evolution of the graviton tower *Run:* `python3 receipts/P10_canonical_time/P10_graviton_lift.py`

**P10R3** `P10_the_scale_factor_factors_out_of_the_free_tower` [OK]
*'PO-23'; 'sec:lock'* (for the FREE tower $\{}omega_n=\{}mu_n/a$, so $\{}sum d_n\{}omega_n=(1/a)\{}sum d_n\{}mu_n$ and **the scale factor factors out of the whole sum identically, for every $a$** — so the leading coefficient's time-independence is exact FACTORISATION rather than insensitivity, and **a non-adiabatic FREE computation cannot move it**; the row's discharge must be the coupled one. Both of the row's numbers are also made identities: $\{}zeta(0)=10$ by a terminating expansion, and the log coefficient $39/4$ as the $1/m$ term of $d(m)\{}mu(m)=2mˆ{3}-11m+(39/4)/m+\{}dots$). *Computes:* the label map to the standard TT-on-$Sˆ3$ form; $Z(0)=-5/2$, $Z(-1)=5/2$; the summand's expansion; a cancellation-free hard cutoff returning $9.74998$ with no zeta function in it; and $R=12/\{}alphaˆ{2}$ constant, so $\{}int Rˆ{2}$ and $\{}int R$ are both $\{}propto V$ *Bound:* bounded: **the coupled and non-adiabatic case is untouched** — what is shown is that the free case cannot be the route to it, and that on this background the one counterterm is degenerate with Einstein–Hilbert so the constant is not OBSERVABLE there *Run:* `python3 receipts/P10_canonical_time/P10_the_scale_factor_factors_out_of_the_free_tower.py`

**P10R4** `P10_the_weyl_invariant_at_second_order_carries_two_more_derivatives` [OK]
*'PO-43'; 'sec:lock''s shear clause* (the Weyl-squared invariant at second order in the shear is $Cˆ{2}=2\{}sum_i(\{}dot\{}sigma_i+H\{}sigma_i)ˆ{2}$ — **two more derivatives than the stated $4\{}sigmaˆ{2}$, and the right dimension $1/Lˆ{4}$** — so for a mode of frequency $\{}omega$ it goes as $\{}omegaˆ{2}\{}sigmaˆ{2}$, which is what makes the counterterm ultraviolet-sensitive at all). *Computes:* the full contraction from the metric with nothing assumed, and the closed form **read off one computation and verified against another** — ratio to the exact $Cˆ{2}$ converging to $1$ linearly in the amplitude ($1.0404$, $1.0040$, $1.00040$); plus the separating case, a constant comoving shear on a static background, giving $Cˆ{2}=0$ where $4\{}sigmaˆ{2}$ would not *Bound:* bounded: **the COEFFICIENT is not computed** — that is 'PO-43''s object and needs the sub-leading heat-kernel coefficients for the tower's own operator. This fixes what they multiply *Run:* `python3 receipts/P10_canonical_time/P10_the_weyl_invariant_at_second_order_carries_tw`

**P10R5** `P10_the_two_replacement_questions_are_one_answered_one_coupled` [OK]
*'PO-43' 'r4545''s two replacement questions* (**neither survives as stated**: the connected-isometry question is answered (the tower's degeneracy $2(n-1)(n+3)$ is exactly $(j,j+2)\{}oplus(j+2,j)$ with $2j+1=n-1$, so the helicities are **inequivalent** irreducibles of $\{}mathrm{SO}(4)=(\{}mathrm{SU}(2)_L and the exchange is the factor swap, orientation-reversing); and the topological-terms question **is not independent** — a topological term contributes no field equation, so its coefficient's only home is the horizon entropy, where the corpus is uncommitted). *Computes:* the branching identity checked at every level $n=2$–$11$ against 'P10''s Peter–Weyl degeneracy, and the two families shown distinct; and the enumeration of where a topological coefficient could be observable *Bound:* bounded: **Q2 is identified, not answered** — it closes only when the de˜Sitter horizon entropy does, and 'P17' declines that number rather than asserting it *Run:* `python3 receipts/P10_canonical_time/P10_the_two_replacement_questions_are_one_answered_one_coupled.py`

**P10R6** `P10_the_thermal_condition_is_helicity_blind_at_the_mode_functions_and_the_parity_odd_entry_is_no` [OK]

*'PO-43''s one unrun check; 'sec:lock''s fibre-by-fibre condition, and 'P17' 'sec:ledger'* (**THE CHECK COMES OUT FOR THE CLAIM, AND NOT BY WORDING: THE TWO HELICITY TOWERS ARE POPULATED ALIKE, SO THE PARITY-ODD ENTRY IS NOT OWED AND THE ROW CLOSES ABOVE THE DECLINED ENTROPY QUESTION.** The condition's entire dependence on the tower is $\{}hat\{}Gamma=\{}gamma+c\{}sum_n\{}hat\{}pi_nˆ{2}$, a sum of squares over a mode set the helicity swap PERMUTES, pairing modes of **equal frequency** — so $[\{}hat\{}Gamma,P]=0$ exactly, and with it $\{}nu=\{}sqrt{\{}hat\{}Gamma+\{}tfrac14}$, the sub-threshold projector and the retained branch $xˆ{1/2+\{}nu}$; and $\{}beta=2\{}pi\{}alpha$ carries no mode index at all, $\{}kappa=1/\{}alpha$ being the **background** horizon's. **AND THE EQUAL FREQUENCIES ARE FORCED BY AN ISOMETRY RATHER THAN BY ARITHMETIC.** $\{}sigma:g\{}mapsto gˆ{-1}$ on $Sˆ{3}=\{}mathrm{SU}(2)$ is $\{}mathrm{diag}(1,-1,-1,-1)$: in $O(4)$ with $\{}det=-1$, **orientation-reversing**, conjugating left translations into right ones so it exchanges $(j_L,j_R)$, commuting with the Laplacian and **anti-commuting with the curl** — one map that preserves every frequency and reverses every handedness. Read off the Casimirs, $\{}muˆ{2}=2(C_L+C_R)-6$ is manifestly symmetric and reproduces 'P10''s own $\{}mu_nˆ{2}=n(n+2)-2$ at every level, with the degeneracy splitting $50/50$ ($(n-1)(n+3)$ each) and the floor a self-dual and an anti-self-dual **five**. ⇒ **The three attacks the order named: (a) and (b) FAIL; (c), the Euclidean continuation, is ANSWERED** — the rotation $x_0\{}mapsto ix_0$ and $\{}sigma$ act on disjoint coordinate blocks and commute, and while a Riemannian four-section does admit a real self-dual split a Lorentzian one does not, **weighting the towers apart needs a parity-odd exponent**, and both candidates are absent: the de Sitter cosmological horizon has $\{}Omega=0$, and assuming a $\{}theta$-term is circular). *Computes:* the degeneracy split symbolically and to eight levels; $\{}muˆ{2}=2(C_L+C_R)-6$ against 'P10''s own eigenvalue; $\{}sigma$'s orthogonality, determinant and the left-to-right conjugation on 200 random quaternion pairs; $[\{}hat\{}Gamma,P]$, $[\{}nu(\{}hat\{}Gamma),P]$ and $[\{}Pi_{\{}Gamma¡3/4},P]$ on a truncated Fock space; $\{}kappa$ and $\{}beta$ DERIVED off $f=1-rˆ{2}/\{}alphaˆ{2}$; the parity-odd expectation in the Fock space and in closed form level by level; and the Wick rotation against $\{}sigma$ — **28 gates, with THREE controls that break as they must**: a helicity-weighted $\{}hat\{}Gamma$, unequal degeneracies or frequencies, and a rotating horizon returning an $\{}Omega$-linear parity-odd expectation *Bound:* **not that the tower is achiral** — 'r4547' stands, it IS chirally capable and what is shown is that the STATE does not use the capacity; nothing about the interacting tower's **ultraviolet definition**, 'P10''s own open frontier, the reach here being that a parity-even bare action and a parity-even state generate no parity-odd divergence under a parity-even regulator (a parity-breaking one is a scheme choice, and the anomaly route needs unbalanced chiral FERMIONS, which the row's own index obstruction excludes — that last step is a READING of a corpus result, not computed); not the declined $S=A/4$ question; and the floor-as-subtraction-point candidate is **deliberately not built**, per the order *Run:* `python3 receipts/P10_canonical_time/P10_the_thermal_condition_is_helicity_blind_at_the_mode_functions_and_t`

**P10R7** `P10_the_remainder_is_the_last_of_three_and_is_not_reachable_by_refining_the_method` [OK]

*'PO-23'; 'P07''s three parts, and the 'r6445' bound* (**the 'r6445' premise cannot be established ahead of the row itself**: it asks what the coupling does to the sum, and **the open item IS the definition of that sum** — so '60''s table is the reach of the free-spectrum method and the remainder is not reachable by refining it. And 'PO-23' is **the last of three named parts**, 'P07' saying two can now be said (the operator is bounded below; its spectrum is computable branch by branch at $\{}tfrac14$ and $\{}tfrac34$)). *Computes:* the triple enumerated with two said and one open; that the premise and the open item name the same object; and that the tempting re-scoping is the one already reversed *Bound:* **NOT a re-scoping to "not CR-specific"** — that was done at 'r1243', found wrong, **restored at 'r1291' and Daryl-flagged**, and is not re-made. The remainder is a genuine CR open and room (1) does not close *Run:* `python3 receipts/P10_canonical_time/P10_the_remainder_is_the_last_of_three_and_is_not_reachable_by_refining`

**P10R8** `P10_the_weyl_squared_coefficient_is_twice_a_scalar` [OK]

*'PO-43'; 'sec:lock'* (**the Weyl-squared coefficient is $1/60$ in units of $1/(4\{}pi)ˆ{2}$ — exactly TWICE a real minimally coupled scalar's** — because 'P10''s own description of the tower

(mass $a^{3}$, frequency $\{}mu_n/a$) matches two such scalars term for term, and because the $C^{2}$ coefficient is a property of the field content rather than the background). *Computes:* the bridging identity $2\{},\{}mathrm{Riem}^{2}-2\{},\{}mathrm{Ric}^{2}=3C^{2}-E_{4}$ verified **algebraically on 200 random Riemann tensors** to $2\{}times10^{-15}$, with the Weyl trace checked to vanish; and a cross-check from the standard anomaly formula $c=(N_0+6N_{1/2}+12N_1)/120$ at $N_0=2$, sharing the physics and none of the algebra *Bound:* **IMPORTED: Gilkey's $a_2$, cited not derived. SCOPED: this is the PHYSICAL-MODE count** — a covariant spin-2 computation gauge-fixes and carries ghosts and is a different object; 'P10' solves the constraints rather than gauge-fixing them, which is an argument from what the construction does and not a theorem, and is where this is attackable. The overall normalisation is convention-dependent; the convention-free content is the RATIO, exactly two *Run:* `python3 receipts/P10_canonical_time/P10_the_weyl_squared_coeffi`

**P10R9** `LIFT_adiabatic_correction` [OK]
*eq:adiabatic-exponent* (THE ADIABATIC CORRECTION. Exact Euclidean suppression for mode n is exp(-int omega_n ds), with omega_n = mu_n / a (P10 sec:lock) and a = |r| along the lift. omega DIVERGES at the branch point (a-¿0). Does the exponent converge? And is the evolution adiabatic (|dw/ds|/w^2 ¡¡ 1)?). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P10_canonical_time/LIFT_adiabatic_correction.py`

**P10R10** `LIFT_gravitational_action` [OK]
*eq:grav-action* (THE CALIBRATION. On-shell S = int p_a da with p_a = (3/4pi) a adot (from P10's H). Continuing tau = i s : S = -i S_E with S_E^grav = (3/4pi) int r r'^2 ds along the SAME trajectory. Compare with the contour action S_E^c = int [ (1/2) r'^2 + f/2 ] ds = 1.489 (alpha=1).). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P10_canonical_time/LIFT_gravitational_action.py`

**P10R11** `LIFT_instanton_action` [OK]
*eq:euclidean-action* (FAMILY 10, final item: does the lift have a variational principle of its own? Claim: S_E = int ds [ (1/2) r'^2 + V(r) ], V = f/2, with Euclidean first integral (1/2)r'^2 - V = -1/2. Test the closed form r(s) = -(2 M a^2)^{1/3} |sin(3s/2a)|^{2/3} against BOTH.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P10_canonical_time/LIFT_instanton`

**P10R12** `P10_the_straddle_does_not_need_a_floor` [OK]
*sec:lock* (**THE STRADDLE DOES NOT NEED A FLOOR, AND THE PARAGRAPH'S TWO STATEMENTS ABOUT Gamma-hat ARE ABOUT DIFFERENT OPERATORS.** The coupled sector promotes the boundary coefficient from the c-number gamma ¡= 1/4 to an operator Gamma-hat on the tower. **The paragraph asserted it 'bounded below by gamma' where it builds the straddle, and listed 'whether it is bounded below' as OPEN four hundred characters later.** Both are right of different operators: the LEADING-ORDER Gamma-hat = gamma + c sum_n pi_n^2 is a sum of squares over a positive coefficient, hence spectrum in [gamma, oo); the COMPLETE one carries the cubic and higher self-interactions, which enter at the same inverse-square order and are not sign-definite. ** AND THE REPAIR COSTS THE ARGUMENT NOTHING, WHICH IS THE POINT: the direct-integral decomposition of -d^2/dx^2 + Gamma-hat/x^2 uses only that the spectrum MEETS BOTH SIDES of 3/4 – limit point at or above, limit circle below – and a spectral projection onto {Gamma-hat ¡ 3/4} exists whether or not the spectrum has a floor.** Exhibited on a two-channel model in which the cubic takes one channel: the union is unbounded below and both spectral subspaces stay non-trivial.). *Computes:* the leading-order infimum asserted equal to gamma and the operator asserted monotone in k; the complete model asserted to break the floor and asserted driven down at large momentum; both spectral subspaces asserted non-trivial in both cases *Bound:* **bound: the model is a diagonal two-channel stand-in for the singular part of the interacting graviton Hamiltonian, not that Hamiltonian. It establishes what the decomposition NEEDS, not what the spectrum IS – and the spectrum is exactly the frontier the paragraph names ('L-165')** *Run:* `python3 receipts/P10_canonical_time/P10_the_straddle_does_not_need_a_floor.py`

**P10R13** `P10_the_straddle_is_computed` [ALL PASS]
*sec:lock* (THE STRADDLE AS A COMPUTED FACT: spec(Gamma-hat)=[gamma,inf) from

the displayed form, and gamma¡=1/4¡3/4 puts spectrum strictly below the threshold while unboundedness puts it strictly above, so both sides are occupied for every admissible gamma; the 3/4 exhibited by solving the indicial equation rather than quoted). *Computes:* r3803 *Bound:* 59 *Run:* `python3 receipts/P10_canonical_time/P10_the_straddle_is_computed.py`

**P10R14** `P10_the_adiabatic_residual_at_low_n_is_bounded_by_the_towers_own_floor` [ALL PASS]
*sec:dissolution* (**THE ADIABATIC RESIDUAL AT LOW n IS BOUNDED BY THE TOWER'S OWN FLOOR, AND IT REACHES NOTHING OBSERVABLE.** The suppression falls monotonically with harmonic index (2.8e-4 at n=2, 5.9e-6 at n=3, ˜2 orders per step after), so the mixing the adiabatic projection neglects can only carry amplitude into MORE-suppressed modes; and the S^3 tensor tower has no mode below n=2, so the least-suppressed mode has no source beneath it. **The bound is therefore one-sided by construction rather than by estimate.** Worst-case transfer of the whole O(eps) fraction out of n=2 leaves that mode MORE suppressed and does not raise the ceiling. Ceiling on transmitted power = unmixed n=2 = **7.9e-8**, and below 1e-3 even on the naive exponent.). *Computes:* r4231 *Bound:* 61 *Run:* `python3 receipts/P10_canonical_time/P10_the_adiabatic_residual_at_low_n_is_bounded_by_the_towers_own_floor.`

**P10R15** `P10_gamma_hat_is_bounded_below` [OK]
*sec:lock* (**THE COMPLETE BOUNDARY OPERATOR IS BOUNDED BELOW, AND THE REASON IS THE DEPARAMETRIZATION ITSELF: THE SCALE FACTOR IS THE SUPERMETRIC'S ONE NEGATIVE DIRECTION.** P10 lists 'the spectrum of Gamma-hat, including whether it is bounded below' among what is open in the interacting tower; that clause is separable and is settled here. **The DeWitt form tr((g pi)^2) - lambda (tr g pi)^2 has exactly ONE negative eigenvalue on the round metric, realised by the identity direction — the conformal one, which is the whole indefiniteness of superspace and the reason Wheeler-DeWitt is hyperbolic.** On g-TRACELESS momenta the trace term drops and the form is exactly tr((g pi)^2), which for L the Cholesky factor of g equals tr(S^2) with S = L^T pi L SYMMETRIC — a sum of squares of real eigenvalues, zero only at pi = 0. **Positive definite for every non-degenerate spatial metric, verified as an IDENTITY on 400 random pairs rather than as a sample.** **So the cubic term's apparent unboundedness is a TRUNCATION ARTEFACT: pi^2(1 - lambda phi + ...) is the expansion of pi^2/(1 + lambda phi), and the full coefficient is positive wherever the spatial metric is non-degenerate.** The deparametrization that produces a true Hamiltonian is the same move that produces a bounded-below one — both are the removal of the conformal direction, and P10 argues the first without noticing it had the second.). *Computes:* the signature asserted one negative; the Cholesky identity and positivity asserted on 400 valid (g, pi) pairs; min Q asserted positive on three metrics including a 625:1 anisotropy; the truncation asserted to go negative where the full expression does not *Bound:* **bound: positivity holds on NON-DEGENERATE spatial metrics. At the degenerate boundary of superspace the inverse metric blows up and the argument says nothing — which is exactly the a -¿ 0 boundary, handled instead by the thermal condition. Two different jobs, neither substituting. NOT TOUCHED: the UV definition of the tower sums, and the closed-form nonlinear solution** *Run:* `python3 receipts/P10_canonical_time/P10_gamma_hat_is_bounded_below.py`

**P10R16** `P10_the_towers_zeta_at_zero_is_ten_and_the_claim_needs_its_scoping` [PASS]
*sec:lock* (**THE TENSOR TOWER'S SPECTRAL ZETA(0) IS 10, SO THE MODE SUMS SPEND ONE DIMENSIONLESS CONSTANT AND THE NO-FREE-CONSTANT CLAIM NEEDS THE SCOPING ITS OWN SENTENCE NAMES.** CR_synthesis sec:ledger turned the claim into a prediction about a computable number – 'that coefficient vanishes' – and it does not. On P10 sec:lock's own spectrum (mu_n^2 = n(n+2)-2, n¿=2, degeneracy 2(n-1)(n+3)) the free static tower gives zeta(0) = 10 EXACTLY, and the zero-point sum's log-scale coefficient Res_{s=-1} zeta = 39/4. Non-zero on either reading of which spectral quantity sec:ledger meant, so the verdict does not turn on that ambiguity. AND THE PAPER NAMES THE COST ITSELF: the quartic leader is absorbed into Lambda, but the logarithmic successor goes with curvature-squared invariants and 'a curvature-squared coupling is not an entry in this framework's ledger'.). *Computes:* the spectral zeta of the deparametrized TT graviton tower: zeta(0) exactly (the asymptotic expansion terminates at s=0, so the continuation has no remainder), the residue at s=-1 by two independent routes, and a hard-cutoff cross-check with no zeta function in it *Bound:* FREE and STATIC: the tower's couplings to the scale-factor sector are not in it and the instantaneous spectrum is used, so a coupled or non-adiabatic computation could still return zero – what is excluded is the

claim holding trivially, on the most favourable case (physical TT modes only, no ghosts, constraints already solved). Does not touch essential self-adjointness of the coupled system, and does not disturb the geometric ledger: c, G, hbar as unit gauges over Lambda are untouched; what is measured is that the QUANTUM sector's mode sums spend one constant. *Run:* `python3 receipts/P10_canonical_time/P10_the_towers_zeta_at_zero_is_ten_and_the_claim_needs_its_scoping.py`

**P10R17** `P10_no_rescaling_discharges_the_log_and_the_one_mass_shift_that_does_is_the_curvature_offset` [PASS]

*sec:lock* (**THE COUPLED ATTEMPT RETURNS 39/4 UNCHANGED IN KIND: NO RESCALING OF THE TOWER'S FREQUENCIES CAN DISCHARGE THE LOG, AND THE ONE MASS-LIKE SHIFT THAT DOES IS EXACTLY THE THREE-SPHERE CURVATURE OFFSET.** The log coefficient is the 1/m term of d(m)mu(m), an exact functional of the spectrum, so which deformations can move it is answerable without perturbation theory in a coupling nobody has. Overall rescale gives L = (39/4)sqrt(1+e), zero only where every frequency is; mass shift gives L = -(d-3)(d+13)/4, zero at d = 3 and -13; a 1/m^2 tail at e = -39/4; linear-in-m at e = +-2sqrt(65)/5. NO RESCALING DISCHARGES IT, which generalises r6411 from the scale factor to any multiplicative renormalisation – the class the leading back-reaction lives in, since the coupling acts through a and through Gamma-hat = gamma + c sum(pi_n^2) whose expectation is (a^2/2)sum(d_n mu_n), the same sum rescaled. At d = 3 the summand is 2m^3-8m, a POLYNOMIAL: the 1/m term does not cancel, it does not exist – and +3 is exactly the curvature offset that makes the frequencies non-integer.). *Computes:* the exact 1/m coefficient of d(m)mu(m) over four one-parameter deformations of the spectrum, symbolically, plus a cancellation-free numerical confirmation summed term by term *Bound:* NOT a solution of the coupled sector and does not claim to be: it states which deformations COULD discharge the row and observes that the leading back-reaction is not among them. Does not compute the coupling's induced shift; if a higher-order effect supplies a mass-like delta this says what it would have to equal, not that it cannot. The non-adiabatic half is untouched. Does not disturb r6435's counterterm reading: on this conformally flat background the log's counterterm is degenerate with Einstein-Hilbert and the cosmological term, so a non-vanishing L is not a new ledger entry here – the result narrows the row rather than reopening it. *Run:* `python3 receipts/P10_canonical_time/P10_no_rescaling_discharges_the_log_and_the_one_mass_shift_that_does_is`

**P10R18** `P10_the_floor_is_forced_as_a_mode_but_the_subtraction_point_is_a_convention_and_the_residue_is_t` [OK]

*'PO-51'; 'sec:lock''s log, and 'P17' 'sec:ledger'* (**THE FLOOR IS FORCED AS A MODE AND THE SUBTRACTION POINT IS A CONVENTION — AND THOSE ARE DIFFERENT STATEMENTS THE ROW'S PREMISE RUNS TOGETHER.** *Forced:* $m=3$ is the lowest transverse-traceless harmonic the three-sphere admits — $j_R=(m-3)/2$ vanishes exactly there and is negative below, the degeneracy $2(m^2-4)$ being zero at $m=2$ — and the tower is **gapped at $\{}mu_3=\{}sqrt6$ with no zero mode and no soft region**, so the infrared is regulated by the geometry. *And the log is real and scheme-independent*: $d\{}mu=2m^3-11m+\{}tfrac{39}{4m}+\{}tfrac{45}{8m^3}+\{}cdots$, and the Dirichlet series $\{}sum d(m)\{}mu(m)m^{-s}$ has a **simple pole at $s=0$ with residue exactly $39/4$** (with $s=4$ and $s=2$ carrying the quartic and quadratic at $2$ and $-11$), ***so zeta regularisation does not evade a subtraction point***. ⇒ **$39/4$ does NOT move with that point** — identical across $m_0=3,5,10,50$ to the summation's own rounding, $25$ orders below the truncation error — while the finite part shifts by **exactly $\{}tfrac{39}{4}\{}ln(m_0'/m_0)$** and nothing else. ***BUT THE DECIDING TEST COMES OUT AGAINST, IN TWO STEPS.*** *First, **a mode number is dimensionless**, so $\{}ln(M/m_0)$ introduces no length for ANY $m_0$ and the one-scale claim — the row's whole motivation — protects every choice equally and **cannot single out the floor**. Second, on the admitted one-parameter family 'sec:lock''s own argument makes the counterterm basis **one-dimensional**, so the log's counterterm is degenerate with the cosmological term on the tower's own background ($R=12/\{}alpha^{2}$, verified), and 'P17' absorbs a constant vacuum energy into the one observed curvature with no bare-versus-vacuum split — **so no observable depends on $m_0$**.* ⇒ *And the floor fixes the constant to nothing: subtracting there leaves $C=-8.51485690643\{}ldots$, neither zero nor named.* ***THE ANSWER TO THE BURDEN IS "CONVENIENT", AND THE ROW CLOSES***). *Computes:* the degeneracy and eigenvalue against 'sec:lock''s own; $j_R\{}ge0$ at the floor; the asymptotic expansion and $L=39/4$ symbolically; the analytic continuation's three poles and residues

to \$10^{-5}\$; \$L\$ extracted at four subtraction points and three cuts; the finite-part shift to \$10^{-12}\$; the Ricci scalar of \$\{}alpha\{}cosh(T/\{}alpha)\$ derived as \$12/\{}alpha^{2}\$; and \$C\$ to \$16\$ digits — **19 gates with THREE controls that break as they must**: a mass shift MOVES \$L\$ (so the invariance is not a tautology), a soft floor diverges in the infrared (so the gap is this spectrum's), and at \$\{}mu^{2}=m^{2}\$ the product is a polynomial with **no \$1/m\$ term at all** (so the machinery detects the log rather than manufacturing it) *Bound:* **not that the one-scale claim fails** — the opposite for the part that matters, a discrete sum needing no second LENGTH for any subtraction point; what fails is the further step that the floor is thereby determined. **Nothing here uses the coupled tower**: every number is the free spectrum's, 'PO-23''s ultraviolet definition of the sums is not reached, and a subtraction point for a log is not a definition of the sum. Where the shear breaks the counterterm degeneracy the Weyl-squared coefficient is a separate entry whose own condition is 'PO-43''s uncosted one — **named, not claimed**. And \$39/4\$ is not discharged; 'r6436' stands *Run:* `python3 receipts/P10_canonical_time/P10_the_floor_is_forced_as_a_mode_but_the_subtraction_point_is_a_convent`

**P10R19** `I9_the_adiabatic_parameter_diverges_where_the_action_integral_converges` [OK]
*sec:lock* (I9 — THE ADIABATIC PARAMETER DIVERGES WHERE THE ACTION INTEGRAL CONVERGES. Integrable-systems field bake, probe I9. On P10's own near-branch-point scaling abs(r) ~ s^(2/3), the adiabaticity parameter abs(dw/ds)/w^2 goes as s^(-1/3) and DIVERGES, so the adiabatic invariant E/omega is not conserved through the branch point; the suppression exponent int omega ds nevertheless converges as S^(1/3), because omega ~ s^(-2/3) is integrable. Two different powers, two different questions: the finiteness P10 verifies comes from the action integral and not from slow variation, so the correction is semiclassical – which is what P15 calls the same object. Two adversarial controls, one for each half.). *Computes:* runs clean; eight verdicts including controls that fire in opposite directions *Bound:* COMPUTES: the scaling abs(r) ~ s^(2/3) is P10's own; A and mu carried symbolically, no value pinned *Run:* `python3 receipts/P10_canonical_time/I9_the_adiabatic_parameter_diverges_where_the_action_integral_converge`

**P10R20** `P10_gamma_is_one_quarter_and_is_the_maximum` [OK]
*sec:lock* (gamma = 1/4 is the MAXIMUM of the natural ordering family, so (1,1) holds for every ordering). *Computes:* reduces T_s exactly, shows the first-derivative term vanishes, and gets Gamma(s) with its maximum; 11 checks *Bound:* NOT-A-PAPER-CLAIM — verification of a standing claim; node 17 withdrew its own caveat *Run:* `python3 receipts/P10_canonical_time/P10_gamma_is_on`

`D1_the_boundary_is_per_fibre_and_the_UV_is_over_fibres` [OK]
*sec:lock* (PO-6 re-weighted not closed: the boundary closure is PER-FIBRE and the UV is OVER-FIBRES, and the clause c54.129 answered is the one the argument does not need). *Computes:* asserts the direct-integral structure at source, computes the per-fibre deficiency across the threshold, and checks the straddle under an unbounded-below operator (18 checks) *Bound:* NOT-A-PAPER-CLAIM — a bounded re-statement of a PROTECTED row; PO-6 stays open *Run:* `python3 receipts/L165_interacting_tower/D1_the_boundary_is_per_fibre_and_the_UV_is_over_fibres.py`

`D2_the_UV_degree_is_quartic_and_the_IR_is_free` [OK]
*sec:lock* (PO-6s UV clause measured: the tower sum diverges QUARTICALLY, the ordinary zero-point degree, and compactness buys the IR rather than the UV). *Computes:* takes the tower and mode action from P10 at source, derives the linear growth of mu_n and ¡pi_n^2¿, and tracks the partial sums against N^4/4 (13 checks) *Bound:* NOT-A-PAPER-CLAIM — a measurement of one PO-6 clause; PO-6 stays open *Run:* `python3 receipts/L165_interacting_tower/D2_the_UV_degree_is_quartic_and`

`S1_the_subleading_log_divergence_is_gauss_bonnet_and_the_ledger_survives` [OK]
— (PO-6 fixed-background half: the SUB-LEADING (log) one-loop divergence on the tower's own de Sitter slicing is the Gauss-Bonnet term (topological in D=4), so it forces NO curvature-squared coupling outside the one-constant ledger — extending L-809's quartic result to the log order; on de Sitter Weyl^2=0, and the on-shell graviton divergence reduces to GB (t Hooft-Veltman/Christensen-Duff)). *Computes:* verifies R^2/Ric^2/Riem^2=144/36/24 per alpha^4 (S50) and Weyl^2=0, that GB=24/alpha^4 is the topological Euler density, the on-shell R=4Lambda, and the ledger-survival conclusion (4 checks) *Bound:* NOT-A-PAPER-CLAIM — the fixed-background half of PO-6; extends L-809; does NOT touch the coupled-sector dark half (54's/definitional) *Run:* `python3 receipts/L816_po6_subleading_survives/S1_the_subleading_log_divergence_is_gauss_bonnet_and`

`S1_the_one_constant_ledger_survives_on_the_running_layer_not_only_fixed_de_sitter` [OK]

*sec:lock* (PO-6 RUNNING-LAYER half (the residue S50 named: does the one-constant basis survive on a background whose curvature RUNS?): on a(T)=sinhˆ{2/3}(3T/2alpha), where R runs (R-¿12/alphaˆ2 only asymptotically), the pure-graviton log divergence reduces to {Gauss-Bonnet (topological) + Lambda-renorm + Newton/ell_P-renorm + field-equation-proportional (field-redef removable)} with NO irreducible curvature-squared coupling outside {Lambda,G} — for ANY one-loop coefficients. Extends L-816 off fixed de Sitter, via background-independent algebraic identities (not R=const)). *Computes:* confirms Weylˆ2=0 and R runs on the sinhˆ{2/3} layer, the topological (53/90)E_GB split, the Lambda=0 identity (7/20)Ricˆ2+(1/120)Rˆ2=E_mn fˆmn exactly, the coefficient-independent reduction of a1 Riemˆ2+a2 Ricˆ2+a3 Rˆ2 to {E_GB,Lamˆ2,LamR,EOM} with remainder 0, and the vacuum-dS consistency with L-816 (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the running-layer half of PO-6; F5 reserves the verdict; pure graviton (matter as bend), does NOT enter the quantised-background sector or claim one-loop finiteness *Run:* `python3 receipts/L818_po6_running_layer/S1_the_one_constant_ledger_survives_on_the_running_layer_not_only_f`

`S1_the_ledgers_one_irreducible_survivor_is_the_conformal_coupling_that_turns_on_at_the_shear` [OK]

*sec:lock* (PO-6: L-818's boundary LOCATED with a coefficient (answers 56's r2764 routing). The graviton log counterterm = (7/40)Weylˆ2 + (149/360)GB + (1/8)Rˆ2; GB is topological and Rˆ2 reduces (L-818), so the ONE irreducible piece is the conformal (7/40)Weylˆ2 — it is 0 on the shear-free FRW layer (which is WHY L-818's remainder was exactly 0 and the ledger survived) and turns on at 2nd order in the shear, where Weylˆ2=4sigmaˆ2 (56 S10/r2743). The shear IS the TT-graviton tower = P10's open interacting tower; so L-818's caveat, r2743's shear and P10's open item are ONE boundary, now carrying (7/40)Weylˆ2). *Computes:* decomposes the graviton b_4 into {Weylˆ2,GB,Rˆ2} (Weylˆ2 coeff 7/40), shows only Weylˆ2 is neither topological nor ledger-reducible, and computes Weylˆ2=0 on the isotropic layer and O(shearˆ2) on an axisymmetric anisotropic model (no linear term) (5 checks) *Bound:* NOT-A-PAPER-CLAIM — locates L-818's boundary and names the open coupling; F5 reserves the verdict; does NOT close the interacting tower (it names it); coefficients cited (t Hooft-Veltman) *Run:* `python3 receipts/L819_shear_boundary_coupling/S1_the_ledgers_one_irreducible_survivor_is_the_conformal_cou`

`S1_the_weyl_magnitude_is_the_graviton_c_anomaly_and_the_parity_odd_partner_is_a_theta_term` [OK]

*sec:lock* (PO-6 OWED #472: the VALUE of the one shear counterterm (Weylˆ2, c54.219) is 7/40 — the graviton's conformal-anomaly c-coefficient (b_4 in the {Weylˆ2,GB,Rˆ2} basis). And its parity-odd partner, the Pontryagin density R˜R that a CIRCULAR polarisation reveals (nonzero, handedness-odd; a linear mode gives zero), is a topological theta-term at constant coefficient (Jackiw-Pi C-tensor prop nabla(coeff)=0; no D=4 gravitational anomaly), so it does NOT enter the field equations — the dynamical count stays ONE). *Computes:* computes R˜R on linear vs circular TT waves by a finite-difference Riemann (0 / +4.5 / -4.5, parity-odd), pins the b_4 Weylˆ2 coefficient 7/40, and cites the topological/theta-term status (Jackiw-Pi, Alvarez-Gaume-Witten) (4 checks) *Bound:* NOT-A-PAPER-CLAIM — OWED #472, one value + one status on the fixed-background shear; the parity-odd loop coefficient is zero for the pure graviton in D=4; interacting tower untouched *Run:* `python3 receipts/L821_weyl_coefficient_value/S1_the_weyl_magnitude_is_the_grav`

`S2_c1_and_c2_are_one_and_the_source_answers_more` [OK]

*sec:euclidean* (C1 and C2 are ONE condition joined by so — the list is six not seven — and the same P10 passage states that the free TT sector is a harmonic oscillator whose Hamiltonian is bounded below, which PO-6 asks for). *Computes:* asserts the joined sentence verbatim, confirms PO-6s row asks whether it is bounded below and does not carry the answer, and quotes the confinement clause (11 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects a count and surfaces a partial answer; joint satisfiability untested *Run:* `python3 receipts/L165_defining_the_sum/S2_c1_and_c2_are_one_and_the_sou`

`S3_c6_is_a_theorem_not_a_condition` [OK]

*sec:lock* (the C6/C7 tension cannot arise: the per-fibre structure is DERIVED from a commutation relation, not a requirement on a measure — so the conditions list is five plus one theorem). *Computes:* asserts D1s commutation, direct-integral and fibre-by-fibre statements verbatim, and S1s misclassification of C6 as a condition (8 checks) *Bound:* NOT-A-PAPER-CLAIM — reclassifies a list entry; joint satisfiability untested *Run:* `python3 receipts/L165_defining_the_sum/S3_c6_is_a_theorem_not`

`S4_the_open_half_is_the_floor` [OK]
*sec:lock* (PO-6s open half is the FLOOR: the direct-integral decomposition needs only that both sides of the threshold are occupied, and P10 leaves whether the complete Gamma-hat is bounded below open in its own voice). *Computes:* asserts P10s uses-only clause, its leaves-open clause, the sum-of-squares form and the structural commutation verbatim, and the PO-6 rows own wording (7 checks) *Bound:* NOT-A-PAPER-CLAIM — narrows a register row and corrects r2611s caveat *Run:* `python3 receipts/L165_defining_the_sum/S4_the_open_half_is_the_floor.py`

`D3_the_floor_does_not_survive_the_cubic` [OK]
*sec:lock* (PO-6s floor question answered NO at the order the paper names: the cubic enters as pi^2 phi, a square times a SIGNED field, so Gamma-hat is unbounded below wherever phi ¡ -c/g3). *Computes:* asserts P10s leaves-open clause, its leading-order form and its naming of the cubic verbatim, and computes the sign structure and divergence (8 checks) *Bound:* NOT-A-PAPER-CLAIM — answers a narrowed dark half; the state question is untouched *Run:* `python3 receipts/L165_interacting_tower/D3_the_floor_does_not_survive_the_cubic.py`

`D4_the_state_fixer_has_its_own_threshold` [OK]
*sec:lock* (the state-fixing mechanism has its own threshold: nu = sqrt(Gamma + 1/4) is real only for Gamma ¿= -1/4, below which both exponents go complex and no regular branch exists — and r2651s operator reaches -47.8). *Computes:* asserts P10s thermal-regularity sentence and its common-to-every-fibre clause verbatim, computes nu across Gammas range, and recomputes r2651s minimum (11 checks) *Bound:* NOT-A-PAPER-CLAIM — sharpens PO-6s successor question; whether the region is realised is untouched *Run:* `python3 receipts/L165_interacting_tower/D4_the_state_fix`

`D50_the_floor_does_survive_and_the_paper_said_so_232_revisions_earlier` [OK]
*sec:deparam* ( **‘D3‘ (r2651) HAS THE FLOOR BACKWARDS — P10 CALLS THE CUBIC’S UNBOUNDEDNESS AN ”ARTEFACT OF TRUNCATION” IN THE SAME PARAGRAPH, AND HAS SINCE r2419.** The cubic is the first-order term of $\{}pi^2/(1+\{}lambda\{}phi)$, whose full coefficient is positive on non-degenerate metrics — so the floor survives and r2652’s sub-$(-1/4)$ question has no object. Lead ‘L-542‘; VEIN ‘L-165‘). *Computes:* verifies termwise to eighth order that the cubic IS the expansion’s first-order term; samples both forms on 200,000 identical points (truncation $-892.6$, resummed never below $\{}gamma$); six source checks in P10 including the affirmative verdict and its interior-only SCOPE; dates the clause by reading r2419’s own copy of the file; and re-runs ‘P10_gamma_hat_is_bounded_below‘, which passes *Bound:* LATENT — no new physics; the answer was in the paper. NOT claimed: that ‘PO-6‘ closes (the ultraviolet definition of the tower sums is open in the same sentence), or that the floor is proved for the full interacting theory. ‘D3‘/‘D4‘ and the ‘PO-6‘ row are NOT touched — 56’s band and a protected row; routed *Run:* `python3 receipts/L165_interacting_tower/D50_the_floor_does_survive_and_the_paper_said_s`

`S50_the_counterterm_basis_is_one_dimensional_because_the_background_family_is` [OK]
*sec:deparam* (**‘PO-6‘’s UV HALF: THE COUNTERTERM BASIS IS ONE-DIMENSIONAL BECAUSE THE ADMITTED BACKGROUND FAMILY IS.** Every curvature invariant on a maximally symmetric background is a pure power of $1/\{}alpha^2$, and the substrates form a one-parameter family, so a divergence of any degree needs exactly one counterterm — which the framework carries. Scoped: the tower lives on the layer, whose $R$ runs. Lead ‘L-543‘; VEIN ‘L-165‘). *Computes:* **AMENDED c54.211 (‘L-544‘): c54.210’s SCOPE IS WITHDRAWN.** It put P15’s observable rate where P10’s slicing belongs; P10’s own $a(T)=\{}alpha\{}cosh(T/\{}alpha)$ has $R=12/\{}alpha^2$ CONSTANT, so the degeneracy holds on the tower’s own background and the result is STRONGER than claimed. The real limit is the coupled sector, where the scale factor is quantized and there is no fixed background. Both geometries computed side by side so the error stays checkable. — computes the quadratic invariants at $D=4,5,6$ and pins their $\{}alpha$-exponent at exactly 4; five standing-sentence checks (p0’s invariant fact, p0’s foliation, ‘L-533‘’s scale-only descent); computes the $\{}sinh^{2/3}$ layer’s Ricci scalar and asserts it NON-constant with a $12H^2$ late limit; and counts ‘counterterm‘ in P10 from git HEAD (0) against now (7) *Bound:* LATENT, and an ASSEMBLY not a discovery. NOT claimed: that ‘PO-6‘ closes, that the sub-leading divergences vanish, that this is renormalisability, or that the layer breaks it — **that is the question** *Run:* `python3 receipts/L165_defining_the_sum/S50_the_counterterm_basis_is_one_dimensio`

`S7_the_successor_was_misaimed` [OK]
*sec:tower* (L-543 INHERITS r2677s WITHDRAWN OBJECT: it asks about a RUNNING background and the free tower sits on P10s own slicing at R = 12/alpha^2, CONSTANT — and the

layers late limit IS that value, so it asymptotes to it). *Computes:* computes both Ricci scalars, shows the layers late limit equals P10s constant under H -¿ 1/alpha, and pins the successors wording and the back-reaction clause from the row (6 checks) *Bound:* NOT-A-PAPER-CLAIM — withdraws a successor this line registered; the back-reaction question stands *Run:* `python3 receipts/L165_interacting_tower/S7_the_successor_was_misaimed.py`

`S8_the_floor_follows` [OK]

*sec:lock* (PO-6 HAS BEEN CARRYING THE WRONG QUESTION: the complete Gamma-hat inherits its floor from the same non-degeneracy r2671 established, so boundedness is answered YES — what remains is WHERE the spectrum sits relative to 3/4). *Computes:* asserts P10s promotion, inverse-square and Weyl-alternative clauses verbatim, quotes the straddle receipts does-not-need-a-lower-bound line, and shows the truncated coefficient goes negative where the full one stays positive (7 checks) *Bound:* NOT-A-PAPER-CLAIM — the spectrum is untouched; P10 explicitly declines the floor and the decomposition survives without it *Run:* `python3 receipts/L165_interacting_tower/S8_the_floor_follows.py`

`S9_the_ordering_decides_it` [OK]

*sec:lock* (PO-6s SPECTRAL QUESTION IS DECIDED BY OPERATOR ORDERING, which P10 never names: normal-ordered puts the vacuum at 1/4 (BELOW 3/4, freedom survives), symmetric puts N=1 exactly AT 3/4 — and 3/4 = 1/4 + 1/2, the free coefficient plus one zero-point quantum). *Computes:* computes both ordering branches against the threshold, verifies the 1/4+1/2 identity, counts normal-order and symmetric-order at zero in the paper, and pins the both-sides-occupied clause (9 checks) *Bound:* NOT-A-PAPER-CLAIM — neither ordering is asserted correct; P10s decomposition survives either *Run:* `python3 receipts/L165_interacting_tower/S9_the_ordering`

`Q1_the_degeneracy_is_conformal_flatness_not_maximal_symmetry_so_no_scale_factor_can_break_it` [OK]

*sec:tower* (PO-6's DARK HALF HAS AN OBJECT: r2677 stated TWO degeneracies as one. The quadratic trio collapses because the deficit is exactly Weylˆ2 and EVERY FRW is conformally flat for EVERY a(T) – not because of maximal symmetry – so no scale factor can break it and back-reaction cannot; the different-dimension half does fail. The real limit is the SHEAR). *Computes:* R, Ricˆ2 and Riemˆ2 computed from the FRW metric with a(T) a free function at k = +1, 0, -1 (Weylˆ2 identically zero); the Gauss-Bonnet density shown an exact total derivative, sqrt(g)(Rˆ2-4Ricˆ2+Riemˆ2) = d/dT[24(a'ˆ3/3+a')]; the two relations solved to give int Ricˆ2 = A/3 - chi/2 and int Riemˆ2 = A/3 - chi; S50's D=4 row reproduced from the metric (144/36/24); P15's running layer recomputed from its own metric with the deficit zero by a second route in u = exp(3Ht/2); and Bianchi I giving Weylˆ2 = 4 sigmaˆ2 + O(sigmaˆ4). CONTROLS: SdS gives Weylˆ2 = 48Mˆ2/rˆ6 so the identity is not vacuous; the layer's Rˆ2 is t-dependent so the different-dimension degeneracy really does fail; and the numeric residual is a PRECISION-SCALING test (3.06e-37 at 40 digits, 2.24e-77 at 80) rather than a fitted threshold *Bound:* not that the coupled sector is renormalizable – this says which functionals a divergence can NEED, not that they are absorbable; no heat-kernel coefficient computed; Bianchi I is a HOMOGENEOUS shear, fixing the ORDER at which conformal flatness fails and not the mode-by-mode statement on the tower; and not a closure – PO-6 stays open *Run:* `python3 receipts/L549_coupled_counterterms/Q1_the_degeneracy_is_conformal_flatness_not_maximal_symmetry_so`

`S10_the_halves_meet_at_the_shear` [OK]

*sec:lock* (PO-6s TWO DECLARED HALVES MEET AT THE SHEAR: the counterterm half ends where conformal flatness ends (Cˆ2 = 4 sigmaˆ2 + O(sigmaˆ4)) and P10 names the tower as the TRANSVERSE-TRACELESS graviton modes — which ARE the shear). *Computes:* checks Cˆ2 vanishes at sigma=0, is quadratic-leading and non-zero away from it, confirms the source is trace-free, and quotes P10s tower identification twice (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the second-order basis is not computed; cc54 derived the coefficient and its structure is checked here *Run:* `python3 receipts/L165_interacting_tower/S10_the_halves_meet_at_the_shear.py`

`S11_the_ordering_question_dissolves` [OK]

*sec:lock* (PO-6s ORDERING QUESTION DISSOLVES: P10 gives deficiency indices (1,1) INDEPENDENTLY of ordering with the coefficient ¡= 1/4 across the natural family, closes the residual via the horizons Hartle-Hawking state, and answers the coupled straddle fibre by fibre). *Computes:* pins all six of P10s own statements — the independence, the family bound, the non-uniqueness, the

Friedrichs selection, the fibre-by-fibre condition, and what remains open (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the interacting tower is still open; the correction is to WHICH problem the row owes *Run:* `python3 receipts/L165_interacting_tower/S11_the_ordering_question_dissolves.py`

**P10R21** `P10_ordering_selection_is_external` [OK]
*sec:lock / P7 frontier:quantum* (**PO-15 THE ORDERING EXHAUSTION: nothing internal to the construction selects the operator ordering (min Gamma-hat = 1/4 normal vs 3/4 symmetric), and the reason is that selecting it would be solving the cosmological-constant problem.** Completes S11 (boundary CLOSURE is ordering-independent) and the r3014 thermal-state elimination (the thermal state acts downstream). ** THE PIVOT: ** the ordering is BULK-INERT (normal vs symmetric differ by the c-number zero-point 1/2 hbar omega_n, a global phase under the deparametrized evolution) and BOUNDARY-PHYSICAL (the difference surfaces only in the horizon coefficient, 1/4 -¿ 3/4 = 1/4 + 1/2, exactly one zero-point quantum), so "which ordering" IS "does the tower's zero-point energy gravitate at the horizon". ** THE EXHAUSTION: ** (1) the thermal state and (3) the seam's characteristic structure act DOWNSTREAM (they close the extension GIVEN Gamma-hat; a regular branch xˆ(1/2+nu) exists for BOTH orderings); (2) the substrate isometry, (5) positivity, (6) the single-scale ledger and (7) covariance under configuration redefinition each respect BOTH orderings; (4) the deparametrization makes normal ordering AVAILABLE (a preferred vacuum) but not mandatory, and by SOLVING the constraint removes the anomaly-freedom lever a Wheeler-DeWitt quantization would have used. The two external naturalness heuristics DISAGREE (continuity-with-free -¿ 1/4, predictivity -¿ 3/4). ** RESULT: ** the choice is EXTERNAL because it IS the cc problem reached from inside – an epistemic gap (one datum: does vacuum energy gravitate), NOT an ontological family, so admissible under the epistemic reading.). *Computes:* the ordering difference asserted = the zero-point c-number 1/2 hbar omega; the c-number-is-a-global-phase fact asserted numerically (identical amplitude moduli to 1e-12, ratio = eˆ-ict); the boundary shift 1/4 -¿ 3/4 asserted to be exactly one zero-point quantum with 3/4 the threshold; the regular branch nu=sqrt(Gamma+1/4) asserted real for BOTH orderings (thermal closure ordering-independent); both floors asserted finite; the seven-candidate enumeration stated with a structural disqualification each and no survivor *Bound:* PO-15 answered as the row asked – the choice is genuinely external, the enumeration is the evidence, the externality is the cc problem localized. NO ordering is picked and none is claimed derivable; the interacting-tower definition (PO-6) stays separately open; builds on S11/r3014, not a re-derivation of the Friedrichs selection *Run:* `python3 receipts/P10_canonical_time/P10_ordering_selection_is_external.py`

`S12_the_boundaries_coincide` [OK]
*sec:lock* (cc54s L-818 BOUNDARY, r2743s SHEAR FINDING AND P10s OPEN ITEM ARE ONE BOUNDARY: an FRW layer is shear-free by construction, the degeneracy ends at the shear, and P10 names the transverse-traceless tower as what remains open). *Computes:* verifies cc54s Step 1 (Weylˆ2 = 0 with R running, all three k) and pins P10s two namings of the tower (5 checks) *Bound:* NOT-A-PAPER-CLAIM — cc54s five sympy steps are not rerun; the coincidence is by identification, each link a corpus statement *Run:* `python3 receipts/L165_interacting_tower/S12_the_boundaries_coin`

`S13_the_ledger_does_not_reach_second_order` [OK]
*sec:lock* (AT SECOND ORDER IN THE SHEAR THE QUADRATIC BASIS IS GENUINELY THREE-DIMENSIONAL: GB and Cˆ2 span rank 2 of 3, Cˆ2 vanishes identically shear-free so L-818 never had to route it, and it has NONE of L-818s three routes). *Computes:* computes the rank and nullspace of {GB, Cˆ2}, confirms Cˆ2 = 0 at all three k on the shear-free layer, and pins p0s one-constant claim as a statement about the FACES (6 checks) *Bound:* BOUNDED NEGATIVE — one order of one expansion; not a verdict on the construction, and the coefficient is not computed *Run:* `python3 receipts/L165_interacting_tower/S13_the_ledger_does_not_reach_second_order.py`

`S1_the_shear_needs_exactly_one_new_counterterm` [OK]
*sec:tower* (PO-6's OWED SHEAR CALCULATION RUN: on a propagating TT mode Cˆ2 is O(hˆ2) and carries FOUR derivatives, but deltaˆ(1)R = 0 makes int sqrt(g) Rˆ2 a shift of Einstein-Hilbert – so the shear costs exactly ONE new counterterm, Weyl-squared, and c54.215's working count of two was one too many). *Computes:* R, Ricˆ2, Riemˆ2 and Cˆ2 built from the corpus's own L801/N1 TT-wave-on-de-Sitter metric and expanded to O(epsˆ2); the derivative content fixed by freezing the oscillatory factors and taking the total degree in (H,k,omega), giving 4 against sigmaˆ2's 2; deltaˆ(1)R shown to vanish and det g shown eps-independent, hence Rˆ2 at second order equals 2 Rbar times R at second order, pointwise; the span solved from

Gauss-Bonnet plus the Weyl definition, giving B and C both in span of A and C^2; and the Pontryagin density computed for linear, circular and opposite-handed polarisation by a finite-difference pipeline cross-checked against a symbolic evaluation to six digits. CONTROLS: the linear mode returns zero for the parity-odd invariant, which is what catches the incomplete basis; the handedness flips its sign; and at eps = 0 everything reduces to c54.215's one dimension *Bound:* no heat-kernel COEFFICIENT is computed – this says which functionals a divergence can need, not with what coefficient; the mode is a plane-wave TT on flat-sliced de Sitter and not P10's S^3 harmonics, so the COUNT is mode-independent while the COEFFICIENT is not; the Pontryagin term's total-derivative status is cited, not computed; its consequences for the corpus's chirality result are not raised; and not a closure – PO-6 stays open *Run:* `python3 receipts/L553_the_shear_counterterm/S1_the_shear_needs_exactly_one_new_counterterm.py`

`D1_the_degeneracy_carrying_the_quartic_was_never_derived_and_its_constant_is_the_component_count` [OK]

*sec:tower* (THE DEGENERACY CARRYING PO-6's QUARTIC WAS NEVER DERIVED: asserted in P10 and asserted-unchecked in D2. Derived from Peter-Weyl on the parallelizable S^3 it is g(n) = 2(n-1)(n+3), n ¿= 2, so the shell contribution is 2n^3 – and the constant is the propagating-component count, which makes D2's "any tensor rank" false). *Computes:* the level-j totals of frame-valued fields on S^3 = SU(2) for frame-spin 0, 1, 2, coming out as 1, 3, 5 times (2j+1)^2 – the component counts, which is the check the decomposition had to pass; the TT part as the two extreme summands, 2(2j+1)^2, verified to be exactly 2/5 of the symmetric-tracefree total; the closed form by eigenvalue with its floor g(2) = 10 matching P10's independently-asserted n ¿= 2; the cumulative sum and its leading coefficient 2/3; and Weyl's law calibrated on the scalar tower, whose degeneracy follows exactly from Peter-Weyl, with the ratio running 1.01505 to 1.00038. CONTROLS: the scalar tower's leading coefficient 1/3 against the tensor tower's 2/3, which is what refutes "any tensor rank"; and a REPORTED NEGATIVE – matching Weyl to subleading order fails to discriminate the candidate closed forms, because on a closed curved manifold that term carries the curvature through the heat kernel *Bound:* not that the eigenvalue assignment is re-derived, mu_n^2 = n(n+2)-2 being P10's and taken as given; no heat-kernel coefficient, still none computed anywhere; this is the FREE spectrum's mode count and not the interacting tower; and not a closure – PO-6 stays open, this replaces an unchecked input with a derived one *Run:* `python3 receipts/L554_tower_degeneracy/D1_the_degeneracy_carrying_the_quartic_was_never_derived_and_its_cor`

**P10R22** `P10_the_parity_odd_density_goes_with_the_relative_phase_of_the_two_channels` [OK]

*The sentence 'r6803' put into 'P10' after node 60's flag: the parity-odd density's dependence stated as a property of the configuration — vanishing a quarter-cycle apart, largest in step — rather than prescribing a helicity. Node 66 asked for a receipt so the sentence can cite one instead of standing on a reading* (**Derived, and the old wording's prescription is the case that vanishes.** On the plane-wave member with an areal factor the density factorises as $\{}psi'(u)\{},\{}omega'(u)$ times a **strictly positive** weight $-8e^{2\{}psi}/(t^{2\{}lvert t\{}rvert)$ carrying no derivative of either channel and no $\{}omega$ at all — so ***the density's sign and zeros are exactly the product's, exactly rather than perturbatively*** with both channels on one wavenumber a phase $\{}delta$ apart, the cycle average at second order is $\{}langle\{}psi'\{}omega'\{}rangle = (k^{2}\{}epsilon^{2}/2)\{}cos\{}delta$: **maximal in step, exactly zero a quarter-cycle apart**, extremal in anti-phase, zero if either channel is silenced **and the parity cancellation is derived rather than quoted**: $\{}omega\{}to-\{}omega$ is $\{}delta\{}to\{}delta+\{}pi$ and flips the sign, so a parity-symmetric ensemble kills it as a property of the integrand, with no statement about occupation numbers). *Computes:* C1 'r6766''s three facts re-run first through machinery 'exec'-imported from 'r6762' and 'r6766' so the three cannot drift — a pure plane wave gives $*RR=0$ however polarised, the areal factor makes it nonzero, and silencing either channel kills it, each channel silenced by **rebuilding** the metric with it constant rather than substituting a symbol and leaving its derivatives standing the weight shown to have no zeros, so the vanishing is the product's and not the weight's *Bound:* **A torus block, not 'P10''s three-sphere layer** — support for the sentence's *form*, not a computation of 'P10''s own member, which is the line 'r6766' draws and this does not blur one wavenumber and equal amplitudes, which is where a relative phase is defined at all; no superposition and no unspecified ensemble average **no chiral state is supplied and 'r6770' is not reopened** — its conclusion rests on the parity argument, which is what is computed here; *what this sharpens is that balanced **populations** do not by themselves kill a **cross**-correlation, so 'r6770''s route

through equal populations is narrower than it reads* nothing bears on ‘PO-7‘ *Run:* `python3 receipts/P10_canonical_time/P10_the_parity_odd_density_goes_with_the_relative_phase_of_the_two_chan`

**P10R23** `P10_the_shear_breaks_conformal_flatness_but_not_the_gauss_bonnet_degeneracy_so_nothing_sees_the` [OK]

*node 66’s work order in ‘FOR_60‘ (‘r6897‘) — **(1) ‘PO-52‘’s second half, which the order made the whole row**. ‘r6894‘ established the entry has a domain (Weyl$ˆ2$ non-vanishing at $O(\{}varepsilonˆ{2})$ on the construction’s own on-shell confined wave), and the order asked the next thing: ***does anything observe the coefficient?*** — with the closing route named in advance, *”if the honest answer is that the admitted second-order shear configurations are not ones anything observes, that closes the row and is the result”*. Forbidden: importing a value for the coefficient* (**The row closes, and NOT by the route the order offered — the configurations *are* on the admitted list and the invariant *is* non-vanishing. What fails is the step from a non-vanishing INVARIANT to a moved OBSERVABLE, because conformal flatness was never the only degeneracy in play.** **(1)** $Cˆ{2}=E_{4}+2(R_{ab}Rˆ{ab}-Rˆ{2}/3)$ is **algebraic in four dimensions** — verified on closed FLRW with $a(t)$ free (where $Cˆ{2}=0$ and both remainders are non-zero, so it reads more than $0=0$) and on SdS (where $Cˆ{2}=48Mˆ{2}/rˆ{6}\{}neq0$) — ***so no order in the shear can break it*** **(2)** the **Lanczos identity** $H_{ab}\{}equiv0$ holds **on the shear configuration itself**, at the amplitudes where $Cˆ{2}\{}neq0$, to $1.6\{}times10ˆ{-30}$ at 30 digits: ***the Gauss–Bonnet half reaches no field equation*** **(3)** so the remainder is Ricci-squared, which the field equations trade for the stress tensor — and on the substrate, an exact vacuum-$\{}Lambda$ space, $Cˆ{2}=E_{4}-8\{}Lambdaˆ{2}/3$ **exactly** ⇒ ***the subtraction-point change is a cosmological-term renormalisation — ‘PO-51‘’s conclusion, reached from an identity rather than from a property of the family, so the shear cannot take it away***). *Computes:* C1 **the $\{}varepsilon=0$ row is a control that fired**: an earlier $H_{ab}$ raised one index of one Riemann factor and not of the other and returned $\{}lvert H_{ab}\{}rvert/\{}mathrm{Riem}ˆ{2}\{}sim0.8$ on pure de Sitter, where the identity is exact — *recorded because the failure was silent everywhere except on the case with a known answer* every index carried down and raised only through the inverse metric, explicitly the profile re-verified on-shell against the de Sitter TT wave equation before use the Ricci residue of the linearised wave **measured** to scale as $\{}varepsilonˆ{2}$ exactly, to $10ˆ{-6}$ in the ratio *Bound:* **ON-SHELL ONLY, and that is the whole restriction**: the reduction uses the field equations to eliminate $R_{ab}$, and off-shell $Cˆ{2}$ is an independent invariant nothing here bounds a **matter-sourced** shear configuration gives a stress-tensor integral rather than a constant$\{}times$volume — still not a new geometric constant, but not the same statement; the sharp form is the vacuum-$\{}Lambda$ one and is labelled so **the frontier’s $1/60$ is not used, not needed and not tested** — the argument is coefficient-independent the linearised wave is Einstein through **first** order only; (2) does not need it to be a solution (an identity is an identity) and (3)’s sharp form is stated on SdS instead $\{}int\{}sqrt{-g}\{},E_{4}$ does shift the on-shell action by a topology-fixed constant — **but at $\{}varepsilon=0$ as much as at $\{}varepsilon\{}neq0$**, so the broken degeneracy is not what produces it nothing bears on $\{}zeta(0)=10$ or $39/4$, which are the degenerate combination’s *Run:* `python3 receipts/P10_canonical_time/P10_the_shear_breaks_conformal_flatness_but_not_the_gauss_bonnet_degene`

**P10R24** `P10_the_degeneracy_needs_r_constant_not_the_cosh_so_the_anomaly_is_what_makes_its_own_constant_` [OK]

*‘PO-23‘; ‘sec:lock‘’s counterterm basis, and ‘P07‘’s third wall* (node 66’s work order in ‘FOR_60‘ (‘r6919‘) — **‘PO-23‘, the register’s last NEVER-ATTEMPTED row.** The interacting tower’s ultraviolet definition: two walls are built (the operator is bounded below; its spectrum is computable branch by branch), the third untouched. The residual freedom is exactly one number, the log-scale, whose coefficient is $\{}zeta(0)=10\{}neq0$ — and the no-free-constant claim survives only because on this background the counterterm is *degenerate* with Einstein–Hilbert and $\{}Lambda$. The order asked the operator question: ***once $a$ is quantized, is $\{}int\{}!\{}sqrt g\{},Rˆ{2}$ still fixed by $\{}int\{}!\{}sqrt g\{},R$ and $\{}int\{}!\{}sqrt g$, or independent?*** — plus (2) what coupling does to $\{}zeta(0)$, *iff* (1) says observable. Guarded: **do not build the interacting theory**, and *a located wall is the attempt’s result*). *Computes:* **INDEPENDENT — and the mechanism is the anomaly, so the degeneracy that would have hidden the one constant is switched off by exactly that constant’s non-vanishing.** **(1) The order’s premise is CORRECTED, and by measurement**: ‘P10‘’s ”part company as soon

as $a$ is not the de^Sitter $\{}cosh$" is not the operative condition. The degeneracy is equivalent to $R$ being $\{}sqrt g$-a.e. CONSTANT — $\{}Lambda$+radiation has $a$ nothing like the cosh and $R=4\{}Lambda$ exactly, and the free tower's *bare* zero-point energy is exactly radiation-like ($E=S/a$, $\{}rho\{}propto a^{-4}$, $p=\{}rho/3$, trace $\{}equiv0$), so the naive back-reaction breaks nothing: **rank 1 to $10^{-43}$ on BOTH null controls** **(2)** what breaks it is the **log**: the pole of $Z$ at $s=-1$ leaves $E=\{}frac1a[C_0+r\{}ln(a\{}mu)]$, spoiling the $a^{-4}$ scaling, giving $\{}Theta=\{}rho-3p=r/(2\{}pi^{2}a^{4})$ — **$\{}mu$-independent**, i.e. an anomaly — hence $R=4\{}Lambda+4Gr/(\{}pi a^{4})$, non-constant **iff $r\{}neq0$**, and $r=\{}frac{39}{4}$ **(3) rank 3** at $s_3/s_1=9.06\{}times10^{-8}$, 35.5 decades above the MEASURED floor; the exact identity is $I_2I_0-I_1^{2}=I_0^{2}\{}mathrm{Var}_{\{}sqrt g}(R)\{}ge0$, so the gap is **second order in $r$** ($/r^{2}$ constant to $7\{}times10^{-4}$ over three decades — the same order 'P10' found for the shear) **(4) (2) answers as a THEOREM**: $\{}zeta(0)=10+\{}frac12B_3'(0)$ exactly, so only the $m^{-1}$ coefficient can move it, through $\{}zeta_R$'s pole at $1$ — **immune to every constant (mass-like) shift, every multiplicative rescaling** (containing 'r6411''s "$a$ factors out" as one member) **and every even $1/m^{2k}$ shift**; it moves only on an ODD power of the mode label ($\{}delta=cm\{}Rightarrow c-c^{3}/3$; $\{}delta=c/m\{}Rightarrow-c$), and local operators give only even powers. ⇒ **the whole even sector is foreclosed** *Bound:* **The wall, located rather than hit.** The back-reaction is **SEMICLASSICAL** — the free tower's regularized stress tensor in the classical constraint — and the interacting theory is **not built**, as ordered. What higher orders could do is the wall, and it is narrow: restoration needs the TOTAL trace exactly constant, i.e. a pure $\{}Lambda$, and an $a^{-4}$ anomaly is not that. **One premise taken, not proved**: that the coupled sector admits states without a definite $\{}alpha$ — the order's own words and 'P10''s; if the physical Hilbert space selected one background the degeneracy would survive, and that is the single way (1) reverses. No value for the coupled $\{}zeta(0)$. **No detectable signal** — the gap is $O(r^{2})$; structural independence is not laboratory observability. **This is NOT 'PO-43'**: that entry is a NEW invariant ($C^{2}$), this is the SAME invariant becoming an independent functional. Nothing touches the ordering ambiguity or 'prop:flat' *Run:* `python3 receipts/P10_canonical_time/P10_the_degeneracy_needs_r_constant_not_the_cosh_so_the_anomaly_is_what`

**P10R25** `P10_the_premise_is_not_load_bearing_because_superselection_forbids_coherence_and_not_the_ensem` [OK]

*'PO-23'; 'sec:lock''s direct integral over $\{}mathrm{spec}\{},\{}hat\{}Gamma$, and $\{}kappa=1/\{}alpha$ *common to every fibre** (node 66's 'r6925' order, on the one premise 'r6920' took and did not prove: ***\{}"Does that Hilbert space carry a superselection rule in $\{}alpha$? Equivalently: is $\{}alpha$ — or whatever operator carries the background's scale — central in the algebra the true Hamiltonian and the tower generate, or does something in that algebra fail to commute with it?\{}"*** with the three outcomes costed: *superselected* reverses 'r6920''s first item and the degeneracy survives; *not superselected* leaves the finding standing on something proved; *neither* is honest if the algebra is not determined until the coupling is, and then **\{}"say which step needs it\{}"**. And the guard is this line's own returned: ***\{}"Whatever you build here, build its controls first — a case where the answer is known to be superselected and one where it is known not to be — because a superselection statement with no measured floor is exactly the shape of an unfalsifiable one\{}"***. Not asked: an interacting theory, a value for the coupled $\{}zeta(0)$, or a detectable signal). *Computes:* **THE THIRD BRANCH, AND FOR A DIFFERENT REASON THAN THE ORDER OFFERED IT: THE TWO BRANCHES ARE NOT ALTERNATIVES.** **(1) Superselecting $\{}alpha$ does NOT restore the degeneracy, so the first branch reverses nothing.** A superselection rule forbids *coherence* between sectors and not an *ensemble* over them; $\{}int\{}!\{}sqrt g\{},R^{k}$ is read off the ensemble **linearly**; so an ensemble over three distinct $\{}alpha$ reaches rank 3 at $s_3/s_1=1.60\{}times10^{-4}$, **39 decades above the measured floor, INSIDE a superselected theory** — and the ensemble is exhibited as a legitimate state ($\{}rho\{}ge0$, $\{}mathrm{tr}\{},\{}rho=1$, commuting with the label) **(2) and 'r6920' took neither route, so the premise was never a dependency.** Two exact rank lemmas separate them: $R$ constant on ONE history gives $M[i,k]=R_0^{k}V[i]$, **rank 1 for any number of regions**; an ensemble over $N$ distinct $\{}alpha$ is a sum of $N$ outer products, **rank exactly $\{}min(N,3)$** — confirmed at $N=2$, where the third singular value sits at $8.2\{}times10^{-72}$, *below* the quadrature floor. 'r6920' reached rank 3 at $N=1$, where mixing caps the rank at 1, and rank 1 is exactly what its $r=0$ control measured **(3) and the question as put to $\{}alpha$ has no object**: $\{}alpha=\{}sqrt{3/\{}Lambda}$

is a *coefficient* of $\{}Hphys$, so $\{}hat\{}alpha=\{}alpha\{}cdot\{}mathbf1$ is central **with a ONE-point spectrum**, against the control's three — and a central operator with one point decomposes nothing, so there is no rule for the degeneracy to survive on; the enlargement that would supply one is a direct sum over *theories*, which solving the constraint does not return **(4)** the operator $R$ actually depends on, $\{}hat a$, is central in nothing: $[\{}hat a,\{}Hphys]=[\{}hat a,\{}hat p_a^{2}]$ **exactly and at every order**, the residual $7.8\{}times10^{-13}$ sitting below a floor of $7.7\{}times10^{-12}$ measured on eighteen members of the $f(\{}hat a)\{}otimes T$ family, because every coupling term \{}S‘lock‘ names is $f(\{}hat a)\{}otimes(\{}text{tower operator})$ and functions of $\{}hat a$ commute with $\{}hat a$ **(5) the theory's ONE genuine superselection label is $\{}hat\{}Gamma$** — central in the radial algebra ($5.3\{}times10^{-18}$) and NOT in the algebra the order names ($2.7\{}times10^{-2}$), with 7 spectral points — **and it labels the boundary condition, not the background scale**, which is the two-sided control showing that centrality is relative to an algebra *Bound:* **AND IT CORRECTS THIS LINE'S OWN LANDED WORK.** ‘r6920‘ \{}S E lists as a wall *\{}"if the physical Hilbert space selected a single background, the degeneracy would survive\{}"*; **that clause is false** — with a single background the rank is 3 at $9.06\{}times10^{-8}$, ‘r6920‘'s own headline number, computed at a single $\{}Lambda$. The wall list is shorter by one item, not longer. **The step that DOES need the coupling is a different question from the one the order posed**: the criterion is $\{}mathrm{Var}_{\{}sqrt g}(R)=0$, so its operator form asks whether any physical state makes $\{}hat R=4\{}Lambda+8\{}pi G\{}hat\{}Theta$ **sharp** — an eigenvector of $\{}hat\{}Theta$, whose ultraviolet definition \{}S‘lock‘ names as the open frontier. A conditional is stated as one: if $\{}mathrm{spec}\{},\{}hat\{}Theta$ is purely continuous, no state makes $\{}hat R$ sharp either. **This receipt does not compute that spectrum.** **The truncated models are models of the ALGEBRA, not of the spectrum** — they settle commutators and spectral-point counts and settle nothing about the half-line operator's spectrum or its self-adjoint extensions, which ‘P10‘ treats and this does not touch No interacting theory, no coupled $\{}zeta(0)$, no signal; \{}S D's finding does not enlarge the gap, which is still $O(r^{2})$ on one history, the larger ensemble number being an effect a single history does not have. Nothing here touches the ordering ambiguity or ‘prop:flat‘ *Run:* `python3 receipts/P10_canonical_time/P10_the_premise_is_not_load_bearing_because_superselection_forbids_coh`

**P10R26** `P10_no_state_makes_the_curvature_sharp_and_the_obstruction_is_the_anomaly_coefficient_itself` [OK]

*‘PO-23‘; \{}S‘lock‘'s open frontier, and the sentence *\{}"the one ultraviolet constant becomes observable once the scale factor is quantized\{}"** (node 66's ‘r6929‘ order, the step this line itself named at ‘r6928‘: ***\{}"whether any physical state makes $\{}hat R$ sharp — whether $\{}operatorname{spec}\{}hat\{}Theta$ has a point spectrum\{}"***, with the two readings costed: *only continuous* ⇒ every state carries a spread in $R$ and the independence is **\{}"a statement about the ensemble the state already is — which is stronger than it sounds and needs saying in those terms\{}"**; *point states* ⇒ the independence is realized on individual states and the counterterm is observable in the ordinary sense. Plus: ***say which of the two ‘P10‘ is entitled to claim now*** — and **\{}"if the answer changes what that sentence may assert, that is a correction to a paper and it comes to me — do not edit ‘P10‘\{}"**. Controls demanded by name: **a case with known point spectrum and one with known continuous spectrum, run through the same arithmetic** Not asked: an interacting theory, a coupled $\{}zeta(0)$, a detectable signal; and ‘PO-23‘ is not expected to close). *Computes:* **THE FIRST READING, AND AS A THEOREM RATHER THAN A MEASUREMENT: NO PHYSICAL STATE MAKES $\{}hat R$ SHARP, AND THE OBSTRUCTION IS THE ANOMALY COEFFICIENT ITSELF.** **(1) $\{}hat\{}Theta$ is a multiple of the identity on the tower at this order**, which is why the question is answerable without the coupling: $\{}omega_n=\{}mu_n/a$ makes the excitation energy $\{}bar S/a$ with $\{}bar S=\{}sum_n\{}mu_n\{}hat N_n$ free of $a$, so $p=\{}rho/3$ and **the excitation trace vanishes IDENTICALLY as an operator** — in every state, not merely the vacuum — while the whole anomaly sits in the renormalized zero point, a c-number. ⇒ $\{}hat\{}Theta=(r/2\{}pi^{2}a^{4})\{}mathbf 1$ and $\{}hat R=4\{}Lambda+\{}kappa\{}hat a^{-4}$ is a function of $\{}hat a$ **alone** **(2) so the spectral question becomes one about multiplication operators, and there it is exact**: $R'=-4\{}kappa a^{-5}¡0$ on the half-line, so $R$ is strictly monotonic hence injective, so every level set above $4\{}Lambda$ is ONE point (and below it, empty) — and multiplication by $f$ has an eigenvector at $\{}lambda$ **iff** $\{}{f=\{}lambda\{}}$ carries positive measure. ⇒ **$\{}hat

R$ has NO eigenvectors; purely continuous spectrum, the range $[4\{}Lambda,\{}infty)$ with $4\{}Lambda$ itself not attained** **(3)** and the instrument agrees, against controls built first: two refinement exponents (median eigenvalue gap, eigenvector participation length) sit at $\{}mathbf{-1.00}$ for $\{}hat R$, matching multiplication-by-$x$ ($-1.00$) and **a full decade of exponent away from the harmonic oscillator's $0.00$** **(4) and the obstruction is QUANTITATIVE and exact**: $[R(\{}hat a),\{}hat p_a]=i\{}hbar R'(\{}hat a)$ gives $\{}operatorname{Var}(\{}hat R)\{}operatorname{Var}(\{}hat p_a)\{}ge\{}frac{\{}hbar^{2}}{4}\{}langle R'(\{}hat a)\{}rangle^{2}$, **saturated to $0.3\{}%$** by narrow admissible states **(5) and the right-hand side vanishes iff $r=0$** — where $\{}hat R=4\{}Lambda\{}mathbf 1$ and the variance sits at the measured floor $1.9\{}times10^{-12}$. ⇒ **the same residue that makes the counterterm observable is what forbids any state from resolving the curvature it is read off** — the row's shape a third time, from a third direction **(6)** and a consistency that was not put in by hand: $\{}langle\{}hat a^{-4}\{}rangle$ converges at the origin **iff $\{}nu¿\{}tfrac13$**, and the horizon's own thermal condition gives $\{}nu=\{}sqrt{\{}gamma+\{}tfrac14}\{}ge\{}sqrt{\{}tfrac12}$ — **the boundary condition that closes the deficiency is the same one that makes $\{}hat R$'s expectation exist** *Bound:* **'P10' IS NOT EDITED, AS ORDERED — the verdict is routed.** The sentence *\{}"becomes observable once the scale factor is quantized\{}"* **stands**, and the mode is now fixed as **distributional, not sharp-valued**: no state assigns $R$ a definite value, so the third integral's independence is a statement about the spread every state already carries — *unconditional rather than weakened, since there are no special states to except.* What the paper is **not** entitled to is the ordinary-observable reading, a measurement returning a sharp $R$ carrying the counterterm as a definite number; **that is excluded by a theorem, not by a missing calculation** **The step that still needs the coupling, and it is STRICTLY SMALLER than the one it replaces**: $\{}hat\{}Theta$ is a c-number on the tower *at this order*; cubic and higher terms put genuine tower operators into it, and then $\{}hat R$ is not a function of $\{}hat a$ alone, so the level-set argument — which uses injectivity of a function of ONE variable — does not survive the promotion. Whether a correlated interacting state can make $\{}hat R$ sharp is **not settled here** *One direction flagged and not computed: the §D bound needs only the commutator, so any interacting $\{}hat\{}Theta$ still containing a non-constant function of $\{}hat a$ inherits a bound of the same form from that piece alone — removing the obstruction would need the $a$-dependence to CANCEL, not merely to be accompanied* **The grid instrument is an instrument, not a proof**: a finite matrix has pure point spectrum by construction, which is exactly why the statistic is a REFINEMENT EXPONENT and not a spectrum; the verdict rests on the level-set theorem **And one methodological catch recorded because it read as a refutation**: the commutator identity integrates by parts, so trial states with amplitude at the grid boundary break it — two of seven widths, and those two are exactly the ones whose ratio fell below $1$ (to $0.26$). *The domain of the identity is part of the statement.* Back-reaction still SEMICLASSICAL; no interacting theory, no coupled $\{}zeta(0)$, no signal; gap still $O(r^{2})$; nothing touches the ordering ambiguity or 'prop:flat' *Run:* `python3 receipts/P10_canonical_time/P10_no_state_makes_the_curvature_sharp_and_the_obstruction_is_the_anoma`

**P10R27** `P10_the_commutator_bound_survives_the_cubic_because_the_scaling_that_could_cancel_it_carries_no` [OK]

*'PO-23'; the promoted $\{}hat\{}Theta$, and \{}S'lock''s ultraviolet frontier* (node 66's 'r6933' order, on the one step 'r6930' left: the level-set argument dies at the cubic **but the commutator may not**, since ***\{}"a commutator is algebra and not dynamics\{}"***. ⇒ ***\{}"Compute $[\{}hat R,p_a]$ at that order and report whether the variance bound survives with the residue as its coefficient.\{}"*** Three outcomes costed: correction $\{}hat a$-independent or carrying the residue as a common factor ⇒ **the bound survives verbatim and the theorem generalizes**; the correction able to **cancel** the residue at some tower state ⇒ **the question needs the interacting theory, but with the cancellation condition in hand**; or the promotion not well posed without a choice ⇒ ***\{}"say which choice... if this lands on the ordering, that is a different row's business and you should say so rather than pick.\{}"*** Guard, this line's own catch returned: ***\{}"an inadmissible case shown with its arithmetic is evidence about the instrument; the same case dropped is a filtered result. Whatever you build for (1), carry that.\{}"*** Not asked: an interacting theory, a coupled $\{}zeta(0)$, a signal; and 'PO-23' is not expected to close). *Computes:* **THE BOUND SURVIVES — AND THE REASON IS THE ROW'S OWN SHAPE A FOURTH TIME: THE ONLY SCALING THAT COULD CANCEL IT IS THE SCALING THAT CARRIES NO TRACE.** **(1) The bound survives verba-

tim in FORM at every order, ordering-blind.** $[f(\{}hat a)\{}otimes\{}hat T,p_a]=i\{}hbar f'(\{}hat a)\{}otimes\{}hat T$ for **any** $\{}hat T$ — commuting or not, ordered any way — because $p_a$ differentiates only the $a$-function and $\{}hat T$ rides through; exact symbolically, and on a grid the residual falls at **second order** ($-1.99$ to $-2.00$ across four powers with an arbitrary non-commuting $\{}hat T$), i.e. it is the central difference and not the identity **(2) And pointwise cancellation is IMPOSSIBLE, by a theorem.** An energy term $h(a)\{}otimes\{}hat T$ contributes $\{}hat\{}Theta=(h+ah')\{}hat T/2\{}pi^{2}a^{3}$ — **exact for any $h$** — so a term is traceless *iff* $h\{}propto1/a$; a power law $c\{},a^{-n}$ gives $\{}partial_a\{}hat\{}Theta\{}propto(n-1)(n+3)a^{-n-4}$, and matching the anomaly's $a^{-5}$ needs $n=1$, **where the coefficient $(1-n)$ is zero and the term contributes nothing to $\{}hat\{}Theta$ at all.** ⇒ *The anomaly exists because $\{}ln a$ breaks the $1/a$ scaling that made the bare sum traceless; cancelling it would need exactly the scaling it broke* **(3) The expectation CAN be tuned to zero — and that is not sharpness.** At $B^{*}$ the coefficient $\{}langle\{}partial_a\{}hat R\{}rangle$ vanishes to $-1.4\{}times10^{-17}$ and the bound collapses to $1.2\{}times10^{-35}$, **vacuous** — but $\{}operatorname{Var}(\{}hat R)=1.56\{}times10^{-6}$ there, **5.9 decades above the measured floor** $1.9\{}times10^{-12}$. *A vacuous bound only stops constraining* **(4) And the level-set argument survives FURTHER than the order expected.** On every fibre where the cubic's tower operators are simultaneously diagonal — all product states, any commuting family — $\{}hat R$ is multiplication by $R_{\{}rm eff}=4\{}Lambda+\{}sum_j c_ja^{-n_j}$, and $\{}partial_aR_{\{}rm eff}\{}equiv0$ has **no solution at all** while $r¿0$; a level set clears to a degree-6 polynomial, so at most 6 roots, measure zero ⇒ **no eigenvector**; and a direct sum over fibres keeps the refinement exponent at $-1$ ($-1.01$, $-1.02$), so stacking fibres does not manufacture a point spectrum **(5) So the residue is NOT the coupling — it is non-commutativity**, measured: the cubic's own factors $\{}pi^{2}$ vs $\{}phi^{2}$ at $9.8\{}times10^{-2}$ and two cubic terms at $2.8\{}times10^{-1}$, against a commuting family's $5.2\{}times10^{-17}$ *Bound:* **The order's third branch does NOT fire, and the choice is named rather than made.** The commutator needs **no** ordering choice — §B differentiates only the $a$-function — so (1) answers without reaching the ordering. Where the ordering *does* enter is one place: it sets $\{}langle\{}hat T_j\{}rangle$, hence $B^{*}$, hence whether §D's tuned cancellation is reachable on a physical state. **‘P10‘ calls the ordering selection external and localizes it as one computable datum (whether the tower's zero-point energy gravitates at the horizon); this receipt does not pick it** **What is still open, narrowed rather than restated: entangled states over a NON-COMMUTING cubic tower family.** Product states and commuting families are covered by (4); the measured non-commutativity is what blocks a common fibre basis in general. Such a state would have to sharpen $R$ by a mechanism neither (2) nor (3) touches — *not settled here* **The power-law form in (2) is an assumption about the cubic and is stated as one**: the theorem is exact for any finite sum of powers, but a cubic term carrying a **second logarithm** would need its own line — and that is a question about the ultraviolet definition, \{}S‘lock‘'s open frontier, not this receipt's. *Flagged, not computed* **The grid is an instrument, not a proof**: §B's verdict rests on the symbolic identity and the exponent is the falsifiable check; comparing MATRICES on a finite-difference grid fails outright, since the commutator sits off-diagonal there — recorded, because it read as the identity failing at a residual of $1.2$ No interacting theory, no coupled $\{}zeta(0)$, no signal; back-reaction still semiclassical; nothing touches the ordering ambiguity or ‘prop:flat‘ *Run:* `python3 receipts/P10_canonical_time/P10_the_commutator_bound_survives_the_cubic_because_the_scaling_that_co`

**P10R28** `P10_the_second_logarithm_holes_the_mechanism_and_not_the_conclusion_because_the_leading_log_is_` [OK]

*‘PO-23‘; \{}S‘lock‘'s ultraviolet frontier, and ‘r6934‘ \{}S C's mechanism sentence* (node 66's ‘r6939‘/‘r6941‘ order, on the one thing ‘r6934‘ flagged and did not compute: ***\{}"Generalise the trace theorem to $h(a)=c\{},a^{-n}\{}ln^{m}a$ and report $\{}hat\{}Theta$ and $\{}partial_a\{}hat\{}Theta$ for general $(n,m)$\{}"*** ($(1)^{a}$); ***\{}"does the tower sum at cubic order generate $m=2$?\{}"*** ($(1)^{b}$); and ***\{}"report whether the bound's right-hand side can be cancelled at any $(n,m)$ the cubic actually generates, as against the $m=0$ answer which is no\{}"*** ($(1)^{c}$). With 66's **own hypothesis offered to be refuted**: *\{}"at $n=1$ the $\{}ln^{m}$ term vanishes and the $m\{}ln^{m-1}$ term does not ... the mechanism (2) rests on has a hole at exactly $m\{}ge1$, which is exactly where the anomaly lives\{}"*, flagged as *\{}"a hypothesis with a poor pedigree\{}"*. Three guards: **never compare the identity as matrices** (this line's own ‘r6934‘ note, made the standing instrument warning); **the ordering stays named and un-

picked** (‘PO-15‘’s); and **any new cancellation gets the second measurement** against the variance floor *\{}”or it will be read as sharpness by the next reader\{}”*. Not asked: an interacting theory, an ordering choice, or any corpus edit). *Computes:* **66’s HYPOTHESIS IS CORRECT AND ‘r6934‘’s MECHANISM IS HOLED — BUT THE CONCLUSION SURVIVES BY A STRICTLY BETTER ARGUMENT THAT DOES NOT DEPEND ON $(1)^{b}$.** **(1) $(1)^{a}$ exactly**: $2\{}pi^{2}\{}hat\{}Theta[c\{},a^{-n}\{}ln^{m}a]=c\{},a^{-(n+3)}[(1-n)\{}ln^{m}a+m\{}ln^{m-1}a]$, confirming the order’s algebra. At $n=1$ the first term dies and the second does not **(2) and the witness is the ANOMALY ITSELF**: $r/(2\{}pi^{2}a^{4})$ is the $(n{=}1,m{=}1)$ entry of that very formula, so ‘r6920‘’s trace was always a member of the family ‘r6934‘ assumed excluded. ⇒ *‘r6934‘ \{}S C’s \{}”the only scaling that could cancel carries no trace\{}” is true for $m=0$ and false for $m\{}ge1$* **(3) $(1)^{c}$ — AND YET CANCELLATION IS STILL IMPOSSIBLE, at ANY $(n,m)$: the LEADING LOGARITHM IS UNPARTNERED.** From an $(1,m)$ term the highest power in $\{}partial_a\{}hat\{}Theta$ is $\{}ln^{m-1}a$ with coefficient **exactly $-4mc$** (checked $m=1..4$); terms chain downward and never upward, so the coefficient system is **lower triangular with diagonal $-4m$** and $\{}partial_a\{}hat\{}Theta\{}equiv0$ forces every $c=0$ — sympy returns only the trivial solution **(4) and this is UNCONDITIONAL**: whatever $m$ the cubic generates, its own top logarithm has no partner, **so $(1)^{c}$ does not wait on $(1)^{b}$** **(5) but PARTIAL cancellation is real, and is a third answer**: an $(1,2)$ term at exactly $c_2=2r$ kills the $\{}ln^{0}$ coefficient the anomaly sits in and leaves $\{}ln^{1}$ at $-16r$ — *the obstruction RELOCATES one logarithm up rather than being removed*, neither \{}”cancels\{}” nor \{}”adds\{}”. And the second measurement: $\{}operatorname{Var}(R)=5.1\{}times10^{-3}$ there, **11.2 decades above the floor** $3.6\{}times10^{-14}$ **(6) $(1)^{b}$ — a second logarithm needs a NESTED SUM**: $m$ is the pole order, and a double pole needs a $\{}ln$ of the mode label *in the summand*, since the Mellin transform of a log-free power asymptotic has only simple poles. The free tower’s $d(m)\{}mu(m)=2(m^{2}-4)\{}sqrt{m^{2}-3}$ expands at large $m$ as a **pure Laurent series — log-free, exactly** — growing as $2m^{3}$, which is the power putting the pole at $s=-1$. So the free sum yields $m=1$ and **never** $m=2$ *Bound:* **This CORRECTS this line’s own landed ‘r6934‘ — the mechanism, not the conclusion**, and the correction came from a limitation that receipt itself recorded. *Sixth self-correction of landed work, second of that kind.* The conclusion ‘r6934‘ reached is re-established on a stronger footing; its stated reason is not general **$(1)^{b}$ is NOT computed, and the datum is named rather than handed off**: it turns on **one property — whether the cubic vertex’s large-mode-label asymptotic carries a $\{}ln m$**. If it does, the cubic reaches $m=2$ and (5)’s partial cancellation becomes physically available at $c_2=2r$; if not, the cubic stays at $m=1$. **Either way (3) holds.** *The order’s point that this row IS the ultraviolet definition is taken — which is why the answer is a named property of the sum and not a hand-off* **The ordering stayed named and unpicked, and this order’s answer does not depend on it**: nothing in the trace formula or the leading-log argument uses an ordering choice, so there was nothing to stop for **Guard 1 carried live, not recited**: the identity’s convergence order is re-measured at $-1.99$ on a smooth state, so a flat residual would be a representation error. And §D’s rejected width is **printed with its arithmetic**, per the standing rule No interacting theory, no coupled $\{}zeta(0)$, no signal, **no corpus edit** — the site in ‘P10‘ and what it should say are routed; back-reaction still semiclassical; nothing touches ‘prop:flat‘, ‘PO-31‘ or ‘PO-15‘ *Run:* `python3 receipts/P10_canonical_time/P10_the_second_logarithm_holes_the_mechanism_and_not_the_conclusion_bec`

**P10R29** `P10_the_second_logarithm_line_closes_because_the_interaction_cannot_reach_the_anomalys_own_powe` [OK]

*‘PO-23‘; \{}S‘lock‘’s ultraviolet frontier, and whether ‘r6942‘’s relocation is physically available*

(node 66’s ‘r6945‘ order, on the one property ‘r6942‘ reduced the second-logarithm question to and declined to guess: ***\{}”the vertex’s large-label asymptotic, to the order that decides the pole structure ... report whether the inner sum produces one and at what order in the label, with the expansion printed rather than characterised\{}”*** $((1)^{a})$; and ***\{}”whichever way it falls, say what it does to the relocation\{}”*** $((1)^{b})$. With the order’s own statement of the stake — *\{}”$(1)^{c}$ is unconditional, so this does not gate the verdict. **What it gates is whether the relocation at $c_2=2r$ is PHYSICALLY AVAILABLE**\{}”* — and its costing that ***either outcome closes a branch rather than opening one***. Three guards: the convergence-order measurement **stays live rather than recited**; the ordering **stays named and unpicked**, with the addition that *\{}”if the vertex asymptotic turns out to depend on an ordering choice, that is a finding and the move is to name the datum and stop\{}”*; and — NEW, about the shape of the

answer — ***\{}"a second completed argument is worth more than three narrowings, so if the vertex computation closes the second-logarithm line outright, say so in those terms\{}"***, with *\{}"not computed\{}"* and *\{}"computed and inconclusive\{}"* named as different states. Not asked: an interacting theory beyond the vertex's own asymptotic, an ordering choice, or any corpus edit). *Computes:* **THE SECOND-LOGARITHM LINE IS CLOSED — TWICE, AND THE STRONGER CLOSURE NEEDS NO PROPERTY OF THE VERTEX AT ALL.** **(1) The relocation lives at a power of $a$ the interaction cannot reach.** ‘r6942‘’s tuned term is $c_2a^{-1}\{}\{}ln^{2}a$ — an $(n{=}1,m{=}2)$ term, because $n=1$ is where the anomaly sits. The cubic's second-order vacuum shift is an $n=3$ term: in closed form $E^{(2)}=-(\{}\{}lambda^{2}\{}\{}kappa/32a^{3})\{}\{}sum G_{123}\{}\{}mu_1\{}\{}mu_2/\{}\{}mu_3(\{}\{}mu_1{+
**matched against direct diagonalization to $2.5\{}\{}times10^{-9}$ relative** (both give $-1/192$ at $a=1$), with the measured log-log exponent $-2.999992$ against the free tower's $-1.000000$, truncation-stable to $4\{}\{}times10^{-6}$ **(2) And the counting makes that general rather than a feature of the cubic**: the only scales are $a$ and $\{}\{}ell_P$, so a term carrying $\{}\{}ell_P^{\{}\{},j}$ carries $a^{-(1+j)}$ — ***$n=1+j$, and $n=1$ is populated at $j=0$ alone, by the FREE tower***, whose summand is log-free exactly (‘r6942‘). ⇒ **At $n=1$ the logarithm is capped at $m=1$ at every order of the coupling, so there is no $(n{=}1,m{=}2)$ coefficient in this theory to tune**; and sympy returns **no solution** when the two families are asked to cancel at two values of $a$ **(3) (1)[a] answered on its own terms, and the answer is also no.** The $S^3$ triple-harmonic overlap has an exact multiplet sum rule $\{}\{}sum_\{}\{}alpha\{}\{}lvert C_{123}\{}\{}rvert^{2}=n_1n_2n_3/2\{}\{}pi^{2}$ on the selection-rule set (triangle inequality **and** $n_1{+}n_2{+}n_3$ odd) — the integral is $\{}\{}pi/2$ on that set and $0$ off it over all $343$ triples against quadrature, and the rule reproduces the $n_3=1$ constant-harmonic case exactly. In **both** soft corners the internal-label dependence is $j(j^{2}-3)^{p}$, $p\{}\{}in\{}\{}tfrac12\{}\{}mathbb{Z}$: an odd polynomial terminating at $j^{+1}$ for integer $p$, an even-power series for half-integer $p$ ⇒ ***the coefficient of $1/j$ is exactly zero at every order in both corners*** — no harmonic number, no $\{}\{}ln$ of the label, only simple poles, **$m=1$ at the cubic's own power of $a$ too** **(4) And the condition is sharp rather than lucky**: give the weight one inverse power and the integer-$p$ branch stops terminating, producing a $1/j$ coefficient of exactly $c(-3)^{p}$ at every odd $k$ — *the instrument returns a harmonic number when one is there, so the zero is a measurement* **(5) The second measurement carried anyway**, because a relocation is not a sharpening: $\{}\{}operatorname{Var}(R)=5.1\{}\{}times10^{-3}$ at $c_2=2r$, **11.2 decades above the floor** $3.6\{}\{}times10^{-14}$ *Bound:* **This DEMOTES ‘r6942‘’s relocation rather than correcting it.** ‘r6942‘ reported it as *\{}"a fact about the formula\{}"* and asked whether it was available; it is not — the tuned state $c_2=2r$ exists in the trace formula's closure and not in this theory's spectrum. *Nothing in ‘r6942‘’s verdict changes: its \{}S C was already unconditional, and it neither asserted the relocation was reachable nor guessed its sign* **ONE STEP NAMED, and it is not load-bearing.** The scalar sum rule's weight $n_1n_2n_3$ is a polynomial exactly, so (3)'s zero is exact there; the transverse-traceless vertex differs by index contractions whose multiplet-summed weight is a rational function of the labels, and **whether that function is polynomial in each label is the one step at which (3) could differ**. The TT zonal kernel is **not computed here and not guessed** — and it decides nothing about (1)–(2), which is where the verdict is **The ordering stayed named and unpicked, and the new guard is answered rather than deferred**: the cubic's reordering difference is measured to be exactly $+iP$ — **linear in the momenta and supported only on coincident labels** (residual below $10^{-15}$ across three truncations), so it is not a cubic term and does not enter the asymptotic **Guard 1 carried live**: the identity's convergence order re-measured at $-1.99$ on a smooth state this revision, so a flat residual would still be a representation error. The inadmissible width is **printed with its arithmetic** The $n=3$ scaling is measured on a three-mode truncation of the reduced system, not on the full tower; the closed form and the $\{}\{}ell_P$ counting are what carry the claim, and the diagonalization is the falsifiable check on them No interacting theory beyond this vertex's own asymptotic, no coupled $\{}\{}zeta(0)$, no signal, **no corpus edit** — the ‘P10‘ site and its sentence are routed; nothing touches ‘prop:flat‘, ‘PO-31‘ or ‘PO-15‘ *Run:* `python3 receipts/P10_canonical_time/P10_the_second_logarithm_line_closes_because_the_interaction_cannot_re`

**P10R30** `P10_the_entangled_case_is_a_singular_pencil_question_and_the_pencil_is_non_singular_at_every_tr` [OK]

*‘PO-23‘; the entangled case — ‘r6934‘’s remainder, which has been this row's whole wall for four*

*revisions* (node 66's ‘r6949‘ order, reformulating that wall from a category into a computation and offering the reformulation **as a hypothesis to be broken** (*\{}"my record on this row is three withdrawn clauses, one of which you withdrew last revision, so weigh it accordingly\{}"*): (1)[a] ***\{}"first, the structural claim, which everything after it rests on: $\{}\{}hat R$ commutes with $\{}\{}hat a$ ... check it rather than assume it — the conformal factor, the measure, and whatever operator ordering the momentum constraint imposes are all places it could fail, and if it fails the rest of this order is void and that is the finding\{}"***; (1)[b] ***\{}"does the family $\{}\{}hat R(a)$ have an eigenvalue branch that is CONSTANT in $a$ on a set of positive measure?\{}"***—*a question about a matrix family rather than about states*; (1)[c] ***\{}"solve it at the smallest truncations that carry two non-commuting tower operators and report whether a solution exists\{}"***, the determinant being a polynomial in the powers of $1/a$ hence analytic, so vanishing on positive measure means vanishing identically. Guards: **if (1)[a] fails, stop there**; ***\{}"do not let the truncation answer for the tower ... if the polynomial system's solvability changes with $N$, that dependence IS the result\{}"***; the ordering stays named and unpicked, *and if (1)[a] depends on it that is a third surfacing and worth flagging as such*; and ***\{}"reformulated and open is progress and not attempted is not\{}"***). *Computes:* **THE PREMISE HOLDS, THE ENTANGLED CASE IS NOW A SINGULAR-PENCIL QUESTION, AND THE PENCIL IS NON-SINGULAR AT EVERY TRUNCATION TESTED.** **(1) (1)[a] is TRUE, and true for a reason rather than by construction.** $\{}\{}hat R$ is **algebraic** in the matter trace—the trace of the field equation gives $R=4\{}\{}Lambda+\{}\{}kappa\{}\{}Theta$ with no derivative of $a$ in it—and ‘r6934‘'s trace formula makes every term a function of $\{}\{}hat a$ times a tower operator on the other factor, so $[\{}\{}hat R,\{}\{}hat a]$ is **exactly zero**. ⇒ *And the control shows it is not vacuous: the variant that keeps a $\{}\{}hat p_a$—what taking $R$ from the kinetic form would do—measures $8.4\{}\{}times10^{-4}$ against that exact zero.* The **measure** does not touch it (self-adjointness residual $10^{-16}$ in a weighted product), and (1)[a] is **ordering-blind** across three orderings of the tower factor ⇒ ***the ordering does NOT surface a third time*** **(2) (1)[b] the branch value is FORCED before any determinant is taken.** Every power in $\{}\{}hat R(a)=4\{}\{}Lambda+\{}\{}sum_k\{}\{}kappa_k a^{-m_k}\{}\{}hat T_k$ is negative, so $\{}\{}hat R(a)\{}\{}to4\{}\{}Lambda$ and a constant branch must equal $4\{}\{}Lambda$ exactly ⇒ **the question collapses to: is $M(a)=\{}\{}sum_k\{}\{}kappa_ka^{-m_k}\{}\{}hat T_k$ singular for almost every $a$?** *No truncation in that step* **(3) (1)[c] and at finite $N$ that is EXACTLY a singular-pencil question, which is $a$-free and coupling-free.** $\{}\{}det(\{}\{}sum x_k\{}\{}hat T_k)$ is a degree-$N$ form and for two distinct powers each monomial lands on its own power of $1/a$ (checked: six monomials, six powers at $N=5$), so $\{}\{}det M(a)\{}\{}equiv0\{}\{}iff$**the pencil is singular**, with extreme coefficients $\{}\{}kappa_k^N\{}\{}det\{}\{}hat T_k$. ⇒ For the cubic's own **non-commuting** factors $\{}\{}hat T_1=\{}\{}hat\{}\{}pi^2$, $\{}\{}hat T_2=\{}\{}tfrac12(\{}\{}hat\{}\{}pi^2\{}\{}hat\{}\{}phi the pencil is **NON-SINGULAR at $N=2\{}\{}ldots8$, in exact integer arithmetic** **(4) At three operators the powers COLLIDE ($6{+}10=8{+}8$), so the pencil equivalence does not carry and the curve determinant is computed directly**—non-zero at $N=3\{}\{}ldots6$ with symbolic couplings, and the full coupling system has **no solution with all three couplings non-zero at $N=3,4,5$** **(5) AND THE ONE $N$-DEPENDENCE IS THE ORDER'S WARNING MADE CONCRETE.** At **odd** $N$ a ladder matrix is singular, so $\{}\{}det\{}\{}hat T_k=0$ and the system *does* acquire non-trivial solutions—**every one of which switches off at least two of the three couplings**, collapsing to a single-operator model that is singular for that reason alone. ⇒ ***The necessary conditions are $N$-parity dependent; the verdict is not. And the bias runs the helpful way: an odd truncation MANUFACTURES a null vector the tower has not got, so the instrument is tilted toward the affirmative and still returns none*** **(6) The instrument is live**: two operators sharing one null vector give a **singular** pencil at $N=4,5,6$, so the zeros are measurements. And the measure-zero refinement: a non-zero Laurent polynomial has finitely many positive roots (2 at $N=4$), so even where the determinant vanishes it vanishes on measure zero and *\{}"for almost every $a$\{}"* fails *Bound:* **THE ONE NAMED STEP IS THE TOWER LIMIT OF THE DETERMINANT CRITERION.** At finite $N$, singularity is $\{}\{}det=0$; in the tower it is a **null eigenvector for almost every $a$**, and the odd-$N$ parity is exactly where the two criteria part company. *What survives dimension is stated separately and is three things: $\{}\{}lambda=4\{}\{}Lambda$ is forced; the condition is a null eigenvector of $M(a)$; and $\{}\{}hat\{}\{}phi$, $\{}\{}hat\{}\{}pi$ have purely continuous spectrum so neither has the null eigenvector the odd-$N$ determinant reports—

making the finite-$N$ proxy strictly more permissive, so a finite-$N$ \{}"no\{}" is the stronger of the two statements.* ⇒ **So this is reformulated and ANSWERED AT EVERY TRUNCATION with the limit named, not a closed statement about the untruncated tower** — the wall is a different object now, *\{}"the tower limit of a singular-pencil condition\{}"* rather than *\{}"entangled states over a non-commuting family\{}"* **The momentum constraint's algebra is NOT derived**; it can only bear on (1)[a] by putting a $\{}\{}hat p_a$ into $\{}\{}hat R$, which is what the control measures, and transverse-traceless modes satisfy it identically **AND A HABIT ADOPTED FROM 'r6947', which is why this receipt has almost no floats in it.** That revision found 'r6946''s one tolerance was the authoring machine's round-off floor. *The repair for a claim of this shape is not a better tolerance but NO tolerance*: the ladder matrices are carried in the basis $D=\{}\{}mathrm{diag}(1/\{}\{}sqrt{n!})$ where **both are integer matrices**—a diagonal similarity, so spectra and determinant-vanishing are untouched—and every load-bearing determinant, Laurent coefficient and coupling system is computed **over the integers**. The two remaining floats are the commutator measurements, each reported against a control rather than a threshold No ordering choice, no interacting theory beyond the cubic's own two tower factors, no signal, **no corpus edit**; nothing touches 'prop:flat', 'PO-31' or 'PO-15' *Run:* `python3 receipts/P10_canonical_time/P10_the_entangled_case_is_a_singular_pencil_question_and_the_pencil_is_`

**P10R31** `P10_the_tower_limit_closes_at_first_order_and_the_wall_moves_into_a_boundary_condition_at_secon`
[OK]
*'PO-23'; the tower limit — the one step 'r6950' named, and the boundary condition it turns out to hide* (node 66's 'r6953' order, handing the named step over as **a hypothesis they had run rather than guessed**: in the momentum representation $M(a)\{}\{}psi=0$ is a **first-order ODE** with closed form $\{}\{}psi=C\{}\{},e^{\{}\{}mathrm{i}c(a)p}/p$, non-normalizable because of the $1/p$ at the origin, *\{}"and that is not normalizable for a reason independent of both couplings and of the scale\{}"*. (1)[a] ***\{}"Confirm or refute the reduction itself ... is the representation I have assumed the one it uses? If the measure is not $\{}\{}mathrm{d}p$, the $1/p$ verdict changes\{}"***; (1)[b] ***\{}"then the domain question, which is the load-bearing one. Is $M(a)$ essentially self-adjoint on the natural domain, or does it admit a family of extensions?\{}"*** — with the conditional that *if the extension selection is the datum 'PO-15' carries, that is the ordering surfacing a third time*; (1)[c] ***\{}"and whether more than two operators enter ... report the order of the equation the actual operator content gives, since that decides whether (1)[a]'s closed form is the answer or the first term of one\{}"***. Guards: ***\{}"do not take my ODE on trust\{}"*** — four named places it can fail (representation, measure, domain, content), *\{}"and the last three are exactly where this row's earlier results came from\{}"*; if it closes, say the row has a **third completed argument** and the wall is gone as an object; **and if the extension family is non-trivial, say the wall has MOVED into the extension choice and name it as a different open thing**; and keep the arithmetic exact). *Computes:* **THE REDUCTION IS CORRECT; THE OPERATOR IS NOT ESSENTIALLY SELF-ADJOINT; THE VERDICT IS UNIFORM OVER THE EXTENSION FAMILY ANYWAY — AND (1)[c] IS NOT A FOOTNOTE, BECAUSE AT SECOND ORDER THE ANSWER REVERSES.** **(1) (1)[a] confirmed, and in TWO representations.** With $\{}\{}hat\{}\{}pi\{}\{}to p$, $\{}\{}hat\{}\{}varphi\{}\{}to\{}\{}mathrm{i}\{}\{}mathrm{d}/\{}\{}mathrm{d}p$ the symmetrised vertex is exactly $\{}\{}hat T_2=\{}\{}mathrm{i}\{}\{},p\{}\{},\{}\{}frac{\{}\{}mathrm{d}}{\{}\ — *which is where the $1/p$ comes from: the symmetrisation's own inner derivative* — it is symmetric with the difference an exact total derivative $\{}\{}mathrm{d}(\{}\{}mathrm{i}p^{2}\{}\{}bar fg)$, and sympy returns 66's closed form unprompted. **The measure IS $\{}\{}mathrm{d}p$** (Plancherel), and the verdict is **scale-invariant**: a dilation carries $1/p$ to $1/\{}\{}lambda p$, so no oscillator mass or frequency rescues it. ⇒ *And the position representation is an independent route to the same place*: $M=-((A+Bx)\{}\{}psi')'$, whose $z=0$ solutions are $c_1+c_2\{}\{}ln\{}\{}lvert A+Bx\{}\{}rvert$ — **neither in $L^2$, and no numerics in either route** **(2) (1)[b] 66's worry is JUSTIFIED: the operator is NOT essentially self-adjoint.** At general $z$, verified by substitution, $\{}\{}psi_z=\{}\{}frac{C}{p}\{}\{}exp(\{}\{}mathrm{i}\{}\{}frac{A}{B}p at $z=\{}\{}mathrm{i}\{}\{}sigma$ the second exponential is **real**, and the near-origin integrals are exact — $\{}\{}int_0^1e^{-k/p}p^{-2}\{}\{}mathrm{d}p=e^{-k}/k$ finite, $\{}\{}int_0^1e^{+k/p}p^{ 2}\{}\{}mathrm{d}p=\{}\{}infty$ — so each half-line carries an $L^2$ solution for one sign of $\{}\{}operatorname{Im}z$ and none for the other: **deficiency indices $(1,1)$, a $U(1)$ family of extensions** **(3) AND YET THE QUESTION IS SETTLED UNIFORMLY OVER THAT FAMILY.** At $z=0$ the exponential is a **pure phase**, $\{}\{}lvert\{}\{}psi_0\{}\{}rvert^{2}=\{}\{}lvert

C\{}\{}rvertˆ{2}/pˆ{2}$, and $\{}\{}int_0ˆ1\{}\{}mathrm{d}p/pˆ{2}=\{}\{}infty$ — **no $Lˆ2$ solution on EITHER side**. A boundary condition selects among $Lˆ2$ solutions and there are none to select from ⇒ ***$0$ is not an eigenvalue of ANY extension: the first-order tower criterion closes WITHOUT choosing one***, which is stronger than closing it by choosing the right one. ⇒ **So the order's conditional answers in the negative — the selection is never needed, so it cannot be 'PO-15''s datum surfacing** **(4) (1)$^{c}$ AND AT SECOND ORDER THE ANSWER REVERSES.** A $\{}\{}hat\{}\{}varphiˆ{2}$ structure makes the equation second order; then $\{}\{}psi=\{}\{}alpha+\{}\{}beta p$ solves it to $O(p)$ at the momentum origin, so **both solutions are regular there and the $1/p$ obstruction is GONE** — the failure mode the order flagged, confirmed. The constraint moves to large scale, where $\{}\{}psi"+\{}\{}psi'/x-(C/B)x\{}\{}psi=0$ is **exactly Bessel** in $\{}\{}xi=\{}\{}tfrac23\{}\{}sqrt{C/B}\{}\{}lvert x\{}\{}rvertˆ{3/2}$: exponentially decaying at one end, oscillatory with amplitude $\{}\{}lvert x\{}\{}rvertˆ{-3/4}$ at the other — the full equation measured at $-0.7511$ against that **exact** value — so $\{}\{}lvert\{}\{}psi\{}\{}rvertˆ{2}\{}\{}sim\{}\{}lvert x\{}\{}rvertˆ{-3/2}$ is integrable ⇒ ***both ends admit $Lˆ2$, a normalizable null vector is NOT excluded, and everything turns on the boundary condition at the degenerate point $x_0=-A/B$ where $A+Bx$ vanishes and the solutions are a constant and a logarithm (limit-circle)*** *Bound:* **NOT a third completed argument — it is one AT A STATED OPERATOR CONTENT**, and the row's status is said in those terms rather than banked. *Closed for the content the cubic's kinetic vertex gives ($\{}\{}hat\{}\{}piˆ{2}\{}\{}hat\{}\{}varphi$, one power of $\{}\{}hat\{}\{}varphi$, hence first order); the first term of an answer otherwise.* **AND THE WALL HAS MOVED RATHER THAN VANISHED, named as the order asked**: at second order it is **the boundary condition at the degenerate point**, which is a *different* open object from a limit of determinants and must not be carried as the same item. *It is also the same KIND of object \{}\{}S'lock' already closes at $a=0$ with the horizon's own thermal state — the one concrete suggestion this receipt makes about where to look* **The operator content is the deciding datum and is NOT established here.** What is established: the FREE tower contributes no tower operator at all ('r6930''s traceless excitation, at fixed occupation), so the content begins at the cubic. **Whether a $\{}\{}hat\{}\{}varphiˆ{2}$ structure enters $\{}\{}hat R$ alongside $\{}\{}hat\{}\{}piˆ{2}\{}\{}hat\{}\{}varphi$ is not computed and not guessed** **Arithmetic discipline carried from 'r6950'/'r6947'**: every load-bearing step is closed form — the reduction, the general-$z$ solution (verified by substitution, not by form-matching), the deficiency integrals, the Bessel reduction. *The single float is the amplitude exponent, reported against the exact Bessel value rather than a threshold* No ordering choice (§C never reaches one), no interacting theory beyond the two cubic operators and the hypothetical $\{}\{}hat\{}\{}varphiˆ{2}$, **no corpus edit**; the TT weight stays where 'r6946' left it; nothing touches 'prop:flat', 'PO-31' or 'PO-15'
*Run:* `python3 receipts/P10_canonical_time/P10_the_tower_limit_closes_at_first_order_and_the_wall_mov`

**P10R32** `P10_the_field_squared_is_already_in_the_trace_at_quadratic_order_and_the_degenerate_point_is_re`
[OK]
*'PO-23'; the one datum the row was reduced to — whether a $\{}hat\{}varphiˆ{2}$ structure enters $\{}hat R$ alongside the kinetic vertex's $\{}hat\{}piˆ{2}\{}hat\{}varphi$* (node 66's 'r6959' order, making that datum the whole row and naming the two outcomes as **not symmetric**: *"if it does not enter, 'r6954''s first-order argument is the answer at the theory's actual content ... and **the wall is gone as an object** rather than moved ... that is the third completed argument, and it should be reported in those words"*; *"if it does enter, the obstruction at the momentum origin is gone, both ends admit square-integrable behaviour, and **the wall sits in a boundary condition at $x_0=-A/B$**"*. (2) *only if (1) says yes*: **report the deficiency indices there and the form the family takes**, say whether *"the same argument that closed the first-order case — no square-integrable solution to select among — has any purchase at all"*, and whether the $a=0$ mechanism reaches that point. The direction, named and not claimed: *"the curvature enters through the trace, so the question is which structures the stress trace carries at cubic order ... a $\{}hat\{}varphiˆ{2}$ in $\{}hat\{}Theta$ is a potential-like term, and the cubic vertex you used is kinetic"*, flagged with its pedigree of four withdrawn clauses. Guards: *"computed and inconclusive with the step named is progress; not attempted is not"*; ***"do not let the kinetic vertex answer for the content"***, and *"if the content turns out to be larger than two structures, say what the equation's order becomes and stop rather than extrapolating"*; keep the arithmetic exact and report any float against an exact predicted value; the ordering stays named and unpicked). *Computes:* **(1) IS YES — AND YES EARLIER IN THE EXPAN-

SION THAN THE ORDER'S DIRECTION LOOKED. THE DIRECTION IS RIGHT AND THE ORDER IS WRONG: $\{}hat\{}varphi^{2}$ DOES NOT WAIT FOR THE CUBIC.** **(1) The answer, with three independent derivations.** 'sec:lock''s own tower Hamiltonian $\{}hat H=\{}sum_n[\{}hat\{}pi_n^{2}/2a^{3}+\{}tfrac12a\{}mu_n^{2}\{}hat\{}varphi_n^{2}]$ under 'r6934''s trace formula $h\{}otimes T\{}mapsto(h+ah')T/2\{}pi^{2}a^{3}$ gives $2\{}pi^{2}\{}hat\{}Theta_{\quad}=-\{}hat\{}pi_n^{2}/a^{6}+\{}mu_n^{2}\{}hat\{}varphi_n^{2}/a^{2}$ — so **$\{}hat\{}varphi^{2}$ enters $\{}hat R$ with coefficient $\{}kappa\{}mu_n^{2}/2\{}pi^{2}$ at the power $a^{-2}$**, from the tower's own **gradient** energy, at $j=0$ and every order above. *Route 2, taking no $\{}partial/\{}partial V$ at all*: $\{}rho-3p=(\{}partial\{}varphi)^{2}=-\{}dot\{}varphi^{2}+\{}mu^{2}\{}v with $\{}dot\{}varphi=\{}pi/a^{3}$, term for term the same. *Route 3, 'r6930''s own instrument*: its 'trace_of' on $E=S(a)/a$ is exactly $S'(a)/2\{}pi^{2}a^{3}$, and with $S(a)=\{}mu(\{}hat N+\{}tfrac12)=(a^{2}\{}mu^{2}\{}hat\{}varphi^{2}+\{}hat\{}pi^{2}/a^{2})/2$ in the **fixed** field operators it reproduces both. $\{}mu_n^{2}=m^{2}-3\{}ge6$, so the coefficient never vanishes; and the structure is **degree two** in the fields, not a cubic-order effect **(2) WHERE THE ROW LOST IT, AND IT IS A SIXTH FACE OF THE STANDING LESSON: THE DOMAIN OF A VANISHING.** 'r6930''s 'trace_of' docstring scopes itself *"at fixed occupation numbers"*; its verdict line drops the scope — *"traceless IDENTICALLY, as an operator ... whatever the occupation numbers"*. Since $M\{}omega=a^{2}\{}mu$, $\{}hat a_n$ depends on $a$: $S$ is free of $a$ for the c-number zero point $\{}sum\{}mu_n/2$ and along fixed occupation, and **not for the operator**. In the adiabatic basis the same operator is exactly $(\{}mu/a^{4})(\{}hat a^{2}+\{}hat a^{\{}dagger2})=\{}hat X^{2}-\{}hat P^{2}$ with $[\{}hat X,\{}hat P]=\{}mathrm i\{}mu/a^{4}$, whose **diagonal is exactly zero at every truncation** — which *is* the harmonic virial theorem $\{}langle n\{}rvert\{}hat K\{}lvert n\{}rangle=\{}langle n\{}rvert\{}hat V\{}lvert n\{}rangle$, verified exactly, and it holds in every mixture of number states too. **But on the squeezed state $\{}lvert0\{}rangle+t\{}lvert2\{}rangle$ the expectation is $2\{}sqrt2\{}mu t/a^{4}(1+t^{2})\{}ne0$**, and normal ordering cannot remove it, $\{}hat a^{2}+\{}hat a^{\{}dagger2}$ carrying no contraction **(3) THE LANDED CONCLUSION SURVIVES, ON STRONGER GROUND.** The residual is an **inverted oscillator**, point spectrum empty: the null equation $\{}psi"+p^{2}\{}psi/\{}mu^{2}a^{4}=0$ is solved by substitution by $\{}sqrt p J_{\{}pm1/4}(p^{2}/2\{}mu a^{2})$, **both regular at the momentum origin and neither square-integrable at infinity** ($\{}lvert\{}psi\{}rvert^{2}\{}sim1/p$, $\{}int_1^\{}infty\{}mathrm dp/p=\{}infty$; envelope exponent $-0.50000$ against the exact $-\{}tfrac12$, and the $-\{}tfrac12$ exact at every real $z$ since the WKB amplitude is $(p^{2}+z)^{-1/4}$). ⇒ ***$\{}hat R$ has no eigenvector at quadratic order — not because it is a function of $\{}hat a$ alone, which it is not, but because an inverted oscillator has no normalizable eigenvector at any real eigenvalue***, which is a warrant that does not depend on a term being absent **(4) (2) AND THE SECOND BRANCH'S OBJECT DOES NOT EXIST: THE DEGENERATE POINT IS A FEATURE OF ONE REPRESENTATION.** One abstract operator $\{}hat M=c_1\{}hat\{}pi^{2}+c_2\{}tfrac12(\{}hat\{}pi^{2}\{}hat\{}varphi+ verified to be one operator by its exact Hermite matrix against the abstract ladder matrix: in **position** it is $-(c_1+c_2x)\{}psi"-c_2\{}psi'+c_3x^{2}\{}psi$, leading coefficient vanishing at $x_0=-c_1/c_2$; in **momentum** it is $-c_3\{}psi"+\{}mathrm ic_2(p^{2}\{}psi'+p\{}psi)+c_1p^{2}\{}psi$, **leading coefficient the constant $-c_3$, vanishing nowhere**. At large $\{}lvert p\{}rvert$ the two behaviours are $\{}psi_1\{}sim p^{-1}e^{\{}mathrm ic_2p^{3}/3c_3-\{}mathrm ic_1p/c_2}$ and $\{}psi_2\{}sim e^{\{}mathrm ic_1p/c_2}$, each verified by an exact limit on its relative residual and both exponents purely imaginary for real couplings, so $\{}lvert\{}psi_1\{}rvert=1/p$ and $\{}lvert\{}psi_2\{}rvert=1$ exactly ⇒ ***exactly one $L^{2}$ solution at each end, both ends limit-point, deficiency indices $(0,0)$, essentially self-adjoint — the family is a single point and there is nothing to select.*** And what distinguishes $x_0$ from $a=0$: $a=0$ is an **endpoint of the physical configuration space** and a physical locus (the horizon, $\{}kappa=1/\{}alpha$), so the freedom is genuine and the thermal state closes it; $x_0$ is an interior point in one representation and **no point at all** in the other, carrying no locus — *the mechanism does not reach it because there is nothing there to reach* *Bound:* **NOT the third completed argument, and two of the order's own guards are why.** *First*, the closure at $c_2=0$ **does not extend by continuity**: at $c_2=0$ neither solution is $L^{2}$, at $c_2\{}ne0$ exactly one is at each end, so switching on the kinetic vertex CREATES a square-integrable solution where there was none. **What remains is a CONNECTION condition** — one analytic equation asking the two ends' one-dimensional subspaces to coincide — **and not a boundary condition**. The scaling $p\{}to\{}lambda q$ leaves one dimensionless parameter $v=c_1c_3^{1/3}c_2^{-4/3}$, constant in $a$ only if the cu-

bic's trace power is exactly $5$ and the corpus's own *"$\{}pi_n^{2}\{}varphi_m/a^{3}$ in kind"* puts it at $6$; the $c_2\{}to0$ endpoint says the condition is not identically satisfied. *A shape, not a verdict, and not claimed either way* *Second*, **the content may be larger than three structures.** The kinetic expansion $\{}hat\{}pi^{2}/(1+\{}lambda\{}hat\{}varphi)$ always carries two powers of momentum; the **potential** sector — the $\{}tfrac12a\{}mu^{2}\{}hat\{}varphi^{2}$ term (1) came from, $\{}sqrt h\{},{}^{(3)}R$ expanded in the transverse-traceless perturbation — gives $\{}hat\{}varphi^{2}$ at quadratic order and $\{}hat\{}varphi^{3}$ at cubic, with no momenta at all. 'sec:lock' named only the kinetic vertex because that is the one singular at $a=0$; the potential cubic is regular there and so was never in view, **but the trace does not care which is singular**. ⇒ **Order of the null equation reported and stopped, per the guard: 2 at the content established here; 3 if $\{}sqrt h\{},{}^{(3)}R$'s cubic enters** **AND ONE LANDED CLAIM DOES NOT SURVIVE AS STATED**: *"the spectrum is purely continuous, the half-line above $4\{}Lambda$ with $4\{}Lambda$ itself not attained"* rested on the same absence, and $\{}hat X^{2}-\{}hat P^{2}$ has spectrum all of $\{}mathbb R$, so the restored content puts spectrum **below** $4\{}Lambda$ too. *Purely continuous survives; two-sided rather than a half-line.* Claimed at quadratic content only, flagged, routed, not extrapolated **The ordering did not surface**: the quadratic operator has no ordering ambiguity at all **One float in the receipt** — the envelope exponent, against an exact predicted value; an earlier draft measured the two §(2) exponents by integration and the exact limit replaced it, the recessive branch being ill-conditioned outward with a phase integral growing as $p^{3}$ No ordering choice, no interacting theory beyond the trace's content at cubic order, **no corpus edit** — three 'P10' sites and their replacement sentences are routed in 'FOR_66_FROM_60.md'; the TT weight stays where 'r6946' left it; nothing touches 'prop:flat', 'PO-31' or 'PO-15' *Run:* `python3 receipts/P10_canonical_time/P10_the_field_squared_is_already_in_the_trace_at_quadratic_order_and_th`

**P10R33** `P10_the_operator_is_an_inverted_quartic_so_the_deficiency_is_two_two_and_the_measure_closes_it_a`
[OK]
*'PO-23'; the row's last two data — the potential sector's cubic, and the connection coefficient 'r6962' reduced the criterion to* (node 66's 'r6965' order, taking the two data in the order that fixes tractability. (1)[a] ***"Expand $\{}sqrt h\{},{}^{3}R$ to cubic order in the perturbation and report whether a $\{}hat\{}varphi^{3}$ structure survives into $\{}hat\{}Theta$, with the coefficient if it does"*** — *"the trace formula is in hand and the quadratic case is worked, so this is the same computation one order up, and the answer is a yes or a no about one structure, not a range"*; (1)[b] ***"and say what the order of the null equation then is"***. (2)[a] ***"Compute the connection coefficient as a function of $v$ and report where it vanishes"*** — with the offer that if it has no zero, *"the criterion closes at the three-structure content and the wall is gone as an object at that content ... report it in those words only if it holds at the content (1) establishes"*; (2)[b] ***"and if it does have a zero, say what it means that $v$ is not constant in the scale factor"*** — *"an eigenvector needs the condition to hold on a set of positive measure in $a$, which is 'r6950''s own criterion returning one level up"*. Guards: **if (1) says the potential cubic enters, take (2) at the three-structure content only and say so**, *"do not extrapolate across the order the way the row just extrapolated across the diagonal"*; exact arithmetic, or a float against an exact prediction; the ordering stays named and unpicked; and the one that fires — ***"the sixth face applies to your own new results ... for each, say what its scope is in the same sentence that states it, since the defect this revision found was exactly a scope that lived in one place and a claim that lived in another"***). *Computes:* **THE GUARD FIRED ON ITS OWN AUTHOR, AND THE WITHDRAWAL IS THE FIRST THING HERE.** **(1) 'r6962''s DEFICIENCY COUNT IS WRONG AND IS WITHDRAWN.** It reported *"exactly one $L^{2}$ solution at each end, both ends limit-point, deficiency indices $(0,0)$, essentially self-adjoint, nothing to select"* — **all four clauses**. The test was an exact limit on an ansatz's *relative residual*, and for the dominant branch that residual vanishes **at order $1/p$**: $e^{\{}mathrm ic_1p/c_2}$ and $p^{-1}e^{\{}mathrm ic_1p/c_2}$ **both pass** ($p\{}cdot$residual $\{}to-\{}beta$ and $\{}to0$ respectively), so ***the check's discriminating order equalled the effect's order*** and the modulus claim — hence the $L^{2}$ count, hence the deficiency, hence "nothing to select" — lived outside what it could see **(2) AND THE REPAIR IS A CLOSED FORM THAT SETTLES EVERYTHING AT ONCE.** The **unimodular** gauge $\{}psi=\{}exp(\{}mathrm ic_2p^{3}/6c_3)\{}chi$, modulus exactly $1$ hence unitary on $L^{2}(\{}mathbb R,\{}mathrm dp)$, removes the first-order term with residual **exactly zero**: $$\{}hat M=c_1\{}hat\{}pi^{2}+c_2\{},\{}mathrm{sym}(\{}hat\{}pi^{2}\{}hat\{}varphi)+c_3\{}hat

c_3\{}partial_p^{2}+c_1p^{2}-\{}frac{c_2^{2}}{4c_3}p^{4},$$ and the quartic coefficient is negative for $c_3¿0$ **whatever the sign of the cubic vertex** — ***the three-structure operator IS an inverted quartic***. ⇒ At every real $z$ both solutions have WKB amplitude $\{}lvert V\{}rvert^{-1/4}\{}sim p^{-1}$ (measured $-0.9985$ against the exact $-1$), so $\{}lvert\{}psi\{}rvert^{2}\{}sim p^{-2}$ is integrable at **each** end: **limit circle at both ends, deficiency $(2,2)$, NOT essentially self-adjoint.** *And the dichotomy is exact at $c_2=0$: $\{}int_1^\{}infty\{}mathrm dp/\{}sqrt{p^{4}}=1$ converges while $\{}int_1^\{}infty\{}mathrm dp/\{}sqrt{p^{2}}=\{}infty$ diverges — so ‘r6962‘’s quadratic-content closure stands, and the discontinuity it flagged is real and sharper than it said* **(3) (1)[a] THE TERM IS IN THE TRACE, AND THEN THE CONTROL SPLIT THE QUESTION.** Exactly: $\{}mathrm{tr}\{},\{}varepsilon^{3}=3\{}det\{}varepsilon$ for **every** traceless symmetric $3\{}times3$ matrix; $\{}sqrt{\{}det(\{}gamma+\{}varepsilon)}=1-\{}tfrac14\{}mathrm{tr}\{}varepsilon^{2}+\{}tfrac16\{}mathrm{tr}\{}varepsilon^{3}+O(\{}varepsilon^{4})$ $R^{(1)}=0$ for transverse-traceless on a maximally symmetric section, so the cross term dies and the cubic order of $\{}sqrt h\{},{}^{3}R$ is $a\{}sqrt\{}gamma[R^{(3)}+\{}mathrm{tr}\{}varepsilon^{3}]$ — **the second piece being the volume element’s own, against the UNPERTURBED $6/a^{2}$, needing no curvature perturbation**. And the trace formula annihilates **exactly one** power, $h\{}propto1/a$ by ‘dsolve‘, while the potential sector’s $h\{}propto a^{+1}$ gives $h+ah’=2h$, so the term reaches $\{}hat\{}Theta$ at $a^{-2}$, one power of $\{}ell_P$ **(4) BUT THE SINGLE-MODE DIAGONAL SITS EXACTLY ON THE DEGENERATE SET, AND THAT IS COMPUTED RATHER THAN FEARED.** Asked for $R^{(3)}$ on a flat section with an explicit transverse-traceless plane wave, the answer is **exactly zero** — and the reason is algebraic: for a single wave vector transversality is $k^{j}\{}varepsilon_{ij}=0$, making $k$ a **null eigenvector**, so the whole polarisation space $[[A,B,0],[B,-A,0],[0,0,0]]$ has $\{}det=0$ and $\{}mathrm{tr}\{},\{}varepsilon^{3}=0$ identically. ***Two*** wave vectors lift it off: $\{}det=f_1f_2(f_1-f_2)$. ⇒ **So the cubic potential is intrinsically a MULTI-MODE term at flat order, and what would put a $\{}hat\{}varphi^{3}$ on ONE excitation factor is the one integral $\{}int\{}sqrt\{}gamma\{}det e_{(n)}$ on $S^{3}$, where transversality is differential and does not force a pointwise null eigenvector** **(5) (1)[b] SO THE ORDER IS TWO AT THE ESTABLISHED CONTENT**, three only if that integral is non-zero **(6) (2)[a] THE CONNECTION COEFFICIENT DOES NOT EXIST AS PUT, AND $0$ IS AN EIGENVALUE OF SOME REALISATION.** The question asked two *one-dimensional* subspaces to coincide; the deficiency space at $z=0$ is **two**-dimensional at each end and globally, the momentum-side operator being regular everywhere. The gauge-transformed equation has **real** coefficients, so it carries a real $L^{2}$ null solution (tail exponent $-0.9985$ against the exact $-1$), and $\{}hat M_{\{}min}+\{}mathrm{span}\{}{\{}chi_0\{}}$ is symmetric and extends. ⇒ ***‘r6954‘’s uniform argument has no purchase here: there are two solutions to select among, not none*** **(7) AND YET (2)[b] CLOSES IT, FOR EVERY REALISATION HELD FIXED ACROSS FIBRES.** The rescaling $\{}lambda^{6}=4c_3^{2}/c_2^{2}$ gives the canonical $\{}hat H(w)=-\{}partial_q^{2}+wq^{2}-q^{4}$ with $w=2^{4/3}c_1c_3^{1/3}\{}lvert c_2\{}rvert^{-4/3}\{}propto v$ and $w\{}propto a^{4m/3-20/3}$, constant only at $m=5$ where the corpus’s own *”in kind”* puts $m=6$, so $w\{}propto a^{4/3}$. **Limit circle at both ends makes every realisation’s resolvent compact, hence its spectrum purely discrete; and Hellmann–Feynman gives $\{}mathrm d\{}lambda_n/\{}mathrm dw=\{}langle\{}psi_n\{}rvert q^{2}\{}lvert\{}psi_n\{}rangle¿0$ STRICTLY for every realisation** — measured against that exact value at two different wall positions, worst relative departure $8.2\{}times10^{-9}$ and $1.4\{}times10^{-8}$, every derivative positive, and the first six eigenvalues all rising from $w=-4$ to $w=+4$. ⇒ ***Each branch crosses zero at most once, so $\{}{w:0\{}in\{}mathrm{spec}\{}}$ is discrete, so $\{}{a\{}}$ is discrete, so the condition cannot hold on a set of positive measure: NO EIGENVECTOR at the three-structure content*** — which is ‘r6950‘’s criterion returning one level up, exactly as the order predicted *Bound:* **NOT the row’s ”the wall is gone as an object”, for the third time** — and this time not because the argument failed to reach but because **a claim of mine had to be withdrawn and what replaces it is an extension family rather than a closure** **THE ONE LOOPHOLE, NAMED AND NOT CLOSED, AND IT LANDS ON AN ITEM ALREADY OPEN**: the measure argument fixes ONE realisation across fibres, and a realisation chosen *fibre by fibre* could track the zero and would supply an eigenvector. *But the extension sits at $p\{}to\{}pm\{}infty$, the tower’s **ultraviolet**, and \{}S‘lock‘ already carries ”the ultraviolet definition of the tower sums” as its open frontier* — so the wall lands on the item the paper already names rather than a new one, and is a different **kind** of object from the $a=0$ extension, which is at a

physical boundary and is already closed by the horizon's thermal state **NOT the VALUE of the $S^{3}$ diagonal overlap** — only that its flat analogue vanishes for an algebraic reason and that the section's curvature is the only thing that could lift it; 'r6946''s selection rule is cited and is **not** used to infer a value **(2) is at the THREE-structure content and is not extrapolated**, per the guard: a third-order operator is not a Schr\{}"odinger operator and the gauge argument does not reach it **WHAT OF 'r6962' IS NOT WITHDRAWN, and why, one by one**: that $\{}hat\{}varphi^{2}$ enters $\{}hat\{}Theta$ at quadratic order (three independent exact derivations, none asymptotic); the sixth face on 'r6930' (an exactly-zero diagonal against a non-zero matrix element); the quadratic-content inverted **oscillator** and its empty point spectrum (the $c_2=0$ case, where the limit-circle integral **diverges**, so there neither solution is $L^{2}$); and the degenerate point's representational status — *strengthened*, the gauge showing the momentum-side operator is unitarily a plain Schr\{}"odinger operator with **no singular point at all** **Two floats**, each against an exact predicted value: the envelope exponent against $-1$, and $\{}mathrm d\{}lambda_n/\{}mathrm dw$ against the Hellmann–Feynman expectation computed in the same arithmetic No ordering choice — the potential cubic is a product of configuration variables and carries none; no realisation chosen; **no corpus edit**, the withdrawn sentences and their replacements routed in 'FOR_66_FROM_60.md'; the TT weight stays where 'r6946' left it; nothing touches 'prop:flat', 'PO-31' or 'PO-15' *Run:* `python3 receipts/P10_canonical_time/P10_the_operator_is_an_inverted_quartic_so_the_deficiency_is_two_two_an`

**P10R34** `P10_the_lowest_tt_harmonics_are_frame_constant_so_the_overlap_is_a_determinant_times_the_volume` [OK]

*'PO-23'; the overlap integral — the last thing between the row and its own object* (node 66's 'r6967' order. (1)[a] ***"Compute it, and report it as a number with its scope"*** — the value of $\{}int\{}sqrt\{}gamma\{},\{}det e_{(n)}$ on the closed section, *"on harmonics the corpus already has. **If it is zero, say whether it is zero for a reason** — the flat case's zero came from an algebraic identity, and a second zero arriving by cancellation is a different fact from one arriving by a selection rule"*; (1)[b] ***"and if it is non-zero, report the order and stop there ... Do not carry the measure argument across the order"***. (2)[a] ***"Write me the one paragraph that says what this row has established, as a single claim with its scope"*** — *"not a list and not a history ... at the strength the seven approaches jointly support and no higher"*; (2)[b] ***"and say plainly what the remaining open thing is, in the same paragraph. If it is the ultraviolet definition and nothing else, say that — because a row whose remainder is its own founding object is in a different state from one carrying a list"***. Guards: ***the class from 'r6966' applied PROSPECTIVELY*** — *"ask, of whatever test decides the integral, whether its resolution is larger than the effect it is resolving"*; exact arithmetic; and *"(2) is not an invitation to overstate — a paragraph that claims more than the seven jointly support would undo all of them, and I would rather have the honest shape than a clean one"*). *Computes:* **THE OVERLAP IS $2\{}pi^{2}\{}det h$ — NON-ZERO, EXACTLY, AND WITH NO INTEGRATION PERFORMED.** **(1) THE GUARD ANSWERED BY THE CHOICE OF INSTRUMENT, BEFORE THE COMPUTATION.** The lowest transverse-traceless harmonics have **CONSTANT components in the left-invariant orthonormal frame**, so the ordered integrand is a constant and the integral is that constant times the volume: no discretisation, no truncation, no quadrature, no asymptotic order, ***and no floats anywhere in this receipt*** — so there is no resolution to compare against the effect and the class cannot bite. *That is the generalisation used prospectively, which is what the order asked for* **(2) AND THEY ARE THE CORPUS'S OWN HARMONICS, IDENTIFIED TWO INDEPENDENT WAYS.** With $\{}mathrm d\{}sigma^{a}=-\{}tfrac12\{}epsilon^{abc}\{}sigma^{b}\{}wedge\{}sigma^{c}$ verified in coordinates and $e^{a}=\{}sigma^{a}/2$, a **general** constant traceless symmetric $h$ gives $\{}gamma^{ij}\{}varepsilon_{ij}=\{}mathrm{tr}\{},h=0$ and $D^{i}\{}varepsilon_{ij}=0$ **exactly** — computed from the Christoffel symbols for all five parameters at once, so it is the whole multiplet and not one lucky member — and the frame algebra gives $-\{}nabla^{2}\{}varepsilon=6\{}varepsilon$. ⇒ ***$\{}mu^{2}=m^{2}-3=6$ at $m=3$ matches the eigenvalue, and $5$ left-invariant $+\{},5$ right-invariant $=10=2(m^{2}-4)=d(3)$ matches the degeneracy*** **(3) (1)[a] THE NUMBER.** The frame being orthonormal, $\{}varepsilon^{i}{}_{j}=e^{-1}h\{},e$ is a **similarity transform**, so $\{}det\{}varepsilon=\{}det h$ *pointwise* and $$\{}int\{}!\{}sqrt\{}gamma\{},\{}det e_{(n)}=2\{}pi^{2}\{}det h .$$ On $h=\{}mathrm{diag}(2,-1,-1)$ that is $4\{}pi^{2}$, and $\{}mathrm{tr}\{},\{}varepsilon^{3}=3\{}det\{}varepsilon=6$ puts $12\{}pi^{2}$ into the cubic term of $\{}sqrt h\{},\{}^{3}R$ **(4) AND THE SCOPE IS WORTH MORE THAN THE

VALUE: IT IS BASIS-DEPENDENT.** $\{}det h$ vanishes on a **codimension-one hypersurface** of the five-dimensional multiplet ($\{}mathrm{diag}(1,-1,0)$ gives zero), so a rotation inside the multiplet trades the diagonal $\{}hat\{}varphi_nˆ{3}$ against the off-diagonal $\{}hat\{}varphi_n\{}hat\{}varphi_m\{}hat\{}varphi_l$: ***the diagonal is non-zero for a generic basis and zero on a measure-zero set of them, and the basis-independent statement is that the cubic potential is a non-vanishing trilinear form on the multiplet*** **(5) AND THIS IS NOT THE FLAT ZERO WITH THE SIGN FLIPPED — IT IS A DIFFERENT KIND OF FACT**, which is the distinction the order asked for. In the flat case the **entire** polarisation space $[[A,B,0],[B,-A,0],[0,0,0]]$ has $\{}det=0$ for every member, transversality there being the algebraic $kˆ{j}\{}varepsilon_{ij}=0$ and making the wave vector a null eigenvector. *A zero forced by an identity, against a non-zero holding generically* **(6) $(1)^{b}$ SO THE ORDER OF THE NULL EQUATION IS THREE, AND THE MEASURE ARGUMENT DOES NOT CARRY**, for three reasons that are facts rather than caution: the gauge that produced the inverted quartic removes a **first**-order term from a **second**-order operator and has no third-order analogue; an odd-order symmetric operator need not have equal deficiency indices, so even the **counting** at the two ends is a different problem; and the Hellmann–Feynman step needs a self-adjoint operator with discrete spectrum and a positive perturbation. *Named, not analysed* **(7) (2) THE PARAGRAPH IS WRITTEN AND ROUTED**, as one claim with a uniform scope: *once the scale factor is quantized the tower's renormalized zero point makes one ultraviolet constant observable and the construction has no route left to hide it again* — the residue being an exact functional of the free spectrum, the logarithm unpartnered, the interaction unable to reach the counterterm's power, and the curvature operator without an eigenvector for a realisation fixed across the fibres — *"every step at the operator content the construction's own expansion supplies, and for a realisation fixed independently of the scale factor"* *Bound:* **$(2)^{b}$ DECLINED, AND DELIBERATELY: THE REMAINDER IS TWO THINGS, NOT ONE.** The order hoped *"if it is the ultraviolet definition and nothing else, say that"*, and **it is not, so I do not say it was** — $(1)^{a}$ makes the content four structures and thereby opens a second item. The two **differ in kind** and the paragraph says which is which: *(i)* **definitional**, and the construction's own founding object — the ultraviolet definition of the mode sums, where the one freedom the measure argument does not fix lives, a realisation chosen fibre by fibre at unbounded momentum, which no calculation of this sort can close at all; *(ii)* **an ordinary calculation**, newly opened by this revision — the criterion at third order. *The clean answer was available: had the integral come back zero the remainder would have been one item. It did not, and the row is one item further from that than it was, not closer.* *In the row's favour anyway: (ii) is a calculation with a stated content and a known shape, which is what this row has spent eight revisions converting its walls into. It is not a category* **NOT the VALUE on any other level of the tower** — the overlap is computed on the lowest, where the frame-constant construction applies, and the receipt says so in the same sentence that states it **NOT a basis for the multiplet chosen or recommended**: the non-zero is generic and the vanishing set is exhibited, and which basis the mode expansion uses is not this receipt's to fix **NOT the third-order asymptotic count**, and explicitly not an extension of ‘r6966‘'s second-order one **No floats at all**, so no tolerance to calibrate and no prediction to hit No ordering choice — the structure is a product of configuration variables and carries none; **no corpus edit**, the paragraph routed for 66 to place, as the order directs; the TT weight stays where ‘r6946‘ left it; nothing touches ‘prop:flat‘, ‘PO-31‘ or ‘PO-15‘ *Run:* `python3 receipts/P10_canonical_time/P10_the_lowest_tt_harmonics_are_frame_constant_so_the_overlap_is_a_dete`

**P10R35** `P10_the_third_order_operator_is_regular_and_its_deficiency_indices_are_equal_so_a_realisation_e` [OK]

*‘PO-23‘; the criterion at third order — the row's last datum, taken as three obstructions one at a time* (node 66's ‘r6969‘ order, with **two theorems named in advance as premises asserted from outside**. (1) ***"Write the operator down at four structures, in the momentum representation, and say what kind of object it is before analysing it"*** — with 66's own reading offered to be refuted (*"the leading coefficient is a CONSTANT, carrying no momentum at all, and there is no singular point anywhere on the line"*) *premise: that $\{}hat\{}varphiˆ{3}$ is formally symmetric in this representation and the operator is formally symmetric as a whole*. (2) ***"Then count, at each end separately, and report the two counts as two numbers"*** *premise, and the load-bearing one: that for an odd-order formally symmetric ordinary differential operator the deficiency indices at the two ends need not be equal* ⇒ ***"if they come out unequal there is no self-adjoint realisation at all, and the question changes in kind rather than being answered

— which is a result and I want it reported as one"***; and *"if that sign decides the count, say which sign the construction gives"*. (3) ***"And count how many dimensionless combinations of the couplings survive rescaling at four structures"*** — *"if four structures give two, the scale factor traces a PATH in a two-parameter family"* *"do not extend the second-order argument by analogy; if the path is not monotone, the honest answer is that this route does not close it, and I would rather have that than a fourth thing to withdraw"*. Guards: **the seventh face used PROSPECTIVELY** — *"before running any asymptotic count, state the order at which the branches you are separating differ, and show the test resolves finer than that. The $(0,0)$ count died because those two numbers were equal"* — and the fifth face on my own new results, *"state each scope in the sentence that states the claim, not in the item beside it"*). *Computes:* **BOTH PREMISES CHECK OUT AS STATED AND NEITHER CONCLUSION FOLLOWS.** **(1) (1) THE OPERATOR, AND 66's READING CONFIRMED.** Built from $\{}hat\{}pi\{}to p$, $\{}hat\{}varphi\{}to\{}mathrm i\{},\{}mathrm d/\{}mathrm dp$ and checked structure by structure: $\{}hat M\{}psi=-\{}mathrm ic_4\{}psi"'-c_3\{}psi"+\{}mathrm ic_2(pˆ{2}\{}psi'+p\{}psi)+c_1pˆ{ whose **leading coefficient is the constant $-\{}mathrm ic_4$** — no momentum in it, so ***no singular point anywhere on the line***, and the object is a constant-coefficient third derivative plus terms whose coefficients grow as $pˆ{2}$ **(2) THE FIRST PREMISE HOLDS**: $\{}hat\{}varphiˆ{3}$ needs no symmetrisation of its own — abstractly a power of one symmetric operator, and in this representation $(-\{}mathrm i\{},\{}mathrm dˆ{3}/\{}mathrm dpˆ{3})ˆ{\{}dagger}$ is itself, the Lagrange difference being an exact total derivative, verified for the cubic alone **and for $\{}hat M$ entire as one bilinear concomitant** **(3) THE SEVENTH FACE ANSWERED BEFORE THE COUNT, WHICH IS WHAT THE ORDER ASKED.** Two separations, not one: the **exponents** differ at order $p$ (integrated, at $pˆ{2}$), and the **prefactor** is the $pˆ{-1}$ from the $O(1/p)$ term in $s$ — ***exactly what 'r6962' could not see, its discriminating order having been the same $1/p$***. So the test is built for the prefactor: $pˆ{2}$ times the relative residual separates the prefactored ansatz from the bare one by exactly $-\{}alpha$ on the constant branch and $2\{}alpha$ on the others, **non-zero where the old test had nothing to see** — plus an independent numerical measurement of the modulus, the instrument whose absence let the earlier claim through **(4) (2) THE THREE BRANCHES, 66's ASYMPTOTIC READING CONFIRMED**: $s\{}simeq\{}sigma p$ with $\{}sigmaˆ{2}=\{}alpha$ for two, $s\{}to\{}mathrm i\{}beta/\{}alpha$ constant for the third; the prefactor exponent is **exactly $-1$** on the constant branch and $-1$ plus a purely imaginary coupling-dependent term on the others, ***so the modulus is exactly $pˆ{-1}$ either way*** **(5) THE TWO COUNTS, AS TWO NUMBERS, IN TWO CASES DECIDED BY ONE SIGN**: at $\{}alpha¡0$ all three branches are oscillatory with modulus $pˆ{-1}$ and every solution is square-integrable at **both** ends, giving $(3,3)$ — limit circle at each end, the maximum a third-order operator admits; at $\{}alpha¿0$, $\{}exp(\{}pm\{}sigma pˆ{2}/2)$ leaves two of three square-integrable at each end and the two two-dimensional subspaces are distinct, giving $(1,1)$ **(6) AND THE LOAD-BEARING PREMISE IS TRUE IN GENERAL AND DOES NOT BITE HERE, WHICH IS THE REPORT.** The phenomenon is real and the control is computed: $\{}mathrm i\{},\{}mathrm d/\{}mathrm dx$ on a half-line has $\{}exp(-x)$ square-integrable and $\{}exp(+x)$ not, indices $(0,1)$, **no self-adjoint extension at all**. But the mechanism needs the spectral parameter to change the asymptotic count between the half-planes, and **adding $z$ leaves the $pˆ{3}$, $pˆ{2}$ AND $pˆ{1}$ coefficients of the Riccati expansion untouched** — $z$ enters neither the characteristic exponent nor the prefactor. ⇒ ***Hence $n_+=n_-$ and a self-adjoint realisation of the third-order operator EXISTS***, a $U(1)$ family at $\{}alpha¿0$ and a $U(3)$ family at $\{}alpha¡0$, at every non-real $z$ and at both ends **(7) (3) TWO INVARIANTS — 66's FEAR CORRECT — BUT THE PATH IS A RAY.** Under $p=\{}lambda q$ the four couplings leave three coefficients and one scaling, so **two** invariants survive, $v_1=\{}gamma\{}lvert\{}alpha\{}rvertˆ{-1/4}$ and $v_2=\{}beta\{}lvert\{}alpha\{}rvertˆ{-5/4}$, plus the discrete sign of $\{}alpha$. But the trace formula's own powers — $c_1\{}sim aˆ{-6}$, $c_2\{}sim aˆ{-6}$, $c_3\{}sim aˆ{-2}$, $c_4\{}sim aˆ{-2}$, hence $\{}alpha\{}sim aˆ{-4}$, $\{}beta\{}sim aˆ{-4}$, $\{}gamma\{}sim aˆ{0}$ — give ***$v_1\{}propto a$ and $v_2\{}propto a$: BOTH LINEAR, so the path is a RAY through the origin traversed strictly monotonically***, and the two-parameter family collapses to one monotone parameter. *And $\{}mathrm{sign}\{},\{}alpha$ is independent of the scale factor, so **whichever count holds, it holds at every scale factor*** *Bound:* **AND THE MEASURE ARGUMENT IS NOT EXTENDED ALONG THAT RAY, WHICH THE ORDER EXPLICITLY

ASKED.** Having the right shape is not having the argument, and two exact reasons say the instrument does not reach: the Hellmann–Feynman step needs $\{}partial\{}hat H/\{}partial w$ to be a **single positive operator**, and along the ray $q^{2}$, the $q^{2}$-weighted first-order term and the third derivative all move together, so the derivative is not sign definite; and at $\{}alpha¡0$ the realisation family is **nine**-parameter rather than one, so *"each branch crosses zero at most once"* has no single branch to be about. ⇒ *What would close it is a monotone quantity along the ray for the actual realisation — a different instrument from ‘r6966‘’s. **The honest answer is that this route does not close it***, which the order said it preferred to a fourth withdrawal **NOT WHICH SIGN $\{}alpha$ HAS, and the order asked.** Two parts are exact — the trace formula supplies an exact relative minus by itself, $(1-n)$ being $-2$ at $h\{}sim a^{-3}$ against $+2$ at $h\{}sim a^{+1}$; and $\{}alpha=c_2/c_4$ is **invariant** under $\{}hat\{}varphi\{}to-\{}hat\{}varphi$, both coefficients being odd, so its sign is **physical** and not an artefact of the mode’s labelling, which matters because the determinant setting $c_4$ is basis-dependent. *What is left is one ratio — the kinetic three-harmonic overlap against the potential determinant overlap — which the row has not computed, so **both counts stand*** **NOT which realisation the construction picks**: the verdict is about the COUNT, stated with that scope in its own sentence **NOT a third-order analogue of the inverted quartic**: the gauge that produced it removes a first-order term from a second-order operator and has no third-order analogue, named and not sought **66’s CORRECTION TO MY ROUTED PARAGRAPH IS ACCEPTED IN FULL** — the scope clause claimed uniformity my own reply contradicted two items later, and *"when a correction lands in a passage, the unit to re-read is the passage"* is carried forward as the habit **One float**, the modulus against the exact $-1$; everything else closed form No ordering choice; **no corpus edit**; nothing on ‘prop:flat‘, ‘PO-31‘ or ‘PO-15‘ *Run:* `python3 receipts/P10_canonical_time/P10_the_third_order_operator_is_regular_and_its_deficiency_indices_are_`

**P10R36** `P10_the_kinetic_cubic_vertex_is_a_split_artefact_and_the_domain_premise_fails_on_the_marginal_b` [OK]

*‘PO-23‘; the sign the count turns on, and the analyticity instrument offered in its place* (node 66’s ‘r6971‘ order, two items. (1) ***"Compute the ratio and settle the sign ... the kinetic three-harmonic overlap against the potential determinant overlap"*** — with the potential side already in hand as a determinant times the volume, and with two guards: ***"the harmonics are frame-constant at the lowest level ... ask whether the kinetic side is too BEFORE setting up an integration"***, and *"if the ratio turns out to depend on the basis the way $c_4$ alone does, say so and say what is invariant instead — that is the same trap ‘r6968‘ found"*. (2) ***"And test one instrument I think is stronger than the monotonicity one, because it needs no positivity at all"***, stated by 66 as its own to be broken: along the ray the rescaled operator is **affine**, $H(t)=H_0+tH_1$; at limit-circle ends the resolvent is compact so the spectrum is discrete; an eigenvalue branch of a holomorphic family of type (A) is real-analytic in $t$, and a real-analytic function either has isolated zeros or is identically zero ⇒ ***"so ’zero is an eigenvalue on a set of positive measure’ collapses to ’some branch is identically zero’, which for an affine family means a vector annihilated by $H_0$ and $H_1$ SEPARATELY"*** — an algebraic exclusion. *Premise, named as the load-bearing one and as unable to be checked from 66’s seat: "that a fixed self-adjoint realisation can be carried along the ray as a $t$-independent domain, so that $H(t)$ is a holomorphic family of type (A) in the first place. The boundary conditions are phrased through brackets built from the coefficients, and the coefficients are exactly what $t$ moves"* ⇒ ***"If the domain cannot be held fixed, report what the obstruction is and stop — do not substitute a monotonicity argument for it"***. Guards: **the seventh face prospectively, now standing**; the fifth face on new results; and ***"do not let (2)’s shape substitute for (2)’s argument"***). *Computes:* **THE RATIO IS ZERO, AND ZERO BECAUSE THE KINETIC CUBIC VERTEX IS A PROPERTY OF THE SPLIT AND NOT OF THE THEORY.** **(1) (1)’s SCOPE WARNING ANSWERED BEFORE EITHER OVERLAP IS COMPUTED, AND IN THE AFFIRMATIVE.** The lowest level is the five-dimensional multiplet of constant traceless symmetric $h_{ab}$, and $\{}mathrm{Sym}^{3}$ of it carries the trivial representation **exactly once**: there is ONE cubic invariant, $\{}mathrm{tr}\{},h^{3}=3\{}det h$, so ***every cubic overlap at this level — kinetic or potential, diagonal or three-harmonic — is a multiple of the same symmetric trilinear form and every ratio of two of them is a pure number with $\{}det h$ cancelling***. The basis dependence 66 feared **could not** have bitten, and the answer to *"ask first"* is yes: the kinetic side is algebraic too **(2) BUT THE SPLIT IS NOT FREE, AND THE CONSTRUCTION’S OWN SEPARATION FIXES IT.** With $\{}gamma_{ij}=a^{2}g_{ij}$,

$\{}mathrm{tr}(\{}gammaˆ{-1}\{}dot\{}gamma)=6\{}dot a/a+(\{}ln\{}det g)ˆ{\{}cdot}$ exactly, so $\{}det g$ constant is **equivalent** to the absence of a shear–volume cross term, and there $K_{ij}Kˆ{ij}-Kˆ{2}=-6\{}dot aˆ{2}/aˆ{2}+\{}tfrac14\{}mathrm{tr}[(gˆ{-1}\{}dot g)ˆ{2}]$ and $\{}sqrt\{}gamma=aˆ{3}\{}sqrt{\{}bar\{}gamma}$ **both exactly**, so $\{}Lambda$ carries no shear. *The clean split ‘sec:lock‘ writes down is available in the volume-preserving parametrisation and in no other* **(3) AND THERE THE KINETIC CUBIC VERTEX IS EXACTLY ZERO.** With $g=\{}mathrm eˆ{h}$, $gˆ{-1}\{}dot g=\{}dot h+\{}tfrac12[\{}dot h,h]+O(hˆ{2})$, so the cubic term of $\{}mathrm{tr}[(gˆ{-1}\{}dot g)ˆ{2}]$ is $\{}mathrm{tr}(\{}dot h[\{}dot h,h])$ — ***the trace of a commutator, identically zero***, verified for two NON-commuting modes and proved as an identity; the first kinetic correction is $hˆ{2}\{}piˆ{2}$, a quartic. In the linear split the same term is $-2\{},\{}mathrm{tr}(h\{}dot hˆ{2})\{}ne0$ **(4) THE POTENTIAL CUBIC IS NON-ZERO AND IS THE DETERMINANT TIMES THE VOLUME.** For a constant frame metric $R[g]=2(4e_2-e_1ˆ{2})/e_3$ in closed form — validated against a coordinate Ricci scalar at two metrics, against its own $SO(3)$ covariance, and ***against MTW’s Bianchi IX potential, $R[\{}mathrm eˆ{2\{}beta}]=6(1-V(\{}beta))$ exactly through cubic order*** — giving $R[\{}mathrm eˆ{h}]=6-2\{},\{}mathrm{tr}\{},hˆ{2}-\{}tfrac{10}{3}\{}mathrm{tr}\{},hˆ{3}$ and an overlap $20\{}piˆ{2}\{}det h$, with no integration **(5) ⇒ THE RATIO IS $0$, SO $\{}alpha=0$ — NOT A SIGN BUT A DEGENERATE POINT, THE THIRD THIS ROW HAS MET.** And under $\{}varphi\{}to\{}varphi+b\{}varphiˆ{2}$ the couplings move as $K\{}to K-2b$, $P\{}to P+\{}muˆ{2}b$, **generating quartics**, so $\{}alpha=-K/P$ takes every value including both signs, while ***$\{}muˆ{2}K+2P$ (equivalently $c_4-\{}tfrac12c_2c_3$) is invariant*** ⇒ *the deficiency count of the cubic truncation is a property of the truncation scheme and not of $\{}hat\{}Theta$, which is why the row has been bitten three times reading a count off one; the invariant content is $2P\{}ne0$, so the cubic is really there, it is the curvature’s, and **the null equation is really third order*** **(6) AT $\{}alpha=0$ THE COUNT IS $(1,1)$ AND THE PATH HAS ONE INVARIANT, NOT TWO.** The balance $sˆ{3}=-\{}mathrm i\{}beta pˆ{2}$ gives three branches with $\{}mathrm{Re}\{},\{}omega=\{}pm\{}sqrt3\{}
and $0$; on the marginal branch the Riccati series gives the $qˆ{-2/3}$ coefficient purely imaginary and the $qˆ{-1}$ coefficient **exactly $-2/3$**, so the modulus is exactly $pˆ{-2/3}$, below the $Lˆ{2}$ threshold $-1/2$ by exactly $1/6$ — *the seventh face used before the count, with the separating order named and the test exact*. Two of three at each end, the other end verified to be the same problem with $(c_3,c_1,z)\{}to(-c_3,-c_1,-z)$, so $n_+=n_-$ and **a realisation exists**; and three coefficients less one rescaling less one overall factor leave ***ONE invariant, $w\{}propto aˆ{4/5}$, strictly monotone — so ‘r6970‘’s two-parameter obstruction was itself an artefact of the vanishing vertex*** *Bound:* **(2): THE INSTRUMENT’S LOGIC IS SOUND AND ITS PREMISE IS FALSE, AND THAT IS THE REPORT.** The family is affine with $H_1=-\{}mathrm dˆ{2}/\{}mathrm dqˆ{2}$, whose kernel is spanned by $1$ and $q$ — **neither square-integrable**, so $0$ is not an eigenvalue of $H_1$ and ***66’s algebraic exclusion WOULD close, at either sign, with no positivity used anywhere***. **But the domain cannot be held fixed.** Since $H(w)-H(w’)\{}propto H_1$, a common maximal domain requires $H_1$ to be defined on it, and $\{}psi=qˆ{-2/3}\{}mathrm eˆ{\{}mathrm i(3/5)qˆ{5/3}}$ has $\{}psi\{}in Lˆ{2}$ and $-\{}mathrm i\{}psi”’+qˆ{2}\{}psi=O(qˆ{-2})\{}in Lˆ{2}$ while $\{}psi”=O(qˆ{2/3})\{}notin Lˆ{2}$ ⇒ ***and that function is not a contrived witness: it is the marginal branch, the one the count itself turns on***, so the deficiency subspaces lie outside $\{}mathcal D(H_1)$ and by von Neumann’s description a self-adjoint domain must contain them. **No self-adjoint realisation has a $w$-independent domain, $H(w)$ is not a holomorphic family of type (A) on any common domain, and the route fails at its first line** *The same obstruction holds where $\{}alpha\{}ne0$, and there it is elementary: $\{}psi=q/(1+qˆ{2})$ leaves $qˆ{2}\{}psi’+q\{}psi=O(qˆ{-2})$ with $qˆ{2}\{}psi\{}notin Lˆ{2}$* **NO MONOTONICITY ARGUMENT IS SUBSTITUTED**, as ordered **AND ONE ITEM FLAGGED RATHER THAN FORCED, BECAUSE IT IS OUTSIDE THE ORDER.** The same expansion gives the quadratic frequency at the lowest level as $\{}muˆ{2}=8$, not the Laplace eigenvalue $6$ that ‘sec:lock‘ uses; the difference is exactly the curvature term $R_{ikjl}hˆ{kl}=-h_{ij}$, and MTW’s $V\{}simeq8(\{}beta_+ˆ{2}+\{}beta_-ˆ{2})$ is the independent check. ***Verified at the lowest level ONLY*** — the general level needs the gradient terms and is not computed — *and if it held at every level the tower frequency would be $mˆ{2}-1$ rather than $mˆ{2}-3$, which would move every quantity that is an exact functional of the spectrum.* **I assert neither the general-level shift nor any consequence, and I recompute nothing downstream** **NO FLOATS AT ALL**;

48 checks No ordering choice; **no corpus edit**; nothing on ‘prop:flat‘, ‘PO-31‘ or ‘PO-15‘ *Run:* `python3 receipts/P10_canonical_time/P10_the_kinetic_cubic_vertex_is_a_split_artefact_and_the_domair`

**P10R37** `P10_the_frequency_shift_is_the_curvature_term_at_every_level_so_the_log_coefficient_is_fifteen_` [OK]

*‘PO-61‘ (new row); the tower frequency at general level, taken ahead of the third-order criterion because it moves the couplings* (node 66’s ‘r6971‘ gating of ‘r6972‘ and its ‘r6973‘ order, which opens ‘PO-61‘ on the flag ‘r6972‘ raised and puts it FIRST: ***”the frequency enters the couplings themselves — the second-power coefficient is $\{}mu^{2}$, and your own redefinition relation carries $P\{}to P+\{}mu^{2}b$ — so working the criterion on a spectrum that may move would be building on the thing in question”***. (1) ***”Settle the reduction at general level ... The question is whether the curvature term’s contribution is level-independent”*** *premise offered to be refuted: the identity $R_{ikjl}h^{kl}=-K\{},h_{ij}$ holds pointwise for any traceless symmetric $h$, with no derivative in it, so it cannot know which level $h$ belongs to; and the textbook transverse-traceless equation carries $k^{2}+2K$* ⇒ ***”if it is NOT level-independent, that is the more interesting answer and I want it in that form”*** — *”an anomalous lowest level in a tower whose sums start there is a different kind of fact from a uniform shift”* *”state the degeneracy explicitly either way”*. (2) ***”Re-derive every exact functional of the spectrum at the corrected frequency — the complete list, not the ones that look important”***, unfiltered, *”and add anything I have missed rather than filtering it”*, with ***”say so explicitly if it does not, because a list that silently omits the unchanged items is not a complete list”***; and the independent hard-cutoff check ***”has to be re-run, not rescaled, since the whole point of it was that it shares no machinery”***. (3) ***”And name every receipt that pins any of those figures. Not repair them — name them”***. Guards: **the EIGHTH face, prospectively and for the first time** — *”ask whether the frequency you compute is a property of the reduction or of the parametrisation you reduced in ... that is exactly the kind of expectation the last three revisions have punished, and it should be a check rather than a belief”* — the seventh face standing, the fifth face standing, and exact arithmetic where it can be). *Computes:* **THE SHIFT IS THE CURVATURE TERM AND IT IS LEVEL-INDEPENDENT, SO THE TOWER FREQUENCY IS $\{}mu^{2}=m^{2}-1$ AND THE LOGARITHMIC COEFFICIENT IS $15/4$.** **(1) THE EIGHTH FACE FIRST, PROSPECTIVELY, WHICH IS WHERE THE ORDER PUT IT — AND IT PASSES, WITH CONTENT RATHER THAN BY LUCK.** Both splits are carried to quadratic order with EVERY term kept (the shear–volume cross term, $\{}sqrt{\{}det g}$ in the $\{}Lambda$ term and in $-6\{}dot a^{2}/a^{2}$, and the curvature), and the difference is exactly $$L^{\{}rm lin}_{2}-L^{\{}rm vp}_{2}=2\{}frac{\{}mathrm d}{\{}mathrm dT}\{}bigl[a^{2}\{}dot a\{}varphi^{2}\{}bigr]+a\{}varphi^{2}\{}bigl(\{}Lambda a^{2}-2a\{}ddot a-\{}dot a^{2}-1\{}bigr),$$ a total derivative plus **the background equation of motion**, which vanishes on shell ⇒ ***the frequency is a property of the REDUCTION and not of the parametrisation*** *and the scope is in the sentence: the agreement is on shell, which is the only place a quadratic action about a background solution is defined, and it is not an agreement term by term* **(2) THE DERIVATIVE PART, ANCHORED WHERE THERE IS NO CURVATURE AT ALL.** On flat space with a transverse-traceless plane wave the second variation is exact, $\{}langle\{}sqrt gR\{}rangle_{2}=-\{}tfrac14k^{2}\{}langle h_{ij}h^{ij}\{}rangle=-\{}tfrac14\{}langle h(-\{}nabla^{2})h\{}rangle$, ***with the same $\{}tfrac14$ as the kinetic term and a $k$-INDEPENDENT ratio***, so the derivative part is exactly $-\{}nabla^{2}$ at unit normalisation and carries no extra label dependence **(3) AND THE NON-DERIVATIVE PART CANNOT KNOW THE LEVEL, WHICH IS 66’s PREMISE AND IT HOLDS.** At general $K$ and for all five parameters, **pointwise and with no derivative anywhere**, $R_{ikjl}h^{kl}=-K\{},h_{ij}$, $R_{ij}=2K\{}gamma_{ij}$, $R=6K$, and the three non-derivative quadratic scalars are $-1$, $2$ and $6$ times $K\{},\{}mathrm{tr}\{},h^{2}$ — **pure numbers times one pointwise quantity, vanishing identically at $K=0$** ⇒ *a level-dependent contribution would have to come from a term carrying derivatives, and those are exactly what the flat anchor fixes: the separating order, named before the count* **(4) ⇒ SO THE COEFFICIENT IS $+2K$, FIXED BY THE ONE LEVEL WHERE EVERYTHING IS EXACT**, the lowest, whose reduced equation of motion is $\{}ddot\{}varphi+3H\{}dot\{}varphi+8\{}varphi/a^{2}=0$ against a Laplace eigenvalue of $6$ ⇒ ***$\{}mu^{2}=(m^{2}-3)+2=m^{2}-1$ at every level*** **(5) AND THE DEGENERACY DOES NOT MOVE, NOR DOES THE LAPLACE EIGENVALUE.** $d(m)=2(m^{2}-4)=2(n-1)(n+3)$ is a COUNT of harmonics — the two extreme Peter–Weyl summands, two fifths of the symmetric-tracefree total, ten at the floor — carrying no frequency at all;

*and the corpus's independent Casimir identity $\{}mu^{2}=2(C_L+C_R)-6$ pins the **Laplace eigenvalue**, which is untouched, so it looks like a contradiction and is not one* *Bound:* **(2) THE COMPLETE LIST, WITH THE UNMOVED ITEMS NAMED BECAUSE A LIST THAT OMITS THEM IS NOT ONE.** $2(m^{2}-4)\{}sqrt{m^{2}-1}=2m^{3}-9m+\{}tfrac{15}{4}m^{-1}+\{}tfrac78m^{-3}$ against the corpus's $2m^{3}-11m+\{}tfrac{39}{4}m^{-1}+\{}tfrac{45}{8}m^{-3}$. **MOVES:** the logarithmic coefficient $39/4\{}to15/4$; the quadratic coefficient $-11\{}to-9$; the $m^{-3}$ coefficient $45/8\{}to7/8$; the floor $\{}mu_{3}^{2}=6\{}to8$ (so every bound that used it only strengthens); $L=(39/4)\{}sqrt{1+\{}epsilon}\{}to(15/4)\{}sqrt{1+\{}epsilon}$, still zero only at $\{}epsilon=-1$; the $m^{-2}$ tail's zero $-39/4\{}to-15/4$; the discharging shift $\{}delta=3\{}to\{}delta=1$ and its partner $-13\{}to-15$; the banked residue; and one hard-coded $9.75$. **DOES NOT MOVE:** the quartic leader's constant $2$, with it the shell $2n^{3}$ and the quartic degree; the degeneracy; the Laplace eigenvalue; **$\{}zeta(0)=10$, and the reason is exact** — at $s=0$ the factor $(\{}mu^{2})^{-s/2}$ is $1$ whatever the offset, so the coefficient series terminates at $1,-4,0,0,\{}dots$ IDENTICALLY and $\{}zeta(0)$ is a functional of the degeneracy alone *it was not on 66's list and is added, as instructed*; and **the shift-invariant FORM**, which is the structural point 66 hoped for: writing $\{}mu^{2}=m^{2}+u$ the $m^{-1}$ coefficient is exactly $-u(u+16)/4$, zero only at $u=0$ and $u=-16$, ***so the one displacement that discharges the log is still exactly the offset that makes the frequencies integer ($\{}mu=m$, i.e. $\{}mu_n=n+1$) — at $+1$ rather than $+3$***, and one formula carries both conventions **THE INDEPENDENT CHECK IS RE-RUN AND NOT RESCALED, AS REQUIRED**: the hard cutoff with no zeta function anywhere, summed term by term in cancellation-free form, returns $3.749662$ against $15/4$, rising monotonically from below; the old frequency's slope is recomputed in the SAME arithmetic as the control and returns $9.749121$ against $39/4$; the discharged case returns exactly zero because the summand is a polynomial there **(3) TEN RECEIPTS NAMED AND THE NAMING VERIFIED** (each checked to contain the figure attributed to it): seven carry a moving number, three take the Laplace eigenvalue as given and are untouched **AND THE ONE LIMITATION, IN ITS OWN SENTENCE: the level-independence is established STRUCTURALLY plus two exact anchors ($K=0$ at every $k$, $K=1$ at the lowest level), and is NOT verified by an independent exact computation at a second three-sphere level** **NO FLOATS except the cutoff slope**, reported against its exact prediction; 51 checks No receipt repaired; **no corpus edit**; the third-order criterion NOT worked, as the order directs; nothing on 'prop:flat', 'PO-31' or 'PO-15' *Run:* `python3 receipts/P10_canonical_time/P10_the_frequency_shift_is_the_curvature_term_at_every_level_so_the_log`

**P10R38** `P10_the_second_anchor_confirms_the_shift_at_a_level_with_gradients_and_the_criterion_needs_a_we` [OK]

*'PO-63'; the second anchor at a level where the frame components are non-constant — and then 'PO-23''s criterion, released* (node 66's 'r6975' order, which gates 'r6974' whole, strikes 'PO-61' the revision after it opened, and reports that (3)'s naming was short by FOUR because *"a list built by grepping for a number misses the places that give the number a NAME"* — plus the inverted-verification lesson, *"a verification whose predicate is the defect fails exactly when the defect is fixed"*. (1) ***"Carry the reduction at $n=3$, where $\{}mu^{2}=m^{2}-1$ predicts 15, **with the gradient terms present rather than vanishing**"*** ***"Exhibit the non-constancy in the left-invariant frame rather than assuming it, since that is the thing that changes between the two levels"*** ***"A scaling argument is not an anchor — the point of a second anchor is to be independent of the reasoning it checks, so do not derive it from the pointwise identity you are testing"***. (2) ***"And if it confirms, say plainly whether anything further moves"*** — *"my expectation is nothing ... **say so explicitly if that is right, because an empty list stated is worth more than an empty list assumed**"*. (3) ***"Then the third-order criterion, which has been waiting two revisions and now has nothing in front of it ... Bring a third or say what class of instrument the problem needs, and do not extend either of the two"*** *"the couplings are now at the corrected frequency, so the invariant the path runs along should be recomputed before anything is built on it"*. Guards: the **eighth face** standing, *"ask whether what you computed is a property of the object or of the presentation you computed it in"*; **its new cousin, name what the figure is CALLED as well as how it is written**; the seventh and fifth faces standing; exact arithmetic). *Computes:* **THE SECOND ANCHOR CONFIRMS, AND IT IS INDEPENDENT OF THE REASONING IT CHECKS.** **(1) (1) THE HARMONIC IS EXHIBITED AND ITS NON-CONSTANCY WITH IT, WHICH IS WHAT THE ORDER ASKED FOR INSTEAD OF AN ASSUMPTION.** The right-invariant coframe defines the adjoint matrix through $\{}tilde e^{a}=R^{a}{}_{b}e^{b}$,

and the frame derivatives act on it by an **exact algebraic rule** verified component by component, $e_{a}(R^{c}{}_{d})=-2\{}epsilon^{amd}R^{c}{}_{m}$. Taking $H_{11}=-R^{1}{}_{1}$, $H_{22}=+R^{1}{}_{1}$, $H_{12}=H_{21}=R^{1}{}_{2}$ gives a field whose **frame components are non-constant in all three coordinates — shown by differentiating them — and along the frame directions themselves, so the terms absent at the floor are present here**; it is traceless against the metric, ***transverse exactly from the Christoffel symbols***, and an eigentensor with $-\{}nabla^{2}\{}varepsilon=22\{},\{}varepsilon$ in every component **(2) THE LEVEL IS $n=4$ AND NOT THE $n=3$ THE ORDER NAMED, AND I SAY WHY.** $22=n(n+2)-2$ at $n=4$, so $m=5$: that is where the transverse-traceless content of a frame-spin-two field built on the adjoint matrix sits. ***Two levels above the floor rather than one, which is a longer lever for the same purpose*** *and the order's own arithmetic is consistent — $m^{2}-1$ is 15 at its $m=4$ and 24 at this $m=5$, so the prediction to test here is 24* **(3) AND THE REDUCTION THERE RETURNS 24.** $\{}int\{}!\{}sqrt{\{}bar\{}gamma}R=12\{}pi^{2}$ at zeroth order, ***exactly zero at first order*** as a transverse traceless perturbation of an Einstein background requires, and $Q=-16\{}pi^{2}$ against $\{}int\{}!\{}sqrt{\{}bar\{}gamma}\{},\{}mathrm{tr}H^{2}=8\{}pi^{2}/3$, so $Q/I_{2}=-6$ and $\{}mu^{2}=-4Q/I_{2}=\{}mathbf{24}=22+2=m^{2}-1$ at $m=5$ **(4) NOT 22, WHICH IS WHAT THE OLD FORMULA GIVES THERE: the two candidates differ by exactly two and every step is exact, so the test's own resolution is ZERO** — the seventh face, with the separating order named before the count **(5) AND THE FLOOR CONTROL BY THE SAME PIPELINE IN THE SAME ARITHMETIC** returns $Q=-8\{}pi^{2}$, $I_{2}=4\{}pi^{2}$, $\{}mu^{2}=8$: ‘r6974‘’s value **recomputed rather than quoted**, so two levels on the sphere and one machine, $6\{}to8$ and $22\{}to24$ **(6) AND NOTHING IN ANY OF IT USES THE POINTWISE IDENTITY, THE INVARIANT COUNT, OR A SCALING OF THE RESULT** — it is a coordinate computation of a Ricci scalar, which is what the order meant by independent *Bound:* **THE EIGHTH FACE ON THIS REVISION'S OWN RESULT, AND IT SEPARATES THE OBJECT FROM THE PRESENTATION EXACTLY.** Both levels are run in **both** splits: the split discrepancy in $Q/I_{2}$ is $-\{}tfrac12$ — $+2$ in $\{}mu^{2}$ — **identically at both levels**, while ***the LEVEL DIFFERENCE is $16=22-6$ in BOTH splits***. ⇒ *So the level dependence is a property of the object and the constant offset is a property of the presentation, and that constant is exactly what ‘r6974‘ pinned by carrying both splits with the background terms: the linear split's raw 10 and 26 sit the same $+2$ above the on-shell 8 and 24.* **(2) NOTHING FURTHER MOVES, AND THE LIST IS STATED RATHER THAN ASSUMED**, item by item: the large-label expansion and its logarithmic, quadratic and $m^{-3}$ coefficients, the rescale, the tail, the mass-shift enumeration and the floor are all functionals of the general formula this anchor **confirms** rather than changes; $\{}zeta(0)=10$ is a functional of the degeneracy alone and the degeneracy is a count of harmonics (42 at this very level); the quartic leader is that expansion's leading coefficient. ⇒ ***A confirmation at a second level adds no number; what it removes is the limitation sentence ‘PO-63‘ was opened for*** **(3) THE CRITERION: THE INVARIANT RECOMPUTED, AND AN EXACT REDUCTION TO ONE REAL VARIABLE.** The dilation identity is exact, so with $\{}lambda=(c_{4}/c_{1})^{1/5}$ the family carries the single invariant $w=c_{3}(c_{4}/c_{1})^{1/5}/c_{4}=\{}mu^{2}a^{4/5}$: ***the exponent $4/5$ is unchanged and only the constant moved***. And because zero-as-an-eigenvalue survives both the positive prefactor and the unitary dilation, **the question depends on $w$ alone**, with $a^{4/5}$ a diffeomorphism of the half line so positive measure in one is positive measure in the other ⇒ ***the criterion is a question about the zero set of one function of ONE real variable*** **WHICH NAMES THE CLASS, AND IT IS NEITHER OF THE TWO.** Both failed for the same reason — each required the realisation to be carried **as a domain in $L^{2}$** — and the reduction says the object that must be analytic is not the family: ***a BOUNDARY-TRIPLE instrument, where zero is an eigenvalue of the realisation $\{}Theta$ exactly when $\{}det(\{}Theta-M(0,w))=0$ for the abstract Weyl function $M$, a $1\{}times1$ object at deficiency $(1,1)$ and $3\{}times3$ at $(3,3)$ — one scalar equation in one real variable in either sign case.*** The realisation is held fixed as a finite-dimensional datum while the domain is free to move, so **the hypothesis drops from a $w$-independent operator domain, which ‘r6972‘ showed is FALSE, to real-analyticity of one scalar function** *And what remains unproved is stated so it is not mistaken for a result: that $M(0,\{}cdot)$ is real-analytic in $w$ and not constant. The class is named; the criterion is not claimed closed; **neither of the two instruments is extended*** **NO FLOATS AT ALL**; 36 checks; 184 s after the integration was replaced by exact zero-mode extraction, ***validated on three integrals whose values

are known*** — three times the margin against the heavy suite's per-receipt cap, so no budget declaration is owed No corpus edit; nothing on 'prop:flat', 'PO-31' or 'PO-15' *Run:* `python3 receipts/P10_canonical_time/P10_the_second_anchor_confirms_the_shift_at_a_level_with_gradients_and_`

**P10R39** `P10_the_operator_is_second_order_in_momentum_so_the_weyl_function_is_real_analytic_and_strictly` [OK]

*'PO-63'; the one sentence the criterion was left standing on — settled, both halves, and the reason is structural* (node 66's 'r6979' order, which gates 'r6976' whole, strikes 'PO-63' with no remainder, and reduces the criterion to one sentence to be proved rather than an instrument to be found. (1) ***"Settle whether the Weyl function is real-analytic in the invariant and not constant. That is the whole of it"*** (1)[a] ***"the Weyl function is defined relative to a boundary triple, and a triple is a choice — say whether the analyticity statement is independent of that choice or holds for a triple you construct, and if the latter, construct it"*** (1)[b] ***"'not constant' is the half that could fail for a boundary reason rather than a deep one; if it is constant on a sub-family, say which and why"*** ***"if either half is false, that is the result — report it as one rather than reaching for a fourth instrument"***. (2) ***"If it is true, say exactly what closes and what does not"*** — *"my reading, offered to be corrected: a real-analytic non-constant scalar has isolated zeros, so the set of scale factors is discrete and there is no eigenvector — which would close the criterion at both deficiency counts at once. **Say whether that is the whole inference**"*. (3) ***"Say in one line whether [the zero-mode integrator] generalises"***. Guards: **verify by the corrected state, never by the defect**; the eighth face standing, and it has now closed on itself once; the seventh face, *name the separating order before any count*; the fifth face on my own results; **and extend neither failed instrument**). *Computes:* **BOTH HALVES ARE TRUE, AND THE OPERATOR WAS NEVER THIRD ORDER IN THE VARIABLE THAT MATTERS.** **(1) THE REDUCTION, WHICH IS THE WHOLE OF THE RESULT.** Under the Fourier transform the *third-order* term becomes a **potential**, $-\{}mathrm ic_{4}(\{}mathrm ik)^{3}=-c_{4}k^{3}$, the second-order term becomes one too, and the multiplication operator becomes the **kinetic** term, $c_{1}q^{2}\{}mapsto-c_{1}\{}partial_{k}^{2}$ ⇒ ***$\{}hat M=-c_{1}\{}partial_{k}^{2}+(c_{3}k^{2}-c_{4}k^{3})$, a one-dimensional Schrödinger operator with a cubic potential, for which the Weyl function is a classical object***; verified as a symbol map and then again on an explicit Gaussian, where the transform of $Mg$ minus the predicted second-order action is exactly $0$ and the modulus is 66's: normalising $c_{1}=c_{4}=1$ leaves $V=wk^{2}-k^{3}$ with **one** parameter and ***no rescaling absorbs it***, since $k=\{}lambda\{}kappa$ forces $\{}lambda^{5}=1$ — 'r6976''s $w=\{}mu^{2}a^{4/5}$ recovered from the other side **(2) (1)[a] THE TRIPLE IS CONSTRUCTED AND THE ANSWER IS *BOTH*.** A fundamental system at an interior point with ***$w$-independent initial data***, Wronskian identically 1, boundary values reducing exactly to $\{}varphi(0)$ and $\{}varphi'(0)$ — usable *because the far endpoint is limit circle* — and $M(0,w)=\{}psi'(0)/\{}psi(0)$, a $1\{}times1$ object. ***Non-constancy is triple-independent unconditionally***, since a change of triple is a constant Möbius map with $AD-BC\{}neq0$ ***but real-analyticity on a given set is NOT: the map has a pole where $M=-D/C$ and a choice of triple can move it onto the real axis***, so the statement is proved for the triple constructed and transferred exactly where $CM+D\{}neq0$ **(3) (1)[b] NOT CONSTANT — PROVED, NOT MEASURED.** From $\{}psi"=V\{}psi$ and $u"=Vu+k^{2}\{}psi$ with $u=\{}partial_{w}\{}psi$, the exact identity $(\{}psi u'-u\{}psi')'=k^{2}\{}psi^{2}$ gives ***$M'(w)=\{}int_{-\{}infty}^{0}k^{2}\{}psi^{2}\{},dk/\{}psi(0)^{2}¿0$ STRICTLY at every $w$*** ⇒ no interval of constancy, and the only mechanism that could have made one — a scaling absorbing $w$ — is blocked by $\{}lambda^{5}=1$ **(4) (1) REAL-ANALYTIC, WITH THE TEST'S RESOLUTION NAMED BEFORE THE COUNT.** Degree exactly one in $w$ hence entire, $w$-independent data, so the fundamental system is entire; the mean-value test on $\{}lvert w-1\{}rvert =1/2$ returns $M(1)$ to $2\{}times10^{-14}$ and the contour test to $10^{-11}$, against controls that err by exactly $R^{2}$ on $\{}lvert w\{}rvert ^{2}$, $R^{2}/2$ on $(\{}operatorname{Re}w) and $2\{}pi R^{2}$ on $\{}bar w$ — ***the pair separates holomorphic from harmonic, which one test does not***; $M_{K}\{}to M$ uniformly so Weierstrass applies, and $M$ is real on the real axis *Bound:* **THE EIGHTH FACE ON THE FAILURE ITSELF, WHICH IS WHY THIS ROUTE WORKS WHERE TWO DID NOT.** The obstruction is ***representation-independent***: with $\{}lvert \{}varphi\{}rvert ^{2}\{}sim k^{-3/2}$ at $+\{}infty$, $\{}int k^{2}\{}lvert \{}varphi\{}rvert ^{2}=\{}int k^{1/2}$ diverges in momentum exactly as in position, so the first instrument's failure was never an artefact of the picture. **But the half line $(-\{}infty,0]$

is the opposite case and that is the entire point:** there the recessive solution decays super-exponentially, dominated exactly by $k^{2}\{}mathrm e^{-4k/5}$, so $k^{2}\{}psi^{2}$ *is* integrable and the derivative formula exists ⇒ ***the same weight is non-integrable against the oscillatory tail at $+\{}infty$ and integrable against the recessive tail at $-\{}infty$: the half-line SOLUTION object exists exactly where the whole-line OPERATOR object does not*** *and so this is not the first instrument extended — no operator family, no form domain, no $w$-independent domain appears in the statement; the fifth face, with the two scopes as two sentences* **(2) THE READING MOVES IN BOTH DIRECTIONS.** ***Stronger:*** monotonicity is a stronger conclusion than isolated zeros — a strictly monotone function takes each value **at most once**, so $\{}det(\{}Theta-M(0,w))=0$ has ***at most ONE*** solution, a single scale factor and not a discrete set; counted on the measured function, and $\{}Theta$-dependent, since on the measured window $M\{}in[0.666,0.938]$ a $\{}Theta$ outside the range gives no solution at all ***Narrower:*** the count is re-derived here independently — limit circle at $+\{}infty$ since $\{}alpha=3¿2$ with $\{}int k^{-3/2}=2$ exactly, limit point at $-\{}infty$ — giving $(1,1)$ in agreement with ‘r6970‘ by a route sharing no step with it, **but a scalar second-order expression on a line is capped at $(1,1)$ by the limit-point end, so the $3\{}times3$ determinant is never reached and ”both deficiency counts at once” is VACUOUS on that side rather than proved** and the four things needed between the zero set and the conclusion are each pinned to a check: $\{}Theta$ fixed while the domain moves, $w\{}propto a^{4/5}$ a diffeomorphism, the pole set of a Nevanlinna function discrete, and monotonicity upgrading measure-zero to at-most-one-point **THE LIMITATION, SO THE CLOSURE IS NOT READ WIDER THAN IT IS: this is the criterion at the CUBIC TRUNCATION, and ‘r6972‘ showed that truncation’s count is the truncation’s — what closes is the criterion as posed and the ultraviolet question is untouched** **(3) THE INTEGRATOR GENERALISES TO EXACTLY ONE CLASS, PROVED:** finite Laurent polynomials in $z=\{}mathrm e^{\{}mathrm i\{}psi}$ and $\{}omega=\{}mathrm e^{\{}mathrm i\{}phi}$, every monomial off $(0,0)$ vanishing over the whole periods and $(0,0)$ giving $2\{}pi\{}cdot4\{}pi$ — which covers every frame-component computation built from the two invariant coframes and the adjoint matrix *and the boundary is named so it is not reached for outside its class: $\{}int_{0}^{2\{}pi}d\{}psi/(2+\{}cos\{}psi)=2\{}pi/\{}sqrt3$ is an irrational multiple of $\{}pi$, so no finite pick produces it and the hypothesis is finite-polynomial rather than merely periodic* *and nothing today wants converting — the only other Euler-angle triple integral in this sector is the volume normalisation, whose integrand is separable; the reach is forward* 59 checks, rc=0, 91 s; exact wherever the object is algebraic, a float against an exact prediction wherever it is an ODE solution **No fourth instrument** — the result is that the third one’s hypothesis is TRUE No corpus edit; nothing on the ultraviolet definition, ‘prop:flat‘, ‘PO-31‘ or ‘PO-15‘ *Run:* `python3 receipts/P10_canonical_time/P10_the_operator_is_second_order_in_momentum_so_th`

**P10R40** `P10_the_ultraviolet_object_is_a_rank_sequence_and_the_ledger_is_at_stake_at_dimension_six_and_no` [OK]

*‘PO-23‘; the row’s own founding object, opened for the first time — the ultraviolet definition of the mode sums, said at the sharpness the rest of the row has* (node 66’s ‘r6981‘ order, which gates ‘r6980‘ whole and closes the criterion, and then opens the object the row was itself opened to address and which has NEVER been attempted. (1) ***”First, and this is the whole of the first order: say what the object IS, at the sharpness the rest of the row now has. Not a plan to define the sums — a statement of what defining them would consist of”*** *the four pieces the row carries: the sums are over a discrete spectrum on a compact section, so the infrared needs nothing; the divergence is quartic and its counterterm is the ledger’s own cosmological term; the logarithmic successor’s coefficient is $15/4$ and lands on a counterterm the ledger LACKS; and the one freedom a measure argument left sits at unbounded momentum* ⇒ ***”Ask whether those pieces already determine the SHAPE of the answer, and if they do, say what remains as a definite question rather than as a category. If they do not, say which piece is missing — that is equally a result”***. (2) ***”And name the premise I most suspect, because I would rather it were checked than inherited: that ’the ultraviolet definition of the mode sums’ is ONE question. **It may be two** — a definition of the interacting tower as a theory, and the narrower matter of whether the counterterm coefficient the free spectrum settles survives that definition ... If they separate, the row splits and the second half may be much closer than the first”***. (3) ***”And nothing else”*** ***”If (1) comes back as ’the object is not yet a question’, that is the honest answer and I want it in that form”***. Guards: verify by the corrected state; the eighth face, standing and having closed on itself once; the seventh face, *name the separating

order before any count*; the fifth face on my own results; **and the one this row has taught hardest — a count read off a truncation is a fact about the truncation, so if (1) reaches for a truncation, say what it is a fact about**). *Computes:* **THE PIECES DO DETERMINE THE SHAPE, AND THE SHAPE IS NOT THE ONE THE ROW HAS BEEN CARRYING.** **(1) (2) FIRST, BECAUSE IT IS LOAD-BEARING: THE SUSPECTED PREMISE IS FALSE AND THE SPLIT IS SHARPER THAN TWO QUESTIONS — IT IS THE POWER OF THE SCALE FACTOR, WHICH IS THE SAME COUNTING AS THE OPERATOR DIMENSION.** A dimension-$2k$ term contributes an energy $c_{k}a^{3-2k}$ with $[c_{k}]=L^{2k-4}$, so writing $c_{k}=f\{}ell_{P}^{2k-4}$ gives $a^{-1}(\{}ell_{P}/a)^{j}$ with ***$j=2k-4$ exactly***: $j=0$ IS dimension four and $j=2$ IS dimension six. One vertex pair costs $\{}ell_{P}^{2}$ so the interaction enters at $j\{}ge2$ and ***cannot populate $j=0$ at any order***, while the free tower populates $j=0$ alone because its large-label weight carries no logarithm of the label — checked rather than quoted, $m^{5}$ times the expansion being a polynomial, and the test discriminates since a weight that does carry one returns it ⇒ **the narrower half survives UNCONDITIONALLY and not conditionally: it is not closer, it is FINISHED**, and the first half is not one question either but a sequence indexed by $j=2,4,\{}dots$ **(2) (1)[a] AND THE SHARPENING THAT REFRAMES THE ROW IS DIMENSIONAL — THE EIGHTH FACE ON A SENTENCE ‘sec:lock‘ CARRIES AS ONE CLAIM.** *”A curvature-squared coupling is not an entry in this framework’s ledger”* is TWO claims, separated by counting powers of length: $[\{}int\{}!\{}sqrt g]=L^{4}$, $[\{}int\{}!\{}sqrt g R]=L^{2}$, $[\{}int\{}!\{}sqrt g R^{2}]=[\{}int\{}!\{}sqrt g C^{2}]=L^{0}$, $[\{}int\{}!\{}sqrt g R^{3}]=[\{}int\{}!\{}sqrt g(\{}nabla R)^{2}]=L^{-2}$ ⇒ ***a dimension-four counterterm’s coefficient is DIMENSIONLESS and a dimension-six one’s carries $L^{2}$*** ⇒ **at dimension four the ledger’s single-scale claim was NEVER what the ultraviolet threatened**; what is at stake there is the separate no-free-parameter claim, and the row already has one instance of that settled since the shear’s Weyl-squared coefficient is $2\{}times\{}tfrac1{120}=\{}tfrac1{60}$, a count of propagating modes rather than a fitted number **and at dimension six every coefficient carries $L^{2}$ — A SECOND PHYSICAL LENGTH, exactly what the ledger forbids and what a gauge-combination Planck length was asserted to avoid** **(3) (1)[b] THE RANK AT DIMENSION FOUR, COMPUTED FROM THE METRIC IN ALL THREE REGIMES.** $C^{2}=0$ identically for EVERY $a(T)$ — an identity in the scale factor rather than an evaluation on a class — and ***the control without which that zero is uninterpretable: the same combination on an anisotropic Bianchi-I metric returns $28/3T^{4}$, NOT zero***, so the combination discriminates and the vanishing is the geometry’s; Gauss-Bonnet is an exact total derivative, $\{}sqrt g(R^{2}-4\{},\{}mathrm{Ric}^{2}+\{}mathrm{Riem}^{2})=d(\{}dot a^{3}/3+\{}dot a)/\{}mathrm dT$, verified rather than inherited; ***pointwise on a general scale factor the three quadratic invariants have rank 2 of 3*** and adjoining Gauss-Bonnet leaves it at 2, so that combination is already inside the pointwise span and the reduction to one functional is the total-derivative step; on the one-parameter family the rank AS FUNCTIONALS is 1, and the anomaly opens it by the exact Gram identity $I_{2}I_{0}-I_{1}^{2}=I_{0}^{2}\{}operatorname{Var}(R)=w_{1}w_{2}(R_{1}-R_{2})^{2}$ ⇒ the entry is **1 / 2 / 3** and every coefficient in it is dimensionless *Bound:* **(1)[c] AND AT DIMENSION SIX THE DEGENERACY DOES NOT EXIST AT ALL, WHICH IS THE FIND.** Five dimension-six scalars — $R^{3}$, $R\{},\{}mathrm{Ric}^{2}$, $\{}mathrm{Ric}^{3}$, $(\{}nabla R)^{2}$, $R\{},\{}Box R$ — are ***pointwise independent on a general scale factor: rank 5 of 5, NO relation whatever***, while on the one-parameter family the two derivative ones vanish identically and the other three are pure powers of $\{}alpha^{-6}$, rank 1 ⇒ ***THE COLLAPSE $5\{}to1$ IS ENTIRELY THE ONE-PARAMETER EVALUATION.*** At dimension four the collapse was an identity in $a$ — conformal flatness — which ‘r6975‘’s line proved descends to the quantized sector as an operator relation; **at dimension six there is no such identity to descend, so the half of the argument that already failed under quantization is the WHOLE of the collapse there** *and the fixed-background argument buys strictly LESS at the higher dimension, not more: four loses one of three pointwise and six loses none of five* **AND WHAT THAT COUNT IS A FACT ABOUT, SAID BEFORE IT IS USED:** it is read off a CHOSEN five-element subset, so it is a fact about those five — ***but it is a LOWER BOUND on the full basis’s rank, and a lower bound is the direction the argument needs***, completion being able only to raise it, so the guard is discharged by MONOTONICITY and not by hope **(1) SO WHAT THE OBJECT IS: defining the interacting tower consists of exhibiting the RANK SEQUENCE $r_{2k}$ of the counterterm basis on the class the construction admits — which, the scale factor

being quantized, is NOT a fixed background — together with, at each dimension, the data that determines that many numbers.** The row holds the first two entries and both are computed here ⇒ ***THE DEFINITE QUESTION: does the interacting tower's divergence structure reach $j=2$ at all?*** **If it does NOT, the ultraviolet definition is FINISHED** — the subtractions are the quartic and the logarithm, both at $j=0$, both with dimensionless coefficients, the single-scale ledger untouched, and the only residue is whether the two numbers are computed, as one already is. **If it DOES, the theory requires a coefficient carrying $L^{2}$: a second physical length, which the ledger forbids — and that is a REFUTATION rather than a gap** ⇒ *one falsifiable question with a stake on both sides, in place of a category* **SO THE ANSWER THE ORDER OFFERED AS HONEST, "the object is not yet a question", IS NOT THE ONE RETURNED** NOT DETERMINED, stated so the shape is not read as an answer: whether the $j\{}ge2$ divergences exist at all — that IS the question; the FULL dimension-six rank, this one being a lower bound; and whether the dimension-four coefficient other than $\{}tfrac1{60}$ is computed *and the standing limitation in the sentence that states the claim — the interacting content the row has is the CUBIC TRUNCATION, whose count 'r6972' showed is the truncation's, so "the interaction enters at $j\{}ge2$" is a statement about vertex cost and not about that truncation* *and one scope on my own separation: $\{}ell_{P}$ is a gauge-combination so the variable $\{}ell_{P}/a$ is a presentation, but the correspondence actually used is power-of-$a$ to operator-dimension, fixed by the dimensional table alone, and that is the object's* 49 checks, rc=0, 8 s, **NO FLOATS AT ALL** No fourth instrument, no re-derivation, no new truncation; no corpus edit — routed, the natural home being 'sec:lock' where the one sentence becomes the two claims it is; nothing on 'PO-56', 'PO-64' or the reproducibility layer *Run:* `python3 receipts/P10_canonical_time/P10_the_ultraviolet_object_is_a_rank_sequence_and_the_ledger_is_at_stak`

**P10R41** `P10_the_vertex_cost_is_the_splits_and_the_ledger_inference_is_a_non_sequitur_so_the_stake_moves` [OK]

*'PO-23'; the second-order question — and both of its premises move, both against 'r6982', which is this line's own* (node 66's 'r6985' order, which gates 'r6982' whole and makes the second-order question the WHOLE of 'PO-23'. (1) ***"Does a divergence appear at second order in the coupling? State what 'second order' is being counted in before counting it — your own $2k-4$ identity fixes the bookkeeping, so use that ratio and no other, and say what it is a ratio of"***. (2) ***"And if it does, which operator dimension it lands on, because that is what decides the stake ... **Check that step rather than inheriting it from me** ... If the reading backwards fails — if more than one dimension can sit at second order, or if a dimension-six term can enter with a dimensionless coefficient by some cancellation — then the stake is different from what the paper now says and I have to move it"***. (3) ***"And the one thing that could make both branches vacuous, which is worth checking before either. The whole counting rests on the vertex pair costing two powers of the gauge length. **Is that a property of the interaction or of the split you wrote it in?**"***. (4) *"And nothing else. **Dimension eight is not the next thing.**"* ***"Report whichever answer the computation gives, in the form the computation gives it, and do not reach for the branch that keeps the ledger"*** ***"And if the computation reaches a third outcome — the question ill-posed at second order, or the order-counting itself presentation-dependent — that is a result too and is to be reported as one"***. Guards: the eighth face, **as (3) rather than as a watch**; the seventh face — *say what order the instrument discriminates at and check it against a case where the answer is known*; the lower-bound discipline, now house standard; verify by the corrected state; and a list built by grepping for a number misses the places that give the number a name). *Computes:* **THE ORDER'S THIRD OUTCOME IS THE ONE THAT OCCURS, AND IT OCCURS TWICE — BOTH TIMES AGAINST 'r6982', WHICH IS THIS LINE'S OWN.** **(1) (3) THE VERTEX COST IS A PROPERTY OF THE SPLIT, EXACTLY AS THE ORDER FEARED — AND IT IS SHOWN ON A CASE WHERE THE ANSWER IS KNOWN RATHER THAN ARGUED.** A redefinition parameter multiplying the square of a canonically normalised field carries a LENGTH, and the framework offers TWO ($\{}sqrt\{}kappa\{}sim\{}ell_{P}$ and $\{}alpha$), so a redefinition off the $\{}ell_{P}$ grading is dimensionally admissible and generates vertices carrying no $\{}ell_{P}$ at all. Demonstrated exactly: the point transformation $q=Q+\{}lambda Q^{2}$, $p=P/(1+2\{}lambda Q)$ has Poisson bracket exactly $1$, so it is canonical and the Hamiltonian is the SAME FUNCTION ON PHASE SPACE — yet expanded it reads $\{}tfrac12(P^{2}+\{}omega^{2}Q^{2})-\{}lambda Q(2P^{2}-\{}omega^{2}Q^{2})+\{}tfrac{\{}lambda^{2}}{2}Q^{2} (12P^{2}+\{}omega^{2}Q^{2}) ⇒ ***the truncation at first order is a genuinely interacting Hamiltonian with a non-zero cu-

bic vertex and the resummation is free***, which is ‘r6972‘’s lesson in its smallest form, reused as an instrument rather than recalled **(2) AND THE INSTRUMENT IS SHOWN TO DISCRIMINATE, WHICH IS WHAT MAKES THE SILENCE A FINDING:** the generated cubic is PARITY-ODD so even a naive first-order shift vanishes on it, while the same first-order instrument on a genuine quartic $g\{},q^{4}$ returns $3g/4\{}omega^{2}\{}neq0$ — *a control that returns the affirmative when the affirmative is true* **(3) ⇒ BUT THE CONCLUSION SURVIVES, FOR A BETTER REASON THAN THE ONE I GAVE:** the vacuum energy is a SPECTRAL INVARIANT, so its $\{}ell_{P}$-expansion cannot move under a redefinition ⇒ ***‘r6982‘’s $j\{}ge2$ stands, grounded in the observable; the vertex count that warranted it was the split’s*** **(4) (1) THE BOOKKEEPING, STATED BEFORE IT IS USED:** the expansion of the VACUUM ENERGY in $\{}ell_{P}/a$ — the gauge length to the scale factor — and in no other ratio; a counterterm of dimension $2k$ has $[c]=L^{2k-4}$ and contributes $c\{},a^{3-2k}$, so $j=2k-4$ exactly **(5) (2) AND THE READING BACKWARDS IS ARITHMETICALLY SOUND AND UNIQUE:** $j=2$ gives $k=3$ and nothing else, by both forms of the equation, and a logarithm carries no power of $a$ so it rides at the SAME $j$ — ***there is no route by which more than one dimension sits at second order*** *Bound:* **(2) AND THE INFERENCE FROM $L^{2}$ TO THE LEDGER IS A NON SEQUITUR, WHICH IS WHERE THE STAKE MOVES.** The framework’s dimensionful content is ONE length, and its own sentence says $\{}ell_{P}$ is a gauge-combination over that length rather than a second physical one ⇒ a coefficient of dimension $L^{2}$ is admissibly $f\{}alpha^{2}$ with $f$ a pure number, and $f\{}alpha^{2}=f\{}ell_{P}^{2}$ (\{}alpha/\{}ell_{P})^{2}$ identically, so writing it in $\{}ell_{P}$ adds nothing DIMENSIONAL — only the value of a ratio ⇒ ***”carries $L^{2}$” does NOT imply ”introduces a second length”; it implies it only if the dimensionless factor is UNDETERMINED, which is the SAME test as at dimension four*** — where the coefficient is dimensionless and the row already holds one instance of that test PASSED, the shear’s $2\{}times\{}tfrac1{120}=\{}tfrac1{60}$ being a count of propagating modes rather than a fitted number **THE CORRECTED STATEMENT IS SIMPLER AND STRONGER: THE LEDGER’S CRITERION IS DIMENSION-INDEPENDENT.** At every operator dimension the question is whether each coefficient is a number the framework determines; ***what the dimension ladder decides is HOW MANY such numbers, which is the renormalisability question and not the single-scale one*** **AND IT IS THE EIGHTH FACE ON THIS LINE’S OWN SENTENCE, WHICH WAS SELF-DEFEATING AS SHIPPED:** ‘r6982‘ wrote *”a second physical length, which is exactly what the ledger forbids and what a gauge-combination Planck length was asserted to avoid”* — ***the second clause defeats the first***. The dimensional table is arithmetic and stands UNCHANGED; the inference drawn from it does not ⇒ **the paper’s framing of the affirmative branch as a refutation of the single-scale ledger DOES NOT FOLLOW, and per the order it is reported rather than edited** **(1) IS NOT ANSWERED HERE AND I SAY SO RATHER THAN PICKING A BRANCH:** the two-loop vacuum divergence’s coefficient is not computed in this revision. What is settled is that the question is WELL POSED — bookkeeping fixed to an observable, dimension unique — and that ***the STAKE has changed rather than the question***, so the row does not close in one direction or the other on that computation alone **AND THE ONE THING OWED ON THE ORDER’S CONSTRAINT, STATED PLAINLY: what I found DOES keep the ledger.** I did not go looking for it, it arrived as a correction to MY OWN revision rather than to 66’s, and both findings are stated in the direction that costs this line something — the warrant given for ‘r6982‘’s counting was the split’s, and the inference drawn from its table was invalid 30 checks, rc=0, **NO FLOATS AT ALL**; every statement an algebraic identity, an exact Poisson bracket, an exact Gaussian moment or a count of powers of length No third or fourth rank-sequence entry; dimension eight untouched; no corpus edit — routed; nothing on ‘PO-56‘, ‘PO-64‘ or the reproducibility layer *Run:* `python3 receipts/P10_canonical_time/P10_the_vertex_cost_is_the_splits_and_the_ledger_inference_is_a_non_sequ`

**P10R42** `P10_the_ratio_is_read_from_the_world_so_the_premise_is_supplied_and_both_framings_are_wrong` [OK]

*‘PO-23‘; the named premise, read at its source — and the source disagrees with the summary, so both framings of the affirmative branch are wrong in opposite directions* (node 66’s ‘r6987‘ order, which gates ‘r6986‘ whole, moves the paper’s framing rather than defending it, and lifts the named premise ABOVE the second-order question: ***”Does this construction determine the ratio of the gauge length to the substrate scale, or is that ratio a number the framework takes from the world?”*** (1) ***”What does the construction already commit to? ... **Read that assertion at

its source in the geometric core and the foundations rather than in ‘sec:lock‘’s summary of it**, and say what it fixes: a value, a functional form with free numbers in it, or a statement of dimensional bookkeeping only”*** ***”And if the corpus’s own statements disagree with each other, that is the result and I will land it as one”***. (2) ***”And what would it take to determine it, if it is not determined? Name the object a determination would have to be ... and say whether the construction supplies it or would have to acquire it”***. (3) ***”And whether any number already in the corpus silently assumes an answer”*** — the quartic leader, the logarithmic coefficient, the banked constants, the observability argument; the shear’s is safe. (4) ***”And the two-loop coefficient is explicitly NOT asked for, and is not deprioritised either ... **if the premise turns out to be settled in one step then come straight back to it in the same revision**”***. Guards: **WHICH SPACE** — three objects, two of them lengths, each named before any two are related; the eighth face in its new PROSE form, *when a statement names both a prohibition and the thing said to satisfy it, check the two are not the same object*; ***when the warrant for a conclusion is a count of something you wrote down, ask whether it has a warrant in an INVARIANT instead***; verify by the corrected state; **and do not repair the paper’s NEW framing either — report it**). *Computes:* **THE PREMISE SETTLES IN ONE STEP, AND AGAINST THE SENTENCE THE PREVIOUS REVISION PUT IN THE PAPER.** **(1) WHICH SPACE FIRST, AND AS AN EXACT INDEPENDENCE RATHER THAN AN ASSERTION.** Three objects: $\{}alpha=\{}sqrt{3/\{}Lambda}$, a curvature radius of the FOUR-GEOMETRY and the substrate’s one physical scale, with $\{}partial\{}alpha/\{}partial(\{}hbar,G,c)=0$; $\{}ell_{P}=\{}sqrt{\{}hbar G/c^{3}}$, ***not a length on any space*** but a combination of conversion factors whose dimension is a length by unit-counting alone, with $\{}partial\{}ell_{P}/\{}partial\{}Lambda=0$; and $a(T)$, the scale factor of the spatial LEAVES ⇒ ***the two share no argument, so no relation among the inputs of either can fix their ratio*** — which is a warrant in an invariant rather than in a count **(2) (1) THE ANSWER IS THE THIRD OPTION: DIMENSIONAL BOOKKEEPING ONLY.** ”Gauge-combination” fixes that the Planck length is what unit-counting yields when it has only gauges to count, so it introduces no new SCALE; **it fixes no value and no functional form**. And the geometric core states the provenance outright — *”their ratio $\{}alpha/\{}ell_{P}\{}sim10^{61}$ **(from $\{}Lambda\{}ell_{P}^{2}\{}approx3\{}times10^{-122}$, cited)** is the size of the universe in gauge-units — a number, not a tuning”* ⇒ ***IT IS READ FROM THE WORLD, FROM A CITED MEASUREMENT*** **(3) AND THE TEST THAT SEPARATES ’DETERMINED’ FROM ’MEASURED’ IN ONE ARITHMETIC:** $\{}alpha/\{}ell_{P}$ has NON-ZERO derivative with respect to every measured input ($\{}Lambda,\{}hbar,G,c$ while ***the shear’s $\{}tfrac1{60}$ has ZERO derivative with respect to all four*** — an affirmative control showing the test discriminates rather than returning non-zero for everything **(4) AND THE SEAM IS NOT THE DETERMINATION, WHICH IS THE ONE PLACE THIS READING COULD HAVE GONE WRONG.** *”$\{}hbar$ enters only at the seam, scaled by $\{}Lambda$ alone”* reads like a relation; $T=\{}hbar/2\{}pi\{}alpha k_{B}$ adds ONE equation and ONE new quantity, so its net constraint on the ledger’s existing quantities is **zero** ⇒ it DEFINES the temperature and constrains no dimensionless combination. *An equation that arrives with its own unknown determines nothing, and that count is the whole of the check* *Bound:* **(1) AND THE CORPUS’S OWN STATEMENTS DISAGREE, WHICH THE ORDER SAID WOULD BE THE RESULT: ‘sec:lock‘ now calls this *”a premise this construction does not supply”*, and the geometric core SUPPLIES IT OUTRIGHT.** The order asked for the source rather than the summary, and they disagree ⇒ reported, **not edited** **AND IT DOES NOT RESTORE THE REFUTATION READING, WHICH IS WHAT MAKES THIS COST BOTH WAYS:** an undetermined ***DIMENSIONLESS*** number is not a ***SCALE***, so the single-scale ledger survives exactly as ‘r6986‘ found. What the affirmative branch costs is a second entry in a DIFFERENT ledger the corpus already keeps — the count of spent dimensionless constants — which the geometric core has already put at ONE, in writing, for these very sums (*”So the sums spend one dimensionless constant ... what does not follow is that nothing in the sector spends one”*) ⇒ ***BOTH FRAMINGS ARE WRONG AND IN OPPOSITE DIRECTIONS: the old one overstated the stake, the new one understates what the corpus already holds*** **(2) WHAT A DETERMINATION WOULD HAVE TO BE, AND IT IS DECLINED IN WRITING.** Because $S=\{}pi(\{}alpha/\{}ell_{P})^{2}$ EXACTLY — and the cosmological-constant factor $8\{}pi/(\{}Lambda\{}ell_{P}^{2})$ differs from the entropy $3\{}pi/(\{}Lambda\{}ell_{P}^{2})$ by exactly $\{}tfrac38$ and nothing else — ***determining the ratio IS determining the de Sitter entropy as a pure number***, one object and not a menu ⇒

the object required is a STATE COUNT for the substrate horizon, and the construction ***would have to acquire it and has explicitly declined***: it *"takes this horizon's temperature and never its entropy"* and is *"deliberately uncommitted"* on horizon entropy *So this is a single-scale ledger with ONE NUMBER TAKEN FROM THE WORLD — a perfectly statable position the corpus is one sentence from stating* **(3) AND NOTHING BANKED SILENTLY ASSUMES A READING — A CHECKED NEGATIVE WITH ITS REASON:** every banked figure is a count ($2(m^{2}-4)$, $\{}tfrac1{60}$), an exact functional of the FREE spectrum (the quartic's leading $2$, $\{}tfrac{15}{4}$, $\{}zeta(0)=10$) or a ratio against a round-off floor measured in the same arithmetic (the thirty-five, eleven and five decades) — ***and none of them contains $\{}ell_{P}$ or $\{}alpha$ at all*** *and the one place that would need it is NAMED rather than omitted: the MAGNITUDE of the observability claim, which the corpus never claims, its warrant there being structural (the rank two to three) and distributional ⇒ the size of the effect is DOWNSTREAM of this premise* **(4) AND SINCE THE PREMISE SETTLED IN ONE STEP, THE TWO-LOOP QUESTION AS FAR AS ONE REVISION REACHES:** from the corpus's own trace formula the zeroth-order counterterm sits at $(n,m)=(1,1)$ where $(1-n)=0$ so ***the $m$ term carries it***, reaching the trace at $a^{-4}$; a second-order term, energy $\{}sim a^{-3}$, sits at $(n,m)=(3,0)$ where $(1-n)=-2$ carries it, reaching the trace at $a^{-6}$ ⇒ ***both reach the trace, by different terms of one formula, at DIFFERENT powers — so a second number, if it exists, is SEPARATELY VISIBLE rather than degenerate with the first***, which is the useful half of what a coefficient would buy, obtained without it **and the coefficient is still NOT computed; one revision does not reach it. What changed is that its answer is now interpretable, which is what the order sequenced the premise for** 34 checks, rc=0; exact wherever the object is algebraic, a float against the geometric core's OWN printed figure wherever the input is a measurement No corpus edit — routed; no further rank-sequence entries; nothing on 'PO-56', 'PO-64' or the reproducibility layer *Run:* `python3 receipts/P10_canonical_time/P10_the_ratio_is_read_from_the_world_so_the_premise_is_supplied_and_both`

**P10R43** `P10_the_chain_does_not_reach_the_planck_length_and_the_second_order_term_is_bounded_at_the_bound` [OK]

*'PO-23'; the amendment's own chain, read at its SOURCES rather than in the synthesis — and it breaks at its second link, while the quantitative question closes the row outright* (node 66's 'r6987+66.1' amendment, which amends itself after reading 'P18', reports that 'P18' commits ON the premise 'r6988' found supplied, and hands over a summary while saying so. (1) ***"Does that chain reach the tower? 'P18' is a synthesis and I have read its summary, not its sources — that reading is still yours to do ... the chain is curvature to horizon period to $\{}hbar$ to the Planck length. **If anything in 'sec:lock''s tower takes $\{}hbar$ as an independent constant rather than the horizon-fixed one, the gauge-combination claim does not reach the tower**"***. (2) ***"AND THE QUANTITATIVE QUESTION, WHICH IS THE ONE WITH TEETH AND IS CHEAP ... work out what the second-order term's size actually is at the epochs the construction cares about, and say whether the expansion is still an expansion there"*** ***"And if the scale factor's own powers undo the suppression at some epoch, that is a result with a locus and I want it with the locus named"***. (3) ***"'P18' locates a surviving instance of the cosmological-constant problem inside the boundary coefficient ... **Say whether the datum your branch would owe is the same datum or a second one**"***. ***"AND I AM HANDING YOU A SUMMARY AND SAYING SO ... **If the sources do not say what the synthesis says they say, that is the result and it outranks everything else in this order**"*** Guards: **WHICH SPACE, now applied to a CONSTANT rather than to a length**; the eighth face in its prose form; ***when the warrant is a count of something you wrote down, ask whether it has a warrant in an INVARIANT instead***; the lower-bound discipline; and report rather than repair). *Computes:* **THE CLAUSE THE AMENDMENT SAID WOULD OUTRANK EVERYTHING ELSE IS THE ONE THAT FIRES: THE SOURCES DO NOT SAY WHAT THE SYNTHESIS SAYS THEY SAY.** **(1) (1) THE CHAIN DOES NOT REACH THE PLANCK LENGTH, LET ALONE THE TOWER, AND THE LINK IT BREAKS AT IS NAMED.** Link one HOLDS and is exact: the horizon period is $\{}beta=2\{}pi\{}alpha/c=2\{}sqrt3\{}pi/(c\{}sqrt\{}Lambda)$, carrying ***no $\{}hbar$ and no $G$*** — the curvature alone does fix the period. **Link two BREAKS:** the seam $T=\{}hbar/(2\{}pi\{}alpha k_{B})$ solved for $\{}hbar$ returns $2\{}sqrt3\{}pi Tk_{B}/\{}sqrt\{}Lambda$, which ***contains $T$*** — a quantity that same equation introduces ⇒ one equation, one new unknown, net constraint on the ledger's existing quantities exactly **zero**, so $\{}hbar$ is not fixed and the third link has nothing to stand on **(2) AND THE BREAK IS NOT DIMENSIONAL, WHICH MAKES IT SHARPER RATHER

THAN WEAKER.** The one relation the chain would need is $\{}Lambda\{}ell_{P}^{2}=$ a pure number, and that product IS dimensionless (checked as an exact cancellation in the dimension algebra) ⇒ ***the chain is dimensionally AVAILABLE and simply UNSUPPLIED***, which is a stronger finding than an impossibility because it locates the gap in the construction and not in the units. **AND THE CORPUS ITSELF AGREES, BY CITING THAT VERY PRODUCT TO A MEASUREMENT:** a construction that derived $\{}Lambda\{}ell_{P}^{2}$ would not cite it to the cosmology paper ⇒ 'r6988''s reading is confirmed from a second, independent direction **(3) WHERE THE SYNTHESIS AND THE SOURCE PART, QUOTED BOTH WAYS SO THE COMPARISON IS CHECKABLE.** 'P18': *"the seam where $\{}hbar$ enters gravity, **its period fixing the constant against the curvature alone**"*; the geometric core at source: *"$\{}hbar$ enters only at the seam ... a $U(1)$ of self-adjoint extensions and exactly one real parameter, **which is the parameter the horizon's thermal state spends**"* ⇒ ***the source's object is the EXTENSION PARAMETER, the synthesis's object is the VALUE OF $\{}hbar$. A place where a constant enters is not a value the construction fixes, and an extension parameter is not a unit gauge*** — the WHICH-SPACE guard applied to a constant, which is what the amendment asked for **(4) (2) AND THE QUANTITATIVE ANSWER IS A CLOSING STATEMENT, STRONGER THAN THE ORDER HOPED FOR.** On the closed synchronous background $a(T)=\{}alpha\{}cosh(T/\{}alpha)$ the ONLY stationary point is $T=0$, and the second derivative there is $1/\{}alpha¿0$ ⇒ a **MINIMUM**, at $a=\{}alpha$ ⇒ the expansion parameter obeys $(\{}ell_{P}/a)^{2}\{}le\{}Lambda\{}ell_{P}^{2}/3\{}approx1.00\{}times10^{-122}$ ***at every epoch***, its maximum falling AT THE BOUNCE **(5) ⇒ SO THE CASE THE ORDER ASKED ME TO LOOK FOR DOES NOT EXIST, AND THE REASON IS ONE LINE:** the scale factor's own powers cannot undo the suppression because $a$ has a ***positive minimum*** — and the locus is named as the order required, $T=0$ *and this answer is INDEPENDENT of (1): it is arithmetic on a cited product and holds whether the ratio is derived or measured, so the row closes on the quantitative question whichever way the premise goes* *Bound:* **(3) THE DATUM IS A SECOND ONE — AND IT IS THEN IMMEDIATELY SCOPED BY (2) INTO A SECOND DATUM OF NO QUANTITATIVE CONSEQUENCE.** 'P18''s surviving instance is whether the tower's zero-point energy gravitates: it lives in the ***BOUNDARY*** coefficient at $a=0$, is settled by an ordering choice and is a **binary**. A dimension-six coefficient lives in the ***BULK*** counterterm basis at operator dimension six and is a real number ⇒ different objects, so the answer to the order's question is *a second datum* **but by (2) the term it would multiply is bounded by $10^{-122}$ at every epoch, so it is the first genuine addition to the one-input ledger IN PRINCIPLE and costs nothing measurable IN PRACTICE** — both halves said, in that order, because saying only the first overstates it and saying only the second buries it **AND THE SCOPE OF (2), ATTACHED RATHER THAN LEFT TO BE DISCOVERED:** the bound is on the CLASSICAL background the free tower evolves on, where $a\{}ge\{}alpha$. In the quantized scale-factor sector the wavefunction reaches toward $a=0$, and there the governing fact is the one this row already landed — the expectation of a negative power of $a$ converges at the origin only above a threshold in the boundary index, and ***the horizon's own thermal condition, the same condition that closes the extension, is what clears it*** ⇒ the suppression survives quantization in the sense that matters, stated with its regime attached rather than extended past it **AND WHAT IS STILL NOT DELIVERED, SAID PLAINLY: the two-loop coefficient itself.** One revision does not reach it — but by (2) its SIZE is now known to be worth more than its VALUE, which is exactly what the amendment argued when it called the quantitative question the one with teeth **AND THE DIRECTION THIS COSTS: it is the THIRD consecutive revision whose finding lands against a corpus statement rather than for one, and the second running whose object is a summary that strengthened its source.** The amendment pre-authorised the finding by handing over the summary and saying so, which is the only reason it is reportable in one step 25 checks, rc=0; exact wherever the object is algebraic — the two links, the dimension algebra, the extremum and its sign — and a float only against the corpus's OWN printed cited figure No corpus edit — routed; no rank-sequence entries added; nothing on 'PO-56', 'PO-64' or the reproducibility layer *Run:* `python3 receipts/P10_canonical_time/P10_the_chain_does_not_reach_the_planck_length_and_the_second_order_ter`

**P10R44** `P10_the_second_order_term_is_the_quartics_deviation_from_the_cubics_own_completion_and_the_coun` [OK]

*'PO-23', the LAST remainder: the two-loop coefficient — and the second-order vacuum energy turns out to be an IDENTITY, while the counterterm it would need collapses to $\{}Lambda^{3}$*

*on the substrate's own background* (node 66's ‘r6991‘ order, which gates ‘r6988‘, ‘r6990‘ and ‘r6990b‘ and comes back to the question the row was opened for, with no premises left in front of it. (1) ***”THE EXISTENCE QUESTION FIRST, BECAUSE IT MAY BE CHEAPER THAN THE VALUE. Does the two-loop vacuum divergence exist at all? **If there is an argument that it vanishes — a symmetry, a parity, a mode count, a total derivative — that closes the row without a computation**, and it is worth looking for before computing”***. (2) ***”AND IF IT EXISTS, THE COEFFICIENT, WITH ITS BOOKKEEPING STATED BEFORE IT IS USED ... **And say which of the two trace-formula terms it arrives through**, since that is what makes it separately visible”***. (3) ***”AND WHETHER IT IS COMPUTED OR FITTED, WHICH IS THE LEDGER'S OWN CRITERION AND THE WHOLE POINT ... **The count of spent dimensionless constants goes from one to two only in the second case** ... State it with the same derivative test you used on the ratio”***. (4) ***”And the row's discharge condition, which I am putting in front of you rather than keeping to myself ... **say in your own terms whether what you have is a definition of the mode sums**”***. Guards: **count the equations a relation adds against the quantities it introduces** (this line's own, from ‘r6990‘); **a bound that survives the cases tried is not a bound** (also this line's); the eighth face as ***”does it move under a field redefinition that generates terms you dropped?”***; ***a warrant in a count is weaker than a warrant in an invariant***; and the sentence-and-arithmetic test from ‘r6990b‘). *Computes:* **(1) THE DIVERGENCE DOES REACH SECOND ORDER, AND THE ANSWER ARRIVES AS AN IDENTITY RATHER THAN AS A FAILED SEARCH FOR A VANISHING ARGUMENT.** **(1) THE IDENTITY, EXACT AND IN A TRUNCATION-FREE BASIS: $\{}Delta E^{(2)}=\{}langle H_{4}-H_{4}^{\{}rm fake}\{}rangle$.** Every cubic is NON-RESONANT — verified as the absence of any energy-diagonal matrix element — so the generator $A$ with $i[A,H_{0}]=H_{3}$ exists elementwise and is checked by SUBSTITUTION; the redefinition it generates has a UNIQUE quartic completion, with $\{}langle H_{4}^{\{}rm fake}\{}rangle=11g_{3}^{2}/8\{}omega^{4}\{}neq0$ and $\{}Delta E^{(2)}=0$ exactly ⇒ ***the cubic's entire contribution to the second-order vacuum energy IS the completion it generates, so what survives is the actual quartic's DEVIATION from it*** **(2) ⇒ AND THAT IS THE EIGHTH FACE DISCHARGED AS AN IDENTITY RATHER THAN AS A CAVEAT.** The order asked whether a two-loop coefficient moves under a field redefinition that generates terms one dropped; the answer is that this quantity is BUILT to be the part that does not, and the instrument is calibrated both ways — exactly zero on two unitary conjugations (both channels individually non-zero and EQUAL at $27/8\{}omega^{2}$) and $(6g_{4}\{}omega^{2}-11g_{3}^{2})/8\{}omega^{4}$ on a genuine pair, which is the textbook cubic-oscillator shift **(3) (1) SO NO SYMMETRY, PARITY OR MODE COUNT CAN REMOVE IT, AND THE IDENTITY SAYS WHY:** the cubic channel is a sum of NON-NEGATIVE numerators over POSITIVE denominators, so there is nothing for such an argument to annihilate term by term; a PURE cubic already gives $-11g_{3}^{2}/8\{}omega^{4}\{}neq0$; the quantity is EVEN in the cubic coupling, so the one thing a parity assignment can do to the vertex leaves it unchanged; and a total derivative changes no spectrum ***What vanishing requires is the codimension-one coincidence $g_{4}=11g_{3}^{2}/6\{}omega^{2}$ — exactly the locus on which the interaction is a redefinition — for which no mechanism is on offer*** **(4) AND THE PARITY ESCAPE IS CLOSED BY THIS ROW'S OWN ARITHMETIC RATHER THAN BY ARGUMENT:** for traceless $h$, $\{}mathrm{tr}\{},h^{3}=3\{}det h$ exactly and $\{}det\{}mathrm{diag}(2,-1,-1)=2\{}neq0$, so ‘r6967‘'s ordered cubic overlap $6\{}pi^{2}\{}det h$ is non-zero and the tower's cubic coupling survives every selection rule **(5) (2) THE BOOKKEEPING, STATED BEFORE USE AND IN THE ONE RATIO THE ROW FIXED:** $j=2k-4$, so $j=2$ admits $k=3$ and nothing else — operator dimension SIX — and it arrives through the $(1-n)$ term of the trace formula at $a^{-6}$ where the zeroth-order counterterm arrives through the $m$ term at $a^{-4}$, the brackets being $-2$ and $1$ respectively **(6) AND THEN THE NEW FACT, WHICH IS THE ONE THAT MOVES THE STAKE: THE SUBSTRATE BACKGROUND IS EXACTLY MAXIMALLY SYMMETRIC, COMPUTED FROM ITS CHRISTOFFEL SYMBOLS RATHER THAN ASSUMED.** All $256$ components satisfy $R_{abcd}=K(g_{ac}g_{bd}-g_{ad}g_{bc})$ with $K=1/\{}alpha^{2}=\{}Lambda/3$, and $R=12/\{}alpha^{2}=4\{}Lambda$ ⇒ every Weyl component vanishes, so ***BOTH cubic Weyl contractions vanish — the pair that carries pure gravity's known two-loop counterterm*** — and all eight ALGEBRAIC dimension-six scalars are exact rational multiples of $\{}Lambda^{3}$: $64$, $16$, $4$, $\{}tfrac{32}{3}$, $4$, $\{}tfrac83$, $\{}tfrac{16}9$, $\{}tfrac89$, each verified homogeneous of degree three by differentiation; and

every derivative invariant vanishes because $R$'s gradient is zero in all four coordinates **(7) ⇒ SO A DIMENSION-SIX COUNTERTERM, EVALUATED WHERE THIS ROW'S EXPANSION LIVES, HAS THE SHAPE OF A CONSTANT TIMES THE VOLUME — A RENORMALISATION OF THE COSMOLOGICAL CONSTANT AND NOTHING ELSE**, stated as a RANK: the eight values have rank ONE of eight here, and adjoining the control's eight raises it to two **(8) AND THE CONTROL IS THIS ROW'S OWN RATHER THAN ONE CHOSEN TO SUCCEED:** on the anisotropic Bianchi-I background $\{}mathrm{diag}(-1,T^{2},T^{4},T^{6})$ at $T=1$ the instrument returns $C^{2}=\{}tfrac{28}{3}$ — ***‘r6982‘’s own landed number*** — both cubic Weyl contractions non-zero, a non-zero curvature gradient, and eight values NOT in the maximally symmetric ratios *Bound:* **(3) AND IT IS COMPUTED RATHER THAN FITTED, FOR AN ARBITRARY COUNTERTERM AND NOT MERELY FOR THE EIGHT ONE AT A TIME.** A general rational combination of the basis has ZERO derivative with respect to $\{}hbar$, $G$ and $c$ and is exactly homogeneous of degree three in $\{}Lambda$ ⇒ ***whatever the two-loop computation returns, its RESIDUE is a rational times $\{}Lambda^{3}$*** — and the same holds ingredient by ingredient: the eight rationals, the tower's frequency $m^{2}-1$, its degeneracy $2(m^{2}-4)$, the overlap $6\{}pi^{2}\{}det h$ and the $\{}tfrac{11}{6}$ of the identity all have zero derivative with respect to every measured input, where $\{}Lambda$ itself and the gauge ratio $\{}alpha/\{}ell_{P}$ both move, which is the affirmative control ⇒ **THE COUNT OF SPENT DIMENSIONLESS CONSTANTS STAYS AT ONE**, the same verdict the shear's $\{}tfrac1{60}$ already carries in this row **AND THE SCOPE THAT DECIDES IT, IN THE SAME SENTENCE AND SHOWN RATHER THAN ASSERTED: what is computed is the RESIDUE.** A logarithmic mode sum regulated at $M$ and at $2M$ has the SAME log coefficient and finite parts $\{}gamma$ and $\{}gamma-\{}ln2$, differing by exactly $\{}ln2$ ⇒ ***the subtraction's finite part is a scheme convention, and nothing here shows the construction supplies a normalisation condition for it*** **(4) AND MY READING ON THE DISCHARGE CONDITION, WHICH IS THAT THIS IS NOT ONE.** The condition reads *”a definition of the mode sums”*; two prescriptions sharing one residue return numbers differing by a cutoff-independent amount, so naming the residue leaves the sum undefined by exactly that much. This characterises the SUBTRACTION — operator dimension, the single direction its value occupies here, the computed status of its residue — which is ***strictly less*** than the condition asks for, and it is said rather than the row read closed *What HAS changed is the remainder's shape: not an obstruction indistinguishable from the general one — the inference a predecessor was struck on — but a named subtraction at a known dimension, degenerate with the one constant on the background it acts on, with a computed residue and a consequence bounded exactly by $\{}Lambda\{}ell_{P}^{2}/3=10^{-122}$, carried from ‘r6990‘ rather than re-measured* **WHAT IS NOT DELIVERED, AND THE IDENTITY ITSELF SAYS SO:** for ANY target value there is a quartic that returns it, so ***the identity fixes what the coefficient IS and constrains its SIZE not at all*** ⇒ the coefficient's VALUE is not delivered, and neither is whether the tower's own double and triple mode sums CONVERGE at this order 44 checks, rc=0, **NO FLOATS AT ALL**; every number a rational, an exact algebraic identity, an exact matrix element verified independent of the truncation, or an exact tensor contraction No corpus edit — routed; no rank-sequence entries added; nothing on ‘PO-56‘, ‘PO-64‘ or the reproducibility layer *Run:* `python3 receipts/P10_canonical_time/P10_the_second_order_term_is_the_quartics_deviation_from_the_cubics_own`

**P10R45** `P10_the_second_order_sum_is_a_double_sum_and_it_diverges_at_the_sixth_power_of_the_label_cutoff` [OK]

*‘PO-23‘; convergence before the coefficient — and the second order is ONE double sum, which DIVERGES at the sixth power of the label cutoff, two powers above anything the free tower's subtractions reach* (node 66's ‘r6997‘ order, which gates ‘r6994‘ whole, accepts its (4) with no strike, and sequences convergence AHEAD of the coefficient on the ground that ***a number extracted from a sum is only defined if the sum converges***. (1) ***”THE CONVERGENCE, AND STATE WHAT CONVERGES BEFORE TESTING WHETHER IT DOES ... Say what the summand is and over what set, with the degeneracies, before any estimate — and then whether it converges, conditionally converges, or diverges. **a double sum may diverge in a way the single-sum subtraction does not reach**, which is the case worth looking for first”***. (2) ***”AND IF SOMETHING DIVERGES, WHETHER THE SUBTRACTION YOU JUST CHARACTERISED COVERS IT ... or does the count of counterterms at this order exceed one? **If it exceeds one, the rank sequence's second entry is the wrong number and I have landed a wrong count — report that as a correction to my prose**”***. (3) ***”AND ONLY THEN THE COEFFICIENT,

IF (1) AND (2) LEAVE IT DEFINED ... **If (1) or (2) consumes the revision, stop there and say so**"***. (4) ***"Does that kill the whole derivative sector at this dimension, or only the Weyl part of it?"***. And one routed item, 'PO-66': a tolerance site in 'r6980', **this line's own**, read TRUE by node 70's sweep and handed back for repair by this line's own 'r6990b' criterion. Guards: **the domain of a symbol** — the whole point of the sequencing; ***prefer an IDENTITY to a vanishing argument***, which is 'r6994''s own lesson turned into a guard; a bound that survives the cases tried is not a bound; and count the equations a relation adds against the quantities it introduces). *Computes:* **(1) IT DIVERGES, AND AT THE SIXTH POWER OF THE LABEL CUTOFF — WHICH IS EXACTLY THE CASE THE ORDER SAID TO LOOK FOR FIRST.** **(1) THE OBJECT FIRST, AS THE ORDER REQUIRED — AND THE IDENTITY REMOVES ONE OF THE TWO SUMS RATHER THAN BOUNDING IT.** The index set is $m\{}ge3$ with $d(m)=2(m^{2}-4)$ and $\{}mu_{m}=\{}sqrt{m^{2}-1}$, and $d(2)=0$ closes the set from below rather than the label being excluded by hand; in the free vacuum $\{}langle\{}phi_{m}^{2}\{}rangle=\{}hbar/2a^{2}\{}mu_{m}$ and $\{}langle\{}pi_{m}^{2}\{}rangle=\{}hbar a^{2}\{}mu_{m}/2$, their product exactly $\{}hbar^{2}/4$ — the minimum-uncertainty identity, used as the check that the two were not scaled independently. ⇒ ***And since 'r6994''s identity reads $\{}Delta E^{(2)}=\{}langle H_{4}-H_{4}^{\{}rm fake}\{}rangle$ with BOTH operators quartic — verified here rather than assumed, $H_{4}^{\{}rm fake}$ reaching four quanta from the vacuum where the cubic reaches an odd three — a free-vacuum expectation of either is a sum over TWO contractions*** ⇒ **the order's triple sum over intermediate states is not an independent object: the identity summed it already, and what is left is ONE double sum over pairs of labels** **(2) THEN THE RATE, FROM THE TOWER'S OWN WEIGHTS AND THE ACTION'S DERIVATIVE COUNT.** The exact large-label expansions are $d\{}mu=2m^{3}-9m+\{}tfrac{15}{4m}+\{}dots$ — ***whose $1/m$ term is this row's own logarithmic coefficient, arriving from the same expansion, so the weight is the one the row already uses*** — and $d/\{}mu=2m-\{}tfrac7m-\{}dots$, giving partial sums $M^{4}$ and $M^{2}$; and a two-derivative structure carries $\{}mu^{2}$ against $\{}langle\{}phi^{2}\{}rangle$, i.e. the SAME weight as $\{}langle\{}pi^{2}\{}rangle$ ⇒ the three structures the Einstein–Hilbert quartic supplies — and it supplies exactly these, because **the action is second order in derivatives** — give $\{}pi^{2}\{}phi^{2}:M^{6}$, $(\{}partial\{}phi)^{2}\{}phi^{2}:M^{6}$, $\{}phi^{4}:M^{4}$ **(3) ⇒ SO THE LEADING DIVERGENCE IS $M^{6}$ AND THE $j=0$ SUBTRACTIONS ARE $M^{4}$ AND A LOGARITHM: a subtraction cannot reach two powers above itself** ⇒ ***(1) ANSWERED, and answered in the direction the order flagged as the one that would change what the subtraction has to be*** **(4) AND THE TWO ROUTES AGREE, WHICH IS A CROSS-CHECK RATHER THAN A RESTATEMENT:** $M^{6}$ in labels is $\{}Lambda_{\{}rm UV}^{6}$ in momentum, and $\{}ell_{P}^{2}\{}Lambda_{\{}rm UV}^{6}$ has the dimension of an ENERGY DENSITY ⇒ **the mode count lands on operator dimension six, where $j=2k-4$ put it by dimensional bookkeeping alone** **(5) AND THE ONE INPUT THAT IS BOUNDED RATHER THAN COMPUTED, NAMED WITH THE EXPONENT THAT COULD MOVE IT:** the disconnected contraction's diagonal four-harmonic overlap is an integral of two non-negative densities, hence positive, and is not evaluated here; ***if it decayed as $m^{-s}$ the rate would be $M^{6-2s}$, so convergence would need $s¿3$*** — a decay no equidistributing family of harmonics has. *Stated with the exponent rather than as a bound over the cases that happened to be tried* **(6) (4) AND THE DERIVATIVE SECTOR VANISHES ENTIRELY, NOT ONLY ITS WEYL PART:** on this background $\{}nabla_{a}R_{bcde}=0$ in **all $1024$ components**, computed from the Christoffel symbols, with metric compatibility verified in all $64$ and $K=1/\{}alpha^{2}$ constant and non-zero ⇒ every dimension-six invariant carrying a derivative is identically zero here, so **the basis's whole non-vanishing content on this background is the eight ALGEBRAIC invariants and the collapse is stronger than 'r6994' stated** *Bound:* **(2) AND ONE NUMBER STILL COVERS IT HERE — AND THE ORDER'S OFFERED CORRECTION IS DECLINED, WHICH IS THE ANSWER THAT COSTS THIS LINE THE FINDING.** Three divergence structures appear, at two distinct rates, so *"one number"* is a statement about their VALUES and not about their count; and those values lie in a **one-dimensional** space — rank one of eight, and with the derivative sector now identically zero that is the rank of the WHOLE basis here ⇒ a single number absorbs all three on this background ***And 'r6982''s $r_{6}\{}ge5$ is NOT a wrong count: it is a rank over the ADMITTED CLASS, not over this background, and 'r6982''s own receipt computes both and reports them separately.*** 66 offered to have a wrong count reported as a correction to his prose; there is no correction owed, and

saying so is worth more than taking the finding **(3) THE COEFFICIENT IS NOT DELIVERED, WHICH THE ORDER LICENSED — AND (1) MAKES THE REASON SHARPER.** A sixth-power divergence has its $M^{6}$, $M^{4}$ and logarithmic parts to name before a finite part exists: **three scheme choices rather than one**, where a logarithm alone needs one and a quartic two ⇒ ***the coefficient is not one number awaiting computation but a number awaiting a scheme at three orders***, which is more than ‘r6994‘’s bookkeeping implied. ‘r6994‘ measured the lowest of those freedoms exactly (a regulator change from $M$ to $2M$ shifts the finite part by $\{}ln2$), so this is the same scheme-dependence one order at a time rather than a new kind of gap **AND ‘PO-66‘ IS REPAIRED IN THE SAME PUSH, WITH A SHARP PART THAT IS AN ARITHMETIC RATHER THAN A JUDGEMENT.** ‘r6980‘’s central difference was pinned at $h=10^{-5}$; the scan it now prints shows the relative difference falling as $h^{2}$ from $10^{-3}$ to $10^{-4}$ (truncation-dominated, the same on every build) and then flattening and going non-monotonic at $10^{-5}$ (round-off) — which is why node 70’s sweep read it TRUE. ***And the old tolerance $10^{-8}$ sits BELOW the honest worst point $5.8\{}times10^{-8}$ and above the round-off reading, so the old check could not have reported the worst step and still passed: pinning the round-off step was not a loose choice but the only one that tolerance permitted.*** Repaired in both parts of the ‘r6990b‘ criterion — the WORST point asserted, against a margin set from what the check DISCRIMINATES (a wrong closed form is wrong by an $O(1)$ factor, so $10^{-4}$ separates by four decades and leaves $1700\{}times$ of room) — and verified where it was flagged: the same three probes and two compares report **0 flagged sites**, the receipt still ‘rc=0‘ at 57 checks *And the same criterion is applied to this revision’s own rate measurements rather than only to the routed one: each doubling ratio is judged against a margin that is an EIGHTH of the distance to the nearest wrong power, not a number chosen above the reading* 31 checks, rc=0; exact in every algebraic step, with the two rate measurements reported against exactly derived predictions rather than against thresholds No corpus edit — routed; ‘sec:lock‘’s closing paragraph on the row not being discharged left exactly as it stands, per the order; nothing on ‘PO-56‘, ‘PO-65‘, or ‘PO-66‘ beyond the one routed site *Run:* `python3 receipts/P10_canonical_time/P10_the_second_order_sum_is_a_double_sum_and_it_diverges_at_the_sixth_po`

**P10R46** `P10_the_overlap_does_not_fall_off_at_all_so_the_sixth_power_is_exact_and_the_collapse_reaches_ev` [OK]

*‘PO-23‘; the overlap’s fall-off, on which the whole divergence turned — and the exponent is **ZERO**, so the sixth power is EXACT rather than conditional, while the collapse turns out to reach every dimension* (node 66’s ‘r6999‘ order, which gates ‘r6998‘ whole, records that its own question had been the wrong shape (there is no independent triple sum), and puts the overlap FIRST because ***the divergence’s existence, and with it the third subtraction just landed in ‘sec:lock‘, was conditional on one uncomputed exponent***. (1) ***”THE EXPONENT, COMPUTED RATHER THAN BOUNDED ... **Say what the object is before evaluating it** — which harmonics, which contraction, and whether the diagonal is the only part the double sum reaches. If an exact form exists ... **that is cheaper than an asymptotic estimate and is to be preferred — and if it is exact, say so, because an exact overlap would make the whole rate exact**”***. (2) ***”AND WHETHER THE DIAGONAL IS ENOUGH, WHICH IS A DOMAIN QUESTION RATHER THAN A RATE ONE ... **Does the double sum reach off-diagonal overlaps as well**, and if so do they change the rate or only the coefficient? An off-diagonal set that grows with the label could beat a decaying diagonal, and that is the kind of thing this row has been caught by before”***. (3) ***”AND ONLY THEN THE COEFFICIENT, IF (1) LEAVES IT NEEDED ... If (1) and (2) consume the revision, **stop and say so**”***. (4) ***”Does that argument reach dimension eight, or does the covariant constancy stop buying anything above six? ... **a statement about the whole tower rather than about one entry**”***. Guards: the domain of a symbol, with (2) being that guard in its own right; ***a bound that states the threshold at which it flips is a different object from a bound*** — this line’s own, from ‘r6998‘; ***prefer an IDENTITY to an estimate***, which has now paid three times on this row; and ***decline a correction that is wrong, including the gate’s own***). *Computes:* **(1) THE EXPONENT IS ZERO — THE OVERLAP DOES NOT FALL OFF AT ALL — AND IT IS AN IDENTITY RATHER THAN AN ASYMPTOTIC ESTIMATE, SO $M^{6}$ IS NOW EXACT.** **(1) THE OBJECT FIRST, AND STATING IT IS MOST OF THE ANSWER.** Because $\{}langle\{}phi_{a}^{2}\{}rangle$ depends on the LEVEL alone, what the rate needs is not one harmonic’s overlap but the **LEVEL-SUMMED** one, $\{}Sigma(m,m')=\{}sum_{a\{}in m}\{}sum_{b\{}in m'}\{}int\{}!\{}sqrt\{}gamma\{},(Y_{a}\{}!\{}cdot\{}!Y_{a})(Y_{b}\{}!\{}cdot\{}!Y_{b}

— ***a different object from the one the order and the previous revision were both discussing, and the one that happens to be exact*** **(2) AND THE LEVEL SUM IS A CONSTANT TENSOR, BY HOMOGENEITY AND SCHUR WITH BOTH HYPOTHESES TRUE HERE** (the section's isometry group is transitive; a level is one irreducible representation) ⇒ $\{}sum_{a\{}in m}Y_{a}^{ij}Y_{a}^{kl}=\{}tfrac{d(m)}{5V}\{}Pi^{ij,kl}$ and $\{}sum_{a\{}in m}\{}lvert Y_{a}\{}rvert^{2} **pointwise**. The completeness relation $\{}sum_{k}h_{k}^{ab}h_{k}^{cd}=\{}tfrac12(\{}delta^{ac}\{}delta^ \{}tfrac13\{}delta^{ab}\{}delta^{cd}$ is verified **exactly in all 81 components**, on an orthonormal basis checked on all twenty-five pairs **(3) AND AT THE ONE LEVEL THIS ROW OWNS THE DENSITY IS A COMPUTATION RATHER THAN A THEOREM:** 'r6967''s lowest harmonics are FRAME-CONSTANT, so $\{}lvert\{}varepsilon\{}rvert^{2}=\{}mathrm{tr}\{},h^{2}$ pointwise; the level sums to $5$ per chirality and $10$ over both — ***exactly $d(3)=10$*** **with an affirmative control: dropping ONE basis element leaves components disagreeing with the projector and a trace of 4 instead of 5** ⇒ the identity is a property of the FULL level and not of the basis **(4) ⇒ SO $\{}Sigma(m,m')=d(m)d(m')\{}times$ A PURE NUMBER OVER $V$, AND $d(m)d(m')$ IS EXACTLY THE DEGENERACY PRODUCT 'r6998' ALREADY COUNTED** ⇒ the residual factor has zero derivative in both labels: ***$s=0$***, where the reversing threshold was $s¿3$ (checked from both sides: at $s=3$ the rate is zero, above it negative) ⇒ **the divergence STANDS, the rate is EXACT rather than conditional, and the third subtraction 'sec:lock' now carries is real** **(5) AND IT CONFIRMS 'r6998''s COUNT RATHER THAN ONLY COMPLETING IT:** the degeneracy enters exactly ONCE, through the harmonic sum itself, so the weights $d\{}mu$ and $d/\{}mu$ were right and were not double counted — *the error this derivation would have exposed had it been made* **(6) (2) AND THE DOUBLE SUM NEVER REACHES A FOUR-LABEL OVERLAP AT ALL, WHICH IS A DOMAIN ANSWER AND NOT A RATE ONE.** Over **all 81** label combinations of a multi-mode free field the quartic vacuum expectation equals the Wick sum with **zero departures**, and is supported on **one or two distinct labels and never three or four** — because the two-point function is diagonal, computed from the mode operators rather than assumed ⇒ ***the three pairings differ in the ARRANGEMENT of indices, not in the number of labels; each submits to the same level-summed identity, with a different pure number*** ⇒ **off-diagonal arrangements change the COEFFICIENT and not the RATE, and the growing four-label set that could have beaten a decaying diagonal does not exist here** *and the two-label term being a PRODUCT of two single-label moments is what makes the rate factorise into two single sums at all — the step 'r6998' assumed, here derived* *Bound:* **(4) AND THE COLLAPSE REACHES EVERY DIMENSION, NOT JUST EIGHT — WHICH IS THE STATEMENT ABOUT THE WHOLE TOWER THE ORDER ASKED FOR.** Two exact halves: any invariant carrying an explicit covariant derivative contains a factor $\{}nabla_{a}R_{bcde}$, which 'r6998' showed vanishes in all $1024$ components here, ***so the derivative sector is identically zero at EVERY dimension***; and at dimension $2k$ the algebraic invariants are degree-$k$ polynomials in $R_{abcd}=K(g_{ac}g_{bd}-g_{ad}g_{bc})$, hence rational multiples of $\{}Lambda^{k}$ Verified on **TEN** dimension-eight invariants — $256$, $64$, $\{}tfrac{128}3$, $16$, $\{}tfrac{64}9$, $\{}tfrac{32}3$, $4$, $16$, $\{}tfrac{64}9$, $\{}tfrac{32}{27}$ times $\{}Lambda^{4}$ — each checked **homogeneous of degree four by differentiation** rather than by reading an exponent, **rank ONE of ten**, and the statement then shown for an ARBITRARY rational combination of the ten (zero derivative in $\{}hbar$, $G$, $c$; exactly degree four in $\{}Lambda$) ⇒ ***whatever a dimension-eight counterterm turns out to be, its value here is a rational times $\{}Lambda^{4}$*** **AND THE SCOPE IN THE SAME SENTENCE, WHICH IS THE DISTINCTION THIS LINE DECLINED A CORRECTION OVER LAST REVISION, NOW APPLIED TO ITS OWN NEW RESULT:** this is a rank over **values on this background**, where 'r6982''s rank sequence is over the **admitted class** — not a fixed background, because the scale factor is quantized — so ***nothing here touches that sequence either*** **(3) THE COEFFICIENT IS STILL NOT DELIVERED, WITH THE THREE CONVENTIONS NAMED AND SEPARATED FROM IT.** A sixth-power divergence needs its $M^{6}$, $M^{4}$ and logarithmic subtractions fixed: **three CONVENTIONS**, where a quartic needs two and a logarithm one, and 'r6994' measured the lowest exactly as a cutoff-independent $\{}ln2$ ⇒ the list is fixed and finite rather than growing, which is what lets it be separated from the coefficient instead of absorbed into it. **What the coefficient needs beyond them is the Einstein–Hilbert quartic's own vertex numbers, which this revision does not compute** — and (1) and (2) were what the order sequenced first 25 checks, rc=0, **NO FLOATS AT ALL**; every number a rational, an exact tensor identity, an exact matrix element, or an exact

contraction No corpus edit — routed; **the paragraph saying the subtractions are three is left exactly as it stands, and (1) says it is right**; nothing on ‘PO-56‘, ‘PO-65‘ or ‘PO-66‘ item two *Run:* `python3 receipts/P10_canonical_time/P10_the_overlap_does_not_fall_off_at_all_so_the_sixth_power_is_`

**P10R47** `P10_two_of_the_three_conventions_reach_nothing_observable_and_the_third_is_the_constant_the_row` [OK]

*‘PO-23‘; **do the conventions reach anything observable** — and TWO of the three reach NOTHING, while the third is the one constant ‘sec:lock‘ already records the mode sums as spending, so the discharge turns on no new number* (node 66’s ‘r7001‘ order, which gates ‘r7000‘ whole, records that the row now holds the whole rank sequence it was opened to ask for, and puts a NEW question because *a renormalisation scheme is defined when what is subtracted, at what order, and in what basis are fixed — and all three now are, at every dimension*. (1) ***”WHAT THE THREE CONVENTIONS ARE, STATED AS OBJECTS BEFORE THEY ARE TESTED ... **Say what freedom each one is** — an additive constant, a scale inside a logarithm, a subtraction point — and which of them the lowest one ‘r6994‘ measured exactly removes. *A convention that is already fixed by a measurement is not a convention, and if one of the three is in that position the count is two*”***. (2) ***”AND THEN THE OBSERVABLE QUESTION, WHICH IS THE ROW ... **Does a change of convention move any of it?** If the dependence cancels, say by what mechanism — a difference of two quantities, a ratio, a derivative — because **the mechanism is what makes it a result rather than a coincidence of the cases tried**”***. (3) ***”AND ONE THING TO CHECK BEFORE CONCLUDING EITHER WAY, WHICH IS THIS ROW’S OWN RECURRING TRAP ... **Is that constant one of the three conventions, or is it downstream of them?** ... *I think that is the single most likely way for the row to close against itself, and I would rather you looked at it first than found it afterwards*”***. (4) ***”And the coefficients only if (1) to (3) leave room, which I doubt ... **stop and say so**”***. Guards: ***ask whether the sum reaches a coarser object that is exact*** (this line’s own, from ‘r7000‘); ***a convention that a measurement fixes is not a convention***; the domain of a symbol; and ***scope in the same sentence as the result***). *Computes:* **(1) THE THREE CONVENTIONS NAMED AS OBJECTS, AND THE COUNT STAYS THREE — THE ORDER’S OFFERED REDUCTION TO TWO IS DECLINED.** **(1) THE OBJECTS.** (a) the finite part of the **dimension-six** counterterm’s coefficient, an additive constant whose value here is a rational multiple of $\{}Lambdaˆ{3}$ by ‘r7000‘’s collapse; (b) the finite part of the **dimension-four** (cosmological) one, an additive constant; (c) the **SCALE inside the logarithm**, equivalently the subtraction point — ***the only one of the three that is a scale rather than an additive constant, and the only one whose action on the energy density carries a power of the scale factor*** **(2) AND ALL THREE ARE NECESSARY RATHER THAN LISTED:** leaving any one of $Mˆ{6}$, $Mˆ{4}$ or the logarithm unsubtracted leaves an infinite limit and subtracting all three leaves exactly the finite part, so the count of conventions IS the count of structures that must be fixed **(3) WHAT ‘r6994‘ MEASURED IS (c)’s ACTION ON (a) AND (b), NOT A VALUE FOR ANY OF THEM.** Re-derived at THIS row’s own weight rather than borrowed — $d(m)\{}mu(m)=2mˆ{3}-9m+\{}tfrac{15}{4m}+\{}dots$, so the logarithm’s coefficient is the weight’s own $1/m$ term — an octave in the subtraction point moves the finite part by exactly $\{}tfrac{15}4\{}ln2$ and the coefficient not at all ⇒ ***and the gap is the SAME for EVERY member of the family, $G(2\{}kappa)-G(\{}kappa)=L\{}ln2$ identically in $\{}kappa$, while $G$ itself moves with $\{}kappa$: a measured DIFFERENCE fixes the group’s action and not a point of it, so it cannot single out a subtraction point*** *and the guard cuts the other way for the coefficient: $L=\{}tfrac{15}4$ is a computed rational with zero derivative in all three, so $L$ is not a convention at all* **(4) (2) THE RANK IS TWO OF THREE, AND THE KERNEL IS THIS ROW’S OWN COLLAPSE.** The three move the energy density along $\{}{q_{6}\{}Lambdaˆ{3},\{},q_{4}\{}Lambdaˆ{2},\{},Laˆ{-4}\{}}$ — the first two pure CONSTANTS here, because every counterterm’s value on this background is a rational multiple of a power of the one input ⇒ the non-trivial triple $(q_{4}\{}Lambdaˆ{2},-q_{6}\{}Lambdaˆ{3},0)$ shifts it by exactly zero, identically in $a$, ***exhibited rather than counted*** ⇒ ** SO THE DIMENSION-SIX CONVENTION — THE ONE THIS ROW ADDED — IS RANK ONE WITH THE COSMOLOGICAL ONE: the third subtraction is new as a SUBTRACTION and not as a number taken from the world** *control: give the dimension-six counterterm a value that does NOT collapse, an $aˆ{-6}$, and the rank is three with no kernel — so the degeneracy is this background’s and not the rank test’s* **(5) AND THE ONE OBSERVABLE THIS SECTION NAMES IS UNTOUCHED, BY A GRADING RATHER THAN BY THE CASES TRIED.** $\{}Theta[c\{},aˆ{-n}]=(4-n)c\{},aˆ{-n}$ exactly, so $aˆ{-4}$ is the WHOLE kernel

of the trace and the zero is at $n=4$ alone; the anomaly is the $a^{-4}$ coefficient of $\{}Theta$, which is exactly $L$ ⇒ ***zero derivative in all three conventions and unit derivative in $L$***, the two power conventions moving only the CONSTANT part of the trace — the cosmological term the framework absorbs into the one observed curvature — and (c) moving nothing at all, being traceless *controls at $a^{-3}$ and $a^{-6}$ return $c\{},a^{-3}$ and $-2c\{},a^{-6}$, so the vanishing is radiation's own property and not the formula's* **(6) AND THE TOWER'S SPECTRUM IS UNTOUCHED AT FIXED SCALE FACTOR**, a convention being a multiple of the identity there: every spacing exactly unchanged while the total energy shifts, against a control that is not a multiple of the identity and moves every spacing **(7) (3) THE TRAP, TAKEN FIRST — AND THE ANSWER IS NEITHER OF THE TWO OFFERED: THE OBSERVABLE CONSTANT *CONTAINS* EXACTLY ONE OF THE THREE.** It is the logarithmic counterterm's coefficient, and it splits into a divergent part fixed at $L$ — an anomaly, scheme-independent — plus a finite part that IS convention (c) ⇒ ***so "the anomaly is scheme-independent" and "the constant is a convention" are both true of the same object, separated by WHERE THEY APPEAR: the $a^{-4}$ trace is blind to (c) and the coefficient is not*** ⇒ **(2) IS THEREFORE YES FOR ONE OF THE THREE, AND IT WAS YES BEFORE THIS REVISION — the closure the order suspected — BUT THE DEPENDENCE IS ON THE CONSTANT ‘sec:lock‘ ALREADY RECORDS THE MODE SUMS AS SPENDING, so nothing NEW is owed** **(8) AND THE SAME NUMBER MAKES THE THIRD CONVENTION AND THE SECOND DIRECTION.** Set $L=0$: the logarithm, its convention and the $a^{-4}$ direction all go at once, the rank falls to ONE, and every convention is absorbed into the one constant — ***which is the predecessor's own conclusion returned, and locates exactly which ingredient retires it*** ⇒ the difference of the two ranks is $1$, so **the anomaly buys exactly ONE observable direction and no more: this section's own shape — the saving mechanism and the thing it saves us from are one number — now holding for the CONVENTIONS and not only for the counterterm** *Bound:* **SCOPE IN THE SAME SENTENCE, AS THE GUARD REQUIRES:** every rank here is a rank over what the conventions do to the **energy density on this background, at the order at which this section's back-reaction is semiclassical**, where ‘r6982‘'s rank sequence is a rank over the **admitted class** — different objects, and nothing here touches that sequence **NOT a determination of the surviving convention**: it is shown to be degenerate with a radiation-like density (rank one against $a^{-4}$) and NOT with the cosmological term (rank two against it), which names what absorbs it rather than fixing it **NOT a claim that the ledger's entry IS this convention** — that identification is what makes "nothing new is owed" true, and it is a statement about what ‘sec:lock‘'s recorded constant is, **which is 66's to place: routed, not edited** **NOT (4)**: the Einstein–Hilbert quartic's own vertex numbers are not computed, and (1) to (3) did not leave room — *which the order predicted and instructed* *and the answer does not wait on them: carried through as an unknown factor on the dimension-six counterterm, the rank and the kernel are unchanged, while the unknown is not inert — it moves the constant part of the trace* NOT a repair of any paragraph: **no corpus edit at all**; nothing on ‘PO-56‘, ‘PO-65‘ or ‘PO-66‘ 34 checks, rc=0, exact everywhere except one rate measurement, reported at its **WORST** point against an exactly derived prediction with the margin set from what the check DISCRIMINATES *Run:* `python3 receipts/P10_canonical_time/P10_two_of_the_three_conventions_reach_nothing_observable_and_the_third`

**P10R48** `P10_the_descent_is_free_at_the_order_the_section_works_at_and_the_kernel_rotates_rather_than_fa`
[OK]
*‘PO-23‘; **the descent** — and it is FREE at the order this section's back-reaction is semiclassical, flips at the NEXT order, and the kernel that frees the new subtraction **rotates rather than failing**, so ‘r7002‘'s conclusion survives with its mechanism changed* (node 66's ‘r7003‘ order, which gates ‘r7002‘ whole, places the identification it routed (the section's one spent dimensionless constant IS the subtraction point's convention), takes its reading of the discharge — ***and then DECLINES THE STRIKE***, on the ground that every rank in ‘r7002‘ is a rank over values on a classical maximally symmetric background while the discharge question lives where the scale factor is an operator: *"‘PO-6‘ was struck on a result that was true where it was computed and read as true where it was wanted, **and that is the identical shape**"*. (1) ***"SAY WHAT THE QUESTION IS BEFORE ANSWERING IT, BECAUSE I MAY HAVE PUT IT WRONG. What does 'the counterterm basis' mean when the scale factor is an operator — an operator-valued basis, a basis of expectation values in a state, or a basis over the admitted family with the family itself quantized? **If my framing conflates two of those, say so and answer the one that bears on the mode sums**"*** *and the order names this an invitation to decline a correction a third

time*. (2) ***"AND THE ONE THING THAT MIGHT MAKE THE DESCENT FREE ... **is the quantized case a superposition over a family on every member of which the collapse holds** — in which case the rank statement descends by linearity — or does quantizing the scale factor take the geometry off that locus altogether?"***. (3) ***"AND IF IT DOES NOT DESCEND, WHAT SURVIVES AND WHAT DOES NOT, ITEM BY ITEM. The rate? The count of subtractions? The count of conventions? The kernel? **They may not fail together**, and a list of which survive is a better result than a verdict"***. (4) *the coefficients stay last; if (1) to (3) consume the revision, stop and say so*. Guards: ***a result true where it was computed is not true where it is wanted*** (the reason for the order); scope in the same sentence; ask whether the sum reaches a coarser object that is exact; ***and decline a correction that is wrong, including the gate's own***). *Computes:* **(1) THE FRAMING CARRIES ONE OBJECT AND NOT THREE — AND THE READING THE ORDER PUT FIRST IS THE ONE WITH NO OBJECT, WHICH IS THIS ROW'S OWN LANDED RESULT.** **(1) EVERY COUNTERTERM VALUE HERE IS A FUNCTION OF THE SINGLE OPERATOR $\{}hat a$**, the curvature radius entering as a number ⇒ (a) the **operator** reading and the **value** reading are the same object: equal ranks and, normalised, the *identical* kernel vector on a four-point spectrum, because a combination of functions of one operator vanishes as an operator exactly where the function vanishes on its spectrum ⇒ ***a rank descends with no linearity argument at all*** **(2)** (b) the **expectation** reading adds nothing: the expectation of a function of $\{}hat a$ is phase-independent, so it sees the spectral measure alone, **against a non-commuting control whose phase derivative is non-zero** ⇒ superposition and ensemble are one object here **(3) AND (c) THE QUANTIZED-FAMILY READING HAS NO OBJECT AS PUT:** the curvature radius is a coefficient of the physical Hamiltonian rather than an observable of it, so its operator has a ONE-POINT spectrum, where every function of it is a multiple of the identity and **the rank is 1 against 2 on a two-point spectrum** — *‘r6982’’s superselection result, exhibited as an arithmetic* ⇒ ***the framing conflates (c) with (a), the question is the SINGLE-RADIUS one, and the correction is declined for the third time on this row — this time changing which question is asked*** **(4) (2) THE DESCENT IS FREE AT THE ORDER THIS SECTION WORKS AT, AND NOT FOR EITHER REASON OFFERED.** The deformation the theory actually makes is not arbitrary: **pure radiation leaves $R=4\{}Lambda$ EXACTLY** (computed from the exact closed-FRW scalars, not assumed — the section's own claim), so the $a^{-4}$ in the curvature scalar is the **logarithm's and nothing else's**, $R=4\{}Lambda+\{}varepsilon\{}nu a^{-4}$ **(5) AND ON THAT DEFORMATION EVERY DIMENSION-SIX INVARIANT SHIFTS ALONG THE SAME FUNCTION:** the four computed here move by $48$, $12$, $8$, $3$ times $\{}Lambda^{2}\{}nu a^{-4}$ against zeroth orders $64$, $16$, $\{}tfrac{32}3$, $4$ times $\{}Lambda^{3}$ — *each exactly three quarters of its own zeroth order over $\{}Lambda$* ⇒ ***RANK ONE AT FIRST ORDER AS WELL AS AT ZEROTH: ‘r7000’’s COLLAPSE DESCENDS*** *the machinery calibrated first on the member itself, where it returns ‘r7000’’s own $4\{}Lambda$, $4\{}Lambda^{2}$, $8\{}Lambda^{2}/3$ and all four dimension-six values from the metric rather than from that receipt's frame* **(6) AND IT FLIPS AT SECOND ORDER, WHICH IS A THRESHOLD AND NOT A VERDICT:** rank **two** there, the breaking carried by $a^{-8}$ terms with $\{}log^{2}a$ structure that the first-order terms do not have ⇒ ***so ‘r6982’’s "no identity to descend" is REAL and does not BITE at first order: it is a statement about an arbitrary deformation*** *control: put an $a^{-6}$ into ONE invariant — a deformation the anomaly does not make — and the rank is two already at first order, so the survival belongs to the deformation and not to the rank machinery* **(7) (3) ITEM BY ITEM, AND THEY DO NOT FAIL TOGETHER.** ***THE RATE: survives exactly and at every order*** — the label weights are harmonic data in the label alone, the whole scale-factor dependence being the single factor $a^{-4}$, so any state enters as ONE moment multiplying the sum and no divergent structure is added or removed (control: a weight made scale-factor-dependent puts an $a$ into the partial sum itself) ***THE COUNT OF SUBTRACTIONS AND OF CONVENTIONS: survive***, each still necessary with the moment in front ***THE PER-DIMENSION RANK: survives at first order, fails at second*** **(8) THE KERNEL: SURVIVES IN NUMBER AND ROTATES IN PLACE, WHICH IS THE FINDING.** Still two directions of three with a one-dimensional kernel — **but with the anomaly on, the free direction necessarily carries the LOGARITHM's convention, where with it off that entry is exactly zero**: releasing the dimension-six convention costs a shift of the logarithm's in the computed ratio $L/(48\{}Lambda^{2}\{}varepsilon\{}nu q_{6})$, which runs away as the anomaly is switched off, *so the classical kernel is the LIMIT and not a different case* ⇒ ***‘r7002’’s "THE

SUBTRACTION THIS ROW ADDED COSTS NO OBSERVABLE AT ALL" SURVIVES THE DESCENT WITH ITS MECHANISM CHANGED: the absorption is into a combination of the cosmological term and the ONE CONSTANT ALREADY SPENT, and the count of numbers taken from the world is unchanged at this order*** *Bound:* **SCOPE IN THE SAME SENTENCE, AS THE GUARD REQUIRES:** every rank here is a rank over the values these invariants take on the **anomaly-carrying deformation of the substrate background, to FIRST order in the back-reaction** — the order at which this section's back-reaction is semiclassical — where at second order it is two and 'r6982''s rank over the admitted class is a third object again **NOT a claim that the descent is free at every order**: it is not, and the receipt computes where it stops **NOT a proof that the discharge is available** — 66's decline of the strike is not overturned: (3) says the count of observable directions is unchanged *at this order*, and the second-order rank is exactly what the row now owes **NOT a statement about a superposition over curvature radii**: that reading is shown to have no object here, and the one-point spectrum is exhibited rather than asserted **NOT (4)**: the Einstein–Hilbert quartic's vertex numbers are not computed, and (1) to (3) consumed the revision — *which the order licensed for the fourth time* *and the answer does not wait on them: an unknown factor on the dimension-six direction leaves the rank and the rotated kernel unchanged* NOT a corpus edit: **none at all**; nothing on the other rows 25 checks, rc=0, **NO FLOATS AT ALL** — exact closed-FRW scalars, exact rational invariants, exact matrix ranks on exact sample points, and the kernel solved symbolically *Run:* `python3 receipts/P10_canonical_time/P10_the_descent_is_free_at_the_order_the_section_works_at_and_the_kernel`

**P10R49** `P10_second_order_is_an_order_the_construction_does_not_reach_and_the_one_new_direction_carries_` [OK]

*'PO-23'; **second order in the back-reaction** — and it is an order this construction DOES NOT REACH (the parameter's maximum is the geometry's, at $9.6\{}times10^{-123}$, so the square is $10^{-245}$, and the second-order source is an independent datum the section does not supply), while the one new value direction **carries no fourth convention**; and the row's four-times coefficient is ONE object with four faces* (node 66's 'r7005' order, which gates 'r7004' whole, records that the ground the strike was declined on at 'r7003' is GONE, and declines again on a narrower one: ***"a definition that holds at one order of an expansion and not the next is a definition with an order attached, and this row's discharge condition does not have one"***. (1) ***"WHAT THE SECOND DIRECTION IS, AS AN OBJECT ... **Is the second rank direction one counterterm or two**, and does it carry a new convention or ride on the three that exist? ⇒ if it brings a fourth convention the whole convention accounting has to be redone at that order; **if it rides on the three, the accounting survives with a rank change and the row may be much closer than it looks**"***. (2) ***"AND WHETHER SECOND ORDER IS REACHED BY THIS CONSTRUCTION AT ALL, WHICH MAY MAKE (1) ACADEMIC AND IS CHEAPER. The section's back-reaction is semiclassical. **Is that a choice, an approximation with a stated domain, or a consequence of something?** ... **That is a scope answer and it would close the row** ... Take (2) first if it is cheaper. I suspect it is"***. (3) ***"AND THE ROTATED KERNEL'S RATIO IS NOW A BANKED NUMBER ... The same coefficient has appeared four times. **Is it the same object each time, or four quantities that happen to share a value?**"***. (4) *the coefficients stay last, for the fifth revision*. Guards: ***a definition that holds at one order and not the next is a definition with an order attached***; ***count the objects before counting the readings*** (this line's own, from 'r7004'); ask whether the sum reaches a coarser object that is exact; ***and decline a correction that is wrong, including the gate's own***). *Computes:* **(2) TAKEN FIRST, AS THE ORDER INSTRUCTED — AND IT IS THE SCOPE ANSWER: THE SEMICLASSICAL TRUNCATION IS A CONSEQUENCE, AND SECOND ORDER IS AN ORDER THIS CONSTRUCTION DOES NOT REACH.** **(1) THE PARAMETER'S MAXIMUM IS A PROPERTY OF THE GEOMETRY, NOT OF AN EPOCH:** the substrate's scale factor has **exactly one** stationary point — the turnover, with second derivative $1/\{}alpha¿0$, so a MINIMUM — hence $\{}ell^{2}/a^{2}$ is largest *there and nowhere else* and equals $\{}ell^{2}\{}Lambda/3$ **exactly**, which is the form 'r6990' bounded rather than a number quoted from it **(2) AND THE SIZE FOLLOWS:** on the measured inputs that parameter is $9.63\{}times10^{-123}$ and its square is $9.27\{}times10^{-245}$ — *the square computed as the square, to machine precision, rather than quoted* ⇒ **second order sits 244 decades below what the row already calls negligible at first** **(3) AND THE TRUNCATION IS NOT A CHOICE: A GENUINE SECOND-ORDER SOURCE IS AN INDEPENDENT DATUM, SHOWN BY PUTTING ONE IN.** With an independent second-order source every

second-order coefficient moves — $R^{3}$'s by $-576\{}Lambda^{2}\{}sigma a^{-8}$ — while **every first-order one is untouched**, and the coefficients ‘r7004‘ computed carry no such amplitude at all ⇒ ***SO ‘r7004‘’s SECOND-ORDER RANK IS THE RANK ON THE ITERATED FIRST-ORDER DEFORMATION: a PART of second order and a LOWER BOUND on it, not this construction’s own second order*** — the correction this revision owes its predecessor, and the reason is that the source is the tower’s energy on the **unperturbed** background **(4) (1) AND THE RANKS ARE NOW THE WHOLE SECTOR’S RATHER THAN FIVE CHOICES’, BY A MECHANISM.** At a maximally symmetric point **the gradient of every realizable invariant is a multiple of the identity**, so $\{}partial P/\{}partial K_{1}=\{}partial P/\{}partial K_{2}$ there — verified on five dimension-six and four dimension-eight invariants, with those gradients non-zero — and therefore *every* invariant’s first variation depends on the perturbation only through $\{}delta R$ *and the realizable cubics are exactly the three-dimensional subspace $3c_{0}+c_{1}-c_{2}-3c_{3}=0$ that the condition cuts out, which the five invariants SPAN* ⇒ ***the rank statements are the sector’s*** *control: the monomial $K_{2}^{3}$ violates the condition and RAISES the rank when adjoined, so a monomial basis would have given the wrong count* **(5) AND THE FIRST-ORDER DEGENERACY IS EXACTLY THE NO-LOGARITHM DIRECTION:** $\{}delta K_{1}+\{}delta K_{2}=\{}nu/6a^{4}$ carries no logarithm where $\{}delta K_{1}-\{}delta K_{2}$ does ⇒ **first-order value-rank ONE at every dimension, by the gradient argument where ‘r7004‘ had four measurements** **(6) AND THE SECOND DIRECTION IS *ONE*, AND IDENTIFIED RATHER THAN COUNTED.** At second order the rank is **two of three** over $\{}{a^{-8},a^{-8}\{}log a,a^{-8}\{}log^{2}a\{}}$, and every one of the five values decomposes exactly into the plain $a^{-8}$ — one invariant giving $36a^{-8}$ with **no logarithm at all** — and the single combination $(3+6\{}log a+4\{}log^{2}a)a^{-8}$ ⇒ ***the second direction IS that squared-logarithm combination*** **(7) AND IT BRINGS NO FOURTH CONVENTION, WHICH IS THE BRANCH THE ORDER SAID WOULD PUT THE ROW MUCH CLOSER THAN IT LOOKS.** The weights carry no scale factor at all, so a deformed geometry enters the mode sum as **one moment multiplying it**: the divergent structures stay the same three with no new power of the cutoff ⇒ **the new direction is a second VALUE the existing three counterterms must cover, and the convention accounting survives with a RANK CHANGE** **(8) (3) AND THE BANKED NUMBER IS ONE OBJECT WITH FOUR FACES.** $L=\{}tfrac{15}4$ is ***the residue of the tower’s own spectral Dirichlet series at the pole the logarithm’s subtraction sits on***: $\{}sum d\{}mu\{},m^{-s}$ has simple poles at $s=4,2,0$ with residues $2$, $-9$, $\{}tfrac{15}4$ — the $s=0$ residue probed against the exact value with the departure falling **linearly** in the probe as $\{}zeta$’s next term requires, and the $s=4$ pole probing to $2$ as a second reading — and the partial sum’s logarithm, the trace’s $a^{-4}$ (the anomaly) and the curvature scalar’s own $a^{-4}$ all return **the same rational as computed objects** *they move together when the weight’s $1/m$ term is changed, and **all four vanish together** on the control weight that has none (‘r6969‘’s own $\{}mu^{2}=m^{2}$)* ⇒ ***ONE object, four faces: name it once and cite it*** *Bound:* **SCOPE IN THE SAME SENTENCE:** *”an order the construction does not reach”* is a statement about what this construction PRODUCES and about the size its own geometry puts on that order — **it is not a proof that nothing at that order could matter**, and the honest form of the answer is the first and not the second **NOT a claim that the second-order rank is wrong**: it is right as arithmetic on the iterated deformation and is a **lower bound** on the count there; what is corrected is its SCOPE, which is ‘r7004‘’s own sentence and this line’s to fix **NOT a strike and NOT an argument for one**: the order-dependence the gate declined on is real, and this revision gives it a size rather than removing it — **the two halves are kept apart because one is a rank and the other is a size** **NOT a computation of the second-order source**: that needs the interacting tower the section says it does not build, which is precisely the scope point **NOT (4)**, for the fifth revision: the Einstein–Hilbert quartic’s vertex numbers are not computed — *and the answers do not wait on them: carry an unknown vertex factor through and both ranks are unchanged while the logarithm’s coefficient is that factor times $L$* NOT a corpus edit: **none at all**; nothing on the other rows 23 checks, rc=0; exact except two residue probes and one size, each reported against an exactly derived prediction rather than against a threshold *Run:* `python3 receipts/P10_canonical_time/P10_second_order_is_an_order_the_construction_does_not_reach_and_the_on`

**P10R50** `P10_the_vertex_numbers_are_exact_at_the_level_this_row_owns_and_the_whole_scheme_is_one_series_p` [OK]

*‘PO-23‘; **the coefficients, at last** — the vertex numbers are EXACT at the level this row owns,*

*taken from a closed form valid to all orders, and they return this row's own $\{}mu^{2}=8$ and degeneracy $10$ as a calibration; the second-order shift is $\{}tfrac{19}{27}\{}kappa\{}hbar^{2}/(Va^{3})$, positive, with an $a^{-6}$ density; the second order is **not reached** rather than unbounded; and **the whole subtraction scheme is one spectral series' pole data**  (node 66's 'r7007' order, which gates 'r7006' whole, records that the ground for a strike was refused in this line's own favour for the third revision running, and then asks for the coefficients because *"they have been last for six revisions and each time the reason was real ... **none of those holds now**"*. (1) ***"THE VERTEX NUMBERS AND WHAT THEY GIVE. The three quartic structures the action supplies, their coefficients, and what those make the dimension-six counterterm's rational multiple"*** *and carry the unknown through symbolically first if that is cheaper*. (2) ***"AND THE SECOND-ORDER RANK FROM ABOVE, WHICH YOUR OWN CORRECTION MADE OWED. You bounded it below by the iterated deformation. **What bounds it above?** ... if it cannot be written down within this construction, **that is the same answer as 'the order is not reached' and it should be said that way rather than left as a bound**"***. (3) ***"AND ONE THING THAT WOULD CHANGE WHAT THE ROW IS, IF IT IS CHEAP ... **Is the whole subtraction scheme expressible in terms of that one residue** — so that the section's three conventions, the rank and the anomaly are all statements about a single spectral quantity? If so, this row's answer is one sentence rather than a section"***. (4) *and if (1) consumes the revision, stop and say so — sixth time*. Guards: ***a rank and a size are different objects***; ***correct your own previous revision's scope when you find it***; ***carry an unknown through rather than waiting for it***; and decline a correction that is wrong). *Computes:* **(1) THE VERTEX NUMBERS ARE EXACT, AND FROM A CLOSED FORM RATHER THAN AN EXPANSION.** 'r6967''s lowest harmonics are **frame-constant**, so a perturbation there is a LEFT-INVARIANT metric on the three-sphere, for which the geometry is known to *all* orders **(1) THE REDUCTION, DERIVED:** $K_{ij}K^{ij}-K^{2}=-6H^{2}+6(\{}dot\{}beta_{+}^{2}+\{}dot\{}beta_{-}^{2})$ exactly, the Misner velocities summing to zero so volume and anisotropy separate with no cross term; its isotropic limit returns the standard closed-FRW Lagrangian **(2) AND THE THREE-CURVATURE EXACTLY:** $R^{(3)}=2[2\{}sum\{}lambda\{}lambda-\{}sum\{}lambda^{2}]$ with $\{}prod\{}lambda=1$, returning $6$ on the round metric ⇒ ***$R^{(3)}=6-48(\{}beta_{+}^{2}+\{}beta_{-}^{2})+160(\{}beta_{+}^{3}\{}beta_{+}\{}beta_{-}^{2})-336(\{}beta_{+}^{2}+\{}beta_{-}^{2})^{2}+\{}dots$***, the cubic exactly the hexagonal $\{}operatorname{Re}(\{}beta_{+}+\{}mathrm i\{}beta_{-})^{3}$ and **the quartic exactly ISOTROPIC, so ONE number carries it** **(3) THE CALIBRATION IS THE CHECK THAT THIS IS THE ROW'S OWN LEVEL:** the ratio of the first two coefficients is $48/6=8=\{}mu^{2}$, which is 'r6998''s $m^{2}-1$ at $m=3$, and the degeneracy $2(m^{2}-4)=10$ is 'r7000''s frame-constant count ⇒ ***two numbers this row carried from the SPECTRUM arrive here from the METRIC*** **(4) AND ONE OF 'r6998''s THREE STRUCTURES IS A PARAMETERISATION AT THIS LEVEL — 'r6994''s IDENTITY ARRIVING FROM THE OTHER SIDE.** In the exponential variable the kinetic term is **exactly** quadratic, no $\{}pi^{2}\{}varphi^{2}$ vertex at any order, where in the linear one $\{}dot\{}beta^{2}=\{}dot h^{2}(1-4h+12h^{2}-\{}dots)$ carries a cubic AND a quartic; the map between them is a point transformation with unit Jacobian ⇒ ***the $\{}pi^{2}\{}varphi^{2}$ structure IS the redefinition-generated "fake" quartic, and in the variables where it is absent the shift is carried by the potential and the cubic alone*** **(5) SO THE SECOND-ORDER SHIFT AT THIS LEVEL IS A NUMBER:** $\{}Delta E^{(2)}=\{}tfrac{4\{}kappa\{}hbar^{2}}{27Va^{3}}\{}cdot\{}t200}{\{}mu^{4}}\{}to\{}tfrac{19}{27}\{}kappa\{}hbar^{2}/(Va^{3})$ at $\{}mu^{2}=8$, **POSITIVE** — the quartic beating the cubic *computed exactly in a two-mode Fock space: the quartic's $8c_{4}s^{2}$, and **the cubic's ONE-quantum vacuum amplitude cancelling identically** so that it reaches the vacuum only at three quanta, which is the non-resonance 'r6994''s identity requires, here as an arithmetic* **and the threshold with the result: the sign flips at $\{}mu^{2}=200/63$, so "positive" is this level's statement and not the tower's** **(6) AND THE DENSITY FALLS AS $a^{-6}$** — ***the dimension-six behaviour the row read off the mode count and the dimensional bookkeeping, now arriving from the vertices themselves*** **(7) (2) AND THE SECOND-ORDER RANK CANNOT BE BOUNDED FROM ABOVE HERE — REPORTED AS THE ORDER ASKED, AS *"THE ORDER IS NOT REACHED"* RATHER THAN A MISSING BOUND.** As an arithmetic rather than a tally: the first-order system determines its one unknown uniquely, where the second-order one — whose source needs the state's own correction — returns a FAMILY with the free state parameter still in the answer; and the

exact reduction has no independent second-order datum of its own, its three eigenvalue logarithms satisfying exactly one relation so that the geometry is exhausted by the scale factor and two anisotropies **(8) (3) AND YES — THE WHOLE SCHEME IS ONE SERIES' POLE DATA, WHICH MAKES THE ROW'S ANSWER ONE SENTENCE.** $\{}sum d\{}mu\{},m^{-s}$ has simple poles at $s=4,2,0$ with residues $2$, $-9$, $\{}tfrac{15}4$ and $\{}sum(d/\{}mu)m^{-s}$ at $s=2,0$ with $2$, $-7$; **each pole contributes its residue over its own location to the corresponding power of the cutoff** (checked pole by pole), **the logarithmic coefficients are the $s=0$ residues exactly** — so the anomaly is one entry of this table rather than a separate object — and **the second-order sum's leading coefficient is a PRODUCT of two residues**, $\{}tfrac24\{}cdot\{}tfrac22=\{}tfrac12$ ⇒ ***the three subtractions, the anomaly and the value-rank are all statements about the pole data of the tower's spectral series*** *and the subleading integer powers MIX, which is Faulhaber's and not the spectrum's — 'r7002''s division between a coefficient and a convention, now visible as the difference between a residue and a partial sum's lower terms* *Bound:* **SCOPE IN THE SAME SENTENCE:** the vertex numbers and the shift are exact **at the frame-constant level**, whose free symbols are exactly $\{}{\{}kappa,\{}hbar,V,a\{}}$ with no level label among them ⇒ **the TOWER-WIDE rational multiple is NOT computed**, and what it now needs is ONE thing: the higher levels' overlap integrals, whose structure 'r7000' fixed **NOT a claim that the quartic is positive for the tower**: the sign flips at $\{}mu^{2}=200/63$ and the statement is made against that threshold **NOT a second-order rank from above**: it is reported as an order the construction does not reach, which is what the order asked for in that case, and **not** as a bound left open **NOT a corpus edit**: none at all — including the closing paragraph that keeps the rank and the size apart, which (2) now speaks to and which is 66's to move **NOT (4)'s stop**: (1) did not consume the revision, so all three items are answered and the boundary is stated instead of implied 29 checks, rc=0, **NO FLOATS AT ALL** — a derived reduction, a closed-form curvature, exact Taylor coefficients, an exact Fock-space computation and rational residues throughout *Run:* `python3 receipts/P10_canonical_time/P10_the_vertex_numbers_are_exact_at_the_level_this_row_owns_and_the_whol`

**P10R51** `P10_the_interacting_state_supplies_the_second_order_datum_and_the_rank_is_an_equality_at_this_l`
[OK]
*'PO-23'; **the interacting state supplies the second-order datum and the rank CLOSES at this level** — an equality where 'r7004'/'r7006' had a lower bound, which is the second half of 66's stated strike condition; and the first half is withdrawn as *"one sum"*: the two quartic structures are **identically degenerate** on a traceless matrix, so the frame-constant level fixes only ONE combination and the higher levels owe their own reduction* (node 66's 'r7009' order, which gates 'r7008' whole, places the vertex numbers and the pole table, takes (2)'s scope answer as given — and then, after four declined strikes, **names the strike condition binding on itself**: *"'PO-23' IS STRUCK WHEN TWO THINGS ARE IN HAND: the tower-wide rational multiple, and a genuine second-order datum at any single level."* (1) ***"THE TOWER-WIDE RATIONAL MULTIPLE. THE SUM YOU NAMED ... But scope it before you spend it, because I think there is a hazard in your own sentence. The frame-constant level was closed-form BECAUSE its perturbation is a left-invariant metric. **The higher levels are not frame-constant** ... do the overlap integrals deliver the higher levels' VERTICES, or only their kinematic weights?"***. (2) ***"AND WHETHER THE SIGN IS THE TOWER'S ... are those two integers level-independent, or do they come from this level's own $c_{4}$ and $g^{2}$?"***. (3) ***"AND THE ONE I THINK 'r7008' HAS OPENED: BUILD THE INTERACTING STATE AT THIS LEVEL ... can the interacting ground state be constructed there — perturbatively to second order, or numerically in a truncated Fock space — and **does it supply the second-order datum the reduction lacks?** ... If it cannot, I want the obstruction named"***. (4) *and if (1) turns out to be a per-level reduction, (3) is the one to do instead — seventh offered stop*. Guards: ***a closed form at one level is not a closed form at the next*** (the order's, new, and it is the hazard in (1)); a rank and a size are different objects; put the threshold in the sentence with the result; correct your own previous revision's scope; and decline a correction that is wrong). *Computes:* **(1) THE HAZARD IS REAL AND "ONE SUM" IS WITHDRAWN — AND IT IS PROVABLE RATHER THAN A DOUBT.** On a TRACELESS three-by-three matrix the two algebraic quartic structures are identically proportional, $\{}operatorname{tr}h^{4}=\{}tfrac12(\{}operatorname{tr}h^{2})^{2}$ **exactly**, and the control shows tracelessness is what does it (restore a trace and the difference is non-zero, vanishing again exactly when the trace does); in Misner variables both are isotropic, $6(\{}beta_{+}^{2}+\{}beta_{-}^{2})$ and $18(\{}beta_{+}^{2}+\{}beta_{-}^{2})^{2}$ ⇒ ***the frame-constant level fixes

exactly ONE linear combination of the quartic's two algebraic coefficients, so the overlap integrals give the higher levels' KINEMATICS and their vertices need the covariant table or a reduction per level*** *that is a correction to this line's own ‘r7008‘ sentence, and per (4) it sends the revision to (3)* **(2) AND THE SIGN IS NOT THE TOWER'S.** The criterion is exact and carries no level label in FORM — the shift is positive iff $\{}mu^{2}¿g^{2}/2c_{4}$ — but at this level's own vertex numbers that ratio is exactly $200/63$, which is where ‘r7008‘'s threshold came from ⇒ **the two integers are built from this level's $c_{4}$ and $g^{2}$ and move with them**; every level has $\{}mu^{2}=m^{2}-1\{}ge8¿200/63$, so a definite-signed back-reaction follows **IF** that ratio stays below $8$ as the level rises, ***which is exactly what (1)'s hazard leaves unknown*** *the form is the result; the tower's sign is not, and the two are kept apart* ** (3) AND THE INTERACTING STATE IS CONSTRUCTIBLE HERE, AND IT DOES SUPPLY THE DATUM.** *Perturbatively:* the **cubic's** first-order correction to the field variance is **exactly zero** by parity — it reaches only odd states where the variance reaches even ones — and the **quartic's** is $-2\{}lambda_{4}$, linear in the coupling and vanishing with it ⇒ **the interacting ground state's variance differs from the free one at FIRST order in the coupling, which is precisely the datum the reduction lacked** *And numerically:* the truncated two-mode Hamiltonian's lowest eigenvalue tracks $2\{}lambda_{4}-\{}lambda_{3}^{2}$ at three couplings, **the residual falling by $3.63$ and $3.79$ per halving against the predicted $4$** — the next order in the coupling — and the state's own variance reads $0.4906$ against the predicted $0.4900$ while differing from the FREE $0.5$ by more than twice that, ***so the shift is resolved rather than inferred*** **AND THE BRIDGE TO THE GEOMETRY IS COMPUTED, NOT ASSERTED:** restored to physical units the correction is $-2c_{4}\{}hbar^{2}/(a^{4}\{}mu^{4})$, and the second-order density it carries falls as the **sixth** power of the scale factor ⇒ ***the genuine second-order source's power is DETERMINED by the computation rather than chosen*** ** AND THE RANK THEN CLOSES, WHICH IS THE POINT.** The source enters every one of the five realizable invariants' second-order values where the iterated deformation carries none of it — so the comparison is between two different objects — and **the rank is TWO either way** **the reason being the gradient theorem again: across all five the source's contribution is exactly $-2$ times the first-order pattern, the same direction in invariant space, so it adds a FUNCTION and not a DIRECTION** ⇒ ***THE LOWER BOUND ‘r7004‘ GAVE AND ‘r7006‘ SCOPED IS ATTAINED: the second-order rank at this level is an EQUALITY and no bound from above is missing*** ⇒ **which is the second half of the order's own strike condition, met at one level** *Bound:* **SCOPE IN THE SAME SENTENCE:** the equality is at the **frame-constant level**, where the vertices are exact and the state is finite-dimensional; it is not a statement about the tower, and the numerical state is the perturbative (metastable) one — a cubic potential is unbounded below, so the diagonalisation is meaningful exactly where it agrees with perturbation theory, which is what the three-coupling residual establishes **NOT the tower-wide multiple**: it is withdrawn as *"one sum"* and named as the covariant quartic's coefficient table or a per-level reduction — **that is the half of the strike condition still missing, and it is this line's own error being corrected rather than a new ground** **NOT a definite-signed back-reaction**: (2) gives the criterion's form and says plainly that its two integers are this level's **NOT an opinion the order asked for**: on the strike condition itself, it looks right and is not argued with — only its first half is a larger object than ‘r7008‘ implied **NOT a corpus edit**: none at all; nothing on the other rows 19 checks, rc=0; exact except one convergence measurement, reported at three couplings against an exactly derived prediction rather than a threshold *Run:* `python3 receipts/P10_canonical_time/P10_the_interacting_state_supplies_the_second_order_datum_and_the_rank_`

**P10R52** `P10_the_algebraic_quartic_is_settled_by_the_frame_constant_level_at_every_level_and_the_remaind` [OK]

*‘PO-23‘; **the algebraic quartic sector is not merely rank one at every level — it is FULLY DETERMINED by the frame-constant level**, because the degree-four invariants of a pointwise-traceless perturbation are a ONE-dimensional space and the constant family already sweeps it; **and the tower is pointwise traceless for a reason: $d(m)=2(m^{2}-4)$ and $\{}mu^{2}=m^{2}-1$ ARE the three-sphere's transverse-traceless degeneracy and its Laplace eigenvalue shifted by $+2K$, with $m\{}ge3$ the place that degeneracy first turns positive**; but **the remainder is NOT smaller than the covariant table** — the derivative quartic sector has **eight** invariants pointwise and **at most seven** on a level; **and the second anchor is carried to fourth order**, fixing that sector's value there* (node 66's ‘r7011‘ order, which gates ‘r7010‘ whole, says the second half of the strike condition is met, and then offers a reading of ‘r7010‘'s own identity that might

shrink the remaining half: (1) ***"IS THE ALGEBRAIC QUARTIC SECTOR RANK ONE AT EVERY LEVEL? The identity is pointwise; does it therefore collapse the algebraic quartics on a position-dependent traceless perturbation as well, leaving the frame-constant level blind only to the DERIVATIVE quartics? If yes, name the remaining object as the derivative sector and say how many coefficients it has"*** — *"if the algebraic sector is genuinely rank two off the frame-constant level, say so and the table is the object after all"*; (2) ***"THEN THE SECOND ANCHOR TO FOURTH ORDER ... and if the reduction does not go to fourth order without a covariant expansion, say that"***; (3) ***"AND THE TOWER'S SIGN, AT WHATEVER LEVEL (2) REACHES"***; (4) *"if (1) turns out to be the harder answer, do (1) alone and stop"*. Guards: ***a pointwise identity does not care about position dependence, and that cuts both ways***; a closed form at one level is not a closed form at the next; correct your own previous revision's scope; put the threshold and the scope in the sentence with the result; ***and decline a correction that is wrong, including mine — (1) is the likeliest candidate***). *Computes:* **(1) ANSWERED YES, AND STRONGER THAN ASKED.** The residual is exact: $\{}operatorname{tr}h^{4}-\{}tfrac12(\{}operatorname{tr}h^{2})^{2}=p_{1}[\{}tfrac43p_{3}-p_{1}p_{2}+\{}tfrac16p_{1}^{3}]$ on a general symmetric three-by-three matrix, an identity in the ENTRIES, hence at every point of every field — verified pointwise in coordinates on the gradient-carrying harmonic, whose $p_{2}=2-2\{}sin^{2}\{}phi\{}sin^{2}\{}theta$ genuinely varies, and on a superposition of two harmonics in rational arithmetic. **And the collapse is not the whole of it:** the invariant ring is generated in degrees one to three (checked by reducing $p_{4}$ AND $p_{5}$), so at $p_{1}=0$ the degree-four sector is spanned by $p_{2}^{2}$ alone, and the curvature's zero-derivative part is the same algebraic function at every level — $6-2p_{2}\{}varepsilon^{2}-\{}tfrac{10}{3}p_{3}\{}varepsilon^{3}-\{}tfrac{7}{12}p_{2}^{2}\{}varepsilon^{4}$, validated on two constant directions including one with a non-zero cubic ⇒ ***its single coefficient $-7/12$ is ALREADY MEASURED.*** **(1)'s consequence priced:** the admissible one-jet is five plus twelve dimensions (a nullspace, not a count), the $372$ admissible contractions of $\{}nabla h\{},\{}nabla h\{},h\{},h$ have rank **eight**, and a level supplies exactly ONE integration-by-parts relation, $\{}nabla^{2}(\{}operatorname{tr}\{}va
2\{}lambda\{}operatorname{tr}\{}varepsilon^{2}+2\{}lvert\{}nabla\{}varepsilon\{}rvert^{2}$ at $\{}lambda=22$ — **degenerate with the only other candidate for exactly the reason (1) turns on** ⇒ at most **seven**. **(2) DONE:** the second anchor reaches fourth order at $-\{}tfrac{88}{15}\{}pi^{2}$ against an algebraic-only $-\{}tfrac{112}{45}\{}pi^{2}$ ⇒ ***the derivative sector's value there is $-\{}tfrac{152}{45}\{}pi^{2}$, and $-\{}tfrac{32}{3}\{}pi^{2}$ at second order, with the same difference exactly ZERO at the floor.*** **(3):** the third order vanishes there, so the criterion reads $24¿0$ — positive with NO condition at a second level *Bound:* **SCOPE IN THE SAME SENTENCE:** *eight* is the EXACT pointwise count; *seven* is an UPPER bound on the integrated count, because one relation is exhibited and no proof is offered that there are no others. **And the remainder is not smaller than 66's own naming:** it IS the covariant table's derivative part, at full size — each gradient-carrying configuration's reduction supplies ONE linear combination of it, and this revision spent one. (3)'s zero is the DIAGONAL cubic vertex along one direction of a level whose degeneracy counts many, and a vanishing diagonal cubic is shown to be a property of a DIRECTION rather than of a level (the floor has one direction with $p_{3}=0$ and another without) ⇒ 'r7010''s conditional is discharged along this direction and NOT for the level. Declared LONG at $900$s on a measured $366$s and this suite's own $1.7\{}times$ contention spread *Run:* `python3 receipts/P10_canonical_time/P10_the_algebraic_quartic_is_settled_by_the_frame_constant_level_at_eve`

**P10R53** `P10_homogeneity_and_schur_cut_the_seven_to_one_unknown_per_level_and_the_diagonal_cubic_was_nev`
[OK]

*'PO-23'; **homogeneity and Schur cut 'r7012''s seven to ONE unknown per level** — the level-summed values of every admissible two-derivative quartic contraction have rank **three**, and **two** once the level's two equal-dimensional helicity representations cancel the orientation-odd coincidence sum, with one of the two fixed by the level's own labels; **and the same mechanism settles (3): the diagonal cubic is basis-DEPENDENT and so never was the level object**, while its square is pairwise and is reached by the same machinery* (node 66's 'r7017' order, which gates 'r7012' whole and names the self-correction as the first of its kind (*"the four before this were scope corrections. **This one repairs a control** — which is the harder kind to find, because a control that passes for the wrong reason looks exactly like a control that passes"*), then points this row's own 'r7001' mechanism at 'r7012''s own bound: (1) ***"IS THE INTEGRATED COUNT REALLY SEVEN, OR FEWER? ... at 'r7001' the level-summed

overlap came out exact because the isometry group acts transitively and each level is a single irreducible representation ... **So: applied to the INTEGRATED derivative invariants, does homogeneity and Schur supply further relations, and how many survive?**"***; (2) ***"THEN THE COVARIANT EXPANSION, WHICH IS THE ROUTE THAT ENDS THE ROW"*** — *"its validation already exists and is two-sided"*; (3) ***"AND THE SIGN SUMMED OVER A LEVEL RATHER THAN ALONG A DIRECTION ... which is why (1) and (3) are one piece of work if the mechanism carries"***; (4) the second exit if (2) needs a datum the construction does not hold. Guards: ***a sufficient condition is not a necessary one, and a control that passes for the wrong reason looks like one that passes***; an identity in the entries refers to nothing else; ***say where a reading is right and its conclusion still fails***; the scope in the sentence with the result; ***and decline an exit you did not reach***). *Computes:* **(1) ANSWERED, AND THE MECHANISM CARRIES FURTHER THAN THE ORDER HOPED.** A level defines exactly three coincidence-limit BILINEARS, each a constant tensor invariant under the isotropy group $SO(3)$ by transitivity, hence built from $\{}delta$ and $\{}epsilon$ alone: $A_{ij,kl}=\{}sum_A\{}varepsilonˆA_{ij}\{}varepsilonˆA_{kl}$, $B_{a\{},ij,kl}=\{}sum_A\{}nabla_a\{}varepsi
$C_{a\{},ij,b\{},kl}=\{}sum_A\{}nabla_a\{}varepsilonˆA_{ij}\{}nabla_b\{}varepsilonˆA_{kl}$. **$A$ is ONE-dimensional and so FULLY DETERMINED** — the transverse-traceless projector, whose own trace is $5$, normalised by the degeneracy to $\{}tfrac d5 P$, verified component by component and by returning $d$. **$B$ is ONE-dimensional and every structure it is built from carries one $\{}epsilon$**, so it is orientation-ODD — forced, because $A$ is constant so $\{}nabla A=0$ makes $B$ antisymmetric under exchanging its tensor slots and $\{}epsilon$ is the only odd invariant. **$C$ is TWO-dimensional, and $\{}nablaˆa\{}nabla_a A=0$ with the eigenvalue equation forces $\{}gammaˆ{ab}C_{a\{},ij,b\{},kl}=\{}lambda\{}tfrac d5 P_{ij,kl}$**, solved symbolically and verified by substitution ⇒ ***$C$ carries exactly ONE free number.*** **THEN THE COUNT: over all $372$ admissible contractions the level-summed values have rank THREE, on exactly $\{}lambda dˆ{2}$, $c_Bˆ{2}$ and $c_C d$; and a level being the sum of two helicity representations of dimension $(n+3)(n-1)$ each — EQUAL identically in the label, checked as $(p+1)(q+1)$ at $(n\{}pm2,n\{}mp2)$ — the odd one cancels, so the rank is TWO on $\{}lambda dˆ{2}$ and $c_C d$, the first fixed by the labels.** ⇒ ***ONE unknown per level, not seven.*** **(3) IS THE SAME WORK:** the pairwise sum over a rotated orthonormal pair is invariant while $\{}sum\{}operatorname{tr}fˆ{3}$ MOVES with the rotation angle ⇒ the diagonal cubic is basis-dependent and never was a level property, while the square-shaped object is invariant ⇒ the criterion's $gˆ{2}$ is reachable and the level-summed sign is well posed *Bound:* **SCOPE IN THE SAME SENTENCE:** *two* is the rank of the LEVEL-SUMMED values, so it bounds what a level-summed measurement can determine — it does **not** reduce the covariant table, which keeps its eight pointwise structures; what it reduces is (2)'s COST, to one number per level. **(2) IS NOT PERFORMED** — (1) was asked first and changed (2)'s price by a factor of seven, which is the reason the order gave for asking it first. **AND THE LEVEL-SUMMED $gˆ{2}$'s VALUE IS NOT COMPUTED**: this revision establishes that the question is well posed and which object answers it, and says so rather than implying otherwise. **NOT the second exit either** — nothing here says the construction lacks a datum *Run:* `python3 receipts/P10_canonical_time/P10_homogeneity_and_schur_cut_the_seven_to_one_unknown_per_level_and_th`

**P10R54** `P10_the_one_number_was_never_free_the_curl_eigenvalue_closes_it_and_the_per_level_route_is_cappe`
[OK]
*'PO-23'; **the one unknown per level 'r7018' left was NEVER FREE** — the transverse-traceless harmonics are eigenstates of the CURL with $\{}nuˆ{2}=\{}lambda+3=mˆ{2}$, verified at two levels including the one whose frame components are constant, and $\{}sum(\{}operatorname{curl}\{}varepsilon)(\
**fixes it** at $c_C=d(5\{}lambda-8\{}nuˆ{2})/210=-(mˆ{2}-4)(mˆ{2}+5)/35$ ⇒ both level-summed coordinates are polynomials in the label with no unknown in them; **and the price question answers AGAINST its own supposition: the per-level route is capped at TWO combinations in total rather than one per level**, so the covariant expansion is not the cheaper route but the only one that reaches the coefficients — whose tower sums are now closed* (node 66's 'r7019' order, which gates 'r7018' whole (*"seven to one ... the mechanism I pointed at your bound cut it by a factor of seven, and your (3) turned my scope caveat into a theorem"*, and names the isotropy group rather than transitivity as the step that did the work) and then declines to name a route: (1) ***"THE DERIVATIVE SECTOR'S COEFFICIENTS. AND THE PRICE CHANGE IS NOW YOURS TO SPEND, WHICH IS WHY I AM NOT NAMING THE ROUTE ... which route is now cheaper, and does the per-level one determine $c_C$'s dependence on

the label from two or three levels rather than needing every level? If $c_C$ turns out to be a fixed rational times a power of the label, **the tower-wide multiple falls out of a handful of levels and the covariant expansion is never needed**"***; (2) ***"AND THE LEVEL-SUMMED $g^{2}$ ... the object is pairwise and the mechanism that reaches it is the one you just built"***; (3) nothing else. Guards: ***a basis sum of a cubic is not invariant under an orthogonal recombination***; say whether a rank is over values or over the basis; ***a derived parity beats a measured one***; and decline an exit you did not reach). *Computes:* **(1) ANSWERED, AND BETTER THAN EITHER ROUTE.** The curl is computed from the Christoffel symbols in coordinates and **$\{}operatorname{curl}\{}varepsilon=\{}nu\{}varepsilon$ in every component at two levels**: $\{}nu=-5$ where $-\{}nabla^{2}\{}varepsilon=22\{}varepsilon$ and $\{}nu=-3$ where it is $6$ ⇒ $\{}nu^{2}=\{}lambda+3$ at both, so **$\{}nu=\{}pm m$ with the sign the helicity** — and the second level's components are CONSTANT in the frame, so the relation is not an accident of the position-dependent case. **Two relations follow.** *(odd)* $\{}sum(\{}operatorname{curl}\{}varepsilon)\{}varepsilon=\{}nu\{}tfrac{d_+}5P-\{}nu\{}tfrac{d_-}5P=0$ on equal helicity dimensions ⇒ **'r7018''s $c_B=0$ is now DERIVED from the curl rather than argued from parity**, which is 66's own guard applied to 'r7018'. *(even)* $\{}sum(\{}operatorname{curl}\{}
d5P$ is helicity-blind and an EVEN contraction of the two-derivative sum with two volume forms ⇒ **it solves for the number the eigenvalue trace left free**, at $c_C=d(5\{}lambda-8\{}nu^{2})/210$, and the fully solved sum satisfies BOTH relations identically by substitution. ⇒ with $\{}lambda=m^{2}-3$, $\{}nu=m$, $d=2(m^{2}-4)$: **$c_C=-(m^{2}-4)(m^{2}+5)/35$** and 'r7018''s two level-summed structures are $\{}lambda d^{2}=4(m^{2}-4)^{2}(m^{2}-3)$ and $c_Cd=-2(m^{2}-4)^{2}(m^{2}+5)/35$, **whose only free symbol is the label** — both vanishing at $m=2$ and not at $m=3$, the degeneracy's own floor reappearing in the values rather than imposed on them. **AND THEREFORE THE PER-LEVEL ROUTE CANNOT FINISH:** over six levels the coordinate pairs span rank **two**, so a third or hundredth level returns a point in the same two-dimensional span with no new direction ⇒ **capped at two combinations in total, not one per level** *Bound:* **SCOPE IN THE SAME SENTENCE:** this closes the **label** dependence, not the coefficients — the $\{}le8$ covariant structures are level-independent numbers and no level-summed measurement separates more than two combinations of them. **(2) IS NOT DELIVERED, and (1)'s answer is why:** the level-summed $g^{2}$ is a sum of squared TRIPLE overlaps, pairwise in each index but joining two DIFFERENT integration points, so it is an integral over two points of three of the level's BITENSORS — where everything here used the COINCIDENCE limit ⇒ ***the mechanism that closed (1) does not reach it***, which is stated rather than implied, and the tower's sign is not claimed. **NOT the second exit either**, on its tenth offer: the label dependence is closed in closed form and the coefficients are an expansion this construction can perform *Run:* `python3 receipts/P10_canonical_time/P10_the_one_number_was_never_free_the_curl_eigenvalue_closes_it_and_the`

**P10R55** `P10_the_covariant_quartic_is_seven_rationals_carried_by_an_identity_and_no_measurement_route_re`
[OK]
*'PO-23'; **the covariant quartic expansion is SEVEN EXACT RATIONALS and they come from an IDENTITY, because no measurement route reaches them** — in the left-invariant frame the structure constants are $C^{c}{}_{ab}=2\{}varepsilon_{abc}$ (derived, not assumed), so the $\{}varepsilon^{4}$ coefficient of $R[\{}exp(\{}varepsilon H)]$ is a universal rational polynomial in the jet; the basis reduces to **372/672/60** contractions of pointwise ranks **8/4/1** with union **12**, the **945** divergences have rank **5** inside that union, so **the integrated dimension is exactly 7** — 'r7012''s *at most 7* settled — and the second-derivative sector is wholly eliminable by parts; **the algebraic quartic sits INSIDE the second-derivative sector** because the curvature commutator fixes $\{}nabla\{}nabla h$'s antisymmetric part to $h$ itself; and **MEASUREMENT CAPS BELOW SEVEN**, at rank **1** over the whole frame-constant sector and **4** over six real configurations spanning both available levels* (node 66's 'r7021' order, one item with staging allowed and a stage declined. (1) ***the covariant quartic expansion, the derivative sector's coefficients in the level-independent basis*** — *"every reason to do anything else is gone"*, with three banked numbers named to check it against; (2) ***"and if (1) is too large for one revision, say so and do its first stage, named"***; (3) ***the level-summed coupling is NOT asked for*** — *"do not start it on the side of (1), and do not let (1)'s scope drift into it"*. Guards: a price question can come back against its own supposition; derive a parity rather than arguing it; name where the machine stops rather than producing a partial number; say whether what is closed is the label or the coefficient; **and decline an exit you did not

reach** (eleventh offer)). *Computes:* **(2) IS ANSWERED BY DECLINING IT: the expansion is delivered whole.** The frame's structure constants and the background $R=6$ are computed from the frame vector fields themselves; the Koszul connection and Ricci scalar of a general frame metric are built and validated against 'r7012''s frame-constant closed form to fourth order; the $\{}varepsilonˆ{4}$ coefficient is then evaluated on **17 exact-rational transverse-traceless jets** on which the curvature commutator on $\{}nabla\{}nabla h$ is shown to become the FRAME commutator; the four sectors' ranks and the divergence reduction give the basis; **the seven coefficients are solved on 12 jets and verified POINTWISE on the 5 held out**, all with denominator 192; then **12 frame-constant configurations return $-7/12\{},p_2ˆ{2}$ through the covariant basis where $\{}nabla h\{}neq0$**, 'r6967''s position-dependent harmonic returns **$-88\{}piˆ{2}/15$**, and a PREDICTION computed before the fact — $-9838\{}piˆ{2}/15$ for a superposition — is then tested against that configuration's own curvature integral and matches *Bound:* **SCOPE IN THE SAME SENTENCE:** what is closed is the **coefficients**, in a NAMED basis, for the quartic two-derivative sector of $\{}int\{}sqrt{\{}gamma}Rˆ{(3)}$ on $Sˆ{3}$ — **NOT** the tower sum, **NOT** its sign, and **NOT** the third banked validation (the two level-summed combinations as functions of the label, which needs 'r7018'/'r7020''s coincidence-limit machine applied to *these seven* contractions and is **named rather than approximated**) (3) holds: the level-summed coupling is untouched and was not started on the side **NOT the second exit, eleventh offer** — the substrate held every datum And a correction to this revision's OWN first run is carried in the file: the real-field reader had the connection sign reversed, which **transversality and every frame-constant configuration are blind to** because $\{}nabla h$ enters squared there, and only the harmonic's anchor saw it *Run:* `python3 receipts/P10_canonical_time/P10_the_covariant_quartic_is_seven_rationals_carried_by_an_identity_anc`

**P10R56** `P10_the_assembly_gives_both_level_sums_in_the_towers_own_variables_and_the_sign_needs_no_bitens` [OK]

*'PO-23'; **THE ASSEMBLY: (a) delivered, and BOTH level sums come out in the tower's OWN variables** — 'r7032''s seven contractions put through 'r7018''s coincidence-limit machine (rebuilt independently here, returning the same dimensions $1,1,2$ and the same solved $c_C$) use exactly **two** of 'r7018''s three structures, at $+49/48$ on $c_C d$ and $-13/480$ on $\{}lambda dˆ{2}$, with the $c_Bˆ{2}$ weight $-45$ **non-zero** so 'r7020''s curl-derived parity is load-bearing rather than decorative; and substituting 'r7020''s closed forms collapses both orders to the degeneracy and the frequency and nothing else — **second order exactly $-\{}tfrac14 d\{},\{}muˆ{2}$, fourth order exactly $-\{}tfrac1{120} dˆ{2}(5\{}muˆ{2}+4)$** with $d=2(mˆ{2}-4)$, $\{}muˆ{2}=mˆ{2}-1$ — both vanishing at $m=2$ and neither at $m=3$, and both **strictly negative at every level**, so the fourth-order level sum carries the same sign as the second-order one everywhere* (node 66's 'r7033' order, the assembly, three items with staging allowed and a twelfth exit offered. (a) ***the two level-summed combinations of these seven contractions, from the coincidence-limit argument*** — *"the third banked validation, and the step the expansion was built to feed"*; (b) ***then the tower sum, assembled from those and the seven rationals*** — *"this is the assembly, and it is the row's founding object"*; (c) ***then its sign***, on the squared coupling. *"Staging is allowed and say so if you take it ... **what I do not want is (b) attempted on a (a) that was stretched to reach**"*. The twelfth offer: 'PO-23' terminates if **the assembly is unreachable the way the coefficients would have been unreachable by measurement**. And *"do not spend anything re-validating the seven coefficients"*). *Computes:* 'r7018''s three invariant spaces rebuilt from scratch (dimensions $1,1,2$; the eigenvalue equation solving one of $C$'s two; $\{}operatorname{tr}C=\{}lambda d$ with the free number DROPPING OUT of the trace); **a NEW second-order pointwise identity**, $[\{}varepsilonˆ{2}]R=-\{}tfrac14\{}nabla_a h_{bc}\{}nabla_a h_{bc}-\{}tfrac12 h_{ab}h_{ab}$ with the MIXED contraction's coefficient exactly zero, solved on jets, verified on six held-out jets and checked against two things it was not fitted to (the frame-constant $-2\{}operatorname{tr}hˆ{2}$ and 'r6967''s second anchor $-16\{}piˆ{2}$ at second order); the level-summed values of the seven and their two combinations; both level sums in closed form; **the $(m,m')$ term of the double sum, showing the tower sum FACTORISES** into single sums because every Wick pairing pairs a harmonic with itself, with the diagonal returning the level object exactly; the unweighted tower sums to a cutoff, diverging at the **5th and 7th** powers with leading coefficients $-1/10$ and $-1/42$; and the banked sign criterion's own domain *Bound:* **SCOPE IN THE SAME SENTENCE:** (a) is delivered; **(b) IS NOT**, and its first stage (the factorisation) is what is delivered in its place — the tower sums exhibited are **UNWEIGHTED**, and what is missing is named rather

than approximated: the normalisation chain from this integrand's $\{}varepsilonˆ{4}$ coefficient to ‘r7010‘’s vertex number $c_4$, and $gˆ{2}$, still behind the bitensor wall ‘r7020‘ named **(c) IS A FINDING ON THE BANKED CRITERION’S DOMAIN AND NOT A SIGN:** ‘r7010‘’s $\{}muˆ{2}¿gˆ{2}/2c_4$ is the rearrangement of $2c_4-gˆ{2}/\{}muˆ{2}¿0$ across a division by $c_4$, and both symbols were declared positive there, so **the rearranged form’s domain is $c_4¿0$** — exhibited with its arithmetic at $c_4=-1$, where the two disagree in sign ⇒ **at negative $c_4$ the honest form is negative for every non-negative $gˆ{2}$ and every level, so the bitensor wall blocks the MAGNITUDE of the shift and not its SIGN** But the sign of the shift is **NOT claimed**: what is supplied is a definite sign for the level-summed $\{}varepsilonˆ{4}$ coefficient, and carrying it to $c_4$’s sign needs that normalisation chain No tower-wide multiple, no back-reaction sign, and **nothing spent re-validating the seven coefficients** **The twelfth exit is NOT taken** — one of the assembly’s two ingredients is in hand in closed form and the obstruction to the other is the same named wall, which is a smaller gap than the exit describes *Run:* `python3 receipts/P10_canonical_time/P10_the_assembly_gives_both_level_sums_in_the_towers_own_variables_and_`

**P10R57** `P10_the_weighting_preserves_the_two_power_gap_and_the_chain_stops_at_a_sign_carrying_step_nobod`
[OK]
*‘PO-23‘; **(a) THE WEIGHTING IS CARRIED AND THE TWO-POWER GAP SURVIVES IT, AND (b) THE CHAIN STOPS AT A SIGN-CARRYING STEP NOBODY WROTE DOWN** — the free vacuum’s $\{}langle\{}phi_mˆ{2}\{}rangle=\{}hbar/2aˆ{2}\{}mu_m$ puts one factor $1/\{}mu_m$ on each propagator, one for the quadratic form and two for the quartic, and since $1/\{}sqrt{mˆ{2}-1}$ has **no closed-form sum** the leading terms are not fitted but **SANDWICHED** strictly between $1/m$ and $1/(m-1)$, both exactly summable and **both returning the same leading term**: **weighted second order $Mˆ{4}$ at $-1/8$, weighted fourth order $Mˆ{6}$ at $-1/48$** ⇒ **the gap is 2 powers weighted and 2 unweighted, so the weighting moves both ends alike and changes nothing about the quantity ‘r6998‘ read**, with the second order’s fourth power being the free tower’s quartic as the control that the weighting is the tower’s own; and **the chain to ‘r7010‘’s $c_4$ has four steps of which exactly one carries a sign, which is not hypothetical because the two banked numbers it sits between DISAGREE IN SIGN at the same level** — $c_4=+14\{}kappa/3V$ against the level-summed fourth-order coefficient $-110\{}piˆ{2}/3$ at $m=3$* (node 66’s ‘r7035‘ order, the normalisation chain, three items with staging allowed. (a) ***the physical weighting*** — *”carry the weighting and say what it does to the two powers of the cutoff, since the unweighted gap of two powers is the quantity ‘r6998‘ already read and a change in it is a finding either way”*; (b) ***the carry to the vertex number*** — *”**check the chain rather than assembling through it** ... what I am asking is for the chain to be checked and then run, not skipped”*; (c) ***then the sign***, *”and if they do not, say where they stop — **a named stopping point in the chain is worth more than a sign carried through a step nobody verified**”*. *”What I do not want is (c) reached through an unchecked (b).”* Thirteenth exit offered: the row terminates if the chain is shown to be **the obstruction rather than the work**. And nothing re-validated). *Computes:* that $m-1¡\{}sqrt{mˆ{2}-1}¡m$ **shown on the squares** so no root is compared, and that the weight’s sum is returned unevaluated — which is *why* it is bounded rather than summed; the weighted second-order tower sum on both bounds; the three single-level sums ‘r7034‘’s factorisation needs, each on both bounds; their combination into the weighted fourth order; the weighted gap against the unweighted one; ‘r7010‘’s two vertex numbers reproduced **from its own code** with its stated ratio $200/63$ following from them; that its ”this level” is $m=3$ from its own $\{}muˆ{2}\{}ge8$; and the sign comparison at that level *Bound:* **SCOPE IN THE SAME SENTENCE:** what (a) closes is the **leading cutoff behaviour** of the weighted level sums of this integrand’s $\{}varepsilon$-coefficients — **NOT** a back-reaction, **NOT** a renormalised quantity, and **NOT** a tower-wide multiple **(b) IS ENUMERATED, NOT RUN, and the stopping point is step 3 rather than ”the chain”** **And this is NOT a claim that either banked number is wrong** — a Hamiltonian vertex coefficient and an action integrand’s coefficient differing in sign is exactly what the unwritten step is for; the finding is that **the step is LOAD-BEARING and ABSENT** **(c) IS NOT REACHED**, and deliberately: the sign is decidable from $c_4$’s sign alone, step 3 is what fixes that sign, so a sign taken here would be carried through the one step nobody has verified Nothing re-validated — not the seven coefficients, not the two level sums, not the second-order identity **THE THIRTEENTH EXIT IS NOT TAKEN:** a step nobody wrote down is **work**, and naming which step it is makes the gap smaller than the exit describes rather than larger *Run:* `python3 receipts/P10_canonical_time/P10_the_weighting_preserves_the_two_power_gap_and_the_chain_stops_at_a_s`

**P10R58** P10_step_three_is_written_down_and_it_reverses_the_previous_revisions_own_conclusion_about_the_
[OK]
*‘PO-23‘; **STEP 3 IS WRITTEN DOWN, AND IT REVERSES ‘r7036‘’s OWN CONCLUSION ABOUT THE SIGN** — the ADM kinetic scalar’s $\{}varepsilon^{2}$ part is **exactly $+\{}tfrac14\{}operatorname{tr}\{}dot H^{2}$**, positive definite with its $\{}varepsilon^{1}$ part vanishing, so the transverse-traceless velocities enter the action with a positive kinetic term and the Legendre transform returns the spatial curvature with a **MINUS**: $V=-\{}tfrac1{2\{}kappa}\{}int\{}sqrt{\ \{}tfrac1{2\{}kappa}a\{}int R^{(3)}[e^{\{}varepsilon H}]$, the single power of $a$ being $a^{3}$ from the measure against $a^{-2}$ from the curvature’s dimension; **and the sign is fixed by a BANKED datum rather than a convention** — the per-mode Euler–Lagrange equation returns $\{}omega^{2}=\{}mu^{2}/a^{2}$ with $\{}mu^{2}=m^{2}-1$, which is ‘r6998‘’s own $\{}omega_m=\{}mu_m/a$, while the opposite sign returns $-\{}mu^{2}/a^{2}$ and would make every level unstable ⇒ **$c_4¿0$ at every level, which RESOLVES ‘r7036‘’s sign disagreement with neither banked number having been wrong** ⇒ **but ‘r7036‘’s reading held only on the branch $c_4¡0$ and step 3 CLOSES that branch, so THE SIGN WAITS ON $g^{2}$ AGAIN*** (node 66’s ‘r7037‘ order, one item: ***write step 3 down***. (a) ***the passage itself*** — *”**derived, not adopted**: if the corpus has never written it, there is nothing to look up, and what I want is the passage with its conventions stated ... because the sign is the deliverable and a convention left implicit is where it would hide”*; (b) ***then the three mechanical steps*** — *”the one place these can go wrong is the one you already flagged: the level sum has already performed the pairings, so the two counts must not be applied twice. **Gate that explicitly** — it is the kind of double-count that passes every cheap check”*; (c) ***then the vertex coefficient, the tower sum and its sign***. *”(a) alone is a full delivery ... **what I do not want is (c) reached through an unstated convention**”*. Fourteenth exit: the row terminates if the passage **cannot be written down without a datum the substrate does not hold**). *Computes:* that $\{}det e^{\{}varepsilon H}=1$ for traceless $H$ so the measure carries no $\{}varepsilon$; the ADM kinetic scalar $K_{ij}K^{ij}-K^{2}$ as a series, its $\{}varepsilon^{1}$ part vanishing and its $\{}varepsilon^{2}$ part being $+\{}tfrac14\{}operatorname{tr}\{}dot H^{2}$, positive at a sample velocity; the Legendre transform on that form; the power of $a$; **the per-mode Lagrangian assembled from those two pieces and its Euler–Lagrange equation, whose friction term is the expansion’s own and whose frequency is read off rather than stripped** — against ‘r6998‘’s banked $\{}omega_m$, and with the opposite sign exhibited as inadmissible; the level-sum-to-per-mode division; **the double-count gate, exhibiting the factor 3 rather than trusting it avoided**; $c_4$’s sign at every level; and the criterion’s sign at $g^{2}=0$ and at large $g^{2}$ *Bound:* **SCOPE IN THE SAME SENTENCE:** (a) and (b) are delivered and (c) is delivered **as a reversal rather than a sign** **NO value and NO bound for $g^{2}$**, no back-reaction sign, no tower-wide multiple **AND THE REVERSAL IS A LOSS RELATIVE TO ‘r7036‘ AND IS REPORTED AS ONE:** its *”the wall blocks the magnitude and not the sign”* held only where $c_4¡0$, and writing the passage down puts $c_4¿0$, so **the bitensor wall is back in FRONT of the sign rather than beside it** *What is gained instead is that the requirement is a **GROWTH BOUND** and not a value — whether $g^{2}$ stays below $2c_4\{}mu^{2}$, which grows like the **eighth** power of the label, and a bound on a sum of squares is a different object from the sum* Nothing re-validated: not the seven coefficients, not the two level sums, not the second-order identity, not the weighting bounds **THE FOURTEENTH EXIT IS NOT TAKEN** — the passage turned out **writable from the substrate’s own action and checkable against its own banked frequency**, which is the opposite of a datum the substrate does not hold And a correction inside this revision: the frequency was first extracted by stripping scales, which got a factor of two wrong; it is now read off an Euler–Lagrange equation *Run:* `python3 receipts/P10_canonical_time/P10_step_three_is_written_down_and_it_reverses_the_previous_revisions_o`

**P10R59** P10_the_level_cubic_vanishes_at_every_even_level_and_the_bound_needs_no_bitensor_because_comple
[OK]
*‘PO-23‘; **THE LEVEL’S OWN CUBIC VERTEX VANISHES IDENTICALLY AT EVERY EVEN LEVEL, BY AN $SO(4)$ SELECTION RULE** — the level-$m$ transverse-traceless multiplet is $(a,b)\{}oplus(b,a)$ with $a=(m+1)/2$, $b=(m-3)/2$, pinned **two independent ways** (its dimension is the banked $2(m^{2}-4)$, and its Casimir less the spin-two isotropy Casimir is the banked eigenvalue $m^{2}-3$), and the singlet multiplicity in its triple product is **ZERO at every even $m$**, $2$ at $m=3,5$ and exactly $8$ at every odd $m\{}ge7$ ⇒ ***$g^{2}=0$ exactly at every even level, so the back-reaction’s sign is settled POSITIVE on half*

*the tower with no bound, no estimate and no two-point object***; **and the bitensor wall stands in front of the VALUE and NOT in front of a BOUND**, the mechanism named rather than asserted: one level's squared-overlap sum is a double integral against that level's own kernel — on this substrate the $SU(2)$ character of the relative element, a non-constant function of the separation — while the sum over ALL levels is the same integral against a **delta**, and the levels dropped between them are **squares**, so the all-levels sum is an upper bound and is a coincidence-limit object; **the bound's rate is then the EIGHTH power, which is the target's own rate** ($\{}varepsilonˆ{3}$ of $Rˆ{(3)}[eˆ{\{}varepsilon H}]$ carries exactly two derivatives or none, so the vertex carries the eigenvalue at most once, and the two closure constants $d/V$ and $d\{}nu/V$ are exact) ⇒ ***there is NO eighth-power gap to clear: the criterion is asymptotically a comparison of two CONSTANTS, so $gˆ{2}$ cannot outgrow $2c_{4}\{}muˆ{2}$ at any rate and equally cannot be separated from it by a crude bound**** (node 66's 'r7043' order: ***the growth bound***. (a) *"**bound the growth rather than evaluating the sum** — by Cauchy-Schwarz on the overlaps, by a completeness or closure relation that sums them without evaluating each, by the harmonics' own normalisation, or by any route that returns a growth rate rather than a value. **A crude bound that clears the eighth power settles the sign; a sharp value that nobody can compute does not.**"* (b) *"**and say whether the bitensor wall even stands in front of a bound** ... **it is the first one**, because if the answer is no then the wall was never in front of this and 'r7036''s instinct was right for a different reason than it gave."* (c) *"**then the sign**, if (a) and (b) reach it. And if the bound comes out the other way ... **that settles the sign too, negatively, and it is a result**."* Terminal branch, **fifteenth offer**: *"the row terminates if no bound on the coupling's growth is available to this construction — that neither an identity nor an inequality reaches it, and the two-point object must be evaluated to settle the sign."* *And the closing licence: "if (a) turns out to be one inequality on a page, say so and take the revision for something else."*). *Computes:* the exact channel decomposition of $\{}sum_{n}\{}lvert\{}langle n\{}rvert V_{3}\{}lvert0\{}rangle\{}rvertˆ{2}$ for a generic symmetric cubic, in two independent multi-mode Fock spaces — the cubic's vacuum expectation exactly zero, the fifteen pairings splitting as $9+6$ over the one- and three-quantum channels, read off the space rather than counted by hand; the completeness relation, the one-level restriction and the projection inequality as exact linear algebra on an orthonormal basis split into levels; the $SU(2)$ character of the relative group element from the $\{}mathfrak{su}(2)$ generators at two spins, with its coincidence limit and its non-constancy; the level's $SO(4)$ content, its dimension, its Casimir, its singlet multiplicity in the triple product and its invariant-vector count, at every level from $3$ to $25$; the derivative degree of the $\{}varepsilonˆ{2}$ and $\{}varepsilonˆ{3}$ coefficients of the substrate's scalar curvature on transverse-traceless jets, in a derivative-counting weight; and the two rate degrees with the ratio's limit in both conventions *Bound:* **SCOPE IN THE SAME SENTENCE:** (b) is answered outright and (a) returns a **rate and not a constant** **NO value for $gˆ{2}$, NO bound CONSTANT, and NO sign at any ODD level** — what is settled is the even half, exactly, and the statement that the sign cannot fail by growth **and the criterion's $gˆ{2}$ is the level's OWN self-coupling as 'r7008' defined it**; the cross-level object is the divergent double sum 'r6997' already banked, and nothing here is a claim about it the second-derivative closure constant enters only through its rate **THREE CORRECTIONS THE ARITHMETIC FORCES, one of them against this line's own last revision:** (i) the order's (a) presumes an eighth-power **gap** and there is none, both sides being of degree eight, so a crude bound cannot settle it; (ii) the order's (c) is not symmetric — an upper bound **above** the target settles nothing, and settling the sign negatively needs a **lower** bound, a different object, exhibited on an admissible set that carries both signs; (iii) 'r7038''s eighth power is in the **level-sum** convention while its own step 1 reduces to a **per-mode** $c_{4}$, where the same quantity is of degree **six** — so the convention-free statement is the ratio's flatness and the bare exponent is not a statement without its convention **the fifteenth exit is NOT taken**, on a fifth distinct ground: all three clauses of its condition fail, and what remains is a **constant** rather than an uncomputable object (eight declined of fifteen) *Run:* `python3 receipts/P10_canonical_time/P10_the_level_cubic_vanishes_at_every_even_level_and_the_bound_needs_no`

**P10R60** `P10_the_same_level_sum_is_orthogonality_rather_than_an_integral_and_the_completeness_bound_was_` [OK]

*'PO-23'; **THE ODD-LEVEL CONSTANT IS NOT AN INTEGRAL AT ALL — IT IS ORTHOGONALITY, AND 'r7044''s COMPLETENESS BOUND WAS LOOSE BY EXACTLY ONE POWER OF THE LABEL.** On this substrate a level's harmonics are matrix elements of one*

*representation, so a triple overlap is a **product of two $3j$ symbols** — exhibited here by direct symbolic integration over the group rather than quoted — and summed over the degeneracy each factor contributes $\{}sum\{}lvert 3j\{}rvert^{2}=1$ **exactly** ⇒ ***the same-level sum carries no integral and no two-point object***; **and a second, independent route through the very kernel ‘r7044‘ called the wall returns the same number**, because that kernel is a **character**, so the double integral is a class function of the relative element alone and collapses to one class integral which is exact by exponential orthogonality: $\{}int\{}chi^{2}=1$ always and $\{}int\{}chi^{3}$ is the **singlet multiplicity** ⇒ ***the law: the same-level sum is $d^{3}/V$ times the number of cubic invariants***, whose **vanishing case at half-integer spin re-derives ‘r7044‘’s even-level selection rule**; ⇒ **so the wall stands in front of a ROUTE and not in front of the OBJECT**, and **the slack between the bound $d^{4}/V$ and the object $d^{3}/V$ is exactly $\{}sqrt{D}$, one power of the label** ⇒ ***the coupling’s own growth is of degree SEVEN against the target’s EIGHT in the level-sum convention, so the rates do NOT tie and the ratio tends to zero: for a bounded channel count the criterion cannot fail at large odd label*** (node 66’s ‘r7047‘ order: ***the odd-level constant***. (a) *”**the invariant channels’ coefficients at odd $m$** — at most eight of them, a finite same-level representation-theoretic object rather than an integral over two points. **The bound’s slack is exactly the levels completeness throws away**, so the question is whether the same-level object closes that slack.”* (b) *”**and the comparison decided there, which is a comparison of constants and not of rates** ... **state which convention and carry it**, since that is the correction you just made.”* (c) *”**and if the constant comes out against it, that is the answer and not a failure** — a back-reaction positive on the even half and negative on the odd half is a stranger and more interesting result than a uniform sign.”* *Staging allowed: ”**(a) alone is a delivery**.”* Terminal branch, **sixteenth offer**: *”the row terminates if the odd-level comparison cannot be decided by any same-level object — that the constant **requires** the two-point value, which your own kernel argument shows the wall does stand in front of.”*). *Computes:* the triple overlap of three matrix elements of one representation, by direct symbolic Haar integration over the Euler ranges, against the product of two Wigner $3j$ symbols at three index assignments including a vanishing one, with the measure’s range and divisor carried together; the sum of $\{}lvert 3j\{}rvert^{2}$ over every projection at seven triples of spins, integer and half-integer; the normalised Haar volume of those ranges; the class integrals of the level character’s square and cube at eleven spins, by **exponential orthogonality with no quadrature**, against the singlet multiplicity in the triple product; the identity of the two routes’ law; the ratio of ‘r7044‘’s completeness bound to the object; and the two growth degrees in the label with the ratio’s limit, in the level-sum convention named with them *Bound:* **SCOPE IN THE SAME SENTENCE, AND IT IS THE LOAD-BEARING PART:** the $d^{3}$ law is derived on the **matrix-element realisation** of a level; the transverse-traceless harmonics add **one spin-two coupling** on the right index (the content ‘r7044‘ pinned two ways) and **that coupling’s factor is NOT evaluated here** ⇒ what carries without it is both routes, the selection rule as the vanishing case, and **the slack being one power — because the slack is a ratio of two sums over the SAME index set, so the factor enters numerator and denominator alike** **NO individual channel coefficient, NO constant, and NO sign at any one odd level**: (b) is answered as a **rate** and (c) is not reached **TWO CORRECTIONS TO ‘r7044‘, this line’s own last revision:** (i) its *”the wall stands in front of the value”* was a property of **one representation** of the value and not of the value, since a second representation carries no two-point object and the kernel route closes here anyway; (ii) its **rate tie was a statement about its BOUND**, and the bound is loose by a power, so the object and the target do **not** tie *and a correction inside this revision, reported as the row reports them: the overlap formula first came out at half its value, from integrating over the $2\{}pi$ range while dividing by the $4\{}pi$ normalisation — **the domain of a MEASURE is part of the statement**, another face of this row’s own standing lesson* **the sixteenth exit is NOT taken**, on a sixth distinct ground: its condition is **falsified** rather than unmet (nine declined of sixteen) *Run:* `python3 receipts/P10_canonical_time/P10_the_same_level_sum_is_orthogonality_rather_than_an_integral_and_the`

**P10R61** `P10_the_odd_residue_is_the_levels_below_a_crossing_linear_in_one_constant_and_a_count_bounds_ter` [OK]

*‘PO-23‘; **Q1 ANSWERED WITH A NUMBER: THE ODD RESIDUE IS THE LEVELS BELOW A CROSSING THAT IS LINEAR IN ONE CONSTANT.** **In the level-sum convention** the ratio of the coupling’s own growth to what it must stay below is exactly $60Kn\{}sqrt{2m^{2}-8}\{},(m^{2}-3)^{2}/(5m^{6}-26m^{4}+25m^{2}-4)$, whose leading behaviour is $12\{}sqrt2\{},Kn/m$ with $K$ the vertex’s own $m$-independent constant and $n\{}le8$ ‘r7044‘’s channel count ⇒*

****the crossing exists and $m_{0}\{}le96\{}sqrt2\{},K$; at $K=1$ and the cap $n=8$ it is located by an exact comparison at $m=136$, so the residue there is the odd levels $m\{}le135$***; **but the channel count alone does NOT reach it, which corrects the order's own Q1** — *a count bounds the NUMBER of terms and not the SIZE of one*, eight channels of coefficient size $S$ contributing $8S^{2}$, unbounded in $S$ at every fixed count ⇒ ***Q1 returns a LIST AS A FUNCTION OF ONE NUMBER rather than a list, and REDUCES TO Q2 quantitatively*** (and it does **not** return "unbounded", so the order's stop condition does not fire: what is missing is $K$'s value, not its finiteness); **and Q2's FEASIBILITY is established, solved for rather than posited** — the frame derivative satisfies $e_{c}D=DM_{c}$ for an $M_{c}$ of the level's own size **in every direction, so a derivative does not leave the level**, and the three solved matrices **close under commutators**, so a derivative inserts a **spin-one** operator: one extra Clebsch–Gordan coupling and nothing more ⇒ ***with 'r7048''s $3j$ orthogonality the whole same-level vertex is finite recoupling data rather than an integral****
(node 66's 'r7049' order: ***make the odd residue a finite list***. **Q1** *"**where does the ratio cross one, and is the crossing bounded without the channel coefficients?** ... the ratio falls like $1/m$ with a constant in front of it that the $\{}le8$ channel count bounds ... Report $m_{0}$ if it exists and **say so plainly if the channel count alone does not reach it — an unbounded crossing is a real answer and is not a failure of this order**."* **Q2** *"the spin-two recoupling factor ... **it does not cancel from the value**. A constant at any one odd level needs it ... **the factor, written down, for the tower's own harmonics — not a bound on it**."* **Q3** *"and if Q1 and Q2 both land, the constant at the smallest undecided odd level."* Required calibration: ***every degree, power and ratio states which convention it is in, in the same sentence as the number***, and the two counts of step 2 must still not be applied twice. Scope, **60's call and not referred up**: *"if Q1 returns 'unbounded', say so and stop there"* *and not a crude bound, not an upper bound offered as a negative settlement, and not the tower sum reopened.* **This order carries NO exit offer — the first in sixteen that does not.**). *Computes:* the two growth degrees with their convention named, from 'r7048''s law and 'r7038''s target; the ratio in closed form and its leading coefficient as an exact limit; the crossing located by exact rational comparison at the channel-count cap, with the bracketing levels on either side of it; the crossing's linearity in the vertex constant; the count-versus-size statement as an unbounded limit in the coefficient size; the partial derivatives of the leading coefficient in the constant, the count and the label; the matrix a frame derivative inserts, **solved for** from the level's own matrix elements in all three directions and tested for closure under commutators, with the constancy pattern reported as a control; the same two degrees in the **per-mode** convention; and what a second application of the pairing count would do to the crossing *Bound:* **SCOPE IN THE SAME SENTENCE, AND EVERY NUMBER CARRIES ITS CONVENTION:** all degrees, the ratio and the crossing are **level-sum**; the per-mode convention moves both degrees down by one (seven→five and eight→six), which is why a bare exponent is not the statement **NO value for $K$, NO constant at any odd level, NO sign at any odd level, and NO crossing number that does not carry $K$** — *the $m=136$ figure is the crossing **at** $K=1$ and is not a claim that $K=1$* **Q3 is NOT reached**, and Q2 is delivered as **feasibility and not value** **ONE CORRECTION TO THE ORDER THAT COMMISSIONED IT:** its Q1 supposes the $\{}le8$ channel count bounds the prefactor, and it does not — the count fixes $n$ and leaves $K$, and the crossing depends on their product *and a correction inside this revision: the Q3 scope gate first failed because it tested 'sp.Symbol("K")' against an expression holding 'sp.Symbol("K", positive=True)' — **different objects in sympy**, so the membership was false about a symbol plainly present; the gate now tests the declared object and, as a control, that the undeclared one is absent* **calibration two honoured by not spending it: this revision re-assembles no vacuum expectation, so step 2's pairing count is applied neither once nor twice** — and a second application would move the crossing from 136 to 408, which is exhibited rather than asserted **no exit is declined because none is offered; nine of sixteen stands unchanged** *Run:* `python3 receipts/P10_canonical_time/P10_the_odd_residue_is_the_levels_below_a_crossing_linear_in_one_consta`

**P10R62** `P10_the_cubic_vertex_is_written_down_and_K_is_not_label_independent_so_the_algebraic_channel_is` [OK]

*'PO-23'; **THE CUBIC VERTEX IS WRITTEN DOWN, AND $K$ IS NOT LABEL-INDEPENDENT.** The recoupling half of $K$ is computed here exactly for the tower's own transverse-traceless harmonics — the degeneracy-summed square of a same-level triple overlap is, per channel, $(2j'_1+1)(2j'_2+1)(2j'_3+1)$ times one $9j$ squared with rows $(j_1j_2j_3)$, $(2\{},2\{},2)$, $(j'_1j'_2j'_3)$, **exhibited by direct summation over every degeneracy label and Clebsch–Gordan coefficient before any $9j$ is*

*named, at four cases including a vanishing one** — and it **MOVES WITH THE LEVEL**: $126/125$ at $m=3$, $444/1715$ at $m=5$, $1441/7875$ at $m=7$, strictly decreasing, maximum at $m=3$ ⇒ ***so ‘r7050‘’s ”the vertex’s own $m$-independent constant” names something that does not exist***; *and the part that could have moved the exponent is checked and does not — the dependence is $O(1/m^{2})$ about a finite non-zero limit identified as $81/640$ from its own tail, so the **exponent is untouched***; **and the order’s premise is false by exactly one object, which this revision closes**: $K$ is *(the $\{}varepsilon^{3}$ pointwise coefficients)* $\{}times$ *(the recoupling)*, and the first factor is not recoupling data at all but the cubic term of the curvature integrand on transverse-traceless jets — the link ‘r7034‘ named as missing — **solved here on jets and verified on six held-out jets from a different seed**: $[\{}varepsilon^{3}]R_{3}[e^{\{}varepsilon H}] = \{}tfrac16\{},\{}mathrm{tr}\{},h^{3} + \{}tfrac16 h_{ab}(e_{a}h_{cd})(e_{b}h_{cd}) - \{}tfrac16 h_{ab}(e_{c}h_{ad})(e_{c}h_{bd}) + \{}tfrac14 h_{ab}(e_{c}h_{ad}) + \{}tfrac16 h_{ab}h_{cd}(e_{a}e_{c}h_{bd})$; ⇒ ***so Q2 is answered for the ALGEBRAIC channel and it never crosses: in the level-sum convention it is of degree THREE against the target’s EIGHT, the exact rational comparison reading $3/440$ already at $m=3$ and falling like the fifth power of the label*** — **five powers clear at every level of the tower, so the whole residue question is the derivative structures and nothing else**  (node 66’s ‘r7051‘ order: ***compute $K$, it is the whole open part of the odd half***. **Q1** *”$K$ itself, for the tower’s own transverse-traceless harmonics — the vertex’s label-independent constant, assembled from the recoupling data ‘r7050‘ identified ...  **it does not cancel from the value and it is now the only piece not written down**.”* **Q2** *”and then the crossing, at that $K$ rather than at one.”* Premise, in the order’s own words: *”**there is no integral, no limit and no coincidence limit left between here and the number**.”* Calibrations: the convention in the sentence with every number; and *”say once, explicitly, where step 2’s count enters the assembly and that it enters once.”* *”Not a bound on $K$ ... the row wants the number”*, and nothing re-validated. Scope, **60’s call**: *”if the assembly turns out to need something the recoupling data does not hold, that is the row’s terminal branch and it is now a sharp one — a claim about a finite algebraic sum rather than about a wall. **Report it as the finding it would be, not as a failure to reach a number**.”* *This order, like ‘r7049‘, carries no exit offer.*). *Computes:* the degeneracy-summed square of the same-level triple overlap written out over every right label, every projection and every Clebsch–Gordan coefficient, against the right dimensions times one $9j$ squared, at four spin assignments including a vanishing control; the eight channels as the $2^{3}$ assignments of the three harmonics to the level’s two irreps, with the six triangle failures at $m=3$ and $m=5$ printed with the triple that fails and all eight non-zero from $m=7$ up; the channel-independence of the product of the three left with the three right dimensions against $(m^{2}-4)^{3}$; the recoupling factor as an exact rational at twelve odd levels with its monotonicity, its maximum and its scaled deviation from $81/640$ at $m=161$ and $m=321$; the $\{}varepsilon^{3}$ coefficient of the curvature scalar on twelve random transverse-traceless jets fitted to nine cubic structures, its rank, its three identities, and the representative checked on six held-out jets from a different seed; the index count that forbids any parity-odd cubic structure; the collapse to a single invariant at frame-constant $h$; and the algebraic channel against $2c_{4}\{}mu^{2}$ by exact rational comparison at five levels with its log-slope across three brackets *Bound:* **SCOPE IN THE SENTENCE WITH EACH RESULT, AND EVERY NUMBER CARRIES ITS CONVENTION:** all degrees and ratios are **level-sum**; in the per-mode convention both drop by one and **the five-power gap is unchanged, because a convention shifts an exponent and not a difference of exponents**. **NO single number $K$** — because $K_{\{}mathrm{rec}}$ is a **function**, given exactly at every odd level; **no crossing for the full vertex, no constant and no sign at any odd level, no closed form in $m$**, and the limit $81/640$ is **identified and checked, not derived**; **Q3 not reached**. **TWO CORRECTIONS TO THIS LINE’S OWN LAST TWO REVISIONS:** (1) ‘r7050‘’s $K$ is not label-independent; (2) ‘r7048‘’s $d=\{}sqrt{D}$ is a **leading-order stand-in and not the level’s size** — a transverse-traceless level is $(a,b)+(b,a)$ with $a\{}neq b$, so it is not of the form $(j,j)$ and has no single $d$; the law is exact in **rate** and off by a label-free factor in the **constant**, which is exactly what Q1 asks for. **AND ‘r7044‘’s CHANNEL COUNT IS RE-DERIVED RATHER THAN RE-VALIDATED**, out of the $9j$ structure: two at $m=3,5$ and eight from $m=7$. *Two controls: no parity-odd cubic structure can exist, by an index count of eleven rather than by an argument; and at frame-constant $h$ the identity collapses to ‘r6971‘’s single invariant.* **Calibration two, said once: step 2’s pairing count enters EXACTLY ONCE, on the target side inside the banked $c_{4}$, and not

at all on the coupling side, whose object is a sum of squares over three independently summed degeneracy labels with no harmonic contracted against itself.** **The terminal branch is NOT taken, on the order's own ground:** the assembly did need something the recoupling data does not hold, and that object turned out to be writable from the substrate's own curvature on jets ⇒ *what remains is recoupling data again — the derivative structures' spin-one insertions, whose coefficients are now written down, so **the crossing is linear in a recoupling sum and no longer in an unknown vertex coefficient***. *Nine declined of sixteen stands; no exit offered.* *Run:* `python3 receipts/P10_canonical_time/P10_the_cubic_vertex_is_written_down_and_K_is_not_label_independent_so_t`

**P10R63** `P10_the_odd_residue_is_five_levels_and_the_sign_is_mixed_because_the_vertex_carries_covariant_n` [OK]

*'PO-23'; **THE SIGN: THE ODD RESIDUE IS FIVE LEVELS AND THE BACK-REACTION'S SIGN IS MIXED.** **In the level-sum convention** the full same-level vertex's degeneracy-summed square is of degree **SEVEN** against the target $2c_{4}\{}mu^{2}$'s **EIGHT**, and the exact rational comparison crosses one between $m=11$ ($104584340/101897367¿1$) and $m=13$ ($96233385/112183214¡1$) ⇒ ***the odd residue is exactly the five levels $m=3,5,7,9,11$, where the sign is NEGATIVE, and it is POSITIVE at every other level of the tower*** — every odd $m\{}ge13$ by this comparison and every even $m$ by 'r7044''s selection rule, used and not re-derived; **and the order's own invariance guard is what made it true, by catching a different error than it was set for: THE VERTEX'S DERIVATIVES ARE COVARIANT AND NOT FRAME DERIVATIVES** — 'to_frame''s own relation says the jet variables $u,w$ are $\{}nabla h$ and $\{}nabla\{}nabla h$, and assembled with the frame derivative the recoupling sum is **not** invariant across 'r7052''s three-parameter family ($120575/6$ against $19775/6$ at $m=3$) while with the covariant one it is invariant in all three directions at every level ⇒ ***this is the corpus's banked "the sign of 'Dframe' vs 'to_frame' is invisible to transversality" made arithmetic, and the reason it is invisible there is exhibited: the connection terms cancel in the divergence, so transversality cannot see the difference and the cubic vertex can***; **the substrate's frame is derived rather than adopted** — of the four combinations of side and generator sign exactly one carries the banked $C^{c}{}_{ab}=2\{}varepsilon_{abc}$, and it is orthonormal for the unit-radius round metric with the banked volume $2\{}pi^{2}$ ⇒ ***the derivative acts on the RIGHT index by a CONSTANT spin-one generator in ALL THREE directions***; **and 'r7044''s multiplet content is re-derived as a KERNEL** — transversality's kernel at left spin $j$ has dimension $4j+2$ and right Casimir $j'(j'+1)$ for $j'=j\{}pm2$ with multiplicities $2j'+1$, at three spins including a half-integer one* (node 66's 'r7053' order: ***the derivative structures' recoupling, which is the odd half***. **Q1** *"the recoupling of the three derivative structures ... what is missing is the constant."* **Q2** *"then the crossing, and it is the first time the full vertex has one ... an exact comparison rather than a bound."* **Q3** *"the sign, at the levels below it. Which is the row's object and has been since 'r3809'."* A GUARD, and it turned out to be the order's load-bearing instruction: *"the recoupling sum must be invariant across that [three-parameter] family too, or the representative was load-bearing after all. **If two members give two different recoupling sums, that is a finding and not a bug**."* Calibrations: the convention in the sentence with every number; and *"Q1 is the first assembly where a derivative structure is squared, so say once whether that changes the [pairing-count] answer — **a derivative insertion is not a contraction of a harmonic against itself, but it is the first place one could hide**."* *This order, like 'r7051' and 'r7049', carries no exit offer.*). *Computes:* the four combinations of invariant side and generator sign against the banked structure constants, read off algebraically rather than by solving; the frame's defining relation as the tracelessness of $g^{-1}\{}partial g$; the induced metric against the round metric and its volume against $2\{}pi^{2}$; the failure of $Z_{a}$ to commute with $g$ in all three directions; the five symmetric traceless basis tensors' orthonormality; the frame generator's scale solved by closure and its Casimir; the connection's frame action against that generator halved, on all five basis tensors; the identical cancellation of the connection terms in the divergence; transversality's kernel, its dimension and its right Casimir with multiplicities at three left spins including a half-integer; the standard multiplet basis built by highest weight and ladder; the vertex amplitude by polarisation of 'r7052''s identity over all six assignments, at nineteen admissible label triples against the $3j$; the amplitude's exact trilinearity under three independent scalings; the recoupling sum at six odd levels under three representatives each, with and without the connection terms; the sum's log-slope across three brackets; and the ratio against $2c_{4}\{}mu^{2}$ at six levels with the crossing located by exact rational comparison *Bound:* **SCOPE IN THE SENTENCE WITH EACH RESULT, AND EVERY

NUMBER CARRIES ITS CONVENTION:** all degrees, the ratio and the crossing are **level-sum**; in the per-mode convention both degrees drop by one and **the difference of exponents is unchanged at one**. **The comparison is in the same normalisation ‘r7044‘, ‘r7048‘ and ‘r7052‘ all used** — the $\{}varepsilonˆ{3}$ level-summed square against $2c_{4}\{}muˆ{2}$ with the label-free factor common to both sides — **and the crossing at $m=13$ is exact IN THAT NORMALISATION**. **NO closed form in $m$**, no value for the ratio’s leading coefficient (only that $m$ times it stays near eleven over the six levels computed), and **no reconstruction of ‘r7050‘’s own setup: its control is corrected, not explained**. **THREE CORRECTIONS TO THIS LINE’S OWN WORK AND ONE TO THE ORDER:** (1) the vertex’s derivatives are **covariant**, which is what the order’s guard caught; (2) ‘r7050‘’s control — *”only the invariant direction returns a constant”* — is wrong: on the frame carrying the corpus’s own $C$ all three do; (3) ‘r7052‘’s **derivative-free channel is not a channel**, because its own null direction $3\{},\{}mathrm{tr}\{},hˆ{3}=-W_{1}+W_{2}+W_{3}$ expresses the derivative-free structure in second-derivative ones, so the split moves under the representative and **only the total has a degree** — which also corrects ‘r7053‘’s *”it can be set aside for good”*; and ‘r7052‘’s $3/440$ at $m=3$ was in a normalisation that is not the vertex’s, where the ratio is $175/22$. *A structural fact the assembly did not put in: the two mirror channels contribute **exactly equally** at every level.* **Calibration two, said once: squaring a derivative structure hides NO pairing** — the amplitude’s exact trilinearity is the arithmetic of it — so step 2’s count still enters exactly once, on the target side inside the banked $c_{4}$. **The terminal branch is NOT taken, and this time the sum it was about is the delivery**; nine declined of sixteen stands, no exit offered *Run:* `python3 receipts/P10_canonical_time/P10_the_odd_residue_is_five_levels_and_the_sign_is_mixed_because_the_ver`

**P10R64** `P10_the_cubic_normalisation_is_written_down_and_it_empties_the_residue_so_the_sign_is_positive_` [OK]

*‘PO-23‘; **THE CUBIC COUPLING’S ABSOLUTE NORMALISATION, WRITTEN DOWN — AND IT EMPTIES THE RESIDUE, SO THE BACK-REACTION’S SIGN IS POSITIVE ON THE WHOLE TOWER.** ‘r7044‘, ‘r7048‘, ‘r7052‘ and ‘r7056‘ all compared the $\{}varepsilonˆ{3}$ level-summed square against $2c_{4}\{}muˆ{2}$ **with a factor assumed common to both sides and never computed**; this revision computes it. Assembled through ‘r7038‘’s own per-mode Lagrangian the criterion is ***$(-\{}mathrm{q4lev})\{},\{}muˆ{2} ¿ 2\{}sum_{ABC}T_{ABC}ˆ{2}$, with $\{}kappa$ and the scale factor cancelling EXACTLY*** — and the left side is ‘r7038‘’s target $2c_{4}\{}muˆ{2}$ *identically*, which is the chain closing on itself rather than on a coincidence ⇒ in ‘r7056‘’s own printed variables ***the criterion is $\{}mathcal{G}(m)/\{}mathrm{target} ¡ 18V = 36\{}piˆ{2}\{}approx355.31$*** against ‘r7056‘’s **largest** ratio $175/22\{}approx7.95$ at $m=3$, smaller by a factor above forty-four, and every further level further below ($131$, $206$, $277$, $346$, $414$ at $m=5,7,9,11,13$) ⇒ ***THE ODD RESIDUE IS EMPTY AND THE SIGN IS POSITIVE AT EVERY LEVEL OF THE TOWER***: every odd level by this comparison and every even level by ‘r7044‘’s selection rule; **A CORRECTION TO ‘r7056‘ AND TO ‘r7057‘ AS LANDED, this line’s own** — its *five odd levels $\{}{3,5,7,9,11\{}}$, sign MIXED* and its crossing at $m=13$ were the comparison with this factor set to **one**, so **the row’s object since ‘r3809‘ has a UNIFORM answer and it is the positive one**, which the order pre-authorised as *the expected outcome, not a failure*; and **‘r7058‘’s own in-flight line did NOT survive**: it flagged the factor as degeneracy bookkeeping that would move the *degree* — the degrees are ‘r7056‘’s, seven against eight, **unchanged**, and it is the pure number that was wrong* (node 66’s ‘r7057‘ order, registered under STANDING ORDER ‘r7013‘ (3) as **an order and not a row**. **Q1** *”write the cubic coupling’s absolute normalisation.”* Plus the order’s **pre-authorisation of the outcome**: *”say plainly if fixing it moves the crossing off $m=13$: that is the expected outcome, not a failure, and the five-level count is a statement in the current normalisation rather than a prediction the corpus is now committed to.”* Plus a **second ask, one paragraph and not a revision’s work**: *”with the sign banked, is the definition of the sums now closer than it was, or the same distance and better furnished?”* — *”if the answer is ’no closer’, that is worth landing too.”*). *Computes:* every factor of the chain derived on oscillator matrix elements rather than adopted — $\{}langle 0\{}vert Qˆ{2}\{}vert 0\{}rangle=S$ fixing the scale, then $\{}langle 0\{}vert Qˆ{4}\{}vert 0\{}rangle=3Sˆ{2}$, $\{}lvert\{}langle 3\{}vert Qˆ{3}\{}vert 0\{}rangle\{}rvertˆ{2}=6Sˆ{3}$, the polarisation factor of a distinct cross-term $=6T_{ABC}$ on random symmetric $T$ at three **and** four modes, and the multi-mode three-quantum norm $\{}sum\{}lvert\{}mathrm{amp}\{}rvertˆ{2}=6Sˆ{3}\{}sum_{ABC}Tˆ{2}$ with both diagonal and off-diagonal parts of $T$ present; the occupied numbers of a cubic form checked

to be exactly $\{}{1,3\{}}$, so the energy denominator $3\{}hbar\{}omega$ is a property of the channel; ‘r7034’’s $\{}varepsilonˆ{2}$ level sum **rebuilt from its own pointwise identity** plus ‘r7044’’s gradient closure, returning the banked $-D\{}muˆ{2}/4$ identically **with no volume anywhere**, against the triple product of orthonormal harmonics carrying $Vˆ{-1/2}$; $M$ and $\{}omega$ read off ‘r7038’’s two Lagrangian coefficients and $\{}sigma=2\{}kappa\{}hbar/(aˆ{2}\{}mu)$ forced by them; the assembled criterion checked to have free symbols $\{}{\{}mathrm{q4lev},\{}sum Tˆ{2},\{}mu\{}}$ **and nothing else**; three of ‘r7056’’s six ratios cross-checked against its own banked recoupling sums at $m=3,5,13$; and ‘r7010’’s anchor substituted into the same chain, leaving a $\{}kappa$ standing by exactly $4\{}kappa$. **28 checks, all pass, exact throughout** *Bound:* **SCOPE IN THE SENTENCE WITH EACH RESULT.** The criterion compared is the **ground-state** back-reaction at second order in the cubic and first in the quartic — ‘r7010’’s own comparison — at **one level of the tower at a time**, in ‘r7038’’s passage, on the unit-radius round substrate of banked volume $2\{}piˆ{2}$. **What is corrected is the THRESHOLD and nothing upstream of it:** ‘r7056’’s degrees, its six exact ratios, its covariant-derivative finding and its representative-invariance all stand **exactly as filed**, and the ratio’s decay like one over the label is unaffected. **No closed form in $m$ and no leading coefficient.** **AND THE CHAIN DOES NOT CLOSE ON ‘r7010’’s BANKED ANCHOR, which is named and not settled here:** ‘r7010’’s variance carries no $\{}kappa$ where ‘r7038’’s passage forces one, the two differing by exactly $4\{}kappa$, so ‘r7010’’s $200/63$ — a pure number in its convention — becomes $50/(63\{}kappa)$ in ‘r7038’’s, **not** a pure number ⇒ ***writing the chain moves ‘r7010’’s ratio too***, which is ‘r7010’’s row’s to act on. Nothing re-validated: not ‘r7038’’s passage or $c_{4}$, not ‘r7034’’s level sums, not ‘r7044’’s selection rule or gradient closure, not ‘r7008’’s cancellation, not ‘r7052’’s $\{}varepsilonˆ{3}$ identity and not ‘r7056’’s covariant-derivative finding *Run:* `python3 receipts/P10_canonical_time/P10_the_cubic_normalisation_is_written_down_and_it_empties_the`

**P10R65** `P10_sec_lock_quotes_the_superseding_receipt_everywhere_the_two_mode_shift_sits_under_the_wrong_` [OK]

*‘sec:lock‘ — **r7059 Q1 → 70, the back-reaction passage read cold against its seven receipts: it quotes the superseding receipt everywhere two co-cited receipts disagree** (seventh against eighth from r7048 over r7044’s tie; positive on the whole tower against $18V=36\{}piˆ2$ from r7058 over r7056’s mixed sign and crossing at $m=13$; r7050’s $m=136$ and r7053’s ’five powers clear’ quoted nowhere), and every figure carried from an earlier receipt is one that did not move ($175/22$ and r7056’s six ratios, used by r7058 as filed; $120575/6$, $19775/6$; the recoupling sums $126/125$ … $81/640$). **One (ii), below any headline**: the $Rˆ{(3)}$ expansion and the whole two-mode shift ($\{}tfrac{4}{27}(63\{}muˆ2-200)\{}muˆ{-4}$, $19/27$, $200/63$) close under the r7044/r7048 pair, which computes none of them; r7008 computes all of them and is cited ninety lines on, on another sentence. **The four-receipt group is one chain**: r7056’s ratios computed once, r7058’s threshold the only new input (its ratios copied as literals, which no file graph shows), r7050 supporting no number quoted. Note: ’the label times it holding in a narrow band’ is 11.2–13.5 from $m=5$, 23.9 at $m=3$* (node 66’s ‘r7059‘ order Q1 to 70: *does ‘sec:lock‘, as it now stands, say what its cited receipts compute — and is it quoting the superseding receipt where a superseded one disagrees; and does the four-marker group read as four-times-independent when it is one result*). *Computes:* six of the seven cited receipts re-run from their own directories and required to exit 0 (r7056, 210 s, gated on its source); each quantity checked against the receipt that computes it and against the LATER of any two that disagree; the dependency edges among the four read from source, including the literal table r7058 copies; the paper’s wording REPORTED, never required *Bound:* no reconciliation of the two anchors’ conventions (node 60’s); no physics, no re-derivation; no prose edited; no other seat’s receipt touched *Run:* `python3 receipts/P10_canonical_time/P10_sec_lock_quotes_the_superseding_receipt_everywhere_the_two_mode_shi`

**P10R66** `P10_the_regulator_passage_keeps_its_line_but_the_minus_four_thirds_closes_under_a_group_that_do` [OK]

*‘sec:lock‘ — **r7065 → 70, the anchors / divergence / regulator passage read cold, three revisions deep: the prose keeps the receipt’s line** (the count ’counted rather than asserted’, the renormalised value and the third invariant withheld in the same sentence as the receipt withholds them; the divergence and the definition joined in order, ’no finite value’ → ’must come from a regulator’ → ’the construction’s own regulator reaches it’; one word to weigh, the prose’s ’defined’ against the receipt’s ’reaches’) — **but the passage’s own result closes under the wrong group**: $-\{}tfrac{43}$, the count of three subtractions against two and the constant term $52/15$ are computed by r7065,*

*which is cited in the anchors group a hundred-odd lines earlier, while the four receipts closing the regulator paragraph (r7063, r7058, r7056, r7053) compute none of them. Nothing superseded is quoted (r7063's $39/4$ at the old offset appears nowhere; the paper carries $15/4$), and r7059's (ii) is closed (r7008 now beside the two-mode shift)* (node 66's ‘r7065‘ order to 70: *the ‘r7043‘ method on the passage three revisions deep, and Q2 judged rather than checked — does the prose keep the count where the receipt keeps it, and do the divergence and the definition read as in tension*). *Computes:* five of the six cited receipts re-run and required to exit 0 (r7056 gated on source); each quantity gated on which receipt computes it; every paper-state line REPORTED and never required, so a marker fix by 66 leaves it green *Bound:* no physics, no check of $-4/3$ beyond what the receipt computes; no prose edited; no other seat's receipt touched *Run:* `python3 receipts/P10_canonical_time/P10_the_regulator_passage_keeps_its_line_but_the_minus_four_thirds_clos`

**P10R67** `P10_the_two_anchors_are_already_one_convention_and_they_agree_but_the_candidate_prescription_do` [OK]

*‘PO-23‘; **THE TWO ANCHORS ARE ALREADY ONE CONVENTION AND THEY AGREE EXACTLY — AND THE CANDIDATE SUMMABILITY PRESCRIPTION DOES NOT CLOSE.** **Q1:** the criterion ratio $R=g^{2}/(2c_{4}\{}mu^{2})$ is ***INVARIANT*** under a change of convention, because a convention change rescales the FIELD and the couplings move with it — matching the two kinetic terms forces $q_{10}=q_{38}/(2\{}sqrt\{}kappa)$, hence $c_{4}\{}to c_{4}/(16\{}kappa^{2})$ and $g^{2}\{}to g^{2}/(64\{}kappa^{3})$, and $R$ is unchanged, as a ratio of two ENERGIES must be; **so ‘r7058‘'s section E is WRONG and this corrects this line's own last revision** — it swapped the VARIANCE and kept ‘r7010‘'s couplings, a **hybrid** and not a convention, and its $4\{}kappa$ is the artefact of that ⇒ ***‘r7010‘'s banked $200/63$ does NOT move***, its own constants giving a ratio already free of $\{}kappa$ *and* of $V$; and the restatement is **derived with no fitted parameter** — matching the PER-MODE quadratic ($-48$ against ‘r7034‘'s $-D\{}mu^{2}/4$ over the degeneracy) ***forces*** $\{}lambda^{2}=1/24$, which then returns ‘r7010‘'s banked $c_{4}$ numerator $14/3$ *exactly* and, from its own cubic's SOLVED symmetric $T$, $\{}sum_{ABC}T^{2}=200/27$ ⇒ ***‘r7058‘'s criterion evaluated on ‘r7010‘'s own two-mode sector returns $R=25/63$, IDENTICAL to ‘r7010‘'s banked ratio***; and what separates the two NUMBERS is the **object**, not a convention — two frame-constant Misner modes against the degeneracy-summed level, the factor between them $88\{}pi^{2}/49$ at $m=3$ and **DRIFTING** across the tower, *and a convention factor cannot drift*; **Q2: the summand is written down for the first time** — the quartic energy per level is $(5m^{6}-41m^{4}+88m^{2}-16)/(15(m^{2}-1))$ in units $\{}kappa\{}hbar^{2}/a^{3}$, **positive at every level and growing like $m^{4}$** (the limit of $\{}langle V_{4}\{}rangle/m^{4}$ is exactly $1/3$), and with $0¡R¡1$ everywhere the net is bounded below by $(39/40)\{}langle V_{4}\{}rangle$, so the partial sum is bounded below by $\{}sum k^{4}/3$ ⇒ ***THE SUM DIVERGES AND THE CANDIDATE DOES NOT CLOSE***, which is ‘r7059‘'s stated founding alternative taken; **and a second correction to ‘r7058‘: its closing paragraph read the uniform sign as REMOVING an obstruction — it is the other way round**, a uniform sign FORBIDS the conditional convergence an alternating sign would have permitted, and the one-over-label decay is the decay of the RATIO and not of the summand* (node 66's ‘r7059‘ order, **two questions in a FIXED order**. **Q1** *”the two anchors into one convention, FIRST ... with each banked anchor restated in it and the powers of $\{}kappa$ tracked explicitly. **If reconciling them moves a banked number, that is the finding.**”* — ordered first for the reason this line gave, that *a definition cannot be written across two*. **Q2** *”and then the row's own object ... **does that candidate close?** A prescription resting on a one-over-label decay and a uniform sign, stated as a measure and a domain — **or a demonstration that it does not, which is the row's founding alternative. Either is the row.**”* Calibration named: *”the pairing count enters once, on the target side, inside the banked $c_{4}$ ... say so if [Q1] does [disturb it].”*). *Computes:* ‘r7010‘'s mass **solved from its own variance** as $M=a^{3}$ against ‘r7038‘'s $a^{3}/(4\{}kappa)$, not assumed; the field rescaling forced by matching the kinetic terms and the couplings transformed with it, $R$ then checked invariant *exactly*; the hybrid checked to be exactly $4\{}kappa R$, i.e. the number ‘r7058‘ reported; ‘r7010‘'s banked ratio checked to have free symbols $\{}{\{}mu\{}}$ and nothing else; $\{}lambda^{2}$ **solved** from the per-mode quadratic and then $14/3$ recovered with no further freedom; the cubic's symmetric $T$ **solved** ($T_{000}=160$, $T_{011}=-160$) and verified to reproduce ‘r7010‘'s cubic identically; the criterion on the two-mode sector giving $112/3¿400/27$ ⇒ $25/63$; the level-to-sector quartic factor $55/7$ checked to differ from $D/2=5$; the drift checked to take six distinct values; the per-level energy checked positive for every $m$ from 3 to 59 and its growth

taken as an **exact limit** against $m^{3}$, $m^{4}$ and $m^{5}$ because ‘sp.degree‘ raises on a rational function; and the lower-bounding sum’s divergence taken as a limit. **24 checks, all pass, exact throughout** *Bound:* **SCOPE IN THE SENTENCE WITH EACH RESULT.** Q1’s results are at $m=3$, **the only level at which ‘r7010‘’s anchor is defined** — its vertex numbers are label-INDEPENDENT constants and ‘r7036‘ pinned its level. Q2 is the ground-state back-reaction at second order in the cubic and first in the quartic, summed over the tower’s levels in ‘r7038‘’s passage, with $R$ taken from ‘r7058‘ at the six odd levels it computed. **THE PAIRING COUNT IS NOT DISTURBED, on the order’s own question and said once:** the reconciliation uses ‘r7010‘’s own two-mode Gaussian factor and ‘r7034‘’s level value as filed and applies no count a second time. **No measure is asserted and no regulator proposed** — the claim is that *this* candidate fails, not that none could succeed. The drift is exhibited at six levels and given no closed form. **Nothing ‘r7058‘ closed is reopened:** the threshold $18V=36\{}pi^{2}$, the cancellation, the volume calibration and the positive sign are USED as filed, and what is corrected — section E and the closing paragraph — sits BESIDE those results rather than under them. ‘r7010‘’s reduction is used as filed: its $-48$, $+160$ and $-336$ are read from its own receipt, not recomputed *Run:* `python3 receipts/P10_canonical_time/P10_the_two_anchors_are_already_one_convention_and_they_agree_but_the_c`

**P10R68** `P10_the_construction_own_regulator_reaches_the_interacting_sum_and_spends_one_counterterm_more` [OK]

*‘PO-23‘; **THE CONSTRUCTION’S OWN REGULATOR REACHES THE INTERACTING SUM, GIVES $-4/3$, AND SPENDS ONE COUNTERTERM MORE THAN THE FREE CASE.** ***No regulator is invented***: the one used is the free tower’s own spectral sum $W(\{}sigma)=\{}sum_{m\{}ge3} 1)^{-\{}sigma}$, evaluated by the same terminating binomial expansion whose Pochhammer factor kills the tail at each integer point, and **calibrated against that tower’s banked $\{}zeta(0)=10$ before being applied**. In the regulator’s own variable $x=\{}mu^{2}=m^{2}-1$ the interacting summand is **exact and finite** — $x^{2}/3-(26/15)x+7/5+(12/5)/x$, three polynomial terms and ONE simple pole, the pole at $x=0$ lying outside the tower, which starts at $m=3$ ⇒ ***$Z_{\{}mathrm{int}}(0)=\{}tfrac13W(-2)-\{}tfrac{26}{15}W(-1)+\{}tfrac75W(0)+\{}tfrac{12}{5}W(1)=- 4/3$ exactly***, each of the four pieces separately finite so no cancellation of infinities hides in the total; **and the $m^{4}$ growth is NOT the obstruction the order expected** — what is new against the free case is that the summand is *rational*, so its large-$m$ expansion does not terminate, but **that whole tail IS the single pole**, worth $W(1)=5/12$ absolutely convergent, ***so the third outcome — ”it does not reach”, named as the row’s terminus — is RULED OUT, by exhibiting the continuation rather than by an estimate***; **but the cost is strictly higher, counted rather than asserted**: the subtractions a spectral regularisation spends are its summand’s non-negative powers, and the free $d(m)\{}mu(m)$ has $m^{3},m$ while $U(m)$ has $m^{4},m^{2},m^{0}$ ⇒ ***THREE against TWO, one more than the free case, whose own already lands on curvature-squared invariants the ledger does not hold*** ⇒ **the landing is ‘r7063‘’s SECOND outcome and not its first**; and the new subtraction is the $m^{0}$ one, which the free summand **cannot** have, being odd in $m$ to every order* (node 66’s ‘r7063‘ order, **one question**. **Q1** *”whether a regulator available to this construction defines the interacting sum ... the sharp question is whether that same regularisation reaches the interacting summand you have just written down. **A spectral zeta on a summand growing as $m^{4}$ is a different object from one on $\{}mu_{n}\{}sim n$, and whether it exists is the question.**”** *”the honest possibilities are three, not two, so **name which one you land on**: it reaches and gives a value; it reaches and the value spends a constant the ledger does not hold, as the free case does; or it does not reach. **The third is the row’s terminus and would be a result.**”* Not asked for: *”a regulator invented for the purpose and then defended.”*). *Computes:* $W(1)$ **telescoped** rather than regularised, $\{}tfrac12\{}sum(1/(m-1)-1/(m+1))=5/12$; $W(0)=-5/2$, $W(-1)=-5/2$, $W(-2)=-19/2$ each from the Pochhammer truncation, with the loop asserting it terminates; the free tower’s $\{}zeta(0)=2W(-1)-6W(0)$ rebuilt from this $W$ and **nothing added**, returning the banked $10$ — the calibration that the regulator is the construction’s own; the summand’s $x$-form verified to rebuild ‘r7060‘’s expression identically; the non-terminating tail verified to be **exactly** the pole term, by series subtraction; the $39/4$ and $15/4$ coefficients both computed, at their own two offsets; the offset’s irrelevance at $s=0$ exhibited; the subtraction counts read off the two expansions; the free summand checked to carry **no** $m^{0}$ or $m^{2}$ term; and every $W$ value checked to be separately rational. **18 checks, all pass, exact throughout** *Bound:* **SCOPE IN THE SENTENCE WITH THE RESULT.** The summand regularised is ‘r7060‘’s quartic energy per level — the ground-state back-reaction’s first-order quar-

tic term in ‘r7038‘’s passage — on the tower’s own spectrum $\{}mu^{2}=m^{2}-1$ with degeneracy $2(m^{2}-4)$ from $m=3$ up. **NO VALUE IS CLAIMED FOR THE RENORMALISED SUM:** $Z_{\{}mathrm{int}}(0)$ is the continuation’s value at zero, i.e. the coefficient a subtraction scheme must absorb, and **which geometric invariant carries the third subtraction is left OPEN rather than guessed** — the claim is the count, three against two. **No regulator is invented and none defended.** Whether the regularisation commutes with the evolution is **not** established here, exactly as it is not for the free tower. Nothing ‘r7058‘ or ‘r7060‘ closed is reopened; $\{}zeta(0)=10$ is used, rebuilt only as the calibration that reports (4b) *Run:* `python3 receipts/P10_canonical_time/P10_the_construction_own_regulator_reaches_the_interacting_sum_and_spe`

**P10R69** `P10_the_subtraction_is_at_operator_dimension_six_pinned_three_ways_and_it_is_not_degenerate_so_` [OK]

*‘PO-23‘; **THE SUBTRACTION IS AT OPERATOR DIMENSION SIX, PINNED THREE INDEPENDENT WAYS, AND IT IS NOT DEGENERATE — SO THE LEDGER COST IS REAL.** **(i)** By the section’s own order rule: the interacting quartic energy is checked to be ***exactly*** $(\{}ell_{P}/a)^{2}$ times the free tower’s scale, so it sits at **second** order, and inverting *”a counterterm of operator dimension $2k$ contributes at order $2k-4$ exactly”* with *”second order admits one operator dimension and no other”* gives $k=3$ ⇒ ***dimension SIX, forced and unique***. **(ii)** By the power of the scale factor and no order counting at all: with $\{}sqrt{-g}\{}sim a^{3}$ and $R\{}sim a^{-2}$ an energy sits at $a^{3-2n}$, so the free tower’s $a^{-1}$ is $n=2$ — **which is why its banked counterterm is the curvature-squared one** — and the interacting $a^{-3}$ is $n=3$, the same answer by a different route. **(iii)** And the section had **already placed it**: it records this cubic’s density falling as the sixth inverse power, *”which is the operator dimension the mode count and the dimensional bookkeeping already put it at”* — an energy at $a^{-3}$ over a volume at $a^{3}$ ⇒ the same dimension a third time, **what was missing being the LINK from the constant term to it**; **and the order’s framing is corrected twice, neither about the physics**: there is **no separate invariant for ”the third” subtraction**, since the whole per-level energy is $\{}kappa\{}hbar^{2}/a^{3}$ times a function of the *label* alone, so **all three sit at one power of $a$** and what distinguishes the third is its power of the label; and **it is NOT a volume term** — a volume term is the cosmological one at $a^{+3}$ and this sits at $a^{-3}$, ***six powers apart***, so the order’s disjunction resolves to its **second** branch, a new entry; **and it is not degenerate with the free case’s, computed rather than cited**: on the admitted class (the scale factor *quantised*, so not a point) degeneracy would need one functional to be a multiple of the other, and $a^{-1}$ and $a^{-3}$ are distinct monomials with only the trivial vanishing combination ⇒ ***the count is three AND the ledger cost is three***, the section’s own asymmetry supplying the reason (at dimension four the collapse is an identity **pointwise** in $a$ and descends as an operator relation; at dimension six *”there is no such identity to descend”*); **the row’s terminus is NOT taken and its condition is exhibited false** — an invariant of the admitted family does carry it, so what the row now owes at this dimension is a **value** and not a basis* (node 66’s ‘r7065‘ order, **one object**. **Q1** *”which geometric invariant carries the third subtraction. The constant-term counterterm the interacting tower spends and the free one cannot ... **So this is an identification inside a basis the corpus holds, not a search.**”* Two things ordered stated with it: *”**which dimension it has** — a constant-term subtraction in a spectral sum is a volume term at the dimension the summand’s units give, and whether that is the cosmological term the ledger already carries or a new entry is the whole of what the count costs”*; and *”**whether it is degenerate with the free case’s** ... if the third subtraction is degenerate too, the count is three and the ledger cost is not.”* *”And if no invariant of the admitted family can carry it, **that is the row’s terminus and it is a statement about the counterterm basis rather than about the sum**.”*). *Computes:* the per-level energy checked to have the scale factor factor out completely, its free symbols confirmed to exclude $a$, $\{}hbar$ and $\{}kappa$; the three label powers $[4,2,0]$ read off the expansion; the ratio to the free scale **solved** to $(\{}ell_{P}/a)^{2}$ under $\{}kappa\{}hbar=\{}ell_{P}^{2}$; the order rule **inverted** rather than read off, and its solution checked unique; the $a^{3-2n}$ counting applied at $n=0,1,2,3$ and checked to **reproduce the free tower’s banked dimension four before being used on an unknown** — so the rule is calibrated, not assumed; the density checked to be $a^{-6}$; the volume gap checked at exactly six powers; the non-degeneracy **computed** as the triviality of the only vanishing combination of $a^{-1}$ and $a^{-3}$; **and a control that separates the regimes rather than always returning independence** — at fixed background the same two functionals *are* degenerate, which is the

section's own "one-dimensional at fixed background". **19 checks, all pass, exact throughout, no floats at all** *Bound:* **SCOPE IN THE SENTENCE WITH THE RESULT.** The subtraction is 'r7064''s, on 'r7060''s quartic energy per level, in 'r7038''s passage, on the tower's own spectrum from $m=3$ up. The dimension is fixed **on the section's own admitted class** — the scale factor quantised, so not a class of fixed backgrounds — and the non-degeneracy is a statement **on that class**, not on a fixed background, where everything collapses to one dimension. **NO VALUE for the counterterm's coefficient, and NO CHOICE OF REPRESENTATIVE among the dimension-six scalars:** the identification is of the operator **dimension**, which the order rule and the power counting both fix, and which of the five scalars carries it is fixed by neither and **is not guessed**. The dimension-six rank is **used as filed and not re-derived** ('P10_the_ultraviolet_object_is_a_rank_sequence_and_the_ledger_is_at_stake_at_dimension_six_and_not_four'), as are the order rule, the free tower's $a^{-1}$ scale, 'r7064''s count and 'r7060''s summand. **No claim that the ledger's entry is absent or present in the wider corpus** — only that this subtraction is not degenerate with the free case's, so it is a **second** scoping of the no-free-constant claim in this sector rather than the same one *Run:* `python3 receipts/P10_canonical_time/P10_the_subtraction_is_at_`

**P10R70** `P10_the_coefficient_is_zero_because_the_weights_parity_misses_every_pole_so_the_ledger_spends_n`
[OK]
*'PO-23'; **THE COEFFICIENT IS ZERO, BY A SELECTION RULE ON THE WEIGHT'S PARITY IN $\{}mu$ — SO THE LEDGER SPENDS NOTHING AT DIMENSION SIX.** The corpus's own criterion for a counterterm is a **logarithm**, and the free tower needs one because its spectral function has a **pole** at the physical point. ***$W(\{}sigma)=\{}sum_{m\{}ge3}(\{}mu^{2})^{-\{}sigma}$ has poles ONLY at half-integer $\{}sigma=\{}tfrac12-k$***, since $\{}zeta$'s single pole at argument one is reached at $2\{}sigma+2k=1$ ⇒ **a half-integer argument is reached only by an ODD power of $\{}mu$**. The free weight $d(m)\{}mu(m)$ carries one and hits a pole; ***the interacting summand is a rational function of $\{}mu^{2}$ — EVEN in $\{}mu$ — and misses every one***, so at $s=0$ every $W$ argument is an **integer**, where $W$ is regular, and **each term's residue vanishes SEPARATELY rather than four summing to nothing** ⇒ ***total residue exactly zero: no pole, no logarithm, no scale dependence, nothing for a counterterm to absorb***; **and the machinery is calibrated against a banked number before being used on an unknown one** — the same residue formula on the free tower's own weight at *its* physical point returns $2\{}cdot2(3/16)+(-6)\{}cdot2(-1/4)=15/4$ exactly, 'r6975''s banked logarithmic coefficient, rebuilt from the Pochhammer coefficients and nothing else; **and the test discriminates** — a hypothetical **odd** power of $\{}mu$ *does* produce a pole ($-1/4$ for $\{}mu$, $3/16$ for $\{}mu^{3}$), so the zero is a property of the summand's parity and **not of the instrument**; ⇒ **the representative is not fixed by this route and does not need to be**: a zero coefficient is zero for every dimension-six scalar, so it is representative-independent *completely* and ***the ledger entry is well defined without the choice***; **and the cost is neither of the order's two branches** — not two dimensionless constants in this sector and not one at two dimensions, but ***one constant at ONE dimension, exactly as before***, the free tower's at dimension four; **and that CORRECTS 'r7066', this line's own last revision**: its *"the subtractions a spectral regularisation spends are its summand's non-negative powers"* is the **cutoff's** count, traded for the analytic continuation rather than paid ⇒ ***the count of three stands as a fact about the summand and the cost claim does not***, and the inference that falls is the one from subtractions to entries — **the dimension it was drawn at is still right and the entry there is identified and EMPTY**** (node 66's 'r7067' order, **one number**. **Q1** *"the coefficient of the dimension-six subtraction, and with it the representative ... **the coefficient is what turns the count into a ledger entry**."* Two things ordered stated with it: *"**whether the representative is fixed at all by this route** — if the coefficient is representative-independent, say so, because then the ledger entry is well defined without the choice"*; and *"**what the entry costs in the ledger's own terms** ... state whether the ledger now spends **two dimensionless constants in this sector or one constant at two dimensions**, which are different claims."* *"And if the coefficient is not computable from the construction's own data, **that is the terminus and it is now the smallest it has ever been** — one number at a named dimension."*). *Computes:* the residue written as a function of the argument from the binomial expansion's Pochhammer coefficient, not quoted; $W$ checked **regular at six integer arguments** and its residues checked **non-zero at three half-integers**, so the poles are real and not formal; the free tower's $15/4$ **rebuilt** from that formula at its own physical point — the calibration, run before the unknown; the four interacting arguments checked to be integers and each residue checked zero **separately**; **a control at

two hypothetical odd powers of $\{}mu$ confirming the pole appears**, so the instrument discriminates and returns the affirmative when the affirmative is true; representative-independence checked against five symbolic scalars; and ‘r7066‘’s non-negative-power count re-read off the same decomposition to separate what stands from what falls. **18 checks, all pass, exact throughout, no floats at all** *Bound:* **SCOPE IN THE SENTENCE WITH THE RESULT.** The coefficient is the residue of the interacting sum’s own spectral function at **its own** physical point $s=0$, on the tower’s spectrum $\{}mu^{2}=m^{2}-1$ with degeneracy $2(m^{2}-4)$ from $m=3$ up, in ‘r7038‘’s passage. **NO CLAIM THAT THE INTERACTING SUM NEEDS NO RENORMALISATION AT ALL** — only that it spends **no free constant at dimension six**, which is what a pole at the physical point would have cost. No choice of representative is made or defended, and none is needed. No claim about the ledger’s entries outside this sector. Whether the regularisation commutes with the evolution is **not** established here, exactly as it is not for the free tower. The free tower’s $15/4$ is rebuilt **only as the calibration** and is used, not reopened; so are ‘r7064‘’s regulator and continuation, ‘r7060‘’s summand and ‘r7066‘’s dimension *Run:* `python3 receipts/P10_canonical_time/P10_the_coefficient_is_zero_because_the_weights_parity_misses_every_pol`

**P10R71** `P10_the_parity_is_structural_in_shape_but_order_by_order_in_conclusion_and_the_cubic_half_of_sec` [OK]

*‘PO-23‘; **THE PARITY IS STRUCTURAL IN ITS SHAPE BUT ORDER-BY-ORDER IN ITS CONCLUSION — AND CHECKING ITS OWN PREMISE NARROWS ‘r7068‘.** **(A)** A contribution from $N$ vertices of total degree $P$ carries $\{}sigma^{P/2}$ and $N-1$ energy denominators, so its frequency power is $-P/2-(N-1)$; with the order’s own bookkeeping $j=\{}sum(p_i-2)=P-2N$ that is ***$1-j/2-2N$, whose $N$-dependence is exactly $-2N$*** ⇒ **even, so it cannot change the parity: the parity is a function of the ORDER ALONE**, checked on four different assemblies of one order (one sextic, two quartics, cubic+quintic, four cubics) returning the same parity each way; **and the rule reproduces the known case rather than being fitted to it** — at $j=0$, the free tower with no vertices, the power is $+1$, ODD, which is the half-integer argument giving the banked $15/4$; **(B)** but the parity ***ALTERNATES*** with $j/2$ — odd at $j=0$, even at $j=2$, ***ODD AGAIN AT $j=4$***, even at $6$, odd at $8$ ⇒ **so the corpus carries ”no free constant at SECOND order” and not ”no free constant”**, the distinction ‘r7069‘ asked for, resolved against the general statement, **and the terminus is NOT taken since the parity follows from the vertex counting and not from any higher vertex’s value**; **(C)** the order rule inverts to $2k=j+4$ with **one solution per order**, so ***fourth order is operator dimension EIGHT***; **(D) AND THE PREMISE — that the vertex coefficients are even in the frequency — IS VERIFIED FOR THE QUARTIC AND NOT FOR THE CUBIC, WHICH SITS AT THE SAME ORDER.** ‘r7060‘’s summand is a function of $m^{2}$ alone, hence even; but ‘r7056‘’s recoupling sum grows at degree **seven** in $m$ (log-slopes $7.137$, $7.086$, $7.059$) and **an even function of $\{}mu$ has only EVEN degree in $m$**, while a scan of $\{}mu^{p}m^{q}\{}mathcal{G}$ against polynomials in $\{}mu^{2}$ over $p,q\{}in[-4,4]$ and degrees $\{}le3$, fitted on four levels and **tested on two held out**, closes for **no form at all** ⇒ ***‘r7068‘’s zero is established for the QUARTIC’s contribution and not for the total second-order energy***; if the cubic’s summand is odd it carries a pole at $j=2$ and **the ledger pays at dimension six after all**. This **narrows** ‘r7068‘ rather than reversing it — its machinery, its $15/4$ calibration and its control all stand, and its conclusion stands *for the summand it regularised*; **what was over-claimed is the step from ”the quartic’s summand has no pole” to ”the interacting sum spends nothing at dimension six.”** (node 66’s ‘r7069‘ order. **Q1** *”is there another dimension at which this sum meets a pole? ... whether the parity argument is **exhaustive** or whether it is a statement about this summand at this order. **Two sub-questions and they are not the same**: does the interacting summand stay even in the frequency at **higher orders of the coupling**, where new structures enter; and does the **order rule** admit any other dimension once the order is higher than second?”* *”If the parity is a structural fact about this construction’s vertices rather than an accident of the quartic, say so and the row closes with a general statement. **If it is order-by-order, say that too** — the corpus would then carry ’no free constant at second order’ rather than ’no free constant’, and those are different claims the ledger reads differently.”* *”And if the question is not posable ... that is the row’s terminus.”*). *Computes:* the frequency power derived symbolically and the $N$-dependence **solved** to $-2N$ rather than read off; the same order assembled **four** ways and each parity computed independently; the free tower recovered at $j=0$ as the calibration, its power $+1$ matching the pole that gives $15/4$; the parity table computed at five orders; the order rule **inverted** and its solution count checked

to be one at each order; ‘r7060‘’s summand substituted to $y=m^{2}$ and checked **free of $m$**, so its evenness is verified and not assumed; the cubic’s log-slopes recomputed from ‘r7056‘’s six banked sums and checked to lie strictly between 7 and 7.2; **and a 324-form scan ($p,q\{}in[-4,4]$, degrees $\{}le3$) fitted on four levels and tested on two held out, returning zero hits** — with the vacuity guard stated, that six values cannot fix eight coefficients, **so no four-point fit could have been trusted even had one closed**. **18 checks, all pass, exact throughout** *Bound:* **SCOPE IN THE SENTENCE WITH THE RESULT.** The counting is of ground-state perturbative contributions in ‘r7038‘’s passage on the tower’s own spectrum, with frequency $\{}mu/a$ and variance $\{}hbar/(2M\{}omega)$; the parity statement is about the summand’s power of $\{}mu$, which is what decides whether a half-integer argument is reached. **NO CLAIM THAT THE CUBIC’S SUMMAND IS ODD** — only that its evenness is **unverified** and the banked data does not settle it, which is why (D) is a narrowing and not a reversal. **No value for any higher-order coefficient, no construction of the fourth-order summand, and no claim about dimension eight’s rank or representative.** The parity table is **conditional on the same evenness premise at each order**, stated there. Used as filed: ‘r7056‘’s sums and growth, ‘r7060‘’s summand, ‘r7064‘’s regulator, ‘r7068‘’s pole criterion and parity rule, and ‘sec:lock‘’s order rule *Run:* `python3 receipts/P10_canonical_time/P10_the_parity_is_structural_in_shape_but_order_by_order_in_conclusion_`

**P10R72** `P10_the_cubic_summand_is_odd_over_m_cubed_so_it_reaches_the_pole_and_the_ledger_pays_at_dimensio`
[OK]
*‘PO-23‘; **THE CUBIC SUMMAND IS ODD — AN EVEN POLYNOMIAL OVER $m^{3}$ — SO IT REACHES THE POLE AND THE LEDGER PAYS AT OPERATOR DIMENSION SIX.** **(A)** ‘r7056‘’s recoupling sum is given a **closed form**: its eight channels collapse to two orbits, an EXTREME pair and a MIXED sextet each internally degenerate, and ***$A\{},m^{3}=9(m^{2}-1)(9m^{2}-25)(9m^{2}-1)^{3}/65536$*** for the pair with ***$B\{},m^{3}=(m^{2}-25)(m^{2}-9)(m^{2}-1)^{2}(9m^{2}-1)/65536$*** for the sextet, so ***$\{}mathcal{G}=2A+6B=N(m^{2})/m^{3}$ with $N$ EVEN of degree TEN*** ⇒ **$\{}mathcal{G}$ is ODD in $m$**, and ‘r7056‘’s measured *degree seven* is now exact rather than a log-slope: ten less the denominator’s three; **(B)** **and the two-channel levels are not a second regime** — the channel count is TWO at $m=3,5$ and EIGHT above, but $(m^{2}-9)(m^{2}-25)$ **vanishes at exactly those two levels**, so one form carries all six banked values with no case split and ***the domain boundary is removable by the answer itself***; **(C)** the assembly’s own factor $\{}sigma^{3}/\{}omega$ is $\{}mu^{-4}$, an **integer** power of $x=\{}mu^{2}$, so the parity rides entirely on $\{}mathcal{G}$, and the summand $S=N(m^{2})/(m^{3}(m^{2}-1)^{2})$ decomposes into powers of $m$ that are ***all ODD — $m^{-3},m^{-1},m^{1},m^{3}$*** ⇒ the continued sum meets $\{}zeta$ at **odd integer** arguments, one of which is $\{}zeta$’s only pole, and ***the coefficient of $m^{-1}$ is $-15405/8192$, which is NOT zero*** ⇒ **a logarithm, so a counterterm is required**, with residue $-15405/32768$ over the cubic’s own odd levels and $-15405/16384$ over all of them; **(D)** **and this is the OPPOSITE verdict to ‘r7068‘’s for the other half of the same order, by the same selection rule read in the other parity** — ‘r7060‘’s quartic summand has powers of $m$ equal to $0,2,4$, all EVEN, so argument one is unreachable and its residue is zero, **recovered here as a CONTROL rather than cited** ⇒ ***second order carries a pole after all, contributed by the cubic and not the quartic, and ”the entry at dimension six is empty” is FALSE for the total***; **(E)** and the even levels vanish **for a reason exhibited rather than cited**: the channels are not empty there (eight at $m=8,10$), but the right labels are half-integers and ***three half-integers cannot sum to zero***, so no admissible label triple exists — ‘r7044‘’s selection rule with its mechanism; **and it CORRECTS ‘r7070‘, this line’s own last revision, in its INSTRUMENT and not its verdict**: that scan capped the degree at **three** and the form that closes is of degree **five**, so it could not have been found, and its vacuity guard understated the obstruction — six coefficients are needed and ‘r7056‘ had exactly six levels, so the predicate could only cap below the answer and find nothing or saturate it and test nothing ⇒ ***breaking it required NEW LEVELS***, which is why the order’s numerical route was the one that settled it; ‘r7070‘’s *”unverified”* **stands and is now discharged**  (node 66’s ‘r7071‘ order, **one parity**. **Q1** *”THE CUBIC SUMMAND’S PARITY IN THE FREQUENCY, AT SECOND ORDER. Take whichever of the two you judge cheaper — ‘r7056‘’s own recoupling machinery run at enough further odd levels to fix the degree-seven polynomial, or the sum’s parity read off the recoupling algebra directly.”* The order stated the outcome it was **not** hoping for in advance, *”if the summand is odd, the ledger pays at dimension six and **that is a result**”*, and named a third possibility — *”a sum with both parities in it”* — as the most interest-

ing. *"Fourth order and operator dimension eight are **located** by your counting and **not opened**."* And the route was chosen **from measurement** as the order asked: a level costs 8 s at $m=7$ and 46 s at $m=19$ in this machinery, so the numerical route is cheap and what it buys is a **closed form** rather than a bound). *Computes:* 'r7056''s vertex machinery **loaded from its own source** rather than restated, its seventeen loaded gates checked to pass, and the assembly **calibrated** by reproducing 'r7056''s six banked recoupling sums exactly before any new level is used; nine levels computed from $m=3$ to $m=19$ with the orbit degeneracies **asserted** rather than assumed; both closed forms **fitted on six levels and TESTED on three held out**, two of them in the other channel regime, with the non-vacuity stated (six coefficients on six levels, so the three predictions had nothing left to fit them); **a discriminating control at the ODD ansatz of the same coefficient count, which FAILS on the held-out levels**, so the even fit closing is a fact about the sum and not about six points always being fittable; $N$ checked even and of degree ten and $\{}mathcal{G}$ checked odd; the sextet's zeros checked to sit at exactly $m=3,5$; the even-level label triples **counted** and the affirmative control run at odd levels so the count is discriminating; the summand decomposed by 'apart' and its exponents checked all odd; the residue stated **with the summation set it is over**; the quartic's powers recomputed so 'r7068' is recovered as a control; the $x$-regulator route taken as a **second derivation** and its exponents checked half-integer; and 'r7070''s **own predicate re-run** at its own cap and at the true degree. **26 checks, all pass, exact throughout, no floats at all** *Bound:* **SCOPE IN THE SENTENCE WITH THE RESULT.** Every degree, residue and ratio is in 'r7056''s **LEVEL-SUM convention**, and the summand is the **same-level** cubic's at second order in the cubic coupling, on the tower's spectrum $\{}mu^{2}=m^{2}-1$ in 'r7038''s passage. **THE CLOSED FORMS ARE ESTABLISHED ON NINE LEVELS — six fitted, three held out — AND ARE NOT PROVED FROM THE RECOUPLING ALGEBRA**, which is the second route the order named and which is not taken here; so the parity conclusion rests on that form. **The pole's EXISTENCE needs only $\{}mathcal{G}$'s parity and that $\{}sigma^{3}/\{}omega$ carry an integer power of $x$; the residue's VALUE rests further on 'r7060''s assembly as filed** — a different power there moves the number and not the non-vanishing. **No value for the counterterm's coefficient in the ledger's units, no choice of representative among the dimension-six scalars, and no claim about dimension eight or fourth order**, which this revision does not touch. Used as filed: 'r7056''s machinery and banked sums, 'r7060''s summand and assembly, 'r7064''s regulator, 'r7068''s pole criterion, 'r7044''s selection rule and 'sec:lock''s order rule *Run:* `python3 receipts/P10_canonical_time/P10_the_cubic_summand_is_odd_over_m_cubed_so_it_reaches_the_pole_and_the`

**P10R73** `P10_the_assembly_has_two_invariants_and_the_cubic_is_traceless_but_the_invariant_cannot_be_name`
[OK]
*'PO-71'; **THE ASSEMBLY HAS TWO INVARIANTS AND THE CUBIC IS TRACELESS, SO THE PARITY STANDS AND THE COEFFICIENT'S STRUCTURE DOES NOT — BUT THE INVARIANT CANNOT BE NAMED ON THIS BACKGROUND, WHICH IS THE ROW'S OWN STATED TERMINUS.** **(A)** 'r7060''s assembly **re-derived** rather than inherited: for a totally symmetric cubic on degenerate modes the exact second-order shift is ***$\{}delta E = -(\{}sigma^{3}/\{}hbar\{}omega)\{},[\{},2\{}sum T^{2} + 9\{}sum t^{2}\{},]$ with $t_{C}=\{}sum_{A}T_{A* — calibrated on one mode against the textbook $-11g^{2}\{}sigma^{3}/\{}hbar\{}omega$, verified on a **generic** two-mode symmetric $T$ with four free coefficients, and separated by a **traceless** three-mode control that keeps only the first term ⇒ ***there are TWO invariants, at weights 2 and 9, where 'r7010' filed a single $8g^{2}$***, the three-quantum channel riding the full norm at $3\{}hbar\{}omega$ and the one-quantum channel the TRACE at $\{}hbar\{}omega$; **(B)** **and the FORM is what 'r7074' needed, so its parity stands untouched** — both channels carry the same $\{}sigma^{3}/\{}omega = \{}mu^{-4}$, an **integer** power of $x=\{}mu^{2}$, so the parity rides entirely on the recoupling sum and ***the cubic summand is still odd, still reaches the pole, and the ledger still pays at dimension six***; **(C)** **the trace channel is absent at EVERY level by a singlet count**: the $3j$ pair contraction vanishes unless the **free** label is zero (checked at three spins with the affirmative control non-zero exactly there), and the same identity on the left indices makes a trace need a **full** singlet — right label zero AND left spin zero — which ***no channel at any level is***, since above $m=3$ no label is zero at all and at $m=3$ the two never coincide ⇒ $\{}sum t^{2}=0$ identically and the assembly is $2\{}sum T^{2}\{}sigma^{3}/\{}hbar\{}omega$; **and 'r7060''s criterion is untouched for a computed reason** — its 8 sits in both the cubic and the quartic and **cancels in the ratio**, so the number that was wrong was never load-bearing there and the crossing and 'r7056''s five*

*negative levels all stand; **(D)** **the INVARIANT CANNOT BE NAMED, and rank one is why rather than how**: ‘r7062’’s five dimension-six scalars are pointwise independent on a general scale factor (**rank 5 of 5**) and collapse on the one-parameter family, where the two derivative scalars vanish identically and the other three are pure numbers times $\{}alpha^{-6}$ (**rank 1**) ⇒ ***matching one number there is ONE equation on THREE non-vanishing directions, whose solution set carries TWO free coefficients, and EACH of $R^{3}$, $R\{},\{}mathrm{Ric}^{2}$, $\{}mathrm{Ric}^{3}$ ALONE reproduces the residue after one rescaling*** — exhibited, not counted ⇒ **the family cannot distinguish the representatives it is rank one across, so rank one is the degeneracy that BLOCKS the naming and not the lever that forces it**, and on the class the ledger uses ‘sec:lock‘ says in terms that *”at dimension six there is no such identity to descend”*; **and the defect is THIS SEAT’S, in the paragraph that opened the row** — ‘r7077‘ asked for a reading and this seat answered that the rank one meant the five collapse *”on the admitted class, so the representative is forced up to normalisation”*, and ***both halves are wrong in the same way: a rank read off one domain and applied on another, and a degeneracy read as a determination*** — the row’s own **domain of an EQUIVALENCE**, committed by the seat enforcing it* (node 66’s ‘r7079‘ order, opening ‘PO-71‘ on this seat’s own reading. **Q1** *”**(a)** re-derive ‘r7060‘’s $\{}sigma^{3}/\{}omega$ rather than inheriting it as filed, and say plainly if it moves — **if it moves, the residue moves with it and that is the finding**, not a setback. **(b)** then the invariant: named, with its normalisation matched to ‘r7074‘’s residue on the re-derived assembly.”* The terminus the order wrote down: *”If the one-parameter family turns out not to distinguish the representatives it is rank one across, **that is the row’s stated terminus** — it would make the naming need a datum outside this background, which is a statement about the background’s reach and not about the coefficient. Say so in those terms if that is where it lands.”* *”And what is **not** in this row: the renormalised value ... and fourth order, which stays ordinary order-by-order bookkeeping.”*). *Computes:* the second-order shift computed by **explicit oscillator algebra in a truncated Fock space**, not by a quoted Wick rule — the ladder operators built from the occupation basis and the matrix elements read off, at one mode, at two modes with a generic symmetric $T$, and at three with a traceless one; the one-mode case checked against the textbook $-11$ **before** the unknown, so the instrument is calibrated; the two channels’ weights 2 and 9 **derived** and the traceless control confirming which invariant each carries; the $\{}sigma^{3}/\{}omega$ factor checked to be exactly $x^{-2}$; the $3j$ pair contraction **computed** at three spins and four free labels with the affirmative control returning non-zero exactly at a zero free label; every channel at six levels enumerated for a full-singlet trace slot and none found, with the zero-right-label slots printed beside their non-zero left spins; ‘r7060‘’s 8 checked to **cancel** in the ratio symbolically; ‘r7062‘’s rank machinery and its five scalars **loaded from its own source** with nothing in the loaded prefix failing; both of its ranks recovered (5 pointwise, 1 on the family); **and the matching the order asked for actually performed and found degenerate** — its free-coefficient count solved, and each of the three non-vanishing scalars shown ALONE sufficient; with the affirmative half kept, that the two derivative scalars are excluded as sole carriers. **17 checks, all pass, 8 s, exact throughout, no floats at all** *Bound:* **SCOPE IN THE SENTENCE WITH THE RESULT.** (A)’s assembly is exact for a totally symmetric cubic on modes of **one common frequency** — the same-level case ‘r7074‘ used — and not the cross-level one; the traceless finding is about the **same-level** vertex’s own trace on ‘r7056‘’s channels. **Every rank is ‘r7062‘’s, over a CHOSEN five-element subset, hence a LOWER bound on the full basis** — which is the direction this negative needs, since completing the basis can only **raise** the pointwise rank and so only **worsen** the family’s degeneracy. **NO renormalised value and NO representative chosen** — the point is that this background cannot choose one; **and the residue’s absolute scale is now $2\{}sum T^{2}$ rather than ‘r7010‘’s $8g^{2}$**, which is what (a) was for. Nothing touched at fourth order or dimension eight. Used as filed: ‘r7074‘’s residue and parity, ‘r7056‘’s channels, ‘r7062‘’s ranks and ‘sec:lock‘’s dimension-six prose *Run:* `python3 receipts/P10_canonical_time/P10_the_assembly_has_two_invariants_and_the_cubic_is_traceless_but_the_`

**P10R74** `P10_the_parity_is_the_channel_lists_mirror_symmetry_so_it_is_proved_rather_than_fitted_but_the_c` [OK]

*__THE PARITY IS THE CHANNEL LIST’S MIRROR SYMMETRY, SO IT IS PROVED RATHER THAN FITTED — AND THE CLOSED FORMS ARE NOT, WHICH IS SAID PLAINLY BECAUSE THE ORDER ASKED FOR THAT.** **(A)** ***The map $m\{}to-m$ is the Casimir-preserving reflection $j\{}to-j-1$ applied to BOTH labels, and it EXCHANGES them***: $a(-m)=-(b+1)$ and $b(-m)=-(a+1)$ with $a=(m+1)/2$, $b=(m-3)/2$, the Casimir invariant both*

*ways; **(B)** ***the channel list is CLOSED under the induced $a\{}leftrightarrow b$ relabelling, and so is each ORBIT*** — the swap sends the $a$-count $k$ to $3-k$, so $\{}{0,3\{}}$ and $\{}{1,2\{}}$ are exactly its blocks and the orbits are the swap's own rather than a grouping put in by hand; **(C)** ***the recoupling coefficient is INVARIANT and the degeneracy factor is ODD*** — every $2j+1$ is $m\{}pm2$ and $(-m)+2=-(m-2)$, so $d_{X}(-m)=-d_{\{}mathrm{swap}(X)}(m)$ for all four shapes, **with a control showing that $2j+1$ itself is NOT a function of the Casimirs and is therefore exactly where the sign comes from**; ⇒ **(D)** ***the term $d\{}lvert c\{}rvert^{2}$ at $-m$ is MINUS the swapped channel's at $m$, so summing over an orbit closed under the swap gives $S(-m)=-S(m)$: EVERY ORBIT SUM IS ODD IN $m$, hence so is the recoupling sum*** — checked with the orbit coefficients left as FREE SYMBOLS, so it is the symmetry and not a property of any value, and **with no fit, no levels and no closed form used**; **and it is UNCONDITIONAL — it does not use the mirror equality**: giving the two members of an orbit different coefficients destroys the oddness and restoring the swap-relation recovers it, so what the argument needs is the orbit's CLOSURE and not 'r7056''s observed equality, which is identified as the **stronger separate fact** (the symmetry forces the pair to be each other's reflection; equality says each is its own) ⇒ **so 'r7074''s load-bearing fact — the cubic summand odd, the pole reached, the ledger paying at dimension six — no longer rests on the fit at all, and the $m^{3}$ is explained with it as the only odd factor the degeneracies can supply**; **(E) BUT THE CLOSED FORMS THEMSELVES DID NOT CLOSE, and the order's stop condition is taken rather than stretched**: the a-priori **degree bound** — which would have turned the six-level fit into an exact determination — needs the vertex contraction evaluated, which is what the bound was meant to avoid; and the **Racah recognition** on the factored shape, though the rewrite into consecutive-integer products in $a$, $b$ and $3a$ is an exact identity recorded here, yielded no derivation ⇒ ***the honest answer the order invited is given: the closed forms are not recognisable from the algebra this construction has, by this seat, after a real attempt***, and they stand as 'sec:lock' already states them; **and one self-check is reported because it is the shape that cost 'PO-71'**: the rewrite reads naturally as *"every factor is a shift of $J=3a$"*, but $3a$ is the sum of left spins **only in the extreme channel** — the mixed ones' own sums are $3a-2$ and $3a-4$ — so that reading would be a quantity taken off one domain and applied on another, **one revision after the same error terminated a row**, and the rewrite is kept as an identity and not presented as recoupling structure* (node 66's 'r7081' order, after 'PO-71''s termination. **Q1** *"THE CLOSED FORMS PROVED FROM THE RECOUPLING ALGEBRA RATHER THAN FITTED. ... The corpus's parity conclusion rests on two orbit forms established by a six-level fit tested on three held-out levels. **A proof removes that dependency**, and you have already said where it starts."* The stop condition, written into the order and accepted in writing **before** this was attempted: *"if it is not a recognition after a real attempt, say so and stop ... an honest 'the closed form is not recognisable from the algebra this construction has' is a complete answer and would be landed as one. **Nothing here asks you to keep going until it closes.**"* *"And what is not ordered: the datum that separates $R^{3}$, $R\{},\{}mathrm{Ric}^{2}$ and $\{}mathrm{Ric}^{3}$ is not a row and not an order."*). *Computes:* the reflection checked as an exact identity in both directions and the Casimir checked invariant both ways; the exchange checked NON-vacuous, the two labels differing by two at every level; the channel list enumerated at four levels and checked closed under the swap, and **each orbit checked closed separately**, with the block structure derived from the swap's action on the $a$-count rather than asserted; all four degeneracy shapes checked to satisfy $d_{X}(-m)=-d_{\{}mathrm{swap}(X)}(m)$; **a control separating what the Casimir-rationality does** — $2j+1$ is not a function of the Casimirs and flips sign where $j(j+1)$ does not; the orbit sums' oddness verified **with the coefficients as free symbols**, so no value enters; **and a discriminating control on the unconditionality** — unequal coefficients break the oddness and the swap-relation restores it, which is what shows the mirror equality is not an input; the factored rewrite verified as an exact identity; the three channels' own sums of left spins computed and checked to differ, which is the self-check; and both failed routes named with the reason each stopped. **18 checks, all pass, 0 s, exact throughout, no floats at all** *Bound:* **SCOPE IN THE SENTENCE WITH THE RESULT.** The parity proof is about the **same-level** cubic's channel list on 'r7056''s decomposition, and it uses only (a) that recoupling coefficients are rational in the Casimirs, (b) that the list and each orbit are closed under $a\{}leftrightarrow b$, and (c) the degeneracies' form — **so it holds at every level and is not a statement at the nine computed ones.** **IT PROVES THE PARITY AND NOTHING ELSE: no closed form, no degree bound, no value.** The fitted forms are used **only as a consistency check ON the proof and never as evidence FOR it**,

which is why the receipt computes the symmetry with free coefficients first. **The closed forms' own warrant is UNCHANGED** — six levels fitted, three held out — and ‘sec:lock‘ states it as a limit. Nothing touched at fourth order, dimension eight, or the renormalised value *Run:* `python3 receipts/P10_canonical_time/P10_the_parity_is_the_channel_lists_mirror_symmetry_so_it_is_proved_rat`

**P10R75** `P10_the_payoff_count_of_one_is_right_because_P1_reaches_the_existence_occurrence_distinction_fr` [OK]

*‘P1‘‘s existence/occurrence sentence, its reification clause, its maximal-extension sentence, ‘prop:causal-alignment‘‘s hypotheses, and the ‘P10‘ clause ‘r7168‘ rewrote* ( **NODE 66’s ‘r7168‘ ASKS THIS SEAT ONE THING – WAS ‘P10‘’s PAYOFF COUNT RIGHT? DOES THE ONTOLOGICAL CORRECTION DO INDEPENDENT WORK ON THE HORIZON INFERENCE THAT ‘P1‘’s CAUSAL ARGUMENT DOES NOT ALREADY DO? ⇒ NO, THE COUNT OF ONE IS RIGHT — AND THE REASON IS STRONGER THAN THE ORDER’S OWN “I COULD NOT FIND IT IN ‘P1‘”.** (1) ***‘P1‘ DOES NOT MERELY FAIL TO NEED THE CORRECTION; IT CARRIES THE SAME DISTINCTION IN ITS OWN VOICE***: its text names the *existence/occurrence distinction this paper turns on — that the horizon OCCURS without ever EXISTING on a finite exterior slice* — and says in the same sentence *the present paper reaching it from standard general relativity alone*. It also names the reification error in the register ‘P10‘ names it in, *off a reified coordinate ... caught in a verb tense*. **So the refusal ‘P10‘ supplies in the canonical sector is, in the horizon sector, something ‘P1‘ DERIVES, and a count of two was counting one move twice.** (2) ***AND THE ONE SITE WHERE THE CORRECTION COULD HAVE DONE INDEPENDENT WORK IS WHERE ‘P1‘ IS STRONGEST***: the appeal to the maximally extended geometry is not a claim about exterior time at all, and an ontological refusal would have to DENY EXISTENCE to the extension — where ‘P1‘ instead says the stacked horizon points *merely display the causal ordering of distinct manifold events that are nevertheless metrically coincident*. ** The correction there would be a DOWNGRADE, not a duplicate.** (3) ***AND THAT SENTENCE’S CONTENT IS COMPUTED HERE RATHER THAN QUOTED***: the horizon’s induced metric in ingoing Eddington–Finkelstein is DEGENERATE of rank two with the two-sphere block exactly ‘$r_h^2\{}dd\{}Omega^2$‘ and the generator direction annihilated, in a chart whose four-metric is non-degenerate at the root — so two events on one generator are separated by zero in every invariant the induced metric has; and the approach is a LIMIT at the surface gravity, ‘$\{}dot\{}delta\{}simeq-2\{}kappa\{}delta$‘ so ‘$\{}delta\{}propto e^{-2\{}kappa t}$‘ with the local blueshift ‘$\{}propto e^{\{}kappa t}$‘, both derived from ‘$f$‘ with ‘$\{}kappa$‘ obtained and not supplied. ⇒ **WHAT SURVIVES IS THE DIAGNOSIS-IDENTITY, AND ‘P1‘ ENDORSES IT RATHER THAN MERELY PERMITTING IT**: ‘P1‘ says the companion sets *the same distinction at the root of the cosmological ontology*, so ‘P10‘’s shape-claim is backed by the cited paper instead of asserted over it — firmer footing than the count ever gave it.). *Computes:* **18 checks, all pass, 0.4 s.** **(A)** seven READ-FIRST gates on ‘P1‘’s CURRENT source, each clause located exactly once and the ordering of two of them checked: the existence/occurrence sentence, the general-relativity-alone clause following it, the reification clause, the maximal-extension sentence, the realised-worldtube sentence that places the extension’s curvature singularity the same way, ‘prop:causal-alignment‘’s own quantifier over ANY smooth Cauchy temporal function together with its two named hypotheses, and the single occurrence of the phrase the order warns against — measured, and shown to sit inside the sentence that DISCLAIMS the dependence rather than in any hypothesis. **(B)** the two premises computed: the root and surface gravity derived; the chart’s four-metric checked non-degenerate at the root first, so the degeneracy belongs to the hypersurface and not the coordinates; the induced metric’s determinant, rank, annihilated direction and exact angular block; the linearised infall rate against ‘$-2\{}kappa\{}delta$‘ with the exponential solved rather than asserted; and the blueshift’s ‘$\{}delta^{-1/2}$‘ coefficient and divergence. **(C)** the adjudication gated on the conjunction of ‘P1‘’s three sentences, the downgrade gated on the computation, the diagnosis-endorsement located, the ‘P10‘ clause read with the receipt’s reading ENUMERATED over the states that clause may take, and the forbidden phrase confirmed gone from ‘P10‘. *Bound:* **IT DOES NOT ADJUDICATE WHETHER ‘P1‘’s THREE ROUTES ARE SOUND.** *It reads what their hypotheses ARE and finds no foliation premise among them — the order’s own finding re-established on the current source, not a review of the proofs.* *It computes nothing about Hawking radiation, the information paradox or cosmic censorship, and takes no position on ‘PO-85‘, which is ‘70‘’s.* *It does not touch ‘P1‘ and proposes no paper edit: the correction is already in print, and the only thing offered is an OPTIONAL clause for ‘P10‘ citing ‘P1‘’s own sentence,

which is 66's to take or leave.* *‘$2M=1$‘ is a gauge for the two scalings and no result is pinned to a numeric value; the induced-metric result is symbolic in ‘$M$‘.* *It does not claim the two errors are the same error — only that the DIAGNOSIS is shared, which is what ‘P10‘ keeps.* *Run:* `python3 receipts/P10_canonical_time/P10_the_payoff_count_of_one_is_right_because_P1_reaches_the_existence_o`

**P10R76** `P10_the_clause_66_wrote_on_this_seats_offer_says_exactly_what_its_receipt_supports_and_no_more_s` [OK]

*‘P10‘’s theory-choice close — the three-sentence clause ‘r7170‘ wrote on this seat's offer, its surviving shared-diagnosis sentence, and its scope clause — against ‘P1‘’s own two sentences* ( **‘r7170‘ ASKS ONE THING: READ THE CLAUSE 66 WROTE INTO ‘P10‘ ON THIS SEAT'S ‘r7178‘ OFFER AND SEND IT BACK IF THE VOICE IS WRONG. ⇒ IT IS NOT WRONG — AND SAYING SO IS NOT THE ANSWER.** ***THE ANSWER IS THAT THE CLAUSE SAYS EXACTLY WHAT ITS RECEIPT SUPPORTS AND NO MORE, WHICH IS CHECKABLE AND IS CHECKED.*** **The class ‘r7168‘ opened was a citing site asserting a grade its owning paper does not prove. This runs that check by hand on the citing site ‘r7168‘ ITSELF produced — so the newest instance of the class is the first one tested rather than the first one trusted.** (1) *The attribution is the owner's own words: both halves located in ‘P1‘ verbatim and in the clause's order, so the site QUOTES where the class would have it grade.* (2) *The defender's move is reported and not conceded, **checked as GRAMMAR rather than as tone** — ‘may grant‘ and ‘hold instead‘ carry the attribution, which is the difference between stating an objection and adopting it as a premise.* (3) ***And the strongest sentence is the one this seat did not write***: where ‘r7178‘ said the refusal would be a downgrade, the clause says the stacked points *display the causal ordering of events the metric assigns no separation between, so there is nothing there for an ontology to withhold existence from*. **Sharper than the offer it was made from, and not a reach** — ‘no metric separation‘ is ‘P1‘’s own phrase for the object, and the degeneracy it reports is RECOMPUTED here rather than carried over from ‘r7178‘. (4) ***AND THE ONE OVERSTATEMENT THE SITE MAKES AVAILABLE IS CHECKED FOR AND ABSENT***: the clause could have said the correction does no work, and says instead that what the reading supplies THERE is the account of why the error was available rather than the result that removes it — **the scope on the site and not on the correction**, with the shared diagnosis surviving in the sentence before it. ⇒ **So it goes back unchanged, and the reason is a measurement rather than assent.**). *Computes:* **11 checks, all pass, 0.3 s.** **(A)** the three sentences located in ‘P10‘, each once and in the stated order, and the receipt shown to be cited AT the third sentence — the one resting on a computation — rather than at the paragraph. **(B)** claim one checked against ‘P1‘’s two verbatim halves in their own order; claim two checked on its hedging verbs; the owner's ‘no metric separation‘ phrase located; and claim three's content recomputed — the chart's four-metric verified non-degenerate at the root FIRST, then the induced metric's vanishing determinant, rank two, annihilated generator direction and exact ‘$r_h^2\{}dd\{}Omega^2$‘ block. **(C)** the scope clause located once, the shared-diagnosis sentence shown to survive and to precede the new clause, the phrase ‘r7168‘’s header note forbids confirmed still absent while the exterior-time form is present, and the verdict gated on the conjunction rather than on any one of them. *Bound:* **IT DOES NOT RE-ADJUDICATE THE PAYOFF COUNT** — *‘r7178‘ did that and ‘r7170‘ struck it; this reads only the clause that replaced it.* *It does not review ‘P10‘’s surrounding prose, which is 66's, and it touches neither ‘P1‘ nor ‘P10‘.* *It judges what the clause CLAIMS against what its receipt established — it does not re-derive ‘P1‘’s three routes and does not rule on whether the defender's move is a good objection, only on whether the clause reports it as one.* ***And it is NOT ‘PO-85‘***: *that row wants an instrument for this class and is ‘70‘’s after its ‘C1‘ work. This is one hand-run instance on one clause — what a seat can do while the instrument is somebody else's, and it is offered to that row as evidence rather than as a claim on it.* *Run:* `python3 receipts/P10_canonical_time/P10_the_clause_66_wrote_on_this_seats_offer_says_exactly_what_its_recei`

# Appendix L The Knowledge Ledgers

Each claim marked ⌞ in the text rests on a field bake or the figure–theorem bake recorded in the named ledger, which ships with this work and carries the probes, the receipts, and the three registers kept apart — what bit, what bounced, and where the boundary is. A ledger is working apparatus, not a publication, and is cited here rather than in the bibliography. This appendix is generated from the ledger index, not maintained by hand.

**L10** `FUNCTIONAL_ANALYSIS_LEDGER.md` [field bake]
The functional-analysis / unitarity field-bake ledger — the field that bounced, and the one routing fact it returned. Third of the four fields 'L-272''s re-survey left outstanding. 'OWED' 622

**L17** `QUADRIC_GEOMETRY_LEDGER.md` [field bake]
projective geometry of quadrics against the CR substrate — the CK metric identification, and the ladder gap it exposes

**L20** `VARIATIONAL_LEDGER.md` [field bake]
variational / action against CR — the field the corpus uses and never names

**L14** `INTEGRABLE_SYSTEMS_LEDGER.md` [field bake]
The integrable-systems field-bake ledger — what bit, what bounced, and where the boundary is